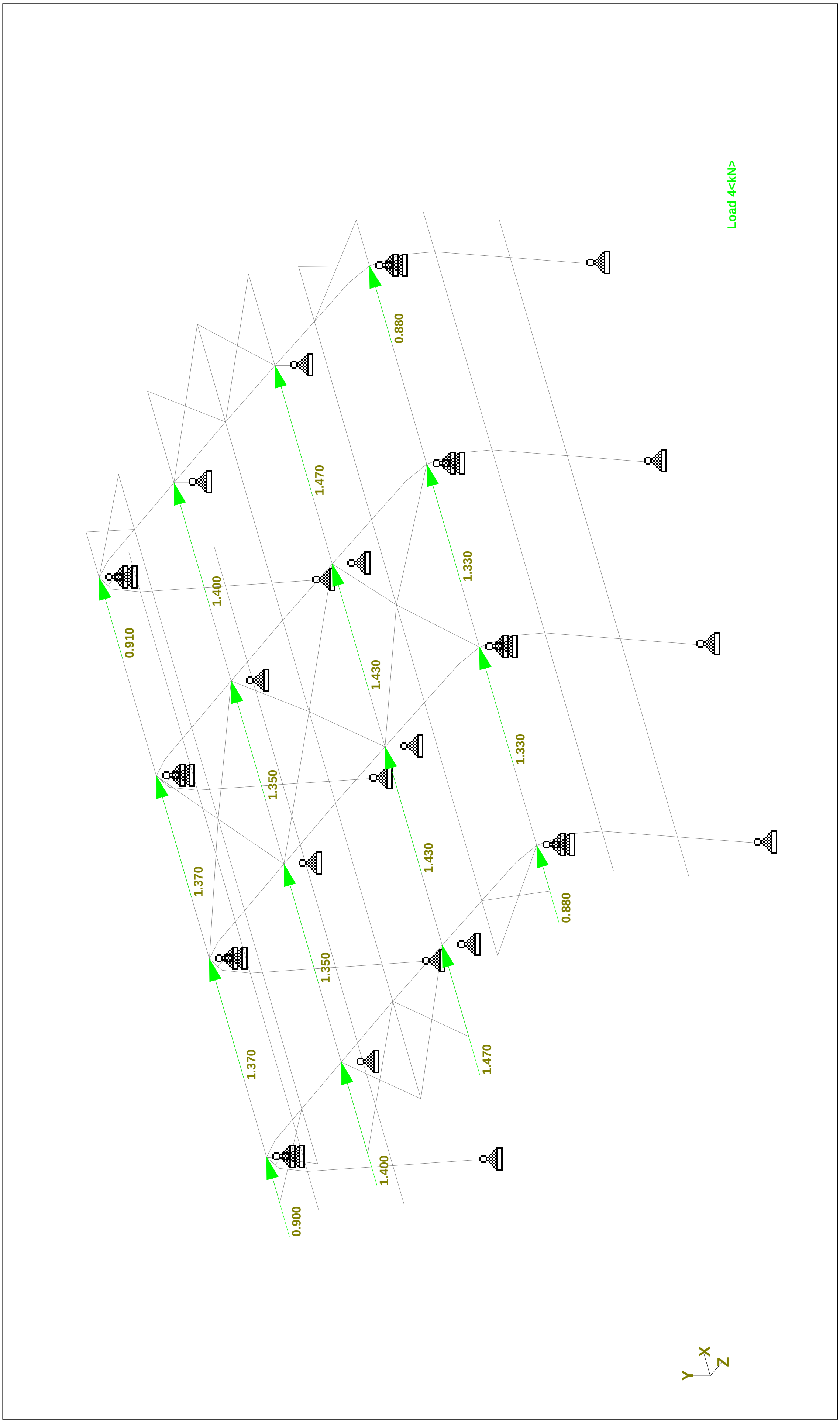
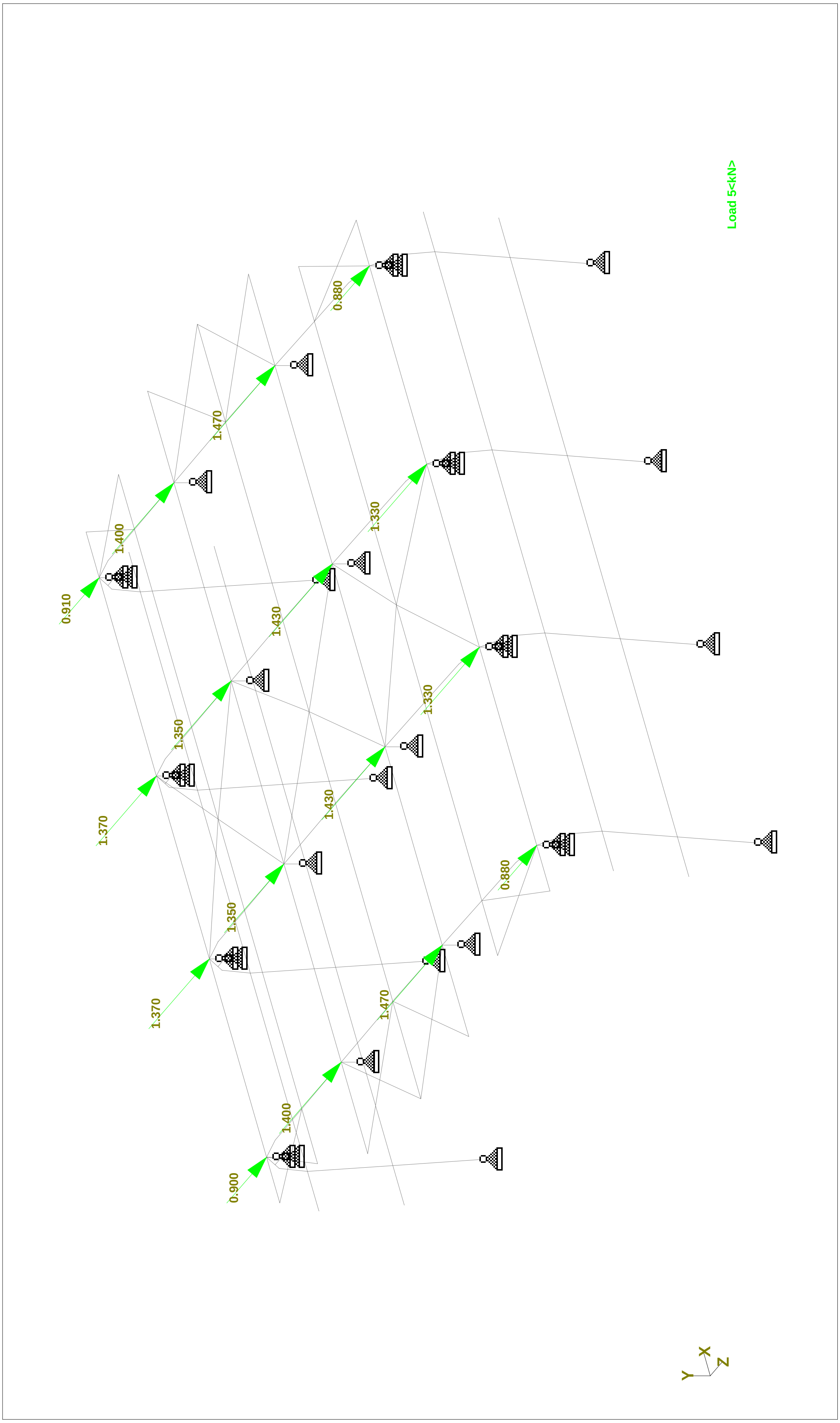
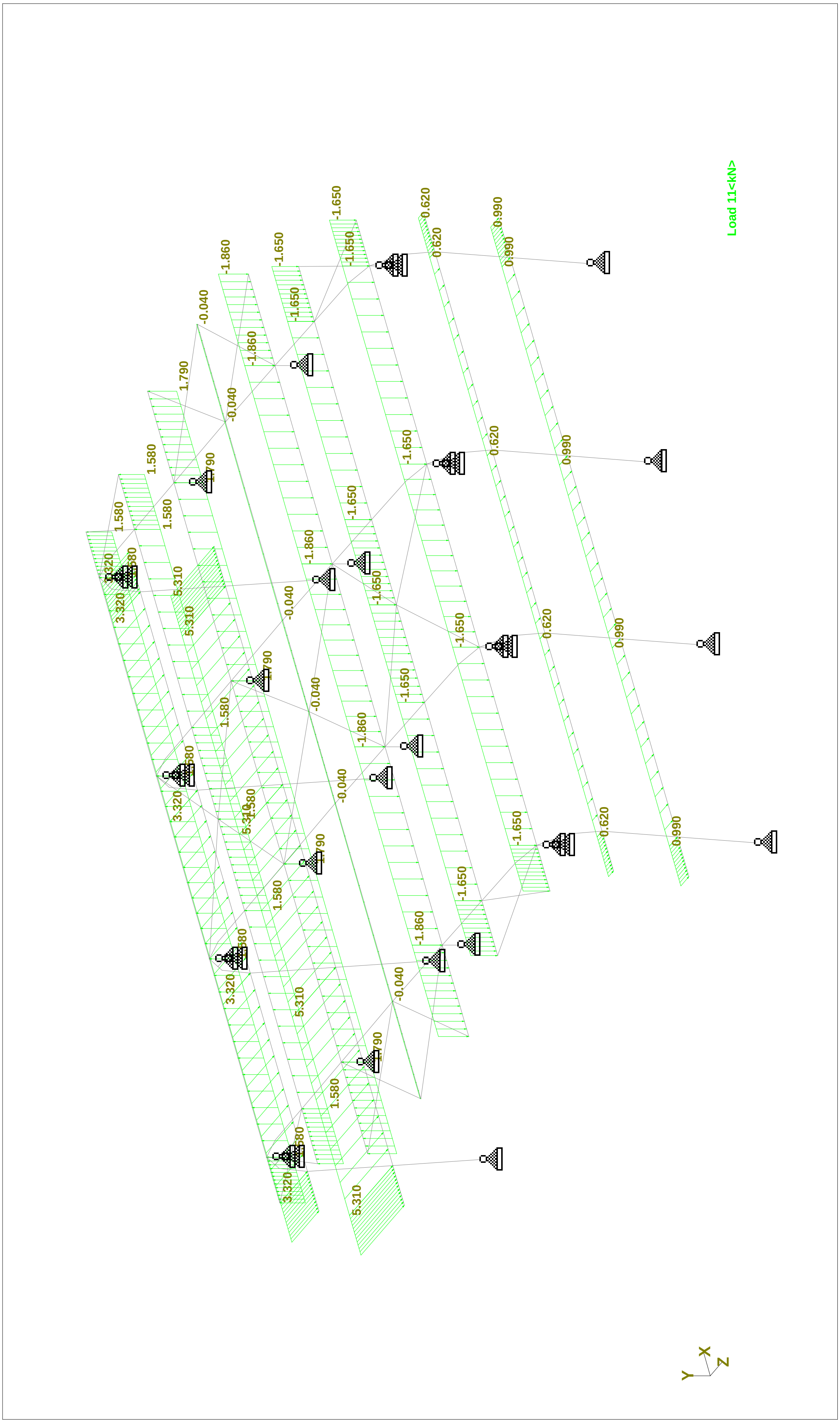


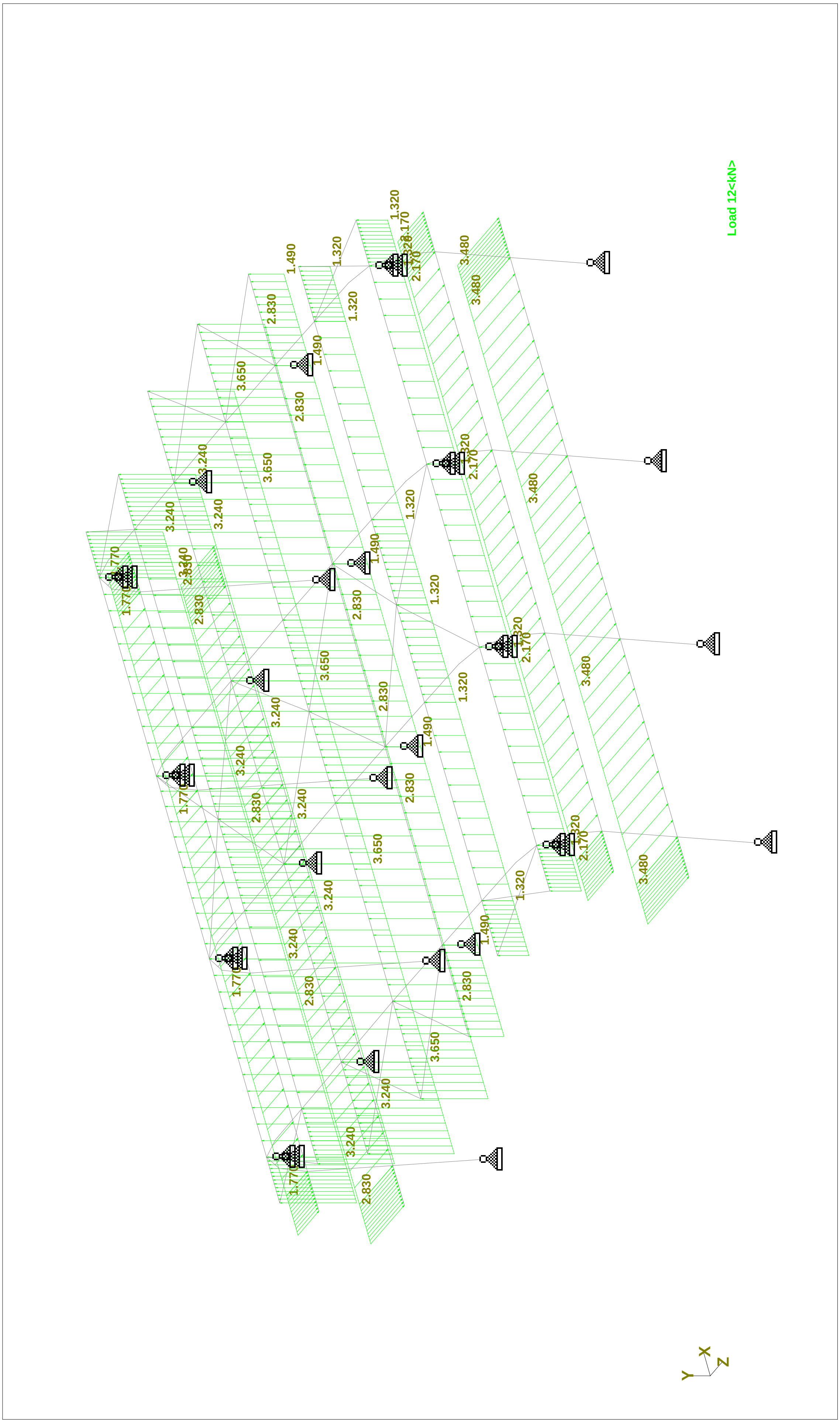

*Load 4 PH(X-DIR)*



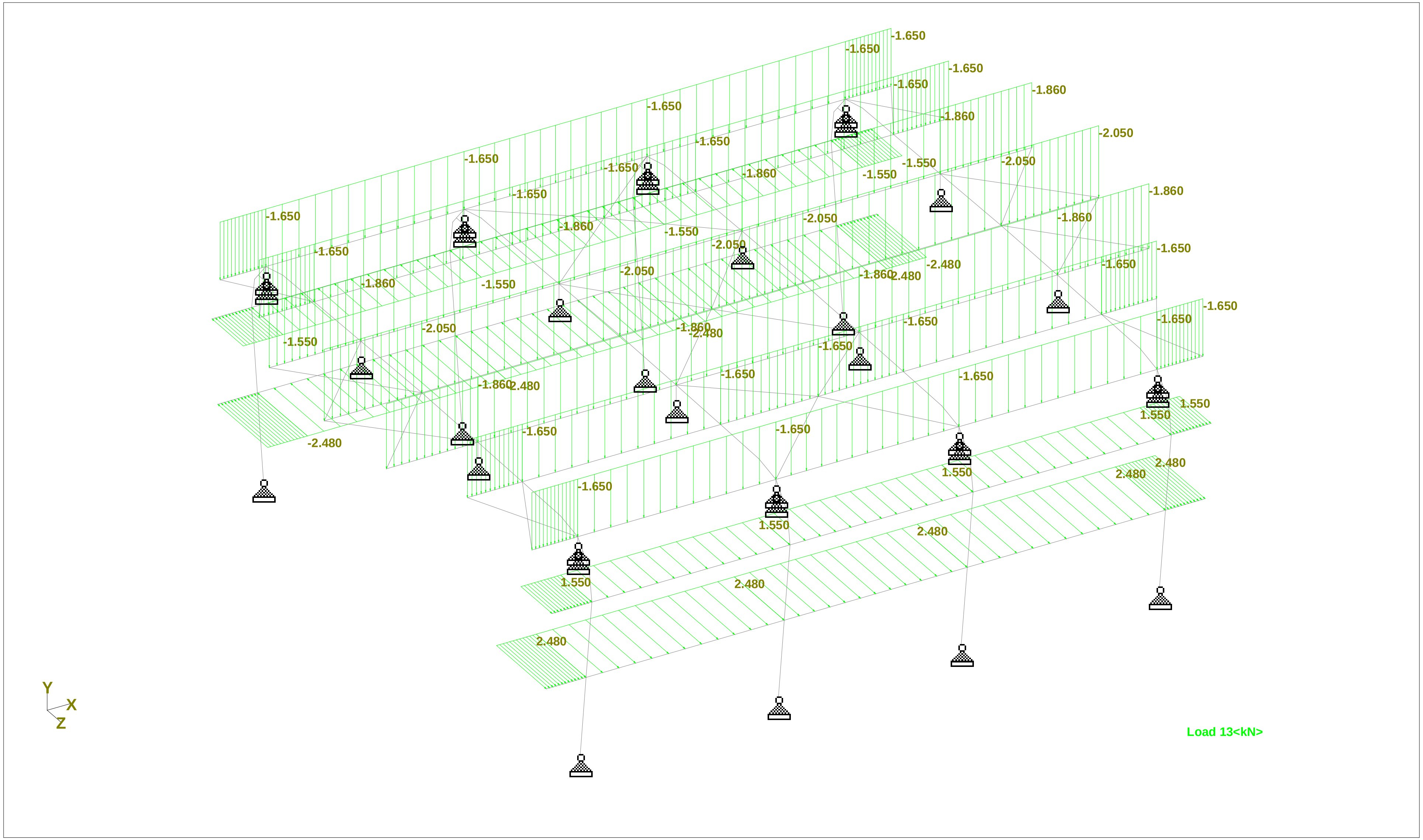

Load 5 PH ( Z - dir)



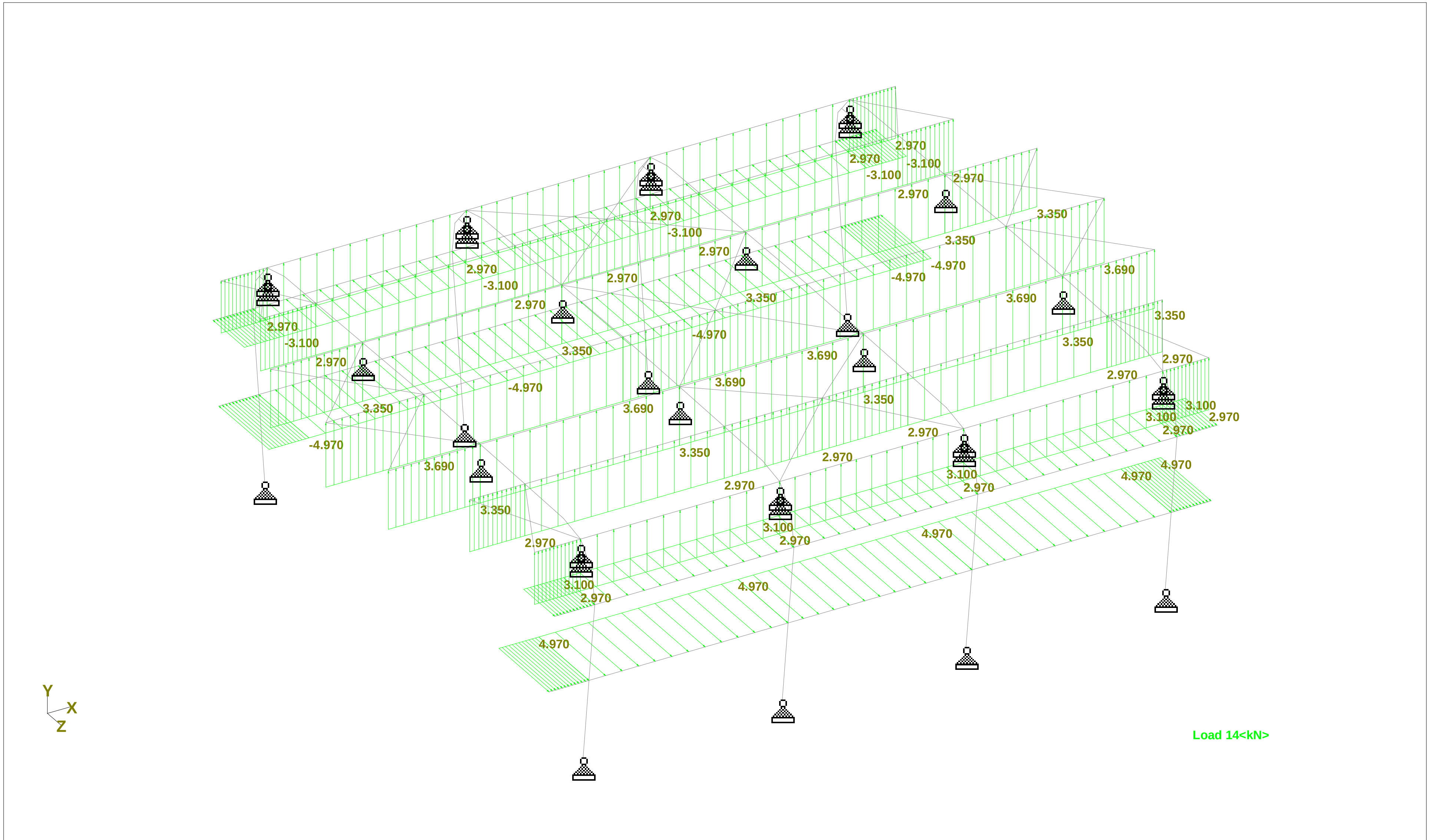

*Load 11 WL1 (0 DEG,INTERNAL SUCTION (CPI=-0.3)*



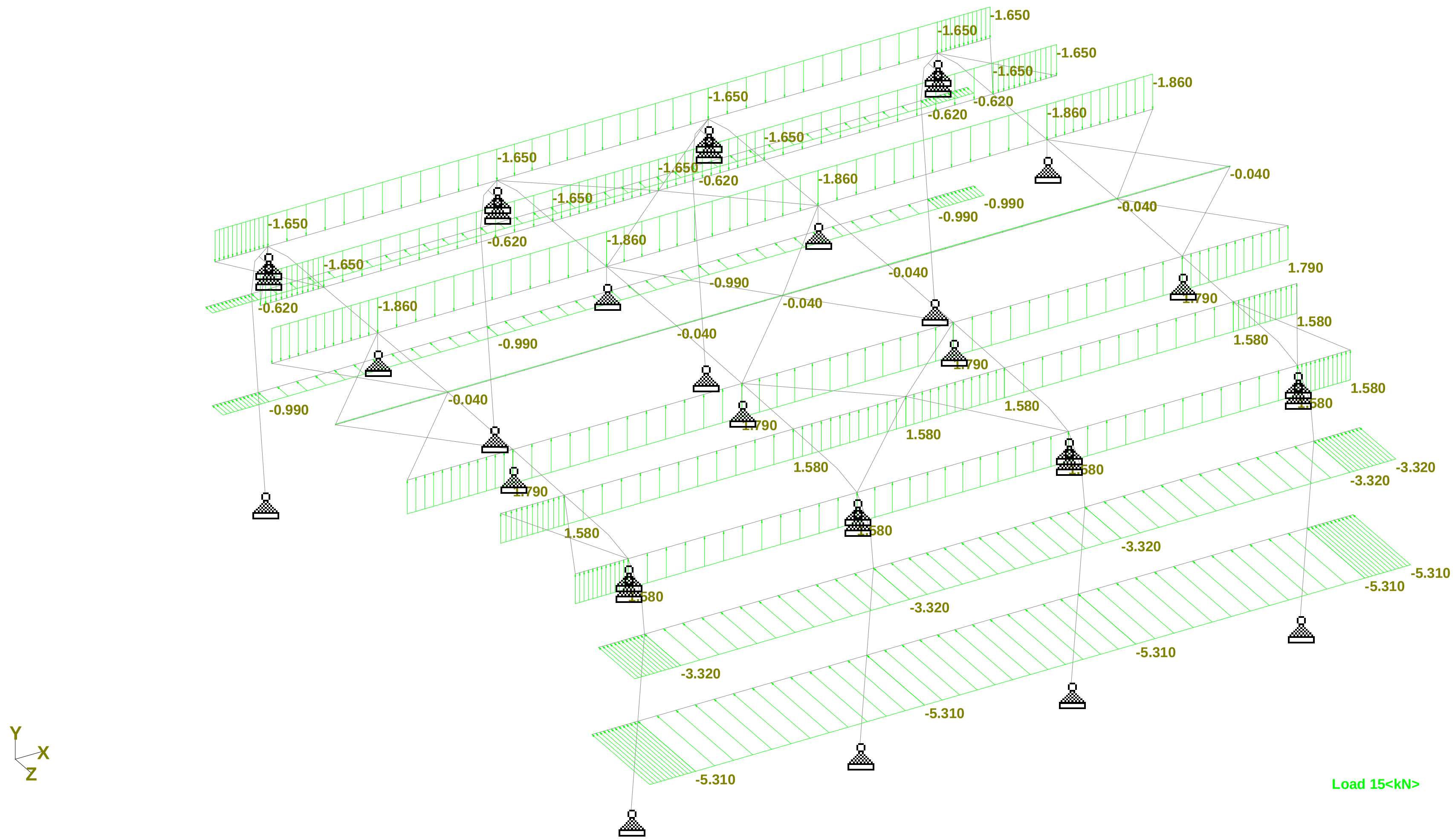

*Load 12 WL2 (0 DEG ,INTERNAL PRESSURE CPI=+0.2)*



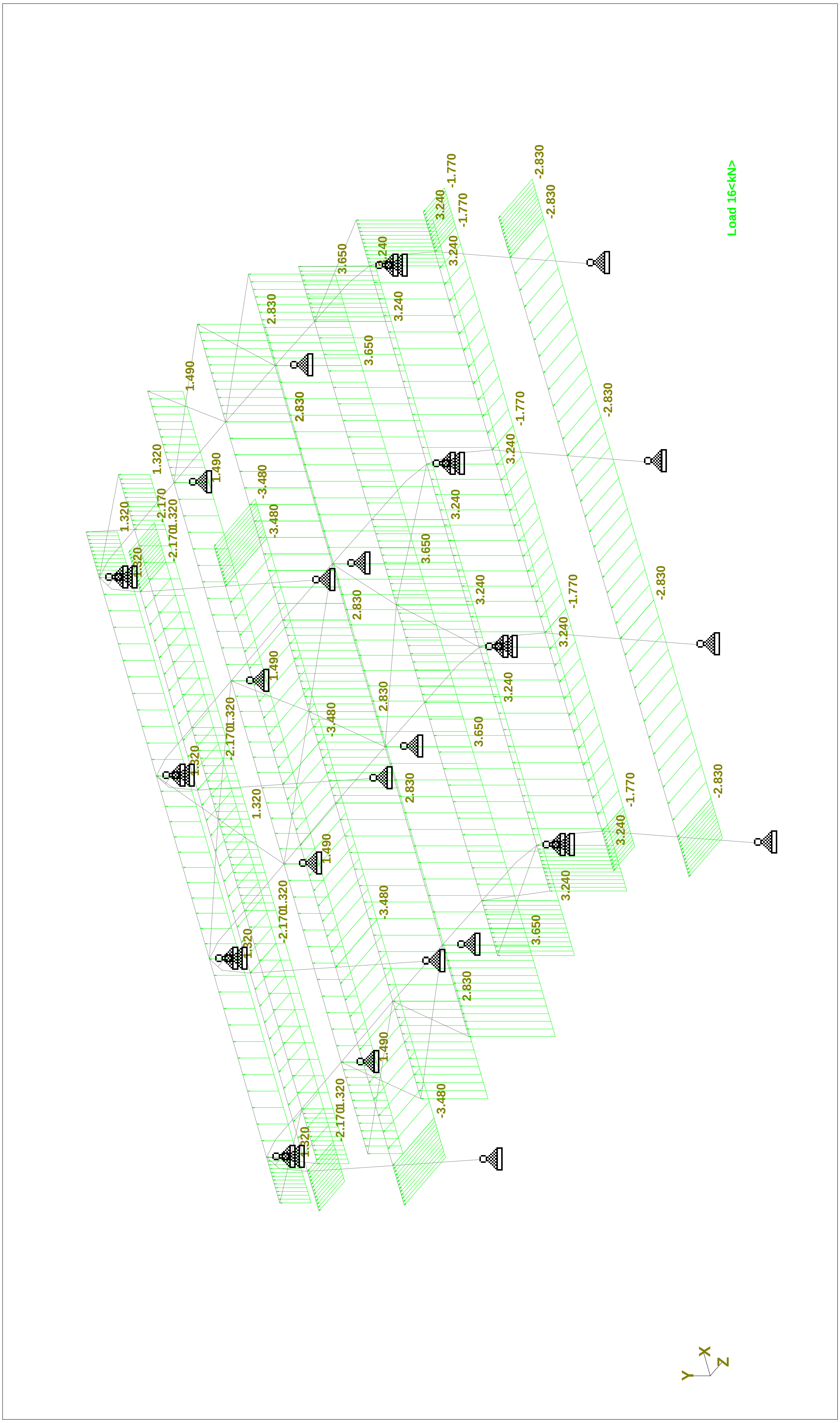

Load 13 WL3 (90 DEG ,INTERNAL SUCTION CPI=-0.3, WW&LW CPE=+0.2)



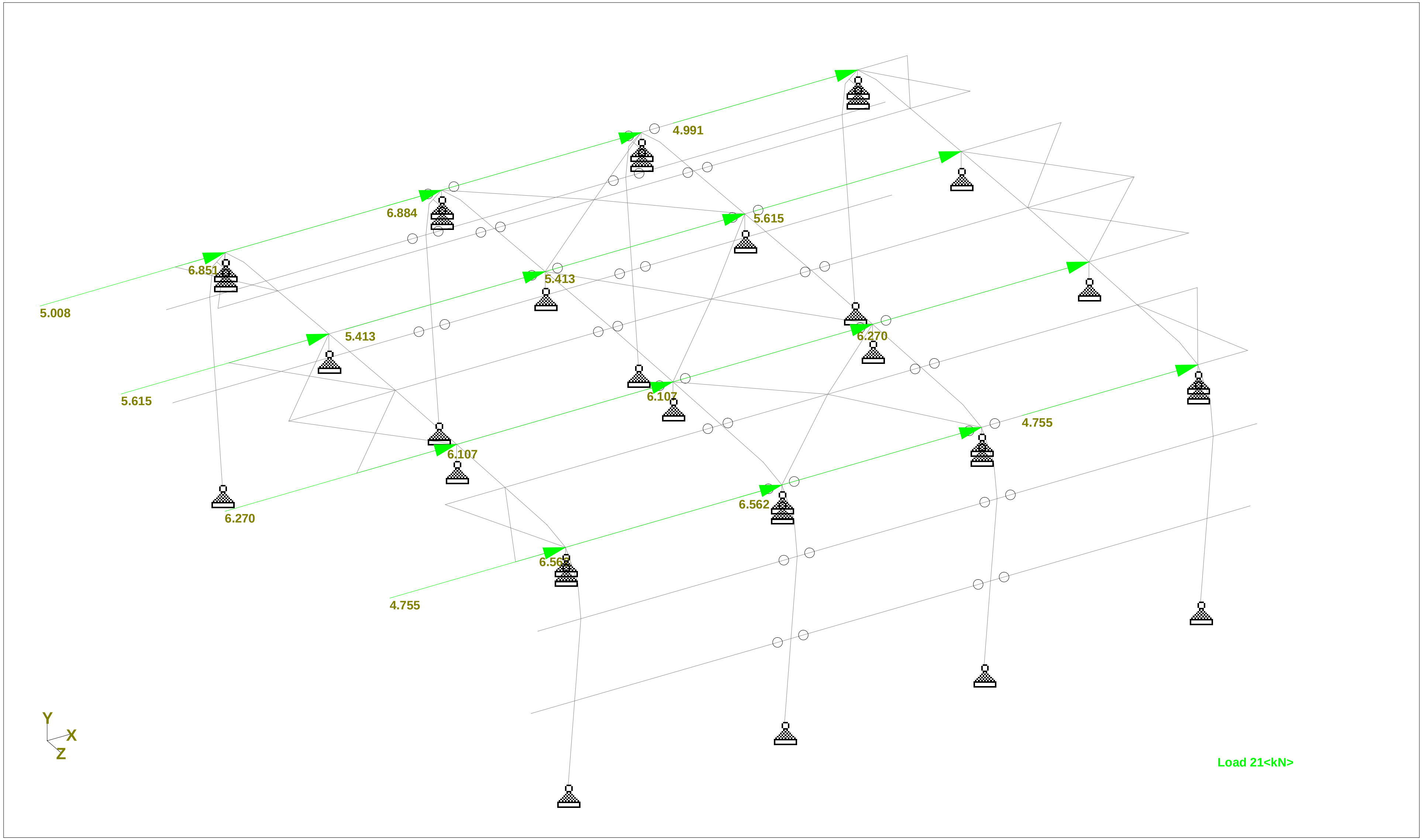

Load 14 WL4 (90 DEG ,INTERNAL PRESSURE CPI=+0.2, WW&LW CPE=-0.7)



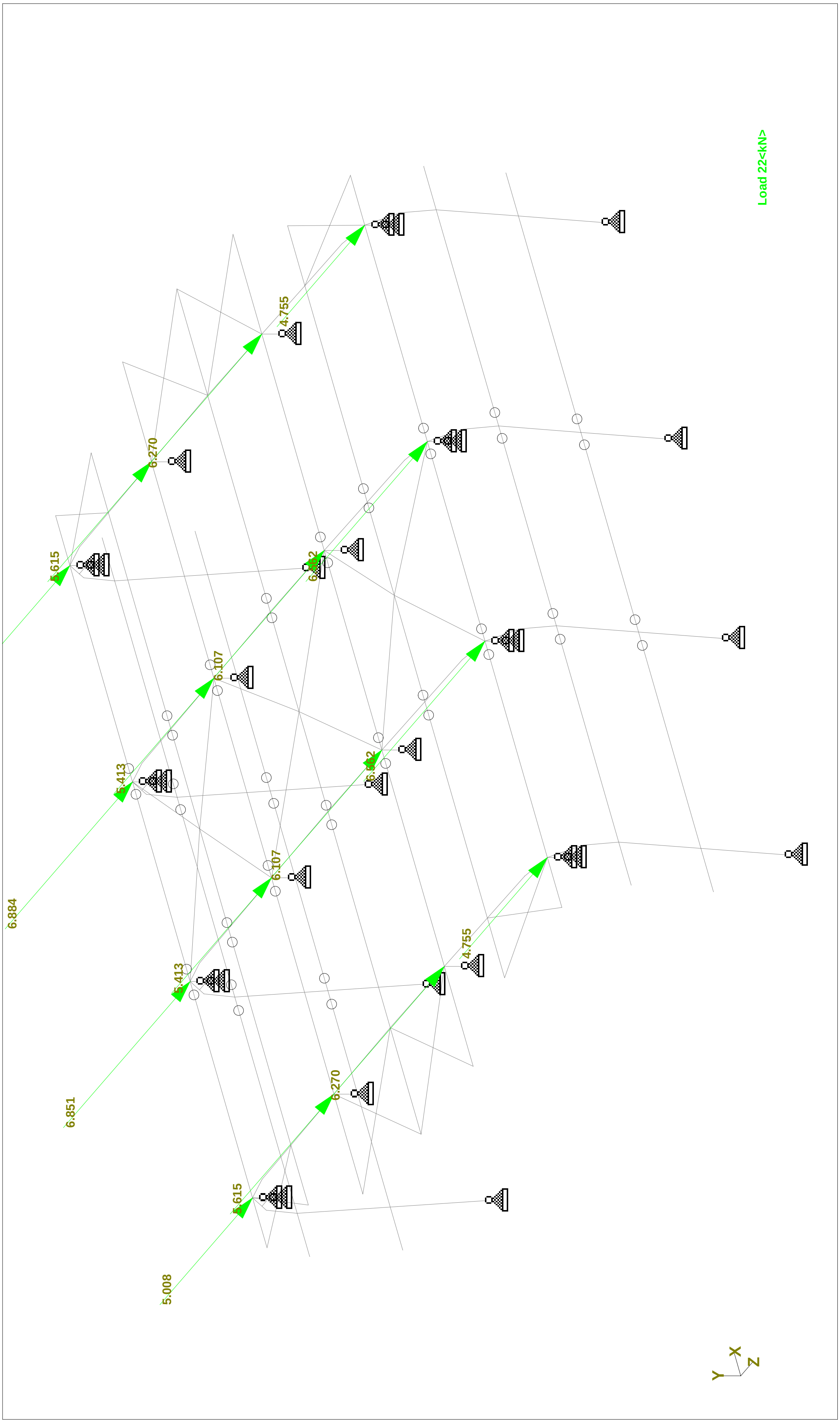

Load 15 WL5 (0 DEG ,INTERNAL SUCTION CPI=-0.3, LEE WARD -0.7, REVERSE)



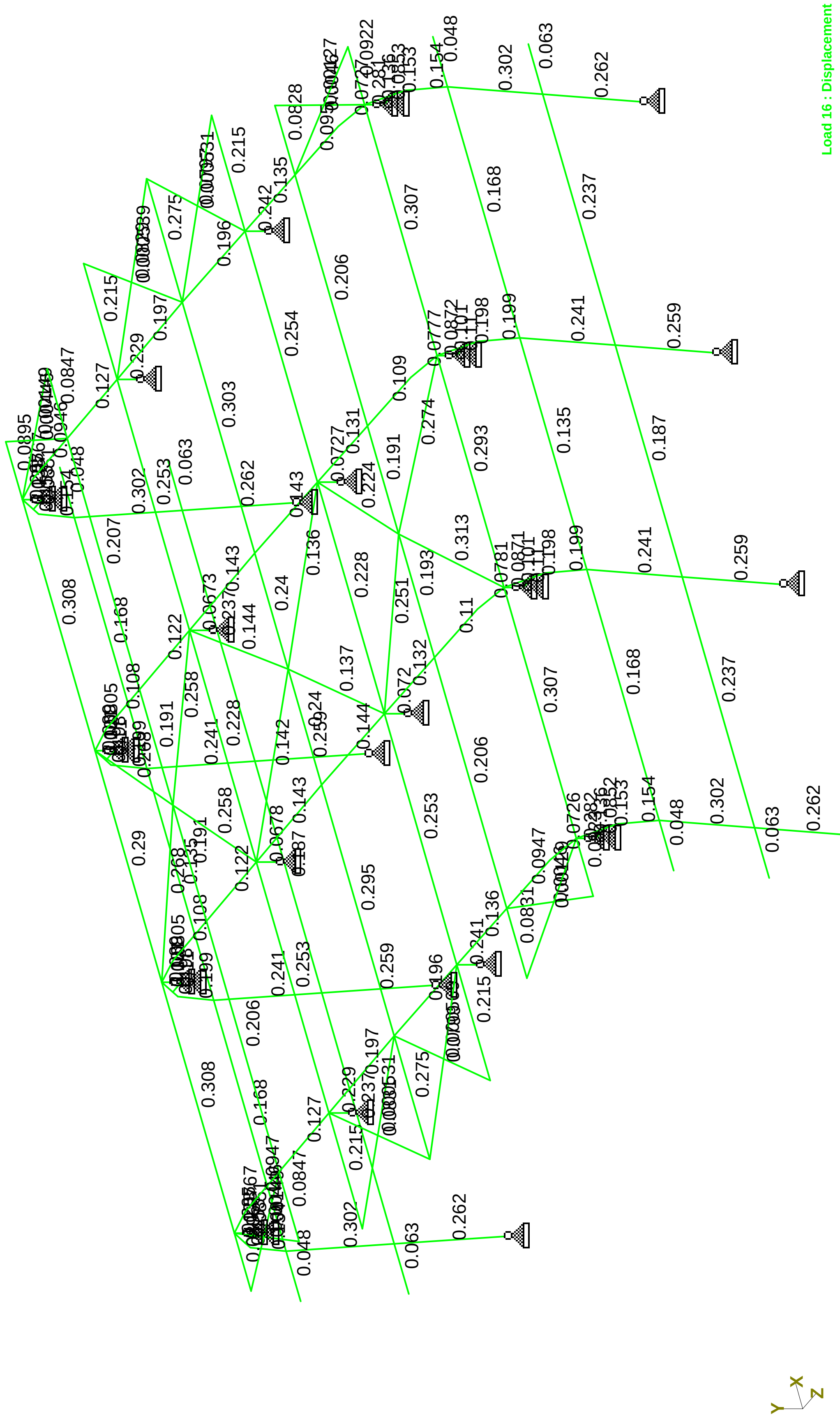

Load 16 WL6 (0 DEG ,INTERNAL PRESSURE CPI=+0.2, LEE WARD -0.2, REVERSE)




*Load 21 EQX (EARTHQUAKE LOAD )*




Load 22 EQZ (EARTHQUAKE LOAD )




*Utilization Ratio*

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****************************************************
*                                                  *
*          STAAD.Pro                               *
*          Version  2006    Bld 1002.US            *
*          Proprietary Program of                  *
*          Research Engineers,  Intl.              *
*          Date=    DEC 25, 2007                   *
*          Time=    15:43:34                       *
*                                                  *
*     USER ID: roark                               *
****************************************************
```

```
     1. STAAD SPACE BLDG -27( HEAT EXCHANGE FACILITY)
INPUT FILE: BLDG-27 STEEL Structure.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 29-OCT-07
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. ***********************************************************************
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 0 5.312 0.662; 3 0 6.317 0.503; 4 0 7.056 -0.14; 5 0 7.253 -1.111
    10. 6 0 7.317 -5.809; 7 0 7.347 -8.999; 8 0 7.306 -12.459; 9 0 7.253 -16.887
    11. 10 0 7.056 -17.856; 11 0 6.317 -18.5; 12 0 5.312 -18.659; 13 0 0 -17.997
    12. 14 6.5 0 0; 15 6.5 5.312 0.662; 16 6.5 6.317 0.503; 17 6.5 7.056 -0.14
    13. 18 6.5 7.253 -1.111; 19 6.5 7.317 -5.809; 20 6.5 7.347 -8.999
    14. 21 6.5 7.306 -12.459; 22 6.5 7.253 -16.887; 23 6.5 7.056 -17.856
    15. 24 6.5 6.317 -18.5; 25 6.5 5.312 -18.659; 26 6.5 0 -17.997; 27 12.5 0 0
    16. 28 12.5 5.312 0.662; 29 12.5 6.317 0.503; 30 12.5 7.056 -0.14
    17. 31 12.5 7.253 -1.111; 32 12.5 7.317 -5.809; 33 12.5 7.347 -8.999
    18. 34 12.5 7.306 -12.459; 35 12.5 7.253 -16.887; 36 12.5 7.056 -17.856
    19. 37 12.5 6.317 -18.5; 38 12.5 5.312 -18.659; 39 12.5 0 -17.997; 40 19 0 0
    20. 41 19 5.312 0.662; 42 19 6.317 0.503; 43 19 7.056 -0.14; 44 19 7.253 -1.111
    21. 45 19 7.317 -5.809; 46 19 7.347 -8.999; 47 19 7.306 -12.459
    22. 48 19 7.253 -16.887; 49 19 7.056 -17.856; 50 19 6.317 -18.5
    23. 51 19 5.312 -18.659; 52 19 0 -17.997; 53 20.5 7.056 -0.14
    24. 54 20.5 7.056 -17.856; 55 -1.5 7.056 -0.14; 56 -1.5 7.056 -17.856
    25. 57 22 7.317 -5.809; 58 22 7.306 -12.459; 59 -3 7.317 -5.809
    26. 60 -3 7.306 -12.459; 61 22.2 7.347 -8.999; 62 -3.2 7.347 -8.999
    27. 63 0 7.27428 -15.1088; 64 6.5 7.27428 -15.1088; 65 12.5 7.27428 -15.1088
    28. 66 19 7.27428 -15.1088; 67 0 7.28269 -3.2908; 68 6.5 7.28269 -3.2908
    29. 69 12.5 7.28269 -3.2908; 70 19 7.28269 -3.2908; 71 20.8 7.27428 -15.1088
    30. 72 -1.8 7.27428 -15.1088; 73 20.8 7.28269 -3.2908; 74 -1.8 7.28269 -3.2908
    31. 75 0 2.67927 0.333901; 76 6.5 2.67927 0.333901; 77 12.5 2.67927 0.333901
    32. 78 19 2.67927 0.333901; 79 20.3 2.67927 0.333901; 80 -1.3 2.67927 0.333901
    33. 81 0 2.67927 -18.3309; 82 6.5 2.67927 -18.3309; 83 12.5 2.67927 -18.3309
    34. 84 19 2.67927 -18.3309; 85 20.3 2.67927 -18.3309; 86 -1.3 2.67927 -18.3309
    35. 87 20.3 5.312 0.662; 88 20.3 5.312 -18.659; 89 -1.3 5.312 0.662
    36. 90 -1.3 5.312 -18.659; 91 9.7 7.28269 -3.2908; 93 9.5 7.347 -8.999
    37. 95 9.5 7.27428 -15.1088; 96 0 6.817 -5.809; 97 0 6.806 -12.459
    38. 98 6.5 6.817 -5.809; 99 6.5 6.806 -12.459; 100 12.5 6.817 -5.809
    39. 101 12.5 6.806 -12.459; 102 19 6.817 -5.809; 103 19 6.806 -12.459
    40. 104 0 6.856 -0.14; 105 0 6.856 -17.856; 106 6.5 6.856 -0.14
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    41. 107 6.5 6.856 -17.856; 108 12.5 6.856 -0.14; 109 12.5 6.856 -17.856
    42. 110 19 6.856 -0.14; 111 19 6.856 -17.856; 112 0 6.556 -0.14
    43. 113 0 6.556 -17.856; 114 6.5 6.556 -0.14; 115 6.5 6.556 -17.856
    44. 116 12.5 6.556 -0.14; 117 12.5 6.556 -17.856; 118 19 6.556 -0.14
    45. 119 19 6.556 -17.856; 128 0 6.556 -18.2917; 129 6.5 6.556 -18.2917
    46. 130 12.5 6.556 -18.2917; 131 19 6.556 -18.2917; 132 0 6.556 0.295047
    47. 133 6.5 6.556 0.295047; 134 12.5 6.556 0.295047; 135 19 6.556 0.295047
    48. ***********************************************************************
    49. MEMBER INCIDENCES
    50. 1 1 75; 2 2 3; 3 3 132; 4 4 5; 5 5 67; 6 6 7; 7 7 8; 8 8 63; 9 10 9; 10 11 128
    51. 11 12 11; 12 13 81; 13 14 76; 14 15 16; 15 16 133; 16 17 18; 17 18 68
    52. 18 19 20; 19 20 21; 20 21 64; 21 23 22; 22 24 129; 23 25 24; 24 26 82
    53. 25 27 77; 26 28 29; 27 29 134; 28 30 31; 29 31 69; 30 32 33; 31 33 34
    54. 32 34 65; 33 36 35; 34 37 130; 35 38 37; 36 39 83; 37 40 78; 38 41 42
    55. 39 42 135; 40 43 44; 41 44 70; 42 45 46; 43 46 47; 44 47 66; 45 49 48
    56. 46 50 131; 47 51 50; 48 52 84; 52 63 9; 53 64 22; 54 65 35; 55 66 48; 56 67 6
    57. 57 68 19; 58 69 32; 59 70 45; 95 75 2; 96 76 15; 97 77 28; 98 78 41; 99 81 12
    58. 100 82 25; 101 83 38; 102 84 51; 201 53 43; 202 43 30; 203 30 17; 204 17 4
    59. 205 4 55; 206 73 70; 207 70 69; 208 69 91; 209 68 67; 210 67 74; 211 57 45
    60. 212 45 32; 213 32 19; 214 19 6; 215 6 59; 216 61 46; 217 46 33; 218 33 93
    61. 220 7 62; 221 58 47; 222 47 34; 223 34 21; 224 21 8; 225 8 60; 226 71 66
    62. 227 66 65; 228 65 95; 229 64 63; 230 63 72; 231 54 49; 232 49 36; 233 36 23
    63. 234 23 10; 235 10 56; 236 91 68; 238 93 20; 240 95 64; 241 20 7; 301 79 78
    64. 302 78 77; 303 77 76; 304 76 75; 305 75 80; 306 85 84; 307 84 83; 308 83 82
    65. 309 82 81; 310 81 86; 311 87 41; 312 41 28; 313 28 15; 314 15 2; 315 2 89
    66. 316 88 51; 317 51 38; 318 38 25; 319 25 12; 320 12 90; 501 55 67; 502 4 74
    67. 503 59 7; 504 6 62; 505 62 8; 506 7 60; 507 72 10; 508 63 56; 509 43 73
    68. 510 53 70; 511 45 61; 512 57 46; 513 46 58; 514 61 47; 515 71 49; 516 54 66
    69. 522 17 91; 523 91 32; 524 30 91; 525 91 19; 530 19 93; 531 93 34; 532 32 93
    70. 533 93 21; 538 34 95; 539 95 23; 540 36 95; 541 95 21; 542 97 8; 543 96 6
    71. 544 99 21; 545 98 19; 546 101 34; 547 100 32; 548 103 47; 549 102 45
    72. 550 105 10; 551 107 23; 552 109 36; 553 111 49; 554 104 4; 555 106 17
    73. 556 108 30; 557 110 43; 558 113 128; 559 115 129; 560 117 130; 561 119 131
    74. 562 128 10; 564 129 23; 566 130 36; 567 131 49; 568 118 135; 569 116 134
    75. 570 114 133; 571 112 132; 572 132 4; 573 133 17; 574 134 30; 575 135 43
    76. START GROUP DEFINITION
    77. MEMBER
    78. _COLUMN 1 TO 3 10 TO 15 22 TO 27 34 TO 39 46 TO 48 95 TO 102 562 564 566 567 -
    79. 572 TO 575
    80. _RAFTER 4 TO 9 16 TO 21 28 TO 33 40 TO 45 52 TO 59
    81. _SEC_BEAM_ROOF 201 TO 218 220 TO 236 238 240 241
    82. _SIDE_BEAM 301 TO 320
    83. _BRACING 501 TO 516 522 TO 525 530 TO 533 538 TO 541
    84. _STUBCOLUMN 542 TO 561 568 TO 571
    85. END GROUP DEFINITION
    86. ***********************************************************************
    87. DEFINE MATERIAL START
    88. ISOTROPIC STEEL
    89. E 2.05E+008
    90. POISSON 0.3
    91. DENSITY 78
    92. ALPHA 1.2E-005
    93. DAMP 0.03
    94. END DEFINE MATERIAL
    95. MEMBER PROPERTY BRITISH
    96. ************************MAIN COLUMN**************************************
```

```
1 12 13 24 25 36 37 48 TAPERED 0.3 0.006 0.35 0.2 0.01 0.2 0.01
95 TO 102 TAPERED 0.35 0.006 0.4 0.2 0.01 0.2 0.01
2 3 10 11 14 15 22 23 26 27 34 35 38 39 46 47 562 564 566 567 572 TO 574 -
575 TAPERED 0.4 0.006 0.4 0.2 0.01 0.2 0.01
*************************MAIN RAFTER **************************************
4 TO 9 16 TO 21 28 TO 33 40 TO 45 52 TO 58 -
59 TAPERED 0.4 0.006 0.4 0.2 0.01 0.2 0.01
*************************SEC BEAM ****************************************
201 TO 218 220 TO 236 238 240 241 TAPERED 0.3 0.006 0.3 0.15 0.01 0.15 0.01
*************************SIDE BEAM **************************************
301 TO 320 TAPERED 0.25 0.006 0.25 0.25 0.012 0.25 0.012
*************************STUB COLUMN **************************************
542 TO 549 TAPERED 0.3 0.006 0.3 0.2 0.008 0.2 0.008
550 TO 561 568 TO 571 TAPERED 0.3 0.006 0.3 0.2 0.008 0.2 0.008
*550 TO 561 568 TO 571 PRIS YD 0.4 ZD 0.01
*************************BRACING *****************************************
501 TO 516 522 TO 525 530 TO 533 538 TO 541 TABLE ST UA70X70X6
**************************************************************************
CONSTANTS
BETA 90 MEMB 301 TO 320 542 TO 557
MATERIAL STEEL ALL
************************************************************************
SUPPORTS
1 13 14 26 27 39 40 52 PINNED
96 TO 119 PINNED
*4 6 8 10 17 19 21 23 30 32 34 36 43 45 47 49 PINNED
*4 6 8 10 17 19 21 23 30 32 34 36 43 45 47 49 128 TO 134 PINNED
************************************************************************
MEMBER RELEASE
203 204 208 209 213 214 218 223 224 228 229 233 234 241 303 304 308 309 313 -
314 318 319 START MZ
202 203 207 212 213 217 222 223 227 232 233 236 238 240 302 303 307 308 312 -
313 317 318 END MZ
MEMBER TRUSS
501 TO 516 522 TO 525 530 TO 533 538 TO 541
DEFINE UBC LOAD
ZONE 0.15 I 1 RWX 4.5 RWZ 4.5 STYP 4 CT 0.0853 PX 0.381 PZ 0.381 NA 1 NV 1
JOINT WEIGHT
4 WEIGHT 39.57
6 WEIGHT 50.32
8 WEIGHT 45.126
10 WEIGHT 41.672
17 WEIGHT 54.606
19 WEIGHT 49.012
21 WEIGHT 43.508
23 WEIGHT 57.012
30 WEIGHT 54.606
32 WEIGHT 49.012
34 WEIGHT 43.508
36 WEIGHT 57.29
43 WEIGHT 39.57
45 WEIGHT 50.32
47 WEIGHT 45.126
49 WEIGHT 41.537
LOAD 21 EQX (EARTHQUAKE LOAD )
UBC LOAD X
```





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LOAD 22 EQZ (EARTHQUAKE LOAD )
UBC LOAD Z
LOAD 1 SELFWEIGHT
SELFWEIGHT Y -1
LOAD 2 DL
**DEAD LOAD = 0.8 KN/M^2
**BEAM SPACING = 3.3 M , 3.0 M & 2.7 M
**DL1 = 3.3X0.8 =2.66=2.7 KN/M
**DL2 = 3.0X0.8 =2.4 KN/M
**DL3 = 2.7X0.8 =2.16=2.2 KN/M
MEMBER LOAD
216 TO 218 220 238 241 301 TO 310 UNI GY -2.7
211 TO 215 221 TO 225 UNI GY -2.4
206 TO 210 226 TO 230 236 240 UNI GY -2.2
201 TO 205 231 TO 235 311 TO 320 UNI GY -2.2
**************************************
LOAD 3 LL
MEMBER LOAD
**LIVE LOAD = 0.6 KN/M^2
**BEAM SPACING = 3.3 M , 3.0 M & 2.7 M
**LL1 = 3.3X0.6 =1.98=2.0 KN/M
**LL2 = 3.0X0.6 =1.8 KN/M
**LL3 = 2.7X0.6 =1.62=1.65 KN/M
216 TO 218 220 238 241 UNI GY -2
211 TO 215 221 TO 225 UNI GY -1.8
201 TO 210 226 TO 236 240 UNI GY -1.65
**********A/C EQUIPMENT S/W = 5 KN ( POINT LOAD )
*JOINT LOAD
*20 33 FY -5
LOAD 4 PH(X-DIR)
************************
**PH =1.5% (1.4DL+1.6LL)
JOINT LOAD
4 FX 0.88
6 FX 1.47
8 FX 1.4
10 FX 0.9
17 FX 1.33
19 FX 1.43
21 FX 1.35
23 FX 1.37
30 FX 1.33
32 FX 1.43
34 FX 1.35
36 FX 1.37
43 FX 0.88
45 FX 1.47
47 FX 1.4
49 FX 0.91
LOAD 5 PH(Z-DIR)
```

```
************************
**PH =1.5% (1.4DL+1.6LL)
JOINT LOAD
4 FZ 0.88
6 FZ 1.47
8 FZ 1.4
10 FZ 0.9
17 FZ 1.33
19 FZ 1.43
21 FZ 1.35
23 FZ 1.37
30 FZ 1.33
32 FZ 1.43
34 FZ 1.35
36 FZ 1.37
43 FZ 0.88
45 FZ 1.47
47 FZ 1.4
49 FZ 0.91
*********************************************************
LOAD 11 WL1 (0 DEG,INTERNAL SUCTION (CPI=-0.3)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF=0.2)
**CPE(WINDWARD WALL)=0.77,CPE(LEEWARD WALL=-0.5)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77+0.3)X4.0 =5.31 KN/M
**WL2(WINDWARD WALL)= 1.24(0.77+0.3)X2.5 =3.32 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5+0.3)X4.0 =0.99 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.5+0.3)X2.5 =0.62 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78+0.3)X3.0 =1.79(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.78+0.3)X2.66 =1.58(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL8(LEEWARD  ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
MEMBER LOAD
306 TO 310 UNI GZ 5.31
221 TO 225 UNI Y 1.79
211 TO 215 UNI Y -1.86
301 TO 305 UNI GZ 0.99
316 TO 320 UNI GZ 3.32
311 TO 315 UNI GZ 0.62
226 TO 235 240 UNI Y 1.58
201 TO 210 236 UNI Y -1.65
216 TO 218 220 238 241 UNI Y -0.04
**********************************************************
LOAD 12 WL2 (0 DEG ,INTERNAL PRESSURE CPI=+0.2)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF)=-0.2
**CPE(WINDWARD WALL)=+0.77,CPE(LEEWARD WALL)=-0.5
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77-0.2)X4.0 =2.83 KN/M
**WL2(WINDWARD WALL)= 1.24(0.77-0.2)X2.5 =1.77 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5-0.2)X4.0 =3.48 KN/M
```



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**WL4(LEEWARD  WALL)= 1.24(-0.5-0.2)X2.5 =2.17 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78-0.2)X3.0 =3.65(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.78-0.2)X2.66 =3.24(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(-0.2-0.2)X3.0 =1.65(UP) KN/M
**WL8(LEEWARD  ROOF)= 1.24(-0.2-0.2)X2.66 =1.49(UP)KN/M
MEMBER LOAD
306 TO 310 UNI GZ 2.83
301 TO 305 UNI GZ 3.48
316 TO 320 UNI GZ 1.77
311 TO 315 UNI GZ 2.17
221 TO 225 UNI Y 3.65
211 TO 215 UNI Y 1.49
226 TO 235 240 UNI Y 3.24
201 TO 210 236 UNI Y 1.32
216 TO 218 220 238 241 UNI Y 2.83
********************************************************
LOAD 13 WL3 (90 DEG ,INTERNAL SUCTION CPI=-0.3, WW&LW CPE=+0.2)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=0.2,CPE(LEEWARD ROOF=0.2)
**CPE(WINDWARD WALL)=-0.8,CPE(LEEWARD WALL=-0.8)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(-0.8+0.3)X4.0 =2.48 KN/M
**WL2(WINDWARD WALL)= 1.24(-0.8+0.3)X2.5 =1.55 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.8+0.3)X4.0 =2.48 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.8+0.3)X2.5 =1.55 KN/M
**WL5(WINDWARD ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL6(WINDWARD ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN) KN/M
**WL7(LEEWARD  ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL8(LEEWARD  ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
MEMBER LOAD
306 TO 310 UNI GZ -2.48
301 TO 305 UNI GZ 2.48
316 TO 320 UNI GZ -1.55
311 TO 315 UNI GZ 1.55
221 TO 225 UNI Y -1.86
211 TO 215 UNI Y -1.86
226 TO 235 240 UNI Y -1.65
201 TO 210 236 UNI Y -1.65
216 TO 218 220 238 241 UNI Y -2.05
*********************************************************
LOAD 14 WL4 (90 DEG ,INTERNAL PRESSURE CPI=+0.2, WW&LW CPE=-0.7)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.7,CPE(LEEWARD ROOF=-0.7)
**CPE(WINDWARD WALL)=-0.8,CPE(LEEWARD WALL=-0.8)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(-0.8-0.2)X4.0 =4.97 KN/M
**WL2(WINDWARD WALL)= 1.24(-0.8-0.2)X2.5 =3.1 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.8-0.2)X4.0 =4.97 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.8-0.2)X2.5 =3.1 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.7-0.2)X3.0 =3.35(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.7-0.2)X2.66 =2.97(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(-0.7-0.2)X3.0 =3.35(UP) KN/M
```

```
**WL8(LEEWARD  ROOF)= 1.24(-0.7-0.2)X2.66=2.97(UP)KN/M
MEMBER LOAD
306 TO 310 UNI GZ -4.97
301 TO 305 UNI GZ 4.97
316 TO 320 UNI GZ -3.1
311 TO 315 UNI GZ 3.1
221 TO 225 UNI Y 3.35
211 TO 215 UNI Y 3.35
226 TO 235 240 UNI Y 2.97
201 TO 210 236 UNI Y 2.97
216 TO 218 220 238 241 UNI Y 3.69
*******************************************************************
LOAD 15 WL5 (0 DEG ,INTERNAL SUCTION CPI=-0.3, LEE WARD -0.7, REVERSE)
MEMBER LOAD
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF=0.2)
**CPE(WINDWARD WALL)=0.77,CPE(LEEWARD WALL=-0.5)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77+0.3)X4.0 =5.31 KN/M
**WL2(WINDWARD WALL)= 1.24(0.77+0.3)X2.5 =3.32 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5+0.3)X4.0 =0.99 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.5+0.3)X2.5 =0.62 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78+0.3)X3.0 =1.79(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.78+0.3)X2.66 =1.58(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL8(LEEWARD  ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
306 TO 310 UNI GZ -0.99
301 TO 305 UNI GZ -5.31
316 TO 320 UNI GZ -0.62
311 TO 315 UNI GZ -3.32
221 TO 225 UNI Y -1.86
211 TO 215 UNI Y 1.79
226 TO 235 240 UNI Y -1.65
201 TO 210 236 UNI Y 1.58
216 218 220 238 UNI Y -0.04
241 UNI Y -0.04
217 UNI Y -0.04
217 UNI Y -0.04
********************************************************************
LOAD 16 WL6 (0 DEG ,INTERNAL PRESSURE CPI=+0.2, LEE WARD -0.2, REVERSE)
MEMBER LOAD
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF)=-0.2
**CPE(WINDWARD WALL)=+0.77,CPE(LEEWARD WALL)=-0.5
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77-0.2)X4.0 =2.83 KN/M
**WL2(WINDWARD WALL)= 1.24(0.77-0.2)X2.5 =1.77 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5-0.2)X4.0 =3.48 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.5-0.2)X2.5 =2.17 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78-0.2)X3.0 =3.65(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.78-0.2)X2.66 =3.24(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(-0.2-0.2)X3.0 =1.65(UP) KN/M
```



```
**WL8(LEEWARD  ROOF)= 1.24(-0.2-0.2)X2.66 =1.49(UP)KN/M
306 TO 310 UNI GZ -3.48
301 TO 305 UNI GZ -2.83
316 TO 320 UNI GZ -2.17
311 TO 315 UNI GZ -1.77
221 TO 225 UNI Y 1.49
211 TO 215 UNI Y 3.65
226 TO 235 240 UNI Y 1.32
201 TO 210 236 UNI Y 3.24
216 218 220 238 UNI Y 2.83
241 UNI Y 2.83
217 UNI Y 2.83
217 UNI Y 2.83
*********************************************
**           SERVICEBILITY*                 **
*********************************************
LOAD COMB 101 1.0DL+1.0LL
1 1.0 2 1.0 3 1.0
LOAD COMB 102 1.0DL+0.5LL
1 1.0 2 1.0 3 0.5
LOAD COMB 103 0.8DL+0.8LL+0.8WL1
1 0.8 2 0.8 3 0.8 11 0.8
LOAD COMB 104 1.0DL+1.0WL1
1 1.0 2 1.0 11 1.0
LOAD COMB 105 0.8DL+0.8LL+0.8WL2
1 0.8 2 0.8 3 0.8 12 0.8
LOAD COMB 106 1.0DL+1.0WL2
1 1.0 2 1.0 12 1.0
LOAD COMB 107 0.8DL+0.8LL+0.8WL3
1 0.8 2 0.8 3 0.8 13 0.8
LOAD COMB 108 1.0DL+1.0WL3
1 1.0 2 1.0 13 1.0
LOAD COMB 109 0.8DL+0.8LL+0.8WL4
1 0.8 2 0.8 3 0.8 14 0.8
LOAD COMB 110 1.0DL+1.0WL4
1 1.0 2 1.0 14 1.0
LOAD COMB 111 0.8DL+0.8LL+0.8WL5
1 0.8 2 0.8 3 0.8 15 0.8
LOAD COMB 112 1.0DL+1.0WL4
1 1.0 2 1.0 15 1.0
LOAD COMB 113 0.8DL+0.8LL+0.8WL6
1 0.8 2 0.8 3 0.8 16 0.8
LOAD COMB 114 1.0DL+1.0WL6
1 1.0 2 1.0 16 1.0
*********EQX  &  EQZ  **************************************
**DL+EQL/1.4 = 1.0DL+0.72EQX+0.22 EQZ, EQL=EQX+0.3EQZ
**0.9 DL+EQL/1.4 =0.9 DL+0.72 EQX+0.22EQZ
**DL 0.75(LL+(EQL/1.4)) =1.0 DL+0.75 LL+0.54 EQX+0.17EQZ
***********************************************************
LOAD COMB 115 DL+0.72EQX+0.22EQZ
1 1.0 2 1.0 21 0.72 22 0.22
LOAD COMB 116 DL+0.72EQX-0.22EQZ
1 1.0 2 1.0 21 0.72 22 -0.22
LOAD COMB 117 DL-0.72EQX+0.22EQZ
1 1.0 2 1.0 21 -0.72 22 0.22
LOAD COMB 118 DL-0.72EQX+0.22EQZ
```

```
1 1.0 2 1.0 21 -0.72 22 -0.22
LOAD COMB 119 DL+0.72EQZ+0.22EQX
1 1.0 2 1.0 22 0.72 21 0.22
LOAD COMB 120 DL+0.72EQZ-0.22EQX
1 1.0 2 1.0 22 0.72 21 -0.22
LOAD COMB 121 DL-0.72EQZ+0.22EQX
1 1.0 2 1.0 22 -0.72 21 0.22
LOAD COMB 122 DL-0.72EQZ-0.22EQX
1 1.0 2 1.0 22 -0.72 21 -0.22
LOAD COMB 123 0.9DL+0.72EQX+0.22EQZ
1 0.9 2 0.9 21 0.72 22 0.22
LOAD COMB 124 0.9DL+0.72EQX-0.22EQZ
1 0.9 2 0.9 21 0.72 22 -0.22
LOAD COMB 125 0.9DL-0.72EQX+0.22EQZ
1 0.9 2 0.9 21 -0.72 22 0.22
LOAD COMB 126 0.9DL-0.72EQX+0.22EQZ
1 0.9 2 0.9 21 -0.72 22 -0.22
LOAD COMB 127 0.9DL+0.72EQZ+0.22EQX
1 0.9 2 0.9 22 0.72 21 0.22
LOAD COMB 128 0.9DL+0.72EQZ-0.22EQX
1 0.9 2 0.9 22 0.72 21 -0.22
LOAD COMB 129 0.9DL-0.72EQZ+0.22EQX
1 0.9 2 0.9 22 -0.72 21 0.22
LOAD COMB 130 0.9DL-0.72EQZ-0.22EQX
1 0.9 2 0.9 22 -0.72 21 -0.22
LOAD COMB 131 1.0DL+0.75LL+0.54EQX+0.17 EQZ
1 1.0 2 1.0 3 0.75 21 0.54 22 0.17
LOAD COMB 132 1.0DL+0.75LL+0.54EQX-0.17 EQZ
1 1.0 2 1.0 3 0.75 21 0.54 22 -0.17
LOAD COMB 133 1.0DL+0.75LL-0.54EQX+0.17 EQZ
1 1.0 2 1.0 3 0.75 21 -0.54 22 0.17
LOAD COMB 134 1.0DL+0.75LL-0.54EQX-0.17 EQZ
1 1.0 2 1.0 3 0.75 21 -0.54 22 -0.17
LOAD COMB 135 1.0DL+0.75LL+0.54EQZ+0.17 EQX
1 1.0 2 1.0 3 0.75 22 0.54 21 0.17
LOAD COMB 136 1.0DL+0.75LL+0.54EQZ-0.17 EQX
1 1.0 2 1.0 3 0.75 22 0.54 21 -0.17
LOAD COMB 137 1.0DL+0.75LL-0.54EQZ+0.17 EQX
1 1.0 2 1.0 3 0.75 22 -0.54 21 0.17
LOAD COMB 138 1.0DL+0.75LL-0.54EQZ-0.17 EQX
1 1.0 2 1.0 3 0.75 22 -0.54 21 -0.17
****************************************************
**              STEEL DESIGN                     **
***************************************************
LOAD COMB 200 1.4DL+1.6LL
1 1.4 2 1.4 3 1.6
*******WL1*****************************************
LOAD COMB 201 1.2DL+1.2LL+1.2WL1
1 1.2 2 1.2 3 1.2 11 1.2
LOAD COMB 202 1.4DL+1.4WL1
1 1.4 2 1.4 11 1.4
LOAD COMB 203 1DL+1.4WL1
1 1.0 2 1.0 11 1.4
*********WL2***********************
LOAD COMB 204 1.2DL+1.2LL+1.2WL2
1 1.2 2 1.2 3 1.2 12 1.2
```



```
LOAD COMB 205 1.4DL+1.4WL2
1 1.4 2 1.4 12 1.4
LOAD COMB 206 1DL+1.4WL2
1 1.0 2 1.0 12 1.4
*******WL3************************
LOAD COMB 207 1.2DL+1.2LL+1.2WL3
1 1.2 2 1.2 3 1.2 13 1.2
LOAD COMB 208 1.4DL+1.4WL3
1 1.4 2 1.4 13 1.4
LOAD COMB 209 1DL+1.4WL3
1 1.0 2 1.0 13 1.4
*******WL4************************
LOAD COMB 210 1.2DL+1.2LL+1.2WL4
1 1.2 2 1.2 3 1.2 14 1.2
LOAD COMB 211 1.4DL+1.4WL4
1 1.4 2 1.4 14 1.4
LOAD COMB 212 1DL+1.4WL4
1 1.0 2 1.0 14 1.4
*******WL5************************
LOAD COMB 213 1.2DL+1.2LL+1.2WL5
1 1.2 2 1.2 3 1.2 15 1.2
LOAD COMB 214 1.4DL+1.4WL5
1 1.4 2 1.4 15 1.4
LOAD COMB 215 1DL+1.4WL5
1 1.0 2 1.0 15 1.4
*******WL6************************
LOAD COMB 216 1.2DL+1.2LL+1.2WL6
1 1.2 2 1.2 3 1.2 16 1.2
LOAD COMB 217 1.4DL+1.4WL6
1 1.4 2 1.4 16 1.4
LOAD COMB 218 1DL+1.4WL6
1 1.0 2 1.0 16 1.4
*********EQX  &  EQZ  *************************
LOAD COMB 219 1.2DL+0.5LL+1.0EQX+0.3 EQZ
1 1.2 2 1.2 3 0.5 21 1.0 22 0.3
LOAD COMB 220 1.2DL+0.5LL+1.0EQX-0.3 EQZ
1 1.2 2 1.2 3 0.5 21 1.0 22 -0.3
LOAD COMB 221 1.2DL+0.5LL-1.0EQX+0.3 EQZ
1 1.2 2 1.2 3 0.5 21 -1.0 22 0.3
LOAD COMB 222 1.2DL+0.5LL-1.0EQX-0.3 EQZ
1 1.2 2 1.2 3 0.5 21 -1.0 22 -0.3
LOAD COMB 223 1.2DL+0.5LL+1.0EQZ+0.3 EQX
1 1.2 2 1.2 3 0.5 22 1.0 21 0.3
LOAD COMB 224 1.2DL+0.5LL+1.0EQZ-0.3 EQX
1 1.2 2 1.2 3 0.5 22 1.0 21 -0.3
LOAD COMB 225 1.2DL+0.5LL-1.0EQZ+0.3 EQX
1 1.2 2 1.2 3 0.5 22 -1.0 21 0.3
LOAD COMB 226 1.2DL+0.5LL-1.0EQZ-0.3 EQX
1 1.0 2 1.2 3 0.5 22 -1.0 21 -0.3
LOAD COMB 227 0.9DL+1.0EQX+0.3EQZ
1 0.9 2 0.9 21 1.0 22 0.3
LOAD COMB 228 0.9DL+1.0EQX-0.3EQZ2
1 0.9 2 0.9 21 1.0 22 -0.3
LOAD COMB 229 0.9DL-1.0EQX+0.3EQZ
1 0.9 2 0.9 21 -1.0 22 0.3
LOAD COMB 230 0.9DL-1.0EQX+0.3EQZ
```

```
539. 1 0.9 2 0.9 21 -1.0 22 -0.3
540. LOAD COMB 231 0.9DL+1.0EQZ+0.3EQX
541. 1 0.9 2 0.9 22 1.0 21 0.3
542. LOAD COMB 232 0.9DL+1.0EQZ-0.3EQX
543. 1 0.9 2 0.9 22 1.0 21 -0.3
544. LOAD COMB 233 0.9DL-1.0EQZ+0.3EQX
545. 1 0.9 2 0.9 22 -1.0 21 0.3
546. LOAD COMB 234 0.9DL-1.0EQZ-0.3EQX
547. 1 0.9 2 0.9 22 -1.0 21 -0.3
548. LOAD COMB 235 1.0DL+1.0LL+1.0PHX
549. 1 1.0 2 1.0 3 1.0 4 1.0
550. LOAD COMB 236 1.0DL+1.0LL+1.0PHZ
551. 1 1.0 2 1.0 3 1.0 5 1.0
552. ******DEFLECTION******************************
553. LOAD COMB 301 SELF+DL
554. 1 1.0 2 1.0
555. LOAD COMB 302 LL
556. 3 1.0
557. LOAD COMB 303 0.8(LL+WL1)
558. 3 0.8 11 0.8
559. LOAD COMB 304 0.8(LL+WL2)
560. 3 0.8 12 0.8
561. LOAD COMB 305 0.8(LL+WL3)
562. 3 0.8 13 0.8
563. LOAD COMB 306 0.8(LL+WL4)
564. 3 0.8 14 0.8
565. LOAD COMB 307 0.8(LL+WL5)
566. 3 0.8 15 0.8
567. LOAD COMB 308 0.8(LL+WL6)
568. 3 0.8 16 0.8
569. LOAD COMB 309 0.8(LL+EQX/1.4+0.3EQZ/1.4)
570. 3 0.8 21 0.57 22 0.17
571. LOAD COMB 310 0.8(LL+EQX/1.4-0.3EQZ/1.4)
572. 3 0.8 21 0.57 22 -0.17
573. LOAD COMB 311 0.8(LL-EQX/1.4+0.3EQZ/1.4)
574. 3 0.8 21 -0.57 22 0.17
575. LOAD COMB 312 0.8(LL-EQX/1.4-0.3EQZ/1.4)
576. 3 0.8 21 -0.57 22 -0.17
577. LOAD COMB 313 0.8(LL+EQZ/1.4+0.3EQX/1.4)
578. 3 0.8 22 0.57 21 0.17
579. LOAD COMB 314 0.8(LL+EQZ/1.4-0.3EQX/1.4)
580. 3 0.8 22 0.57 21 -0.17
581. LOAD COMB 315 0.8(LL-EQZ/1.4+0.3EQX/1.4)
582. 3 0.8 22 -0.57 21 0.17
583. LOAD COMB 316 0.8(LL-EQZ/1.4-0.3EQX/1.4)
584. 3 0.8 22 -0.57 21 -0.17
585. ************************************************
586. PERFORM ANALYSIS
```



```
        P R O B L E M   S T A T I S T I C S
        -----------------------------------

   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =   125/   182/    32
   ORIGINAL/FINAL BAND-WIDTH=   119/    15/     81 DOF
   TOTAL PRIMARY LOAD CASES =   13, TOTAL DEGREES OF FREEDOM =    654
   SIZE OF STIFFNESS MATRIX =       53 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     13.0/  22202.7 MB


 **WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
 OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
 UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS & CHANGE.


  **********************************************************
  *                                                        *
  *  X  DIRECTION : Ta =  0.381 Tb =  0.115 Tuser =  0.381  *
  *  T =  0.381, LOAD FACTOR =  1.000                      *
  * UBC TYPE = 97                                          *
  * UBC FACTOR V = 0.1222 x        761.80 =      93.11 KN   *
  *                                                        *
  **********************************************************


  **********************************************************
  *                                                        *
  *  Z  DIRECTION : Ta =  0.381 Tb =  0.028 Tuser =  0.381  *
  *  T =  0.381, LOAD FACTOR =  1.000                      *
  * UBC TYPE = 97                                          *
  * UBC FACTOR V = 0.1222 x        761.80 =      93.11 KN   *
  *                                                        *
  **********************************************************


  587. LOAD LIST 101 TO 138
  588. PRINT SUPPORT REACTION
 SUPPORT  REACTION
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| 1 | 101 | 0.92 | 15.43 | 1.24 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.91 | 16.23 | 1.34 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.74 | 3.75 | -1.67 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.90 | 6.29 | -1.88 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.68 | -11.74 | -7.55 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.84 | -13.07 | -9.22 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.72 | -6.74 | -5.35 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.88 | -6.83 | -6.47 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.65 | -21.01 | -11.06 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.80 | -24.66 | -13.62 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.79 | 51.90 | 14.39 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.98 | 66.47 | 18.19 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.75 | 35.33 | 8.38 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.92 | 45.77 | 10.69 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.94 | 17.45 | 1.50 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.94 | 16.68 | 1.40 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.87 | 17.39 | 1.49 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.87 | 16.61 | 1.40 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.92 | 18.31 | 1.61 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.90 | 18.29 | 1.61 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.92 | 15.77 | 1.29 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.90 | 15.75 | 1.29 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.85 | 15.75 | 1.36 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.85 | 14.97 | 1.26 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.78 | 15.68 | 1.35 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.78 | 14.91 | 1.25 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.83 | 16.61 | 1.47 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.81 | 16.59 | 1.46 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.83 | 14.07 | 1.14 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.81 | 14.05 | 1.14 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.94 | 16.15 | 1.33 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.94 | 15.55 | 1.26 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.89 | 16.11 | 1.32 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.89 | 15.51 | 1.25 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.92 | 16.79 | 1.41 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.91 | 16.78 | 1.41 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.92 | 14.89 | 1.17 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.91 | 14.87 | 1.17 | 0.00 | 0.00 | 0.00 |
| 13 | 101 | 0.92 | 15.32 | -1.22 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.91 | 16.15 | -1.33 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.79 | 51.80 | -14.37 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.98 | 66.41 | -18.19 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.75 | 35.30 | -8.38 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.92 | 45.78 | -10.69 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.72 | -6.86 | 5.36 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.88 | -6.91 | 6.48 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.65 | -20.98 | 11.06 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.80 | -24.56 | 13.60 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 111 | 0.74 | 3.63 | 1.69 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.91 | 6.20 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.68 | -11.75 | 7.55 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.84 | -13.03 | 9.21 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.94 | 16.62 | -1.40 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.94 | 17.40 | -1.50 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.87 | 16.56 | -1.39 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.87 | 17.34 | -1.49 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.92 | 15.72 | -1.28 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.90 | 15.70 | -1.28 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.92 | 18.26 | -1.60 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.90 | 18.24 | -1.60 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.85 | 14.92 | -1.25 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.85 | 15.70 | -1.35 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.78 | 14.86 | -1.24 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.78 | 15.64 | -1.34 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.83 | 14.02 | -1.14 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.81 | 14.00 | -1.14 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.83 | 16.56 | -1.46 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.81 | 16.55 | -1.46 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.94 | 15.46 | -1.24 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.94 | 16.06 | -1.32 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.89 | 15.41 | -1.24 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.89 | 16.01 | -1.31 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.93 | 14.79 | -1.16 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.91 | 14.77 | -1.16 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.93 | 16.70 | -1.40 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.91 | 16.68 | -1.40 | 0.00 | 0.00 | 0.00 |
| 14 | 101 | 0.02 | 22.29 | 1.74 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.02 | 23.14 | 1.86 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.01 | 6.92 | -2.04 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.02 | 10.35 | -2.32 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.02 | -13.72 | -9.77 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.03 | -15.45 | -11.99 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.01 | -6.64 | -6.83 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.02 | -6.60 | -8.31 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.02 | -26.02 | -14.39 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.03 | -30.83 | -17.77 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.02 | 68.96 | 18.83 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.04 | 87.90 | 23.77 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.03 | 47.10 | 10.96 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.04 | 60.58 | 13.92 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.05 | 23.94 | 1.96 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.05 | 23.03 | 1.85 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.00 | 24.95 | 2.09 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.00 | 24.04 | 1.98 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.03 | 25.33 | 2.14 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.02 | 25.63 | 2.18 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 121 | 0.03 | 22.35 | 1.76 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.02 | 22.65 | 1.80 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.05 | 21.55 | 1.77 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.05 | 20.64 | 1.65 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.00 | 22.55 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.00 | 21.64 | 1.78 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.03 | 22.93 | 1.94 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.02 | 23.23 | 1.98 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.03 | 19.95 | 1.57 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.02 | 20.25 | 1.60 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.04 | 22.69 | 1.80 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.04 | 21.99 | 1.71 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.01 | 23.44 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.01 | 22.74 | 1.80 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.03 | 23.71 | 1.93 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.02 | 23.95 | 1.96 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.03 | 21.48 | 1.65 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.02 | 21.72 | 1.67 | 0.00 | 0.00 | 0.00 |
| 26 | 101 | 0.02 | 22.07 | -1.72 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.02 | 22.96 | -1.83 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.02 | 68.75 | -18.81 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.04 | 87.73 | -23.75 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.03 | 47.02 | -10.95 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.04 | 60.57 | -13.92 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.01 | -6.87 | 6.86 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.02 | -6.79 | 8.33 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.02 | -26.01 | 14.39 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.03 | -30.72 | 17.75 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.01 | 6.72 | 2.06 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.02 | 10.19 | 2.34 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.02 | -13.78 | 9.78 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.03 | -15.43 | 11.99 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.05 | 22.92 | -1.83 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.05 | 23.83 | -1.95 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.00 | 23.89 | -1.96 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.00 | 24.80 | -2.07 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.03 | 22.22 | -1.75 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.02 | 22.52 | -1.78 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.03 | 25.21 | -2.12 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.02 | 25.50 | -2.16 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.05 | 20.53 | -1.64 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.05 | 21.45 | -1.75 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.00 | 21.51 | -1.76 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.00 | 22.42 | -1.88 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.03 | 19.84 | -1.55 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.02 | 20.13 | -1.59 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.03 | 22.82 | -1.93 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.02 | 23.12 | -1.96 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 131 | 0.04 | 21.80 | -1.68 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.04 | 22.50 | -1.77 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.01 | 22.53 | -1.78 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.01 | 23.23 | -1.86 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.03 | 21.28 | -1.62 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.02 | 21.51 | -1.65 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.03 | 23.52 | -1.90 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.02 | 23.75 | -1.93 | 0.00 | 0.00 | 0.00 |
| 27 | 101 | -0.02 | 22.29 | 1.74 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.02 | 23.14 | 1.86 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.01 | 6.92 | -2.04 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.02 | 10.35 | -2.32 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.02 | -13.72 | -9.77 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.03 | -15.45 | -11.99 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.01 | -6.64 | -6.83 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.02 | -6.60 | -8.31 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.02 | -26.02 | -14.39 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.03 | -30.83 | -17.77 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.02 | 68.96 | 18.83 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.04 | 87.89 | 23.77 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.03 | 47.08 | 10.95 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.04 | 60.55 | 13.92 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.00 | 24.94 | 2.09 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.00 | 24.03 | 1.97 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.05 | 23.95 | 1.97 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.05 | 23.04 | 1.85 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.02 | 25.63 | 2.18 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.03 | 25.33 | 2.14 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.02 | 22.65 | 1.80 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.03 | 22.35 | 1.76 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.00 | 22.54 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.00 | 21.63 | 1.78 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.05 | 21.55 | 1.77 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.05 | 20.64 | 1.65 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.02 | 23.23 | 1.98 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.03 | 22.93 | 1.94 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.02 | 20.25 | 1.60 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.03 | 19.95 | 1.57 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.01 | 23.44 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.01 | 22.74 | 1.80 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.04 | 22.70 | 1.80 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.04 | 21.99 | 1.71 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.02 | 23.95 | 1.96 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.03 | 23.72 | 1.93 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.02 | 21.72 | 1.67 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.03 | 21.48 | 1.65 | 0.00 | 0.00 | 0.00 |
| 39 | 101 | -0.02 | 22.09 | -1.72 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.02 | 22.98 | -1.84 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 103 | -0.02 | 68.78 | -18.81 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.04 | 87.76 | -23.75 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.03 | 47.04 | -10.95 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.04 | 60.58 | -13.92 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.01 | -6.84 | 6.85 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.02 | -6.77 | 8.33 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.02 | -26.00 | 14.39 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.03 | -30.72 | 17.75 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.01 | 6.73 | 2.06 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.02 | 10.20 | 2.34 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.02 | -13.84 | 9.79 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.03 | -15.51 | 12.00 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.00 | 23.91 | -1.96 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.00 | 24.82 | -2.07 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.05 | 22.94 | -1.84 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.05 | 23.85 | -1.95 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.02 | 22.53 | -1.79 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.03 | 22.24 | -1.75 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.02 | 25.52 | -2.16 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.03 | 25.23 | -2.13 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.00 | 21.52 | -1.76 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.00 | 22.43 | -1.88 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.05 | 20.55 | -1.64 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.05 | 21.46 | -1.76 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.02 | 20.14 | -1.59 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.03 | 19.85 | -1.55 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.02 | 23.13 | -1.97 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.03 | 22.84 | -1.93 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.01 | 22.55 | -1.78 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.01 | 23.25 | -1.87 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.04 | 21.82 | -1.69 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.04 | 22.53 | -1.78 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.02 | 21.53 | -1.65 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.03 | 21.30 | -1.62 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.02 | 23.77 | -1.93 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.03 | 23.55 | -1.90 | 0.00 | 0.00 | 0.00 |
| 40 | 101 | -0.92 | 15.43 | 1.24 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.91 | 16.23 | 1.34 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.74 | 3.75 | -1.67 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.91 | 6.29 | -1.88 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.68 | -11.74 | -7.55 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.84 | -13.07 | -9.22 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.72 | -6.74 | -5.35 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.88 | -6.83 | -6.47 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.65 | -21.01 | -11.06 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.80 | -24.66 | -13.62 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.79 | 51.90 | 14.39 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.98 | 66.47 | 18.19 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 113 | -0.75 | 35.32 | 8.38 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.92 | 45.75 | 10.69 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.87 | 17.39 | 1.49 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.87 | 16.61 | 1.40 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.94 | 17.45 | 1.50 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.94 | 16.68 | 1.40 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.90 | 18.29 | 1.61 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.92 | 18.31 | 1.61 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.90 | 15.75 | 1.29 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.92 | 15.77 | 1.29 | 0.00 | 0.00 | 0.00 |
| | 123 | -0.78 | 15.69 | 1.35 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.78 | 14.91 | 1.25 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.85 | 15.75 | 1.36 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.85 | 14.97 | 1.26 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.81 | 16.59 | 1.46 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.83 | 16.61 | 1.47 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.81 | 14.05 | 1.14 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.83 | 14.07 | 1.15 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.89 | 16.11 | 1.32 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.89 | 15.51 | 1.25 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.94 | 16.15 | 1.33 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.94 | 15.55 | 1.26 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.91 | 16.78 | 1.41 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.93 | 16.79 | 1.41 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.91 | 14.87 | 1.17 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.93 | 14.89 | 1.17 | 0.00 | 0.00 | 0.00 |
| 52 | 101 | -0.92 | 15.32 | -1.22 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.91 | 16.15 | -1.33 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.79 | 51.80 | -14.37 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.98 | 66.41 | -18.19 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.75 | 35.30 | -8.38 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.92 | 45.78 | -10.69 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.72 | -6.86 | 5.36 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.88 | -6.91 | 6.48 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.65 | -20.98 | 11.06 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.80 | -24.56 | 13.60 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.74 | 3.63 | 1.69 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.91 | 6.20 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.68 | -11.81 | 7.56 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.84 | -13.10 | 9.23 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.87 | 16.56 | -1.39 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.87 | 17.34 | -1.49 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.94 | 16.62 | -1.40 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.94 | 17.40 | -1.50 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.90 | 15.70 | -1.28 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.92 | 15.72 | -1.28 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.90 | 18.24 | -1.60 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.92 | 18.26 | -1.60 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 123 | -0.78 | 14.86 | -1.24 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.78 | 15.64 | -1.34 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.85 | 14.92 | -1.25 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.85 | 15.70 | -1.35 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.81 | 14.00 | -1.14 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.83 | 14.02 | -1.14 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.81 | 16.54 | -1.46 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.83 | 16.56 | -1.46 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.89 | 15.41 | -1.24 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.89 | 16.01 | -1.31 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.94 | 15.46 | -1.24 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.94 | 16.06 | -1.32 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.91 | 14.77 | -1.16 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.93 | 14.79 | -1.16 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.91 | 16.68 | -1.40 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.93 | 16.69 | -1.40 | 0.00 | 0.00 | 0.00 |
| 96 | 101 | 3.25 | 70.18 | -13.12 | 0.00 | 0.00 | 0.00 |
| | 102 | 2.51 | 57.16 | -10.78 | 0.00 | 0.00 | 0.00 |
| | 103 | 3.83 | 71.13 | -5.22 | 0.00 | 0.00 | 0.00 |
| | 104 | 3.32 | 62.86 | -1.86 | 0.00 | 0.00 | 0.00 |
| | 105 | 1.64 | 35.40 | -1.47 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.58 | 18.20 | 2.83 | 0.00 | 0.00 | 0.00 |
| | 107 | 3.81 | 77.59 | -14.18 | 0.00 | 0.00 | 0.00 |
| | 108 | 3.29 | 70.93 | -13.07 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.41 | 17.88 | -2.77 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.95 | -3.70 | 1.20 | 0.00 | 0.00 | 0.00 |
| | 111 | 1.42 | 41.84 | -16.39 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.30 | 26.25 | -15.82 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.15 | 17.92 | -8.50 | 0.00 | 0.00 | 0.00 |
| | 114 | -1.27 | -3.65 | -5.97 | 0.00 | 0.00 | 0.00 |
| | 115 | -4.39 | 43.61 | -9.16 | 0.00 | 0.00 | 0.00 |
| | 116 | -4.40 | 43.69 | -7.78 | 0.00 | 0.00 | 0.00 |
| | 117 | 7.96 | 44.57 | -9.13 | 0.00 | 0.00 | 0.00 |
| | 118 | 7.95 | 44.65 | -7.74 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.10 | 43.85 | -10.72 | 0.00 | 0.00 | 0.00 |
| | 120 | 3.67 | 44.14 | -10.71 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.11 | 44.12 | -6.19 | 0.00 | 0.00 | 0.00 |
| | 122 | 3.66 | 44.42 | -6.18 | 0.00 | 0.00 | 0.00 |
| | 123 | -4.57 | 39.20 | -8.32 | 0.00 | 0.00 | 0.00 |
| | 124 | -4.58 | 39.28 | -6.93 | 0.00 | 0.00 | 0.00 |
| | 125 | 7.78 | 40.16 | -8.28 | 0.00 | 0.00 | 0.00 |
| | 126 | 7.77 | 40.24 | -6.90 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.28 | 39.43 | -9.88 | 0.00 | 0.00 | 0.00 |
| | 128 | 3.49 | 39.73 | -9.87 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.29 | 39.71 | -5.35 | 0.00 | 0.00 | 0.00 |
| | 130 | 3.48 | 40.00 | -5.33 | 0.00 | 0.00 | 0.00 |
| | 131 | -1.75 | 63.28 | -12.50 | 0.00 | 0.00 | 0.00 |
| | 132 | -1.75 | 63.34 | -11.43 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 133 | 7.51 | 64.00 | -12.47 | 0.00 | 0.00 | 0.00 |
| | 134 | 7.51 | 64.06 | -11.40 | 0.00 | 0.00 | 0.00 |
| | 135 | 1.43 | 63.45 | -13.65 | 0.00 | 0.00 | 0.00 |
| | 136 | 4.34 | 63.68 | -13.65 | 0.00 | 0.00 | 0.00 |
| | 137 | 1.42 | 63.66 | -10.25 | 0.00 | 0.00 | 0.00 |
| | 138 | 4.33 | 63.89 | -10.25 | 0.00 | 0.00 | 0.00 |
| 97 | 101 | 3.27 | 66.63 | 12.50 | 0.00 | 0.00 | 0.00 |
| | 102 | 2.53 | 54.28 | 10.32 | 0.00 | 0.00 | 0.00 |
| | 103 | 1.42 | 39.36 | 15.35 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.29 | 24.49 | 14.82 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.20 | 16.82 | 8.01 | 0.00 | 0.00 | 0.00 |
| | 106 | -1.23 | -3.69 | 5.65 | 0.00 | 0.00 | 0.00 |
| | 107 | 3.84 | 73.67 | 13.45 | 0.00 | 0.00 | 0.00 |
| | 108 | 3.32 | 67.37 | 12.44 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.41 | 17.01 | 2.73 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.98 | -3.44 | -0.96 | 0.00 | 0.00 | 0.00 |
| | 111 | 3.86 | 67.86 | 5.27 | 0.00 | 0.00 | 0.00 |
| | 112 | 3.35 | 60.12 | 2.22 | 0.00 | 0.00 | 0.00 |
| | 113 | 1.59 | 33.83 | 1.65 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.51 | 17.57 | -2.30 | 0.00 | 0.00 | 0.00 |
| | 115 | -3.95 | 41.52 | 7.47 | 0.00 | 0.00 | 0.00 |
| | 116 | -3.95 | 41.43 | 8.83 | 0.00 | 0.00 | 0.00 |
| | 117 | 7.52 | 42.41 | 7.44 | 0.00 | 0.00 | 0.00 |
| | 118 | 7.53 | 42.32 | 8.80 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.03 | 41.93 | 5.92 | 0.00 | 0.00 | 0.00 |
| | 120 | 3.54 | 42.20 | 5.91 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.04 | 41.64 | 10.36 | 0.00 | 0.00 | 0.00 |
| | 122 | 3.54 | 41.91 | 10.35 | 0.00 | 0.00 | 0.00 |
| | 123 | -4.13 | 37.33 | 6.66 | 0.00 | 0.00 | 0.00 |
| | 124 | -4.13 | 37.24 | 8.02 | 0.00 | 0.00 | 0.00 |
| | 125 | 7.35 | 38.22 | 6.62 | 0.00 | 0.00 | 0.00 |
| | 126 | 7.35 | 38.13 | 7.98 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.15 | 37.74 | 5.10 | 0.00 | 0.00 | 0.00 |
| | 128 | 3.36 | 38.01 | 5.09 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.14 | 37.44 | 9.55 | 0.00 | 0.00 | 0.00 |
| | 130 | 3.37 | 37.72 | 9.54 | 0.00 | 0.00 | 0.00 |
| | 131 | -1.40 | 60.15 | 10.90 | 0.00 | 0.00 | 0.00 |
| | 132 | -1.40 | 60.08 | 11.95 | 0.00 | 0.00 | 0.00 |
| | 133 | 7.20 | 60.82 | 10.87 | 0.00 | 0.00 | 0.00 |
| | 134 | 7.21 | 60.75 | 11.92 | 0.00 | 0.00 | 0.00 |
| | 135 | 1.54 | 60.46 | 9.75 | 0.00 | 0.00 | 0.00 |
| | 136 | 4.25 | 60.67 | 9.74 | 0.00 | 0.00 | 0.00 |
| | 137 | 1.55 | 60.24 | 13.08 | 0.00 | 0.00 | 0.00 |
| | 138 | 4.26 | 60.45 | 13.07 | 0.00 | 0.00 | 0.00 |
| 98 | 101 | 0.01 | 58.82 | -2.93 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.01 | 47.94 | -2.57 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.01 | 60.31 | 4.06 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.01 | 53.62 | 5.80 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 105 | 0.01 | 30.26 | 3.00 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.01 | 16.05 | 4.46 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.01 | 64.98 | -2.73 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.01 | 59.46 | -2.70 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.01 | 15.22 | -0.24 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.01 | -2.74 | 0.41 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.00 | 34.31 | -9.39 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.00 | 21.12 | -11.03 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.00 | 14.70 | -5.48 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.00 | -3.40 | -6.13 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.01 | 37.52 | -2.57 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.01 | 37.61 | -1.13 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.01 | 36.51 | -3.29 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.01 | 36.60 | -1.85 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.01 | 37.06 | -4.46 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.01 | 36.75 | -4.68 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.01 | 37.36 | 0.26 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.01 | 37.05 | 0.04 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.00 | 33.81 | -2.35 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.00 | 33.90 | -0.91 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.01 | 32.80 | -3.07 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.01 | 32.89 | -1.63 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.01 | 33.35 | -4.24 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.01 | 33.05 | -4.46 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.01 | 33.66 | 0.48 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.01 | 33.35 | 0.26 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.01 | 53.73 | -3.04 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.01 | 53.80 | -1.92 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.01 | 52.97 | -3.58 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.01 | 53.04 | -2.46 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.01 | 53.39 | -4.43 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.01 | 53.15 | -4.60 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.01 | 53.62 | -0.89 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.01 | 53.38 | -1.06 | 0.00 | 0.00 | 0.00 |
| 99 | 101 | 0.01 | 55.53 | 2.59 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.01 | 45.27 | 2.32 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.00 | 32.09 | 8.50 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.00 | 19.59 | 10.10 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.00 | 13.71 | 5.10 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.00 | -3.37 | 5.84 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.01 | 61.35 | 2.30 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.01 | 56.17 | 2.35 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.00 | 14.43 | 0.26 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.00 | -2.48 | -0.20 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.01 | 57.20 | -3.73 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.01 | 50.99 | -5.19 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.01 | 28.77 | -2.72 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.01 | 15.45 | -3.93 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 115 | 0.00 | 35.55 | 0.85 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.00 | 35.45 | 2.27 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.01 | 34.58 | 1.85 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.01 | 34.48 | 3.27 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.01 | 35.33 | -0.41 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.01 | 35.03 | -0.10 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.01 | 35.00 | 4.22 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.01 | 34.71 | 4.53 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.00 | 32.05 | 0.65 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.00 | 31.95 | 2.06 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.01 | 31.08 | 1.65 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.01 | 30.98 | 3.06 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.01 | 31.83 | -0.61 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.01 | 31.53 | -0.31 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.01 | 31.50 | 4.02 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.01 | 31.20 | 4.32 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.01 | 50.81 | 1.53 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.01 | 50.73 | 2.63 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.01 | 50.08 | 2.28 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.01 | 50.00 | 3.38 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.01 | 50.64 | 0.60 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.01 | 50.41 | 0.84 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.01 | 50.40 | 4.07 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.01 | 50.17 | 4.31 | 0.00 | 0.00 | 0.00 |
| 100 | 101 | -0.01 | 58.81 | -2.78 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.01 | 47.94 | -2.45 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.01 | 60.30 | 4.21 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.01 | 53.61 | 5.94 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.01 | 30.25 | 3.08 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.01 | 16.06 | 4.53 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.01 | 64.97 | -2.58 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.01 | 59.45 | -2.56 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.01 | 15.23 | -0.19 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.01 | -2.73 | 0.43 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.00 | 34.37 | -9.40 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.00 | 21.20 | -11.08 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.00 | 10.26 | 0.67 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.00 | -8.93 | 1.51 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.01 | 36.50 | -3.20 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.01 | 36.60 | -1.76 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.01 | 37.51 | -2.47 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.01 | 37.61 | -1.03 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.01 | 36.75 | -4.58 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.01 | 37.06 | -4.36 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.01 | 37.05 | 0.13 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.01 | 37.36 | 0.36 | 0.00 | 0.00 | 0.00 |
| | 123 | -0.01 | 32.80 | -2.99 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.01 | 32.89 | -1.55 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 125 | 0.00 | 33.81 | -2.26 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.00 | 33.90 | -0.82 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.01 | 33.04 | -4.37 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.01 | 33.35 | -4.15 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.01 | 33.35 | 0.35 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.01 | 33.66 | 0.57 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.01 | 52.96 | -3.45 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.01 | 53.03 | -2.33 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.01 | 53.72 | -2.90 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.01 | 53.79 | -1.79 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.01 | 53.14 | -4.47 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.01 | 53.38 | -4.30 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.01 | 53.37 | -0.93 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.01 | 53.61 | -0.76 | 0.00 | 0.00 | 0.00 |
| 101 | 101 | -0.01 | 55.53 | 2.66 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.01 | 45.27 | 2.38 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.00 | 32.09 | 8.59 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.00 | 19.59 | 10.18 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.00 | 13.71 | 5.14 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.00 | -3.38 | 5.86 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.01 | 61.35 | 2.39 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.01 | 56.17 | 2.43 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.00 | 14.43 | 0.28 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.00 | -2.48 | -0.21 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.01 | 57.26 | -3.61 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.01 | 51.06 | -5.07 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.01 | 24.70 | -8.53 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.01 | 10.36 | -11.22 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.01 | 34.58 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.01 | 34.48 | 3.31 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.00 | 35.55 | 0.90 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.00 | 35.45 | 2.31 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.01 | 35.03 | -0.06 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.01 | 35.33 | -0.37 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.01 | 34.70 | 4.57 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.01 | 35.00 | 4.27 | 0.00 | 0.00 | 0.00 |
| | 123 | -0.01 | 31.08 | 1.68 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.01 | 30.98 | 3.10 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.00 | 32.05 | 0.69 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.00 | 31.95 | 2.10 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.01 | 31.53 | -0.27 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.01 | 31.82 | -0.58 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.01 | 31.20 | 4.36 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.01 | 31.50 | 4.06 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.01 | 50.07 | 2.34 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.01 | 50.00 | 3.44 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.01 | 50.81 | 1.60 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.01 | 50.73 | 2.69 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 135 | -0.01 | 50.41 | 0.90 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.01 | 50.64 | 0.67 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.01 | 50.17 | 4.37 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.01 | 50.40 | 4.14 | 0.00 | 0.00 | 0.00 |
| 102 | 101 | -3.32 | 70.19 | -13.12 | 0.00 | 0.00 | 0.00 |
| | 102 | -2.57 | 57.16 | -10.78 | 0.00 | 0.00 | 0.00 |
| | 103 | -3.91 | 71.13 | -5.22 | 0.00 | 0.00 | 0.00 |
| | 104 | -3.39 | 62.86 | -1.86 | 0.00 | 0.00 | 0.00 |
| | 105 | -1.68 | 35.40 | -1.47 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.60 | 18.20 | 2.83 | 0.00 | 0.00 | 0.00 |
| | 107 | -3.89 | 77.59 | -14.18 | 0.00 | 0.00 | 0.00 |
| | 108 | -3.37 | 70.94 | -13.07 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.43 | 17.88 | -2.77 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.96 | -3.70 | 1.20 | 0.00 | 0.00 | 0.00 |
| | 111 | -1.45 | 41.91 | -16.48 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.32 | 26.33 | -15.93 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.19 | 13.48 | -2.40 | 0.00 | 0.00 | 0.00 |
| | 114 | 1.26 | -9.20 | 1.66 | 0.00 | 0.00 | 0.00 |
| | 115 | -8.01 | 44.57 | -9.13 | 0.00 | 0.00 | 0.00 |
| | 116 | -8.01 | 44.66 | -7.74 | 0.00 | 0.00 | 0.00 |
| | 117 | 4.36 | 43.61 | -9.16 | 0.00 | 0.00 | 0.00 |
| | 118 | 4.36 | 43.70 | -7.78 | 0.00 | 0.00 | 0.00 |
| | 119 | -3.72 | 44.14 | -10.71 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.06 | 43.85 | -10.72 | 0.00 | 0.00 | 0.00 |
| | 121 | -3.71 | 44.42 | -6.18 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.07 | 44.13 | -6.19 | 0.00 | 0.00 | 0.00 |
| | 123 | -7.83 | 40.16 | -8.28 | 0.00 | 0.00 | 0.00 |
| | 124 | -7.83 | 40.24 | -6.90 | 0.00 | 0.00 | 0.00 |
| | 125 | 4.54 | 39.20 | -8.31 | 0.00 | 0.00 | 0.00 |
| | 126 | 4.55 | 39.28 | -6.93 | 0.00 | 0.00 | 0.00 |
| | 127 | -3.54 | 39.73 | -9.87 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.24 | 39.44 | -9.88 | 0.00 | 0.00 | 0.00 |
| | 129 | -3.53 | 40.01 | -5.34 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.25 | 39.71 | -5.35 | 0.00 | 0.00 | 0.00 |
| | 131 | -7.59 | 64.00 | -12.47 | 0.00 | 0.00 | 0.00 |
| | 132 | -7.59 | 64.07 | -11.40 | 0.00 | 0.00 | 0.00 |
| | 133 | 1.69 | 63.28 | -12.50 | 0.00 | 0.00 | 0.00 |
| | 134 | 1.69 | 63.35 | -11.43 | 0.00 | 0.00 | 0.00 |
| | 135 | -4.41 | 63.69 | -13.64 | 0.00 | 0.00 | 0.00 |
| | 136 | -1.49 | 63.46 | -13.65 | 0.00 | 0.00 | 0.00 |
| | 137 | -4.40 | 63.89 | -10.25 | 0.00 | 0.00 | 0.00 |
| | 138 | -1.48 | 63.66 | -10.25 | 0.00 | 0.00 | 0.00 |
| 103 | 101 | -3.29 | 66.63 | 12.50 | 0.00 | 0.00 | 0.00 |
| | 102 | -2.54 | 54.28 | 10.32 | 0.00 | 0.00 | 0.00 |
| | 103 | -1.44 | 39.36 | 15.35 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.31 | 24.49 | 14.82 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.21 | 16.82 | 8.01 | 0.00 | 0.00 | 0.00 |
| | 106 | 1.23 | -3.69 | 5.65 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN   METE    STRUCTURE TYPE = SPACE
-----------------

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 107 | -3.86 | 73.67 | 13.45 | 0.00 | 0.00 | 0.00 |
| | 108 | -3.33 | 67.37 | 12.44 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.41 | 17.02 | 2.73 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.98 | -3.44 | -0.96 | 0.00 | 0.00 | 0.00 |
| | 111 | -3.87 | 67.92 | 5.36 | 0.00 | 0.00 | 0.00 |
| | 112 | -3.35 | 60.19 | 2.33 | 0.00 | 0.00 | 0.00 |
| | 113 | -1.62 | 29.75 | -4.19 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.53 | 12.48 | -9.60 | 0.00 | 0.00 | 0.00 |
| | 115 | -7.53 | 42.41 | 7.44 | 0.00 | 0.00 | 0.00 |
| | 116 | -7.54 | 42.32 | 8.80 | 0.00 | 0.00 | 0.00 |
| | 117 | 3.94 | 41.52 | 7.47 | 0.00 | 0.00 | 0.00 |
| | 118 | 3.94 | 41.43 | 8.83 | 0.00 | 0.00 | 0.00 |
| | 119 | -3.55 | 42.20 | 5.91 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.04 | 41.93 | 5.92 | 0.00 | 0.00 | 0.00 |
| | 121 | -3.55 | 41.91 | 10.35 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.05 | 41.64 | 10.36 | 0.00 | 0.00 | 0.00 |
| | 123 | -7.35 | 38.22 | 6.63 | 0.00 | 0.00 | 0.00 |
| | 124 | -7.36 | 38.13 | 7.98 | 0.00 | 0.00 | 0.00 |
| | 125 | 4.12 | 37.33 | 6.66 | 0.00 | 0.00 | 0.00 |
| | 126 | 4.12 | 37.24 | 8.02 | 0.00 | 0.00 | 0.00 |
| | 127 | -3.37 | 38.01 | 5.09 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.14 | 37.74 | 5.10 | 0.00 | 0.00 | 0.00 |
| | 129 | -3.37 | 37.72 | 9.54 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.13 | 37.45 | 9.55 | 0.00 | 0.00 | 0.00 |
| | 131 | -7.22 | 60.82 | 10.87 | 0.00 | 0.00 | 0.00 |
| | 132 | -7.22 | 60.75 | 11.92 | 0.00 | 0.00 | 0.00 |
| | 133 | 1.39 | 60.15 | 10.90 | 0.00 | 0.00 | 0.00 |
| | 134 | 1.39 | 60.08 | 11.95 | 0.00 | 0.00 | 0.00 |
| | 135 | -4.27 | 60.67 | 9.74 | 0.00 | 0.00 | 0.00 |
| | 136 | -1.56 | 60.46 | 9.75 | 0.00 | 0.00 | 0.00 |
| | 137 | -4.27 | 60.45 | 13.07 | 0.00 | 0.00 | 0.00 |
| | 138 | -1.56 | 60.24 | 13.08 | 0.00 | 0.00 | 0.00 |
| 104 | 101 | 22.34 | 44.19 | -7.93 | 0.00 | 0.00 | 0.00 |
| | 102 | 17.34 | 37.81 | -7.34 | 0.00 | 0.00 | 0.00 |
| | 103 | 25.70 | 50.61 | -8.97 | 0.00 | 0.00 | 0.00 |
| | 104 | 22.12 | 50.50 | -10.04 | 0.00 | 0.00 | 0.00 |
| | 105 | 10.82 | 42.47 | -8.21 | 0.00 | 0.00 | 0.00 |
| | 106 | 3.52 | 40.32 | -9.08 | 0.00 | 0.00 | 0.00 |
| | 107 | 25.40 | 56.05 | -8.64 | 0.00 | 0.00 | 0.00 |
| | 108 | 21.75 | 57.30 | -9.62 | 0.00 | 0.00 | 0.00 |
| | 109 | 2.52 | 38.06 | -7.52 | 0.00 | 0.00 | 0.00 |
| | 110 | -6.85 | 34.81 | -8.22 | 0.00 | 0.00 | 0.00 |
| | 111 | 11.22 | 2.22 | -1.33 | 0.00 | 0.00 | 0.00 |
| | 112 | 4.02 | -9.99 | -0.49 | 0.00 | 0.00 | 0.00 |
| | 113 | 2.72 | 2.81 | -2.37 | 0.00 | 0.00 | 0.00 |
| | 114 | -6.60 | -9.26 | -1.79 | 0.00 | 0.00 | 0.00 |
| | 115 | 2.48 | 30.73 | -7.95 | 0.00 | 0.00 | 0.00 |
| | 116 | 2.47 | 31.51 | -5.63 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN   METE    STRUCTURE TYPE = SPACE
-----------------

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 117 | 22.21 | 31.34 | -7.88 | 0.00 | 0.00 | 0.00 |
| | 118 | 22.20 | 32.13 | -5.56 | 0.00 | 0.00 | 0.00 |
| | 119 | 9.35 | 30.05 | -10.56 | 0.00 | 0.00 | 0.00 |
| | 120 | 15.38 | 30.24 | -10.54 | 0.00 | 0.00 | 0.00 |
| | 121 | 9.31 | 32.61 | -2.97 | 0.00 | 0.00 | 0.00 |
| | 122 | 15.33 | 32.80 | -2.94 | 0.00 | 0.00 | 0.00 |
| | 123 | 1.25 | 27.58 | -7.27 | 0.00 | 0.00 | 0.00 |
| | 124 | 1.24 | 28.37 | -4.95 | 0.00 | 0.00 | 0.00 |
| | 125 | 20.98 | 28.20 | -7.20 | 0.00 | 0.00 | 0.00 |
| | 126 | 20.97 | 28.98 | -4.88 | 0.00 | 0.00 | 0.00 |
| | 127 | 8.11 | 26.91 | -9.89 | 0.00 | 0.00 | 0.00 |
| | 128 | 14.14 | 27.10 | -9.86 | 0.00 | 0.00 | 0.00 |
| | 129 | 8.07 | 29.47 | -2.29 | 0.00 | 0.00 | 0.00 |
| | 130 | 14.10 | 29.66 | -2.27 | 0.00 | 0.00 | 0.00 |
| | 131 | 12.45 | 40.46 | -8.56 | 0.00 | 0.00 | 0.00 |
| | 132 | 12.44 | 41.07 | -6.77 | 0.00 | 0.00 | 0.00 |
| | 133 | 27.25 | 40.93 | -8.51 | 0.00 | 0.00 | 0.00 |
| | 134 | 27.24 | 41.53 | -6.71 | 0.00 | 0.00 | 0.00 |
| | 135 | 17.53 | 39.96 | -10.49 | 0.00 | 0.00 | 0.00 |
| | 136 | 22.19 | 40.11 | -10.48 | 0.00 | 0.00 | 0.00 |
| | 137 | 17.50 | 41.89 | -4.80 | 0.00 | 0.00 | 0.00 |
| | 138 | 22.16 | 42.03 | -4.78 | 0.00 | 0.00 | 0.00 |
| 105 | 101 | 22.47 | 45.52 | 8.24 | 0.00 | 0.00 | 0.00 |
| | 102 | 17.44 | 38.86 | 7.58 | 0.00 | 0.00 | 0.00 |
| | 103 | 11.29 | 2.91 | 1.51 | 0.00 | 0.00 | 0.00 |
| | 104 | 4.05 | -9.69 | 0.56 | 0.00 | 0.00 | 0.00 |
| | 105 | 2.71 | 3.02 | 2.39 | 0.00 | 0.00 | 0.00 |
| | 106 | -6.67 | -9.55 | 1.66 | 0.00 | 0.00 | 0.00 |
| | 107 | 25.55 | 57.55 | 9.03 | 0.00 | 0.00 | 0.00 |
| | 108 | 21.87 | 58.61 | 9.96 | 0.00 | 0.00 | 0.00 |
| | 109 | 2.52 | 38.28 | 7.57 | 0.00 | 0.00 | 0.00 |
| | 110 | -6.91 | 34.52 | 8.14 | 0.00 | 0.00 | 0.00 |
| | 111 | 25.85 | 52.11 | 9.26 | 0.00 | 0.00 | 0.00 |
| | 112 | 22.24 | 51.81 | 10.25 | 0.00 | 0.00 | 0.00 |
| | 113 | 10.88 | 43.14 | 8.34 | 0.00 | 0.00 | 0.00 |
| | 114 | 3.54 | 40.59 | 9.10 | 0.00 | 0.00 | 0.00 |
| | 115 | 2.31 | 32.27 | 5.77 | 0.00 | 0.00 | 0.00 |
| | 116 | 2.32 | 31.49 | 8.13 | 0.00 | 0.00 | 0.00 |
| | 117 | 22.49 | 32.90 | 5.70 | 0.00 | 0.00 | 0.00 |
| | 118 | 22.50 | 32.12 | 8.06 | 0.00 | 0.00 | 0.00 |
| | 119 | 9.31 | 33.38 | 3.07 | 0.00 | 0.00 | 0.00 |
| | 120 | 15.47 | 33.57 | 3.05 | 0.00 | 0.00 | 0.00 |
| | 121 | 9.34 | 30.82 | 10.78 | 0.00 | 0.00 | 0.00 |
| | 122 | 15.51 | 31.01 | 10.76 | 0.00 | 0.00 | 0.00 |
| | 123 | 1.07 | 29.05 | 5.08 | 0.00 | 0.00 | 0.00 |
| | 124 | 1.08 | 28.27 | 7.43 | 0.00 | 0.00 | 0.00 |
| | 125 | 21.25 | 29.68 | 5.01 | 0.00 | 0.00 | 0.00 |
| | 126 | 21.26 | 28.90 | 7.37 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 127 | 8.06 | 30.16 | 2.38 | 0.00 | 0.00 | 0.00 |
| | 128 | 14.23 | 30.35 | 2.36 | 0.00 | 0.00 | 0.00 |
| | 129 | 8.10 | 27.60 | 10.09 | 0.00 | 0.00 | 0.00 |
| | 130 | 14.27 | 27.79 | 10.07 | 0.00 | 0.00 | 0.00 |
| | 131 | 12.38 | 42.26 | 7.02 | 0.00 | 0.00 | 0.00 |
| | 132 | 12.39 | 41.65 | 8.84 | 0.00 | 0.00 | 0.00 |
| | 133 | 27.52 | 42.73 | 6.97 | 0.00 | 0.00 | 0.00 |
| | 134 | 27.53 | 42.13 | 8.79 | 0.00 | 0.00 | 0.00 |
| | 135 | 17.56 | 43.08 | 5.03 | 0.00 | 0.00 | 0.00 |
| | 136 | 22.32 | 43.23 | 5.01 | 0.00 | 0.00 | 0.00 |
| | 137 | 17.59 | 41.16 | 10.81 | 0.00 | 0.00 | 0.00 |
| | 138 | 22.35 | 41.31 | 10.79 | 0.00 | 0.00 | 0.00 |
| 106 | 101 | 0.50 | 52.43 | -13.66 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.74 | 45.13 | -12.32 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.03 | 59.87 | -14.13 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.51 | 60.24 | -14.98 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.72 | 52.34 | -11.95 | 0.00 | 0.00 | 0.00 |
| | 106 | 1.37 | 50.82 | -12.25 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.04 | 67.33 | -14.90 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.52 | 69.56 | -15.94 | 0.00 | 0.00 | 0.00 |
| | 109 | 1.10 | 48.42 | -10.83 | 0.00 | 0.00 | 0.00 |
| | 110 | 1.85 | 45.92 | -10.85 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.74 | 0.62 | -4.60 | 0.00 | 0.00 | 0.00 |
| | 112 | 1.39 | -13.83 | -3.06 | 0.00 | 0.00 | 0.00 |
| | 113 | 1.13 | 2.89 | -4.53 | 0.00 | 0.00 | 0.00 |
| | 114 | 1.88 | -10.99 | -2.98 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.14 | 38.05 | -10.75 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.14 | 38.99 | -7.97 | 0.00 | 0.00 | 0.00 |
| | 117 | 1.81 | 36.66 | -13.99 | 0.00 | 0.00 | 0.00 |
| | 118 | 1.81 | 37.61 | -11.21 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.72 | 36.50 | -15.02 | 0.00 | 0.00 | 0.00 |
| | 120 | 1.23 | 36.08 | -16.01 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.72 | 39.58 | -5.95 | 0.00 | 0.00 | 0.00 |
| | 122 | 1.23 | 39.16 | -6.94 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.04 | 34.27 | -9.65 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.04 | 35.21 | -6.88 | 0.00 | 0.00 | 0.00 |
| | 125 | 1.71 | 32.88 | -12.89 | 0.00 | 0.00 | 0.00 |
| | 126 | 1.71 | 33.82 | -10.12 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.62 | 32.72 | -13.92 | 0.00 | 0.00 | 0.00 |
| | 128 | 1.13 | 32.29 | -14.91 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.62 | 35.80 | -4.85 | 0.00 | 0.00 | 0.00 |
| | 130 | 1.13 | 35.37 | -5.84 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.01 | 48.94 | -12.85 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.01 | 49.66 | -10.71 | 0.00 | 0.00 | 0.00 |
| | 133 | 1.25 | 47.90 | -15.28 | 0.00 | 0.00 | 0.00 |
| | 134 | 1.25 | 48.62 | -13.14 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.42 | 47.79 | -16.01 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.82 | 47.46 | -16.78 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 137 | 0.42 | 50.10 | -9.21 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.82 | 49.77 | -9.97 | 0.00 | 0.00 | 0.00 |
| 107 | 101 | 0.55 | 54.09 | 13.69 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.78 | 46.45 | 12.33 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.76 | 1.53 | 4.51 | 0.00 | 0.00 | 0.00 |
| | 104 | 1.41 | -13.36 | 2.91 | 0.00 | 0.00 | 0.00 |
| | 105 | 1.15 | 3.22 | 4.45 | 0.00 | 0.00 | 0.00 |
| | 106 | 1.89 | -11.25 | 2.84 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.09 | 69.17 | 14.98 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.57 | 71.19 | 16.00 | 0.00 | 0.00 | 0.00 |
| | 109 | 1.12 | 48.74 | 10.83 | 0.00 | 0.00 | 0.00 |
| | 110 | 1.86 | 45.65 | 10.80 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.08 | 61.67 | 14.23 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.56 | 61.81 | 15.06 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.75 | 53.18 | 11.96 | 0.00 | 0.00 | 0.00 |
| | 114 | 1.40 | 51.20 | 12.22 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.15 | 39.96 | 8.02 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.15 | 39.02 | 10.83 | 0.00 | 0.00 | 0.00 |
| | 117 | 1.86 | 38.61 | 11.11 | 0.00 | 0.00 | 0.00 |
| | 118 | 1.86 | 37.67 | 13.92 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.75 | 40.56 | 5.90 | 0.00 | 0.00 | 0.00 |
| | 120 | 1.27 | 40.14 | 6.84 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.75 | 37.48 | 15.09 | 0.00 | 0.00 | 0.00 |
| | 122 | 1.27 | 37.07 | 16.03 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.05 | 36.08 | 6.92 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.05 | 35.14 | 9.73 | 0.00 | 0.00 | 0.00 |
| | 125 | 1.76 | 34.73 | 10.01 | 0.00 | 0.00 | 0.00 |
| | 126 | 1.76 | 33.79 | 12.82 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.65 | 36.67 | 4.80 | 0.00 | 0.00 | 0.00 |
| | 128 | 1.17 | 36.26 | 5.74 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.65 | 33.60 | 13.99 | 0.00 | 0.00 | 0.00 |
| | 130 | 1.17 | 33.19 | 14.94 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.02 | 51.14 | 10.77 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.02 | 50.41 | 12.94 | 0.00 | 0.00 | 0.00 |
| | 133 | 1.31 | 50.12 | 13.08 | 0.00 | 0.00 | 0.00 |
| | 134 | 1.31 | 49.40 | 15.26 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.46 | 51.58 | 9.20 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.87 | 51.26 | 9.93 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.46 | 49.27 | 16.09 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.87 | 48.96 | 16.82 | 0.00 | 0.00 | 0.00 |
| 108 | 101 | -0.50 | 52.43 | -13.72 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.73 | 45.13 | -12.37 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.03 | 59.87 | -14.19 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.50 | 60.24 | -15.04 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.72 | 52.34 | -11.98 | 0.00 | 0.00 | 0.00 |
| | 106 | -1.37 | 50.83 | -12.27 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.03 | 67.33 | -14.96 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.51 | 69.56 | -16.00 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 109 | -1.10 | 48.43 | -10.85 | 0.00 | 0.00 | 0.00 |
| | 110 | -1.85 | 45.94 | -10.86 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.73 | 0.62 | -4.61 | 0.00 | 0.00 | 0.00 |
| | 112 | -1.39 | -13.83 | -3.06 | 0.00 | 0.00 | 0.00 |
| | 113 | -1.12 | 3.41 | -5.70 | 0.00 | 0.00 | 0.00 |
| | 114 | -1.88 | -10.34 | -4.43 | 0.00 | 0.00 | 0.00 |
| | 115 | -1.79 | 36.68 | -14.01 | 0.00 | 0.00 | 0.00 |
| | 116 | -1.79 | 37.62 | -11.24 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.14 | 38.05 | -10.80 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.14 | 38.99 | -8.02 | 0.00 | 0.00 | 0.00 |
| | 119 | -1.22 | 36.08 | -16.05 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.72 | 36.50 | -15.07 | 0.00 | 0.00 | 0.00 |
| | 121 | -1.22 | 39.17 | -6.97 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.72 | 39.59 | -5.99 | 0.00 | 0.00 | 0.00 |
| | 123 | -1.70 | 32.89 | -12.91 | 0.00 | 0.00 | 0.00 |
| | 124 | -1.70 | 33.84 | -10.14 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.05 | 34.27 | -9.69 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.05 | 35.21 | -6.92 | 0.00 | 0.00 | 0.00 |
| | 127 | -1.12 | 32.30 | -14.95 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.62 | 32.72 | -13.96 | 0.00 | 0.00 | 0.00 |
| | 129 | -1.12 | 35.38 | -5.87 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.62 | 35.80 | -4.88 | 0.00 | 0.00 | 0.00 |
| | 131 | -1.23 | 47.91 | -15.32 | 0.00 | 0.00 | 0.00 |
| | 132 | -1.23 | 48.63 | -13.18 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.00 | 48.94 | -12.91 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.00 | 49.66 | -10.76 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.81 | 47.47 | -16.83 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.42 | 47.79 | -16.07 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.81 | 49.78 | -10.02 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.42 | 50.10 | -9.26 | 0.00 | 0.00 | 0.00 |
| 109 | 101 | -0.55 | 54.06 | 13.76 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.78 | 46.43 | 12.38 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.76 | 1.51 | 4.59 | 0.00 | 0.00 | 0.00 |
| | 104 | -1.41 | -13.38 | 2.98 | 0.00 | 0.00 | 0.00 |
| | 105 | -1.15 | 3.21 | 4.49 | 0.00 | 0.00 | 0.00 |
| | 106 | -1.89 | -11.26 | 2.86 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.09 | 69.14 | 15.06 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.57 | 71.16 | 16.07 | 0.00 | 0.00 | 0.00 |
| | 109 | -1.12 | 48.73 | 10.84 | 0.00 | 0.00 | 0.00 |
| | 110 | -1.86 | 45.65 | 10.80 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.08 | 61.65 | 14.25 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.56 | 61.79 | 15.06 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.75 | 53.73 | 12.96 | 0.00 | 0.00 | 0.00 |
| | 114 | -1.40 | 51.89 | 13.45 | 0.00 | 0.00 | 0.00 |
| | 115 | -1.86 | 38.60 | 11.14 | 0.00 | 0.00 | 0.00 |
| | 116 | -1.86 | 37.66 | 13.96 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.15 | 39.94 | 8.06 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.15 | 39.00 | 10.88 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 119 | -1.27 | 40.14 | 6.87 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.75 | 40.55 | 5.93 | 0.00 | 0.00 | 0.00 |
| | 121 | -1.27 | 37.05 | 16.09 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.75 | 37.46 | 15.15 | 0.00 | 0.00 | 0.00 |
| | 123 | -1.76 | 34.72 | 10.04 | 0.00 | 0.00 | 0.00 |
| | 124 | -1.76 | 33.78 | 12.85 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.05 | 36.06 | 6.96 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.05 | 35.12 | 9.77 | 0.00 | 0.00 | 0.00 |
| | 127 | -1.17 | 36.26 | 5.77 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.65 | 36.67 | 4.83 | 0.00 | 0.00 | 0.00 |
| | 129 | -1.17 | 33.17 | 14.99 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.65 | 33.58 | 14.04 | 0.00 | 0.00 | 0.00 |
| | 131 | -1.31 | 50.11 | 13.14 | 0.00 | 0.00 | 0.00 |
| | 132 | -1.31 | 49.38 | 15.32 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.02 | 51.12 | 10.83 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.02 | 50.39 | 13.01 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.87 | 51.24 | 9.98 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.46 | 51.56 | 9.25 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.87 | 48.93 | 16.89 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.46 | 49.25 | 16.17 | 0.00 | 0.00 | 0.00 |
| 110 | 101 | -22.26 | 44.19 | -7.93 | 0.00 | 0.00 | 0.00 |
| | 102 | -17.27 | 37.81 | -7.34 | 0.00 | 0.00 | 0.00 |
| | 103 | -25.60 | 50.61 | -8.97 | 0.00 | 0.00 | 0.00 |
| | 104 | -22.03 | 50.50 | -10.04 | 0.00 | 0.00 | 0.00 |
| | 105 | -10.77 | 42.47 | -8.21 | 0.00 | 0.00 | 0.00 |
| | 106 | -3.49 | 40.32 | -9.08 | 0.00 | 0.00 | 0.00 |
| | 107 | -25.30 | 56.05 | -8.64 | 0.00 | 0.00 | 0.00 |
| | 108 | -21.66 | 57.30 | -9.62 | 0.00 | 0.00 | 0.00 |
| | 109 | -2.50 | 38.06 | -7.52 | 0.00 | 0.00 | 0.00 |
| | 110 | 6.84 | 34.81 | -8.22 | 0.00 | 0.00 | 0.00 |
| | 111 | -11.17 | 2.21 | -1.31 | 0.00 | 0.00 | 0.00 |
| | 112 | -4.00 | -10.00 | -0.46 | 0.00 | 0.00 | 0.00 |
| | 113 | -2.68 | 3.31 | -3.55 | 0.00 | 0.00 | 0.00 |
| | 114 | 6.62 | -8.63 | -3.26 | 0.00 | 0.00 | 0.00 |
| | 115 | -22.09 | 31.34 | -7.88 | 0.00 | 0.00 | 0.00 |
| | 116 | -22.08 | 32.12 | -5.56 | 0.00 | 0.00 | 0.00 |
| | 117 | -2.49 | 30.73 | -7.95 | 0.00 | 0.00 | 0.00 |
| | 118 | -2.48 | 31.51 | -5.63 | 0.00 | 0.00 | 0.00 |
| | 119 | -15.30 | 30.24 | -10.54 | 0.00 | 0.00 | 0.00 |
| | 120 | -9.31 | 30.05 | -10.56 | 0.00 | 0.00 | 0.00 |
| | 121 | -15.26 | 32.80 | -2.95 | 0.00 | 0.00 | 0.00 |
| | 122 | -9.27 | 32.61 | -2.97 | 0.00 | 0.00 | 0.00 |
| | 123 | -20.86 | 28.20 | -7.20 | 0.00 | 0.00 | 0.00 |
| | 124 | -20.85 | 28.98 | -4.88 | 0.00 | 0.00 | 0.00 |
| | 125 | -1.27 | 27.58 | -7.27 | 0.00 | 0.00 | 0.00 |
| | 126 | -1.25 | 28.37 | -4.95 | 0.00 | 0.00 | 0.00 |
| | 127 | -14.07 | 27.09 | -9.86 | 0.00 | 0.00 | 0.00 |
| | 128 | -8.08 | 26.91 | -9.88 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 129 | -14.03 | 29.66 | -2.27 | 0.00 | 0.00 | 0.00 |
| | 130 | -8.04 | 29.47 | -2.29 | 0.00 | 0.00 | 0.00 |
| | 131 | -27.12 | 40.92 | -8.51 | 0.00 | 0.00 | 0.00 |
| | 132 | -27.11 | 41.53 | -6.71 | 0.00 | 0.00 | 0.00 |
| | 133 | -12.42 | 40.46 | -8.56 | 0.00 | 0.00 | 0.00 |
| | 134 | -12.41 | 41.07 | -6.77 | 0.00 | 0.00 | 0.00 |
| | 135 | -22.09 | 40.11 | -10.48 | 0.00 | 0.00 | 0.00 |
| | 136 | -17.46 | 39.96 | -10.49 | 0.00 | 0.00 | 0.00 |
| | 137 | -22.06 | 42.03 | -4.78 | 0.00 | 0.00 | 0.00 |
| | 138 | -17.43 | 41.89 | -4.80 | 0.00 | 0.00 | 0.00 |
| 111 | 101 | -22.47 | 45.52 | 8.24 | 0.00 | 0.00 | 0.00 |
| | 102 | -17.44 | 38.86 | 7.58 | 0.00 | 0.00 | 0.00 |
| | 103 | -11.29 | 2.91 | 1.51 | 0.00 | 0.00 | 0.00 |
| | 104 | -4.05 | -9.69 | 0.56 | 0.00 | 0.00 | 0.00 |
| | 105 | -2.72 | 3.02 | 2.39 | 0.00 | 0.00 | 0.00 |
| | 106 | 6.67 | -9.55 | 1.66 | 0.00 | 0.00 | 0.00 |
| | 107 | -25.55 | 57.55 | 9.03 | 0.00 | 0.00 | 0.00 |
| | 108 | -21.87 | 58.61 | 9.96 | 0.00 | 0.00 | 0.00 |
| | 109 | -2.52 | 38.28 | 7.57 | 0.00 | 0.00 | 0.00 |
| | 110 | 6.91 | 34.52 | 8.14 | 0.00 | 0.00 | 0.00 |
| | 111 | -25.85 | 52.10 | 9.25 | 0.00 | 0.00 | 0.00 |
| | 112 | -22.25 | 51.80 | 10.23 | 0.00 | 0.00 | 0.00 |
| | 113 | -10.87 | 43.69 | 9.34 | 0.00 | 0.00 | 0.00 |
| | 114 | -3.53 | 41.29 | 10.35 | 0.00 | 0.00 | 0.00 |
| | 115 | -22.49 | 32.90 | 5.70 | 0.00 | 0.00 | 0.00 |
| | 116 | -22.50 | 32.12 | 8.06 | 0.00 | 0.00 | 0.00 |
| | 117 | -2.32 | 32.27 | 5.77 | 0.00 | 0.00 | 0.00 |
| | 118 | -2.33 | 31.49 | 8.12 | 0.00 | 0.00 | 0.00 |
| | 119 | -15.47 | 33.57 | 3.06 | 0.00 | 0.00 | 0.00 |
| | 120 | -9.31 | 33.38 | 3.08 | 0.00 | 0.00 | 0.00 |
| | 121 | -15.51 | 31.01 | 10.75 | 0.00 | 0.00 | 0.00 |
| | 122 | -9.35 | 30.82 | 10.77 | 0.00 | 0.00 | 0.00 |
| | 123 | -21.25 | 29.68 | 5.01 | 0.00 | 0.00 | 0.00 |
| | 124 | -21.26 | 28.90 | 7.36 | 0.00 | 0.00 | 0.00 |
| | 125 | -1.08 | 29.05 | 5.08 | 0.00 | 0.00 | 0.00 |
| | 126 | -1.09 | 28.27 | 7.43 | 0.00 | 0.00 | 0.00 |
| | 127 | -14.23 | 30.35 | 2.37 | 0.00 | 0.00 | 0.00 |
| | 128 | -8.07 | 30.16 | 2.39 | 0.00 | 0.00 | 0.00 |
| | 129 | -14.27 | 27.79 | 10.06 | 0.00 | 0.00 | 0.00 |
| | 130 | -8.11 | 27.60 | 10.08 | 0.00 | 0.00 | 0.00 |
| | 131 | -27.52 | 42.73 | 6.97 | 0.00 | 0.00 | 0.00 |
| | 132 | -27.53 | 42.13 | 8.79 | 0.00 | 0.00 | 0.00 |
| | 133 | -12.39 | 42.26 | 7.03 | 0.00 | 0.00 | 0.00 |
| | 134 | -12.40 | 41.65 | 8.84 | 0.00 | 0.00 | 0.00 |
| | 135 | -22.33 | 43.22 | 5.02 | 0.00 | 0.00 | 0.00 |
| | 136 | -17.56 | 43.08 | 5.03 | 0.00 | 0.00 | 0.00 |
| | 137 | -22.35 | 41.31 | 10.79 | 0.00 | 0.00 | 0.00 |
| | 138 | -17.59 | 41.16 | 10.80 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| 112 | 101 | -0.01 | 2.30 | 0.41 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.02 | 2.12 | 1.37 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.03 | 5.07 | -5.31 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.00 | 5.98 | -4.71 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.04 | 11.69 | -10.32 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.10 | 14.25 | -10.97 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.02 | 9.27 | -9.30 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.01 | 11.24 | -9.70 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.09 | 15.65 | -13.19 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.15 | 19.21 | -14.56 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.01 | -13.83 | 20.02 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.03 | -17.65 | 26.96 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.04 | -6.93 | 13.04 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.09 | -9.01 | 18.23 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.01 | 2.02 | 1.77 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.01 | 1.84 | 2.91 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.06 | 2.04 | 1.76 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.06 | 1.86 | 2.90 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.03 | 2.23 | 0.47 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.04 | 2.24 | 0.47 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.03 | 1.65 | 4.20 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.04 | 1.65 | 4.20 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.01 | 1.83 | 1.53 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.01 | 1.65 | 2.67 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.05 | 1.85 | 1.53 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.05 | 1.67 | 2.67 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.03 | 2.04 | 0.24 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.04 | 2.04 | 0.23 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.03 | 1.45 | 3.97 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.04 | 1.46 | 3.97 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.01 | 2.27 | 0.45 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.01 | 2.14 | 1.33 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.02 | 2.29 | 0.45 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.02 | 2.15 | 1.33 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.00 | 2.43 | -0.51 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.01 | 2.43 | -0.51 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.00 | 1.99 | 2.29 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.01 | 1.99 | 2.29 | 0.00 | 0.00 | 0.00 |
| 113 | 101 | -0.01 | 2.34 | -0.12 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.01 | 2.15 | -1.15 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.01 | -13.79 | -19.80 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.03 | -17.62 | -26.82 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.04 | -6.91 | -12.99 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.09 | -9.02 | -18.30 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.02 | 9.32 | 9.63 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.01 | 11.27 | 9.97 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.08 | 15.64 | 13.19 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.15 | 19.17 | 14.42 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 111 | -0.03 | 5.11 | 5.59 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.00 | 6.01 | 4.92 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.04 | 11.70 | 10.41 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.09 | 14.24 | 10.95 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.01 | 1.87 | -2.77 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.01 | 2.05 | -1.60 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.06 | 1.88 | -2.76 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.06 | 2.06 | -1.60 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.03 | 1.67 | -4.09 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.04 | 1.67 | -4.08 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.03 | 2.26 | -0.28 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.04 | 2.26 | -0.28 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.01 | 1.67 | -2.55 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.01 | 1.85 | -1.39 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.05 | 1.69 | -2.54 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.05 | 1.87 | -1.38 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.02 | 1.47 | -3.87 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.04 | 1.48 | -3.87 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.02 | 2.06 | -0.06 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.04 | 2.07 | -0.06 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.01 | 2.17 | -1.08 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.01 | 2.31 | -0.19 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.02 | 2.19 | -1.08 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.02 | 2.33 | -0.18 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.00 | 2.03 | -2.06 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.01 | 2.03 | -2.06 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.00 | 2.47 | 0.79 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.01 | 2.47 | 0.80 | 0.00 | 0.00 | 0.00 |
| 114 | 101 | -0.08 | 3.29 | -0.38 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.08 | 3.04 | 1.13 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.08 | 6.78 | -7.32 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.08 | 7.98 | -6.12 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.06 | 15.42 | -13.37 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.06 | 18.78 | -13.69 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.08 | 12.36 | -13.27 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.08 | 14.96 | -13.56 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.05 | 20.62 | -17.13 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.05 | 25.28 | -18.38 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.05 | -17.77 | 25.00 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.05 | -22.71 | 34.28 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.04 | -8.81 | 16.66 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.04 | -11.51 | 23.85 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.11 | 2.78 | 2.77 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.11 | 2.57 | 4.14 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.03 | 3.01 | 1.14 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.03 | 2.80 | 2.51 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.08 | 3.10 | 0.65 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.06 | 3.17 | 0.15 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 121 | -0.08 | 2.41 | 5.13 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.06 | 2.48 | 4.64 | 0.00 | 0.00 | 0.00 |
| | 123 | -0.10 | 2.50 | 2.51 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.10 | 2.29 | 3.88 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.02 | 2.73 | 0.88 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.02 | 2.52 | 2.25 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.07 | 2.82 | 0.39 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.05 | 2.89 | -0.11 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.07 | 2.13 | 4.87 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.05 | 2.20 | 4.37 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.11 | 3.16 | 0.45 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.11 | 3.00 | 1.51 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.05 | 3.33 | -0.77 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.05 | 3.17 | 0.29 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.09 | 3.40 | -1.12 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.07 | 3.45 | -1.50 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.09 | 2.88 | 2.25 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.07 | 2.93 | 1.86 | 0.00 | 0.00 | 0.00 |
| 115 | 101 | -0.08 | 3.31 | 0.55 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.07 | 3.06 | -1.01 | 0.00 | 0.00 | 0.00 |
| | 103 | -0.05 | -17.74 | -24.89 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.05 | -22.68 | -34.23 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.04 | -8.80 | -16.64 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.04 | -11.51 | -23.92 | 0.00 | 0.00 | 0.00 |
| | 107 | -0.07 | 12.38 | 13.46 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.08 | 14.97 | 13.70 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.05 | 20.60 | 17.08 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.04 | 25.24 | 18.24 | 0.00 | 0.00 | 0.00 |
| | 111 | -0.07 | 6.81 | 7.52 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.08 | 8.00 | 6.28 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.06 | 15.41 | 13.40 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.06 | 18.75 | 13.63 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.11 | 2.59 | -4.04 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.11 | 2.80 | -2.64 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.03 | 2.81 | -2.49 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.02 | 3.02 | -1.10 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.08 | 2.42 | -5.08 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.05 | 2.49 | -4.61 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.08 | 3.12 | -0.52 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.05 | 3.19 | -0.05 | 0.00 | 0.00 | 0.00 |
| | 123 | -0.10 | 2.31 | -3.78 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.10 | 2.52 | -2.39 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.02 | 2.52 | -2.23 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.02 | 2.74 | -0.84 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.07 | 2.14 | -4.83 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.05 | 2.21 | -4.35 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.07 | 2.84 | -0.27 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.05 | 2.91 | 0.21 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 131 | -0.11 | 3.02 | -1.35 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.11 | 3.18 | -0.27 | 0.00 | 0.00 | 0.00 |
| | 133 | -0.05 | 3.18 | -0.19 | 0.00 | 0.00 | 0.00 |
| | 134 | -0.05 | 3.35 | 0.89 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.09 | 2.90 | -2.12 | 0.00 | 0.00 | 0.00 |
| | 136 | -0.07 | 2.95 | -1.76 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.09 | 3.42 | 1.30 | 0.00 | 0.00 | 0.00 |
| | 138 | -0.07 | 3.47 | 1.67 | 0.00 | 0.00 | 0.00 |
| 116 | 101 | 0.08 | 3.29 | -0.42 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.08 | 3.04 | 1.10 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.08 | 6.78 | -7.36 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.08 | 7.98 | -6.16 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.06 | 15.42 | -13.40 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.06 | 18.78 | -13.71 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.08 | 12.36 | -13.31 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.08 | 14.96 | -13.59 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.05 | 20.62 | -17.14 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.05 | 25.28 | -18.39 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.05 | -17.77 | 24.99 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.05 | -22.71 | 34.27 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.04 | -8.72 | 15.87 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.04 | -11.39 | 22.88 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.03 | 3.01 | 1.12 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.03 | 2.80 | 2.49 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.11 | 2.79 | 2.74 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.11 | 2.57 | 4.11 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.06 | 3.18 | 0.13 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.08 | 3.11 | 0.62 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.06 | 2.48 | 4.61 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.08 | 2.41 | 5.11 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.02 | 2.73 | 0.86 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.02 | 2.52 | 2.23 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.10 | 2.51 | 2.48 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.10 | 2.30 | 3.85 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.05 | 2.90 | -0.13 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.08 | 2.83 | 0.36 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.05 | 2.20 | 4.35 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.08 | 2.13 | 4.85 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.05 | 3.33 | -0.80 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.05 | 3.17 | 0.26 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.11 | 3.16 | 0.42 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.11 | 3.00 | 1.48 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.07 | 3.45 | -1.53 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.09 | 3.40 | -1.15 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.07 | 2.93 | 1.83 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.09 | 2.88 | 2.21 | 0.00 | 0.00 | 0.00 |
| 117 | 101 | 0.08 | 3.32 | 0.58 | 0.00 | 0.00 | 0.00 |
| | 102 | 0.07 | 3.06 | -0.98 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 103 | 0.05 | -17.73 | -24.85 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.05 | -22.67 | -34.19 | 0.00 | 0.00 | 0.00 |
| | 105 | 0.04 | -8.80 | -16.62 | 0.00 | 0.00 | 0.00 |
| | 106 | 0.04 | -11.51 | -23.91 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.07 | 12.39 | 13.49 | 0.00 | 0.00 | 0.00 |
| | 108 | 0.08 | 14.97 | 13.74 | 0.00 | 0.00 | 0.00 |
| | 109 | 0.05 | 20.60 | 17.09 | 0.00 | 0.00 | 0.00 |
| | 110 | 0.04 | 25.24 | 18.23 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.07 | 6.81 | 7.52 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.08 | 8.00 | 6.27 | 0.00 | 0.00 | 0.00 |
| | 113 | 0.06 | 15.49 | 14.10 | 0.00 | 0.00 | 0.00 |
| | 114 | 0.06 | 18.85 | 14.49 | 0.00 | 0.00 | 0.00 |
| | 115 | 0.02 | 2.81 | -2.47 | 0.00 | 0.00 | 0.00 |
| | 116 | 0.02 | 3.02 | -1.08 | 0.00 | 0.00 | 0.00 |
| | 117 | 0.11 | 2.59 | -4.02 | 0.00 | 0.00 | 0.00 |
| | 118 | 0.11 | 2.80 | -2.62 | 0.00 | 0.00 | 0.00 |
| | 119 | 0.05 | 2.49 | -4.60 | 0.00 | 0.00 | 0.00 |
| | 120 | 0.08 | 2.42 | -5.07 | 0.00 | 0.00 | 0.00 |
| | 121 | 0.05 | 3.19 | -0.02 | 0.00 | 0.00 | 0.00 |
| | 122 | 0.08 | 3.12 | -0.50 | 0.00 | 0.00 | 0.00 |
| | 123 | 0.02 | 2.53 | -2.22 | 0.00 | 0.00 | 0.00 |
| | 124 | 0.02 | 2.74 | -0.82 | 0.00 | 0.00 | 0.00 |
| | 125 | 0.10 | 2.31 | -3.76 | 0.00 | 0.00 | 0.00 |
| | 126 | 0.10 | 2.52 | -2.37 | 0.00 | 0.00 | 0.00 |
| | 127 | 0.05 | 2.21 | -4.34 | 0.00 | 0.00 | 0.00 |
| | 128 | 0.07 | 2.14 | -4.81 | 0.00 | 0.00 | 0.00 |
| | 129 | 0.05 | 2.91 | 0.23 | 0.00 | 0.00 | 0.00 |
| | 130 | 0.07 | 2.84 | -0.24 | 0.00 | 0.00 | 0.00 |
| | 131 | 0.05 | 3.19 | -0.16 | 0.00 | 0.00 | 0.00 |
| | 132 | 0.05 | 3.35 | 0.92 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.11 | 3.02 | -1.32 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.11 | 3.19 | -0.24 | 0.00 | 0.00 | 0.00 |
| | 135 | 0.07 | 2.95 | -1.73 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.09 | 2.90 | -2.10 | 0.00 | 0.00 | 0.00 |
| | 137 | 0.07 | 3.48 | 1.70 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.09 | 3.43 | 1.33 | 0.00 | 0.00 | 0.00 |
| 118 | 101 | 0.01 | 2.30 | 0.41 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.02 | 2.12 | 1.37 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.03 | 5.07 | -5.31 | 0.00 | 0.00 | 0.00 |
| | 104 | 0.00 | 5.98 | -4.71 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.04 | 11.69 | -10.32 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.10 | 14.25 | -10.97 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.02 | 9.27 | -9.30 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.01 | 11.23 | -9.70 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.09 | 15.65 | -13.19 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.15 | 19.21 | -14.56 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.01 | -13.84 | 20.03 | 0.00 | 0.00 | 0.00 |
| | 112 | -0.03 | -17.65 | 26.97 | 0.00 | 0.00 | 0.00 |

SUPPORT REACTIONS -UNIT KN   METE    STRUCTURE TYPE = SPACE
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| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 113 | -0.04 | -6.83 | 12.25 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.09 | -8.90 | 17.24 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.06 | 2.04 | 1.76 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.06 | 1.86 | 2.90 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.01 | 2.02 | 1.77 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.01 | 1.84 | 2.91 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.04 | 2.24 | 0.47 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.03 | 2.23 | 0.47 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.04 | 1.65 | 4.20 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.03 | 1.65 | 4.20 | 0.00 | 0.00 | 0.00 |
| | 123 | -0.05 | 1.85 | 1.53 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.05 | 1.67 | 2.67 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.01 | 1.83 | 1.53 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.01 | 1.65 | 2.67 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.04 | 2.04 | 0.23 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.03 | 2.04 | 0.24 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.04 | 1.46 | 3.97 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.02 | 1.45 | 3.97 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.02 | 2.29 | 0.45 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.02 | 2.15 | 1.33 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.01 | 2.27 | 0.45 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.01 | 2.14 | 1.33 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.01 | 2.43 | -0.51 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.00 | 2.43 | -0.51 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.01 | 1.99 | 2.29 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.00 | 1.99 | 2.29 | 0.00 | 0.00 | 0.00 |
| 119 | 101 | 0.01 | 2.34 | -0.12 | 0.00 | 0.00 | 0.00 |
| | 102 | -0.01 | 2.15 | -1.15 | 0.00 | 0.00 | 0.00 |
| | 103 | 0.01 | -13.79 | -19.80 | 0.00 | 0.00 | 0.00 |
| | 104 | -0.03 | -17.62 | -26.82 | 0.00 | 0.00 | 0.00 |
| | 105 | -0.04 | -6.91 | -12.99 | 0.00 | 0.00 | 0.00 |
| | 106 | -0.09 | -9.02 | -18.30 | 0.00 | 0.00 | 0.00 |
| | 107 | 0.02 | 9.32 | 9.63 | 0.00 | 0.00 | 0.00 |
| | 108 | -0.01 | 11.27 | 9.97 | 0.00 | 0.00 | 0.00 |
| | 109 | -0.08 | 15.64 | 13.19 | 0.00 | 0.00 | 0.00 |
| | 110 | -0.15 | 19.17 | 14.42 | 0.00 | 0.00 | 0.00 |
| | 111 | 0.03 | 5.11 | 5.58 | 0.00 | 0.00 | 0.00 |
| | 112 | 0.00 | 6.01 | 4.90 | 0.00 | 0.00 | 0.00 |
| | 113 | -0.04 | 11.77 | 11.11 | 0.00 | 0.00 | 0.00 |
| | 114 | -0.09 | 14.34 | 11.81 | 0.00 | 0.00 | 0.00 |
| | 115 | -0.06 | 1.88 | -2.76 | 0.00 | 0.00 | 0.00 |
| | 116 | -0.06 | 2.06 | -1.60 | 0.00 | 0.00 | 0.00 |
| | 117 | -0.01 | 1.87 | -2.77 | 0.00 | 0.00 | 0.00 |
| | 118 | -0.01 | 2.05 | -1.61 | 0.00 | 0.00 | 0.00 |
| | 119 | -0.04 | 1.67 | -4.08 | 0.00 | 0.00 | 0.00 |
| | 120 | -0.03 | 1.67 | -4.08 | 0.00 | 0.00 | 0.00 |
| | 121 | -0.04 | 2.26 | -0.28 | 0.00 | 0.00 | 0.00 |
| | 122 | -0.03 | 2.26 | -0.28 | 0.00 | 0.00 | 0.00 |



SUPPORT REACTIONS -UNIT KN   METE    STRUCTURE TYPE = SPACE
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| JOINT | LOAD | FORCE-X | FORCE-Y | FORCE-Z | MOM-X | MOM-Y | MOM Z |
|---|---|---|---|---|---|---|---|
| | 123 | -0.05 | 1.69 | -2.54 | 0.00 | 0.00 | 0.00 |
| | 124 | -0.05 | 1.87 | -1.38 | 0.00 | 0.00 | 0.00 |
| | 125 | -0.01 | 1.67 | -2.55 | 0.00 | 0.00 | 0.00 |
| | 126 | -0.01 | 1.85 | -1.39 | 0.00 | 0.00 | 0.00 |
| | 127 | -0.04 | 1.48 | -3.86 | 0.00 | 0.00 | 0.00 |
| | 128 | -0.02 | 1.47 | -3.86 | 0.00 | 0.00 | 0.00 |
| | 129 | -0.04 | 2.07 | -0.06 | 0.00 | 0.00 | 0.00 |
| | 130 | -0.02 | 2.06 | -0.07 | 0.00 | 0.00 | 0.00 |
| | 131 | -0.02 | 2.19 | -1.08 | 0.00 | 0.00 | 0.00 |
| | 132 | -0.02 | 2.33 | -0.18 | 0.00 | 0.00 | 0.00 |
| | 133 | 0.01 | 2.17 | -1.08 | 0.00 | 0.00 | 0.00 |
| | 134 | 0.01 | 2.31 | -0.19 | 0.00 | 0.00 | 0.00 |
| | 135 | -0.01 | 2.03 | -2.06 | 0.00 | 0.00 | 0.00 |
| | 136 | 0.00 | 2.03 | -2.06 | 0.00 | 0.00 | 0.00 |
| | 137 | -0.01 | 2.47 | 0.79 | 0.00 | 0.00 | 0.00 |
| | 138 | 0.00 | 2.47 | 0.79 | 0.00 | 0.00 | 0.00 |

************** END OF LATEST ANALYSIS RESULT **************

589. LOAD LIST 301 TO 316

590. PRINT JOINT DISPLACEMENTS

JOINT    DISPLACE

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 1 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0003 | 0.0004 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0004 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0008 | -0.0015 | -0.0002 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0006 | -0.0011 | -0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0012 | -0.0021 | -0.0003 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0012 | 0.0023 | 0.0003 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0007 | 0.0012 | 0.0002 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 2 | 301 | 0.0003 | -0.0038 | -0.0011 | 0.0000 | 0.0000 | -0.0005 |
| | 302 | -0.0004 | 0.0006 | 0.0006 | 0.0000 | 0.0000 | 0.0001 |
| | 303 | -0.0008 | 0.0027 | 0.0136 | -0.0001 | -0.0002 | 0.0001 |
| | 304 | -0.0005 | 0.0057 | 0.0440 | -0.0005 | -0.0009 | 0.0000 |
| | 305 | -0.0010 | 0.0050 | 0.0322 | -0.0003 | -0.0006 | 0.0001 |
| | 306 | -0.0004 | 0.0075 | 0.0621 | -0.0007 | -0.0013 | -0.0001 |
| | 307 | 0.0007 | -0.0086 | -0.0670 | 0.0007 | 0.0014 | 0.0000 |
| | 308 | 0.0007 | -0.0051 | -0.0362 | 0.0004 | 0.0007 | -0.0001 |
| | 309 | -0.0174 | 0.0004 | 0.0005 | 0.0000 | 0.0000 | 0.0001 |
| | 310 | -0.0174 | 0.0006 | 0.0005 | 0.0000 | 0.0000 | 0.0001 |
| | 311 | 0.0167 | 0.0004 | 0.0005 | 0.0000 | 0.0000 | 0.0001 |
| | 312 | 0.0167 | 0.0006 | 0.0005 | 0.0000 | 0.0000 | 0.0001 |
| | 313 | -0.0054 | 0.0001 | 0.0004 | 0.0000 | 0.0000 | 0.0001 |
| | 314 | 0.0048 | 0.0001 | 0.0004 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | -0.0054 | 0.0009 | 0.0006 | 0.0000 | 0.0000 | 0.0001 |
| | 316 | 0.0047 | 0.0009 | 0.0006 | 0.0000 | 0.0000 | 0.0001 |
| 3 | 301 | 0.0135 | -0.0028 | -0.0012 | 0.0000 | 0.0002 | 0.0000 |
| | 302 | -0.0159 | 0.0007 | 0.0003 | 0.0000 | -0.0007 | 0.0003 |
| | 303 | -0.0230 | 0.0016 | 0.0015 | -0.0001 | -0.0010 | 0.0004 |
| | 304 | 0.0062 | 0.0015 | 0.0037 | -0.0002 | 0.0002 | -0.0001 |
| | 305 | -0.0193 | 0.0021 | 0.0030 | -0.0002 | -0.0009 | 0.0004 |
| | 306 | 0.0226 | 0.0015 | 0.0050 | -0.0003 | 0.0008 | -0.0003 |
| | 307 | -0.0139 | -0.0021 | -0.0056 | 0.0003 | -0.0005 | 0.0002 |
| | 308 | 0.0052 | -0.0017 | -0.0032 | 0.0002 | 0.0003 | -0.0001 |
| | 309 | -0.0267 | 0.0004 | 0.0002 | 0.0000 | -0.0009 | 0.0002 |
| | 310 | -0.0268 | 0.0007 | 0.0002 | 0.0000 | -0.0009 | 0.0002 |
| | 311 | 0.0013 | 0.0004 | 0.0002 | 0.0000 | -0.0002 | 0.0002 |
| | 312 | 0.0012 | 0.0007 | 0.0002 | 0.0000 | -0.0002 | 0.0002 |
| | 313 | -0.0168 | 0.0001 | 0.0002 | 0.0000 | -0.0006 | 0.0002 |
| | 314 | -0.0085 | 0.0001 | 0.0002 | 0.0000 | -0.0004 | 0.0002 |
| | 315 | -0.0170 | 0.0011 | 0.0002 | 0.0000 | -0.0006 | 0.0002 |
| | 316 | -0.0087 | 0.0011 | 0.0002 | 0.0000 | -0.0004 | 0.0002 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 4 | 301 | 0.0146 | -0.0006 | -0.0001 | -0.0001 | 0.0005 | -0.0009 |
| | 302 | 0.0100 | -0.0003 | -0.0008 | -0.0001 | 0.0002 | -0.0006 |
| | 303 | 0.0159 | -0.0005 | -0.0014 | -0.0001 | 0.0003 | -0.0010 |
| | 304 | 0.0014 | -0.0003 | -0.0003 | 0.0000 | 0.0001 | -0.0001 |
| | 305 | 0.0158 | -0.0006 | -0.0015 | -0.0001 | 0.0003 | -0.0010 |
| | 306 | -0.0066 | -0.0003 | 0.0003 | 0.0000 | 0.0000 | 0.0004 |
| | 307 | 0.0006 | 0.0005 | 0.0003 | 0.0000 | -0.0001 | 0.0000 |
| | 308 | -0.0076 | 0.0004 | 0.0008 | 0.0001 | -0.0002 | 0.0005 |
| | 309 | 0.0145 | -0.0002 | -0.0004 | 0.0000 | 0.0001 | -0.0007 |
| | 310 | 0.0145 | -0.0002 | -0.0009 | 0.0000 | 0.0001 | -0.0007 |
| | 311 | 0.0015 | -0.0002 | -0.0004 | 0.0000 | 0.0002 | -0.0003 |
| | 312 | 0.0015 | -0.0002 | -0.0009 | 0.0000 | 0.0002 | -0.0003 |
| | 313 | 0.0099 | -0.0002 | 0.0001 | 0.0000 | 0.0001 | -0.0006 |
| | 314 | 0.0060 | -0.0002 | 0.0001 | 0.0000 | 0.0002 | -0.0004 |
| | 315 | 0.0099 | -0.0002 | -0.0014 | -0.0001 | 0.0001 | -0.0006 |
| | 316 | 0.0060 | -0.0002 | -0.0014 | -0.0001 | 0.0002 | -0.0004 |
| 5 | 301 | -0.0114 | -0.0153 | -0.0027 | -0.0002 | 0.0000 | 0.0012 |
| | 302 | 0.0064 | -0.0093 | -0.0024 | -0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0086 | -0.0175 | -0.0042 | -0.0002 | 0.0001 | -0.0001 |
| | 304 | -0.0039 | -0.0046 | -0.0010 | 0.0000 | 0.0000 | 0.0003 |
| | 305 | 0.0067 | -0.0161 | -0.0041 | -0.0002 | 0.0001 | 0.0000 |
| | 306 | -0.0110 | 0.0033 | 0.0009 | 0.0000 | -0.0001 | 0.0005 |
| | 307 | 0.0077 | 0.0045 | 0.0009 | 0.0000 | 0.0000 | -0.0005 |
| | 308 | -0.0009 | 0.0099 | 0.0024 | 0.0001 | 0.0000 | -0.0002 |
| | 309 | 0.0197 | -0.0071 | -0.0015 | -0.0001 | 0.0000 | -0.0004 |
| | 310 | 0.0198 | -0.0077 | -0.0022 | -0.0001 | 0.0000 | -0.0004 |
| | 311 | -0.0096 | -0.0071 | -0.0016 | -0.0001 | 0.0001 | 0.0002 |
| | 312 | -0.0095 | -0.0078 | -0.0022 | -0.0001 | 0.0001 | 0.0002 |
| | 313 | 0.0093 | -0.0064 | -0.0008 | -0.0001 | 0.0000 | -0.0002 |
| | 314 | 0.0005 | -0.0064 | -0.0008 | -0.0001 | 0.0000 | 0.0000 |
| | 315 | 0.0096 | -0.0085 | -0.0030 | -0.0001 | 0.0000 | -0.0002 |
| | 316 | 0.0009 | -0.0085 | -0.0030 | -0.0001 | 0.0000 | 0.0000 |
| 6 | 301 | 0.0014 | -0.0022 | -0.0013 | -0.0001 | 0.0000 | -0.0001 |
| | 302 | 0.0011 | -0.0013 | -0.0010 | -0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0017 | -0.0018 | -0.0012 | 0.0000 | 0.0000 | -0.0001 |
| | 304 | 0.0001 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 305 | 0.0018 | -0.0021 | -0.0018 | -0.0001 | 0.0000 | -0.0001 |
| | 306 | -0.0008 | 0.0009 | 0.0006 | 0.0001 | 0.0000 | 0.0001 |
| | 307 | 0.0001 | -0.0003 | -0.0004 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | -0.0005 | 0.0009 | 0.0007 | 0.0000 | 0.0000 | 0.0001 |
| | 309 | 0.0292 | -0.0010 | -0.0003 | -0.0001 | 0.0000 | -0.0004 |
| | 310 | 0.0292 | -0.0010 | -0.0013 | -0.0001 | 0.0000 | -0.0004 |
| | 311 | -0.0274 | -0.0011 | -0.0003 | -0.0001 | 0.0000 | 0.0003 |
| | 312 | -0.0274 | -0.0011 | -0.0014 | -0.0001 | 0.0000 | 0.0003 |
| | 313 | 0.0093 | -0.0010 | 0.0009 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | -0.0076 | -0.0010 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0093 | -0.0010 | -0.0025 | -0.0001 | 0.0000 | -0.0002 |
| | 316 | -0.0075 | -0.0010 | -0.0025 | -0.0001 | 0.0000 | 0.0000 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 7 | 301 | -0.0012 | -0.1451 | 0.0001 | 0.0000 | 0.0000 | 0.0001 |
| | 302 | -0.0006 | -0.0857 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0004 | -0.0692 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0002 | 0.0295 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0010 | -0.1388 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0004 | 0.0592 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | -0.0006 | -0.0713 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0012 | 0.0283 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0276 | -0.0685 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | 0.0276 | -0.0686 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | -0.0286 | -0.0686 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | -0.0286 | -0.0687 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | 0.0079 | -0.0684 | 0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0089 | -0.0685 | 0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0079 | -0.0687 | -0.0016 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | -0.0089 | -0.0687 | -0.0016 | 0.0000 | 0.0000 | 0.0000 |
| 8 | 301 | 0.0012 | -0.0021 | 0.0013 | 0.0001 | 0.0000 | -0.0001 |
| | 302 | 0.0010 | -0.0012 | 0.0011 | 0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0001 | -0.0003 | 0.0003 | 0.0001 | 0.0000 | 0.0000 |
| | 304 | -0.0008 | 0.0008 | -0.0008 | 0.0000 | 0.0000 | 0.0001 |
| | 305 | 0.0016 | -0.0020 | 0.0019 | 0.0001 | 0.0000 | -0.0001 |
| | 306 | -0.0007 | 0.0008 | -0.0006 | -0.0001 | 0.0000 | 0.0001 |
| | 307 | 0.0015 | -0.0017 | 0.0014 | 0.0000 | 0.0000 | -0.0001 |
| | 308 | 0.0004 | 0.0000 | 0.0001 | -0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0271 | -0.0010 | 0.0014 | 0.0001 | 0.0000 | -0.0004 |
| | 310 | 0.0271 | -0.0010 | 0.0004 | 0.0001 | 0.0000 | -0.0004 |
| | 311 | -0.0255 | -0.0010 | 0.0014 | 0.0001 | 0.0000 | 0.0003 |
| | 312 | -0.0255 | -0.0010 | 0.0004 | 0.0001 | 0.0000 | 0.0003 |
| | 313 | 0.0087 | -0.0010 | 0.0025 | 0.0001 | 0.0000 | -0.0002 |
| | 314 | -0.0070 | -0.0010 | 0.0025 | 0.0001 | 0.0000 | 0.0000 |
| | 315 | 0.0086 | -0.0010 | -0.0008 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | -0.0071 | -0.0010 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| 9 | 301 | -0.0092 | -0.0158 | 0.0027 | 0.0002 | 0.0000 | 0.0011 |
| | 302 | 0.0062 | -0.0098 | 0.0025 | 0.0001 | -0.0001 | -0.0001 |
| | 303 | 0.0069 | 0.0043 | -0.0008 | 0.0000 | 0.0000 | -0.0005 |
| | 304 | -0.0012 | 0.0101 | -0.0024 | -0.0001 | 0.0000 | -0.0001 |
| | 305 | 0.0069 | -0.0168 | 0.0042 | 0.0002 | -0.0001 | 0.0000 |
| | 306 | -0.0101 | 0.0037 | -0.0010 | -0.0001 | 0.0001 | 0.0005 |
| | 307 | 0.0086 | -0.0181 | 0.0043 | 0.0002 | -0.0001 | -0.0001 |
| | 308 | -0.0034 | -0.0046 | 0.0010 | 0.0000 | 0.0000 | 0.0003 |
| | 309 | 0.0194 | -0.0081 | 0.0023 | 0.0001 | 0.0000 | -0.0004 |
| | 310 | 0.0193 | -0.0075 | 0.0016 | 0.0001 | 0.0000 | -0.0004 |
| | 311 | -0.0094 | -0.0081 | 0.0023 | 0.0001 | -0.0001 | 0.0002 |
| | 312 | -0.0095 | -0.0075 | 0.0016 | 0.0001 | -0.0001 | 0.0002 |
| | 313 | 0.0094 | -0.0088 | 0.0031 | 0.0001 | 0.0000 | -0.0002 |
| | 314 | 0.0008 | -0.0088 | 0.0031 | 0.0001 | 0.0000 | 0.0000 |
| | 315 | 0.0091 | -0.0068 | 0.0009 | 0.0001 | 0.0000 | -0.0002 |
| | 316 | 0.0005 | -0.0068 | 0.0009 | 0.0001 | 0.0000 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 10 | 301 | 0.0147 | -0.0006 | 0.0001 | 0.0001 | -0.0005 | -0.0009 |
| | 302 | 0.0099 | -0.0003 | 0.0009 | 0.0001 | -0.0002 | -0.0006 |
| | 303 | 0.0006 | 0.0005 | -0.0003 | 0.0000 | 0.0001 | 0.0000 |
| | 304 | -0.0075 | 0.0005 | -0.0008 | -0.0001 | 0.0002 | 0.0005 |
| | 305 | 0.0158 | -0.0006 | 0.0015 | 0.0001 | -0.0003 | -0.0010 |
| | 306 | -0.0066 | -0.0002 | -0.0003 | 0.0000 | 0.0000 | 0.0004 |
| | 307 | 0.0159 | -0.0005 | 0.0014 | 0.0001 | -0.0003 | -0.0010 |
| | 308 | 0.0014 | -0.0003 | 0.0003 | 0.0000 | -0.0001 | -0.0001 |
| | 309 | 0.0146 | -0.0002 | 0.0009 | 0.0000 | -0.0001 | -0.0007 |
| | 310 | 0.0146 | -0.0002 | 0.0004 | 0.0000 | -0.0001 | -0.0007 |
| | 311 | 0.0013 | -0.0002 | 0.0009 | 0.0000 | -0.0002 | -0.0003 |
| | 312 | 0.0013 | -0.0002 | 0.0005 | 0.0000 | -0.0002 | -0.0003 |
| | 313 | 0.0100 | -0.0002 | 0.0015 | 0.0001 | -0.0001 | -0.0006 |
| | 314 | 0.0060 | -0.0002 | 0.0015 | 0.0001 | -0.0002 | -0.0004 |
| | 315 | 0.0099 | -0.0002 | -0.0001 | 0.0000 | -0.0001 | -0.0006 |
| | 316 | 0.0060 | -0.0002 | -0.0001 | 0.0000 | -0.0002 | -0.0004 |
| 11 | 301 | 0.0129 | -0.0027 | 0.0012 | 0.0000 | -0.0002 | 0.0000 |
| | 302 | -0.0158 | 0.0007 | -0.0003 | 0.0000 | 0.0007 | 0.0003 |
| | 303 | -0.0136 | -0.0021 | 0.0056 | -0.0003 | 0.0004 | 0.0002 |
| | 304 | 0.0052 | -0.0017 | 0.0032 | -0.0002 | -0.0003 | -0.0001 |
| | 305 | -0.0193 | 0.0021 | -0.0030 | 0.0002 | 0.0009 | 0.0004 |
| | 306 | 0.0223 | 0.0014 | -0.0050 | 0.0003 | -0.0008 | -0.0003 |
| | 307 | -0.0228 | 0.0016 | -0.0016 | 0.0001 | 0.0010 | 0.0004 |
| | 308 | 0.0060 | 0.0015 | -0.0037 | 0.0002 | -0.0002 | -0.0001 |
| | 309 | -0.0267 | 0.0007 | -0.0002 | 0.0000 | 0.0009 | 0.0002 |
| | 310 | -0.0266 | 0.0004 | -0.0002 | 0.0000 | 0.0009 | 0.0002 |
| | 311 | 0.0013 | 0.0007 | -0.0002 | 0.0000 | 0.0002 | 0.0002 |
| | 312 | 0.0013 | 0.0004 | -0.0002 | 0.0000 | 0.0002 | 0.0002 |
| | 313 | -0.0169 | 0.0011 | -0.0002 | 0.0000 | 0.0006 | 0.0002 |
| | 314 | -0.0086 | 0.0011 | -0.0002 | 0.0000 | 0.0004 | 0.0002 |
| | 315 | -0.0168 | 0.0001 | -0.0002 | 0.0000 | 0.0006 | 0.0002 |
| | 316 | -0.0084 | 0.0001 | -0.0002 | 0.0000 | 0.0004 | 0.0002 |
| 12 | 301 | 0.0002 | -0.0037 | 0.0011 | 0.0000 | 0.0000 | -0.0005 |
| | 302 | -0.0005 | 0.0007 | -0.0007 | 0.0000 | 0.0000 | 0.0001 |
| | 303 | 0.0007 | -0.0085 | 0.0669 | -0.0007 | -0.0014 | 0.0000 |
| | 304 | 0.0007 | -0.0051 | 0.0362 | -0.0004 | -0.0007 | -0.0001 |
| | 305 | -0.0011 | 0.0050 | -0.0323 | 0.0003 | 0.0006 | 0.0001 |
| | 306 | -0.0003 | 0.0075 | -0.0621 | 0.0007 | 0.0013 | -0.0001 |
| | 307 | -0.0009 | 0.0027 | -0.0136 | 0.0001 | 0.0002 | 0.0001 |
| | 308 | -0.0005 | 0.0057 | -0.0439 | 0.0005 | 0.0009 | 0.0000 |
| | 309 | -0.0175 | 0.0006 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0175 | 0.0004 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0167 | 0.0007 | -0.0006 | 0.0000 | 0.0000 | 0.0001 |
| | 312 | 0.0167 | 0.0004 | -0.0005 | 0.0000 | 0.0000 | 0.0001 |
| | 313 | -0.0055 | 0.0010 | -0.0006 | 0.0000 | 0.0000 | 0.0001 |
| | 314 | 0.0047 | 0.0010 | -0.0006 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | -0.0055 | 0.0001 | -0.0005 | 0.0000 | 0.0000 | 0.0001 |
| | 316 | 0.0047 | 0.0001 | -0.0005 | 0.0000 | 0.0000 | 0.0001 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 13 | 301 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0003 | 0.0004 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0012 | -0.0023 | 0.0003 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0007 | -0.0012 | 0.0002 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0006 | 0.0011 | -0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | -0.0012 | 0.0021 | -0.0003 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0004 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0008 | 0.0015 | -0.0002 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 14 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0044 | 0.0006 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0001 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0011 | 0.0003 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0008 | 0.0002 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0015 | 0.0004 | 0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0016 | -0.0005 | -0.0001 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0009 | -0.0003 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| 15 | 301 | 0.0002 | -0.0056 | -0.0011 | 0.0000 | 0.0006 | 0.0002 |
| | 302 | -0.0001 | 0.0007 | 0.0007 | 0.0000 | 0.0002 | 0.0000 |
| | 303 | -0.0002 | 0.0033 | 0.0175 | -0.0002 | 0.0003 | 0.0000 |
| | 304 | -0.0002 | 0.0074 | 0.0573 | -0.0006 | 0.0000 | 0.0000 |
| | 305 | -0.0003 | 0.0062 | 0.0419 | -0.0004 | 0.0003 | 0.0000 |
| | 306 | -0.0001 | 0.0099 | 0.0811 | -0.0009 | -0.0001 | 0.0000 |
| | 307 | 0.0002 | -0.0112 | -0.0873 | 0.0009 | 0.0000 | 0.0000 |
| | 308 | 0.0002 | -0.0065 | -0.0471 | 0.0005 | -0.0002 | 0.0000 |
| | 309 | -0.0172 | 0.0005 | 0.0006 | 0.0000 | 0.0002 | 0.0000 |
| | 310 | -0.0172 | 0.0009 | 0.0006 | 0.0000 | 0.0002 | 0.0000 |
| | 311 | 0.0170 | 0.0002 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0170 | 0.0005 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0052 | 0.0001 | 0.0005 | 0.0000 | 0.0002 | 0.0000 |
| | 314 | 0.0050 | 0.0000 | 0.0005 | 0.0000 | 0.0001 | 0.0000 |
| | 315 | -0.0052 | 0.0011 | 0.0007 | 0.0000 | 0.0002 | 0.0000 |
| | 316 | 0.0050 | 0.0010 | 0.0006 | 0.0000 | 0.0001 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 16 | 301 | -0.0251 | -0.0043 | -0.0017 | 0.0000 | -0.0009 | 0.0004 |
| | 302 | -0.0041 | 0.0007 | 0.0003 | 0.0000 | -0.0002 | 0.0001 |
| | 303 | -0.0068 | 0.0018 | 0.0020 | -0.0001 | -0.0003 | 0.0001 |
| | 304 | -0.0013 | 0.0019 | 0.0048 | -0.0003 | 0.0000 | 0.0000 |
| | 305 | -0.0070 | 0.0024 | 0.0039 | -0.0002 | -0.0003 | 0.0001 |
| | 306 | 0.0017 | 0.0020 | 0.0065 | -0.0004 | 0.0001 | 0.0000 |
| | 307 | 0.0008 | -0.0027 | -0.0073 | 0.0004 | 0.0000 | 0.0000 |
| | 308 | 0.0037 | -0.0021 | -0.0041 | 0.0002 | 0.0001 | -0.0001 |
| | 309 | -0.0244 | 0.0006 | 0.0003 | 0.0000 | -0.0008 | 0.0002 |
| | 310 | -0.0244 | 0.0010 | 0.0003 | 0.0000 | -0.0008 | 0.0002 |
| | 311 | 0.0178 | 0.0002 | 0.0003 | 0.0000 | 0.0005 | -0.0001 |
| | 312 | 0.0178 | 0.0006 | 0.0003 | 0.0000 | 0.0005 | -0.0001 |
| | 313 | -0.0096 | 0.0000 | 0.0003 | 0.0000 | -0.0003 | 0.0001 |
| | 314 | 0.0030 | -0.0001 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0096 | 0.0012 | 0.0003 | 0.0000 | -0.0003 | 0.0001 |
| | 316 | 0.0030 | 0.0011 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
| 17 | 301 | 0.0042 | -0.0008 | 0.0006 | -0.0001 | -0.0002 | -0.0002 |
| | 302 | 0.0028 | -0.0003 | -0.0008 | -0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0044 | -0.0006 | -0.0014 | -0.0001 | 0.0000 | -0.0002 |
| | 304 | 0.0004 | -0.0004 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0044 | -0.0007 | -0.0014 | -0.0001 | 0.0000 | -0.0002 |
| | 306 | -0.0018 | -0.0004 | 0.0002 | 0.0000 | 0.0000 | 0.0001 |
| | 307 | 0.0002 | 0.0006 | 0.0005 | 0.0001 | 0.0000 | 0.0000 |
| | 308 | -0.0021 | 0.0005 | 0.0009 | 0.0001 | 0.0000 | 0.0001 |
| | 309 | 0.0123 | -0.0002 | -0.0006 | 0.0000 | 0.0001 | -0.0006 |
| | 310 | 0.0123 | -0.0003 | -0.0012 | -0.0001 | 0.0001 | -0.0006 |
| | 311 | -0.0079 | -0.0002 | 0.0000 | 0.0000 | 0.0000 | 0.0004 |
| | 312 | -0.0079 | -0.0002 | -0.0006 | 0.0000 | 0.0000 | 0.0004 |
| | 313 | 0.0052 | -0.0002 | 0.0002 | 0.0000 | 0.0000 | -0.0003 |
| | 314 | -0.0008 | -0.0002 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0052 | -0.0003 | -0.0016 | -0.0001 | 0.0000 | -0.0003 |
| | 316 | -0.0008 | -0.0003 | -0.0014 | -0.0001 | 0.0000 | 0.0000 |
| 18 | 301 | 0.0200 | -0.0179 | -0.0024 | -0.0002 | 0.0001 | -0.0012 |
| | 302 | 0.0033 | -0.0114 | -0.0026 | -0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0056 | -0.0207 | -0.0048 | -0.0002 | 0.0000 | -0.0002 |
| | 304 | 0.0013 | -0.0053 | -0.0013 | 0.0000 | 0.0000 | -0.0001 |
| | 305 | 0.0059 | -0.0199 | -0.0046 | -0.0002 | 0.0000 | -0.0002 |
| | 306 | -0.0010 | 0.0039 | 0.0009 | 0.0001 | 0.0000 | 0.0000 |
| | 307 | -0.0011 | 0.0049 | 0.0012 | 0.0000 | 0.0000 | 0.0001 |
| | 308 | -0.0033 | 0.0117 | 0.0027 | 0.0001 | 0.0000 | 0.0001 |
| | 309 | 0.0186 | -0.0091 | -0.0021 | -0.0001 | 0.0000 | -0.0003 |
| | 310 | 0.0186 | -0.0098 | -0.0028 | -0.0001 | 0.0000 | -0.0003 |
| | 311 | -0.0133 | -0.0085 | -0.0014 | -0.0001 | 0.0000 | 0.0001 |
| | 312 | -0.0133 | -0.0092 | -0.0021 | -0.0001 | 0.0000 | 0.0002 |
| | 313 | 0.0074 | -0.0080 | -0.0009 | -0.0001 | 0.0000 | -0.0002 |
| | 314 | -0.0021 | -0.0079 | -0.0007 | -0.0001 | 0.0000 | 0.0000 |
| | 315 | 0.0074 | -0.0104 | -0.0034 | -0.0001 | 0.0000 | -0.0002 |
| | 316 | -0.0021 | -0.0102 | -0.0032 | -0.0001 | 0.0000 | 0.0000 |

JOINT DISPLACEMENT (CM   RADIANS)    STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 19 | 301 | -0.0002 | -0.0018 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | -0.0001 | -0.0011 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0001 | -0.0015 | -0.0014 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0002 | -0.0018 | -0.0014 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0007 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | -0.0001 | -0.0002 | 0.0004 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0003 | 0.0007 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0291 | -0.0009 | -0.0004 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0291 | -0.0009 | -0.0014 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | -0.0292 | -0.0008 | 0.0002 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | -0.0292 | -0.0008 | -0.0009 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0086 | -0.0009 | 0.0011 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | -0.0088 | -0.0009 | 0.0012 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0086 | -0.0009 | -0.0025 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | -0.0088 | -0.0009 | -0.0023 | 0.0000 | 0.0000 | 0.0002 |
| 20 | 301 | -0.0003 | -0.1010 | 0.0003 | 0.0000 | 0.0000 | 0.0001 |
| | 302 | -0.0002 | -0.0595 | 0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0001 | -0.0477 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0001 | 0.0209 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0003 | -0.0962 | 0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0001 | 0.0417 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | -0.0002 | -0.0501 | 0.0011 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0013 | 0.0195 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0281 | -0.0478 | 0.0007 | 0.0000 | 0.0000 | -0.0004 |
| | 310 | 0.0281 | -0.0479 | -0.0004 | 0.0000 | 0.0000 | -0.0004 |
| | 311 | -0.0283 | -0.0473 | 0.0006 | 0.0000 | 0.0000 | 0.0005 |
| | 312 | -0.0283 | -0.0473 | -0.0004 | 0.0000 | 0.0000 | 0.0005 |
| | 313 | 0.0083 | -0.0475 | 0.0019 | 0.0000 | 0.0000 | -0.0001 |
| | 314 | -0.0085 | -0.0474 | 0.0018 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | 0.0083 | -0.0478 | -0.0016 | 0.0000 | 0.0000 | -0.0001 |
| | 316 | -0.0085 | -0.0476 | -0.0016 | 0.0000 | 0.0000 | 0.0001 |
| 21 | 301 | -0.0004 | -0.0017 | 0.0011 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | -0.0002 | -0.0010 | 0.0011 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0001 | -0.0002 | -0.0003 | 0.0001 | 0.0000 | 0.0000 |
| | 304 | 0.0001 | 0.0007 | -0.0011 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0003 | -0.0017 | 0.0018 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0001 | 0.0007 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | -0.0003 | -0.0015 | 0.0018 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0003 | 0.0000 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0270 | -0.0008 | 0.0017 | 0.0000 | 0.0000 | -0.0005 |
| | 310 | 0.0270 | -0.0008 | 0.0007 | 0.0000 | 0.0000 | -0.0005 |
| | 311 | -0.0273 | -0.0008 | 0.0010 | 0.0000 | 0.0000 | 0.0005 |
| | 312 | -0.0273 | -0.0008 | 0.0000 | 0.0000 | 0.0000 | 0.0005 |
| | 313 | 0.0080 | -0.0008 | 0.0027 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | -0.0082 | -0.0008 | 0.0024 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0079 | -0.0008 | -0.0007 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | -0.0083 | -0.0008 | -0.0010 | 0.0000 | 0.0000 | 0.0002 |



JOINT DISPLACEMENT (CM   RADIANS)    STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 22 | 301 | 0.0176 | -0.0186 | 0.0026 | 0.0002 | -0.0001 | -0.0011 |
| | 302 | 0.0032 | -0.0119 | 0.0028 | 0.0001 | 0.0000 | -0.0001 |
| | 303 | -0.0009 | 0.0046 | -0.0011 | 0.0000 | 0.0000 | 0.0001 |
| | 304 | -0.0030 | 0.0119 | -0.0028 | -0.0001 | 0.0000 | 0.0001 |
| | 305 | 0.0055 | -0.0206 | 0.0048 | 0.0002 | 0.0000 | -0.0002 |
| | 306 | -0.0011 | 0.0044 | -0.0010 | -0.0001 | 0.0000 | 0.0000 |
| | 307 | 0.0052 | -0.0213 | 0.0049 | 0.0002 | 0.0000 | -0.0002 |
| | 308 | 0.0011 | -0.0052 | 0.0012 | 0.0000 | 0.0000 | -0.0001 |
| | 309 | 0.0183 | -0.0102 | 0.0029 | 0.0001 | 0.0000 | -0.0004 |
| | 310 | 0.0184 | -0.0095 | 0.0022 | 0.0001 | 0.0000 | -0.0004 |
| | 311 | -0.0133 | -0.0096 | 0.0022 | 0.0001 | 0.0000 | 0.0002 |
| | 312 | -0.0133 | -0.0089 | 0.0015 | 0.0001 | 0.0000 | 0.0002 |
| | 313 | 0.0072 | -0.0108 | 0.0036 | 0.0001 | 0.0000 | -0.0002 |
| | 314 | -0.0022 | -0.0106 | 0.0033 | 0.0001 | 0.0000 | 0.0000 |
| | 315 | 0.0073 | -0.0085 | 0.0011 | 0.0001 | 0.0000 | -0.0002 |
| | 316 | -0.0022 | -0.0083 | 0.0009 | 0.0001 | 0.0000 | 0.0000 |
| 23 | 301 | 0.0041 | -0.0008 | -0.0006 | 0.0001 | 0.0001 | -0.0002 |
| | 302 | 0.0027 | -0.0003 | 0.0008 | 0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0002 | 0.0006 | -0.0004 | -0.0001 | 0.0000 | 0.0000 |
| | 304 | -0.0020 | 0.0006 | -0.0009 | -0.0001 | 0.0000 | 0.0001 |
| | 305 | 0.0043 | -0.0008 | 0.0015 | 0.0001 | 0.0000 | -0.0002 |
| | 306 | -0.0018 | -0.0004 | -0.0002 | 0.0000 | 0.0000 | 0.0001 |
| | 307 | 0.0043 | -0.0006 | 0.0015 | 0.0001 | 0.0000 | -0.0002 |
| | 308 | 0.0004 | -0.0004 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0124 | -0.0003 | 0.0012 | 0.0001 | -0.0001 | -0.0006 |
| | 310 | 0.0124 | -0.0002 | 0.0007 | 0.0000 | -0.0001 | -0.0006 |
| | 311 | -0.0081 | -0.0002 | 0.0006 | 0.0000 | 0.0000 | 0.0004 |
| | 312 | -0.0081 | -0.0002 | 0.0001 | 0.0000 | 0.0000 | 0.0004 |
| | 313 | 0.0052 | -0.0003 | 0.0017 | 0.0001 | 0.0000 | -0.0003 |
| | 314 | -0.0009 | -0.0003 | 0.0015 | 0.0001 | 0.0000 | 0.0000 |
| | 315 | 0.0052 | -0.0002 | -0.0002 | 0.0000 | 0.0000 | -0.0003 |
| | 316 | -0.0009 | -0.0002 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| 24 | 301 | -0.0239 | -0.0042 | 0.0017 | 0.0000 | 0.0008 | 0.0003 |
| | 302 | -0.0040 | 0.0008 | -0.0004 | 0.0000 | 0.0002 | 0.0001 |
| | 303 | 0.0007 | -0.0027 | 0.0072 | -0.0004 | 0.0000 | 0.0000 |
| | 304 | 0.0035 | -0.0021 | 0.0041 | -0.0002 | -0.0001 | -0.0001 |
| | 305 | -0.0068 | 0.0025 | -0.0039 | 0.0002 | 0.0003 | 0.0001 |
| | 306 | 0.0018 | 0.0019 | -0.0065 | 0.0004 | -0.0001 | 0.0000 |
| | 307 | -0.0065 | 0.0018 | -0.0020 | 0.0001 | 0.0003 | 0.0001 |
| | 308 | -0.0012 | 0.0019 | -0.0048 | 0.0003 | 0.0000 | 0.0000 |
| | 309 | -0.0245 | 0.0010 | -0.0003 | 0.0000 | 0.0008 | 0.0002 |
| | 310 | -0.0245 | 0.0006 | -0.0003 | 0.0000 | 0.0008 | 0.0002 |
| | 311 | 0.0181 | 0.0006 | -0.0003 | 0.0000 | -0.0005 | -0.0001 |
| | 312 | 0.0181 | 0.0003 | -0.0003 | 0.0000 | -0.0005 | -0.0001 |
| | 313 | -0.0096 | 0.0013 | -0.0003 | 0.0000 | 0.0003 | 0.0001 |
| | 314 | 0.0032 | 0.0012 | -0.0003 | 0.0000 | -0.0001 | 0.0000 |
| | 315 | -0.0095 | 0.0001 | -0.0003 | 0.0000 | 0.0003 | 0.0001 |
| | 316 | 0.0032 | 0.0000 | -0.0003 | 0.0000 | -0.0001 | 0.0000 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 25 | 301 | 0.0001 | -0.0055 | 0.0011 | 0.0000 | -0.0005 | 0.0002 |
| | 302 | -0.0002 | 0.0007 | -0.0008 | 0.0000 | -0.0002 | 0.0000 |
| | 303 | 0.0002 | -0.0111 | 0.0873 | -0.0009 | 0.0000 | 0.0000 |
| | 304 | 0.0002 | -0.0065 | 0.0471 | -0.0005 | 0.0001 | 0.0000 |
| | 305 | -0.0004 | 0.0063 | -0.0420 | 0.0004 | -0.0003 | 0.0000 |
| | 306 | -0.0001 | 0.0099 | -0.0811 | 0.0009 | 0.0001 | 0.0000 |
| | 307 | -0.0003 | 0.0033 | -0.0176 | 0.0002 | -0.0003 | 0.0000 |
| | 308 | -0.0002 | 0.0074 | -0.0573 | 0.0006 | 0.0000 | 0.0000 |
| | 309 | -0.0173 | 0.0009 | -0.0007 | 0.0000 | -0.0002 | 0.0000 |
| | 310 | -0.0173 | 0.0006 | -0.0006 | 0.0000 | -0.0002 | 0.0000 |
| | 311 | 0.0171 | 0.0006 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0171 | 0.0002 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0053 | 0.0011 | -0.0007 | 0.0000 | -0.0002 | 0.0000 |
| | 314 | 0.0050 | 0.0010 | -0.0007 | 0.0000 | -0.0001 | 0.0000 |
| | 315 | -0.0053 | 0.0001 | -0.0006 | 0.0000 | -0.0002 | 0.0000 |
| | 316 | 0.0050 | 0.0000 | -0.0005 | 0.0000 | -0.0001 | 0.0000 |
| 26 | 301 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0043 | 0.0006 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0016 | 0.0005 | -0.0001 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0009 | 0.0003 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0008 | -0.0002 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | -0.0015 | -0.0004 | 0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | -0.0001 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0011 | -0.0003 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| 27 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0043 | -0.0006 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0001 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0011 | -0.0003 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0008 | -0.0002 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0015 | -0.0004 | -0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0016 | 0.0005 | 0.0001 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0009 | 0.0003 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 28 | 301 | 0.0000 | -0.0056 | -0.0011 | 0.0000 | -0.0006 | -0.0002 |
| | 302 | 0.0002 | 0.0007 | 0.0007 | 0.0000 | -0.0002 | 0.0000 |
| | 303 | 0.0004 | 0.0033 | 0.0175 | -0.0002 | -0.0003 | 0.0000 |
| | 304 | 0.0002 | 0.0074 | 0.0573 | -0.0006 | 0.0000 | 0.0000 |
| | 305 | 0.0005 | 0.0062 | 0.0419 | -0.0004 | -0.0003 | 0.0000 |
| | 306 | 0.0000 | 0.0099 | 0.0811 | -0.0009 | 0.0001 | 0.0000 |
| | 307 | -0.0002 | -0.0112 | -0.0873 | 0.0009 | 0.0000 | 0.0000 |
| | 308 | -0.0003 | -0.0065 | -0.0470 | 0.0005 | 0.0001 | 0.0000 |
| | 309 | -0.0169 | 0.0002 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0169 | 0.0005 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0173 | 0.0005 | 0.0006 | 0.0000 | -0.0002 | 0.0000 |
| | 312 | 0.0173 | 0.0009 | 0.0006 | 0.0000 | -0.0002 | 0.0000 |
| | 313 | -0.0049 | 0.0000 | 0.0005 | 0.0000 | -0.0001 | 0.0000 |
| | 314 | 0.0053 | 0.0001 | 0.0005 | 0.0000 | -0.0002 | 0.0000 |
| | 315 | -0.0049 | 0.0010 | 0.0006 | 0.0000 | -0.0001 | 0.0000 |
| | 316 | 0.0053 | 0.0011 | 0.0007 | 0.0000 | -0.0002 | 0.0000 |
| 29 | 301 | 0.0253 | -0.0043 | -0.0017 | 0.0000 | 0.0009 | -0.0004 |
| | 302 | 0.0043 | 0.0007 | 0.0003 | 0.0000 | 0.0002 | -0.0001 |
| | 303 | 0.0071 | 0.0018 | 0.0020 | -0.0001 | 0.0003 | -0.0001 |
| | 304 | 0.0013 | 0.0019 | 0.0048 | -0.0003 | 0.0000 | 0.0000 |
| | 305 | 0.0073 | 0.0024 | 0.0039 | -0.0002 | 0.0003 | -0.0001 |
| | 306 | -0.0019 | 0.0020 | 0.0065 | -0.0004 | -0.0001 | 0.0000 |
| | 307 | -0.0008 | -0.0027 | -0.0073 | 0.0004 | 0.0000 | 0.0000 |
| | 308 | -0.0038 | -0.0021 | -0.0041 | 0.0002 | -0.0002 | 0.0001 |
| | 309 | -0.0176 | 0.0002 | 0.0003 | 0.0000 | -0.0005 | 0.0001 |
| | 310 | -0.0176 | 0.0006 | 0.0003 | 0.0000 | -0.0005 | 0.0001 |
| | 311 | 0.0244 | 0.0006 | 0.0003 | 0.0000 | 0.0008 | -0.0002 |
| | 312 | 0.0244 | 0.0010 | 0.0003 | 0.0000 | 0.0008 | -0.0002 |
| | 313 | -0.0028 | -0.0001 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0097 | 0.0000 | 0.0003 | 0.0000 | 0.0003 | -0.0001 |
| | 315 | -0.0028 | 0.0011 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0097 | 0.0012 | 0.0003 | 0.0000 | 0.0003 | -0.0001 |
| 30 | 301 | -0.0043 | -0.0008 | 0.0006 | -0.0001 | 0.0002 | 0.0002 |
| | 302 | -0.0028 | -0.0003 | -0.0008 | -0.0001 | 0.0000 | 0.0001 |
| | 303 | -0.0045 | -0.0006 | -0.0014 | -0.0001 | 0.0000 | 0.0002 |
| | 304 | -0.0004 | -0.0004 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0045 | -0.0007 | -0.0014 | -0.0001 | 0.0000 | 0.0002 |
| | 306 | 0.0019 | -0.0004 | 0.0001 | 0.0000 | 0.0000 | -0.0001 |
| | 307 | -0.0002 | 0.0006 | 0.0005 | 0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0021 | 0.0005 | 0.0010 | 0.0001 | 0.0000 | -0.0001 |
| | 309 | 0.0077 | -0.0002 | 0.0000 | 0.0000 | 0.0000 | -0.0004 |
| | 310 | 0.0077 | -0.0002 | -0.0006 | 0.0000 | 0.0000 | -0.0004 |
| | 311 | -0.0122 | -0.0002 | -0.0006 | 0.0000 | -0.0001 | 0.0006 |
| | 312 | -0.0122 | -0.0003 | -0.0012 | -0.0001 | -0.0001 | 0.0006 |
| | 313 | 0.0007 | -0.0002 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0052 | -0.0002 | 0.0002 | 0.0000 | 0.0000 | 0.0003 |
| | 315 | 0.0007 | -0.0003 | -0.0014 | -0.0001 | 0.0000 | 0.0000 |
| | 316 | -0.0052 | -0.0003 | -0.0016 | -0.0001 | 0.0000 | 0.0003 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 31 | 301 | -0.0204 | -0.0180 | -0.0024 | -0.0002 | -0.0001 | 0.0012 |
| | 302 | -0.0036 | -0.0114 | -0.0026 | -0.0001 | 0.0000 | 0.0001 |
| | 303 | -0.0059 | -0.0207 | -0.0048 | -0.0002 | 0.0000 | 0.0002 |
| | 304 | -0.0014 | -0.0053 | -0.0013 | 0.0000 | 0.0000 | 0.0001 |
| | 305 | -0.0063 | -0.0199 | -0.0046 | -0.0002 | 0.0000 | 0.0002 |
| | 306 | 0.0012 | 0.0039 | 0.0009 | 0.0001 | 0.0000 | 0.0000 |
| | 307 | 0.0011 | 0.0049 | 0.0012 | 0.0000 | 0.0000 | -0.0001 |
| | 308 | 0.0034 | 0.0103 | 0.0027 | 0.0001 | 0.0000 | -0.0001 |
| | 309 | 0.0130 | -0.0085 | -0.0014 | -0.0001 | 0.0000 | -0.0001 |
| | 310 | 0.0130 | -0.0092 | -0.0021 | -0.0001 | 0.0000 | -0.0001 |
| | 311 | -0.0187 | -0.0091 | -0.0021 | -0.0001 | 0.0000 | 0.0004 |
| | 312 | -0.0187 | -0.0098 | -0.0028 | -0.0001 | 0.0000 | 0.0003 |
| | 313 | 0.0018 | -0.0079 | -0.0007 | -0.0001 | 0.0000 | 0.0000 |
| | 314 | -0.0076 | -0.0080 | -0.0009 | -0.0001 | 0.0000 | 0.0002 |
| | 315 | 0.0019 | -0.0102 | -0.0032 | -0.0001 | 0.0000 | 0.0000 |
| | 316 | -0.0075 | -0.0104 | -0.0034 | -0.0001 | 0.0000 | 0.0002 |
| 32 | 301 | 0.0005 | -0.0018 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0003 | -0.0011 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0004 | -0.0015 | -0.0015 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0005 | -0.0018 | -0.0014 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | -0.0002 | 0.0007 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0001 | -0.0002 | 0.0003 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0010 | 0.0016 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0295 | -0.0008 | 0.0001 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0295 | -0.0008 | -0.0009 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | -0.0290 | -0.0009 | -0.0004 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | -0.0290 | -0.0009 | -0.0015 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0090 | -0.0009 | 0.0012 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | -0.0085 | -0.0009 | 0.0010 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0089 | -0.0009 | -0.0023 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | -0.0085 | -0.0009 | -0.0025 | 0.0000 | 0.0000 | 0.0002 |
| 33 | 301 | 0.0005 | -0.1010 | 0.0002 | 0.0000 | 0.0000 | -0.0001 |
| | 302 | 0.0003 | -0.0595 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0002 | -0.0478 | -0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | -0.0001 | 0.0209 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0004 | -0.0963 | 0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | -0.0002 | 0.0417 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0002 | -0.0508 | 0.0011 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0015 | 0.0702 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0284 | -0.0473 | 0.0006 | 0.0000 | 0.0000 | -0.0005 |
| | 310 | 0.0284 | -0.0474 | -0.0005 | 0.0000 | 0.0000 | -0.0005 |
| | 311 | -0.0280 | -0.0479 | 0.0006 | 0.0000 | 0.0000 | 0.0004 |
| | 312 | -0.0280 | -0.0479 | -0.0004 | 0.0000 | 0.0000 | 0.0004 |
| | 313 | 0.0086 | -0.0474 | 0.0018 | 0.0000 | 0.0000 | -0.0001 |
| | 314 | -0.0082 | -0.0476 | 0.0018 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | 0.0086 | -0.0477 | -0.0016 | 0.0000 | 0.0000 | -0.0001 |
| | 316 | -0.0082 | -0.0478 | -0.0016 | 0.0000 | 0.0000 | 0.0001 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 34 | 301 | 0.0005 | -0.0017 | 0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0002 | -0.0010 | 0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0001 | -0.0002 | -0.0003 | 0.0001 | 0.0000 | 0.0000 |
| | 304 | -0.0001 | 0.0007 | -0.0011 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0004 | -0.0017 | 0.0018 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | -0.0001 | 0.0007 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0003 | -0.0015 | 0.0018 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0001 | 0.0002 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0274 | -0.0008 | 0.0010 | 0.0000 | 0.0000 | -0.0005 |
| | 310 | 0.0274 | -0.0008 | -0.0001 | 0.0000 | 0.0000 | -0.0005 |
| | 311 | -0.0270 | -0.0008 | 0.0017 | 0.0000 | 0.0000 | 0.0005 |
| | 312 | -0.0270 | -0.0008 | 0.0007 | 0.0000 | 0.0000 | 0.0005 |
| | 313 | 0.0083 | -0.0008 | 0.0024 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | -0.0079 | -0.0008 | 0.0026 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0083 | -0.0008 | -0.0010 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | -0.0079 | -0.0008 | -0.0007 | 0.0000 | 0.0000 | 0.0002 |
| 35 | 301 | -0.0176 | -0.0186 | 0.0026 | 0.0002 | 0.0001 | 0.0011 |
| | 302 | -0.0032 | -0.0119 | 0.0027 | 0.0001 | 0.0000 | 0.0001 |
| | 303 | 0.0009 | 0.0046 | -0.0011 | 0.0000 | 0.0000 | -0.0001 |
| | 304 | 0.0030 | 0.0119 | -0.0028 | -0.0001 | 0.0000 | -0.0001 |
| | 305 | -0.0055 | -0.0206 | 0.0048 | 0.0002 | 0.0000 | 0.0002 |
| | 306 | 0.0011 | 0.0044 | -0.0010 | -0.0001 | 0.0000 | 0.0000 |
| | 307 | -0.0052 | -0.0213 | 0.0049 | 0.0002 | 0.0000 | 0.0002 |
| | 308 | -0.0011 | -0.0066 | 0.0013 | 0.0001 | 0.0000 | 0.0001 |
| | 309 | 0.0133 | -0.0096 | 0.0022 | 0.0001 | 0.0000 | -0.0002 |
| | 310 | 0.0133 | -0.0089 | 0.0015 | 0.0001 | 0.0000 | -0.0002 |
| | 311 | -0.0183 | -0.0102 | 0.0029 | 0.0001 | 0.0000 | 0.0004 |
| | 312 | -0.0184 | -0.0095 | 0.0022 | 0.0001 | 0.0000 | 0.0004 |
| | 313 | 0.0022 | -0.0106 | 0.0033 | 0.0001 | 0.0000 | 0.0000 |
| | 314 | -0.0072 | -0.0107 | 0.0036 | 0.0001 | 0.0000 | 0.0002 |
| | 315 | 0.0022 | -0.0083 | 0.0008 | 0.0001 | 0.0000 | 0.0000 |
| | 316 | -0.0073 | -0.0085 | 0.0011 | 0.0001 | 0.0000 | 0.0002 |
| 36 | 301 | -0.0041 | -0.0008 | -0.0006 | 0.0001 | -0.0001 | 0.0002 |
| | 302 | -0.0027 | -0.0003 | 0.0008 | 0.0001 | 0.0000 | 0.0001 |
| | 303 | -0.0002 | 0.0006 | -0.0004 | -0.0001 | 0.0000 | 0.0000 |
| | 304 | 0.0020 | 0.0006 | -0.0009 | -0.0001 | 0.0000 | -0.0001 |
| | 305 | -0.0043 | -0.0008 | 0.0014 | 0.0001 | 0.0000 | 0.0002 |
| | 306 | 0.0018 | -0.0004 | -0.0002 | 0.0000 | 0.0000 | -0.0001 |
| | 307 | -0.0043 | -0.0006 | 0.0015 | 0.0001 | 0.0000 | 0.0002 |
| | 308 | -0.0004 | -0.0005 | 0.0004 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0081 | -0.0002 | 0.0006 | 0.0000 | 0.0000 | -0.0004 |
| | 310 | 0.0081 | -0.0002 | 0.0001 | 0.0000 | 0.0000 | -0.0004 |
| | 311 | -0.0124 | -0.0003 | 0.0012 | 0.0001 | 0.0001 | 0.0006 |
| | 312 | -0.0124 | -0.0002 | 0.0007 | 0.0000 | 0.0001 | 0.0006 |
| | 313 | 0.0009 | -0.0003 | 0.0015 | 0.0001 | 0.0000 | 0.0000 |
| | 314 | -0.0052 | -0.0003 | 0.0017 | 0.0001 | 0.0000 | 0.0003 |
| | 315 | 0.0009 | -0.0002 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | -0.0052 | -0.0002 | -0.0002 | 0.0000 | 0.0000 | 0.0003 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 37 | 301 | 0.0239 | -0.0042 | 0.0017 | 0.0000 | -0.0008 | -0.0003 |
| | 302 | 0.0040 | 0.0008 | -0.0004 | 0.0000 | -0.0002 | -0.0001 |
| | 303 | -0.0007 | -0.0027 | 0.0072 | -0.0004 | 0.0000 | 0.0000 |
| | 304 | -0.0035 | -0.0021 | 0.0041 | -0.0002 | 0.0001 | 0.0001 |
| | 305 | 0.0068 | 0.0024 | -0.0039 | 0.0002 | -0.0003 | -0.0001 |
| | 306 | -0.0018 | 0.0019 | -0.0065 | 0.0004 | 0.0001 | 0.0000 |
| | 307 | 0.0065 | 0.0018 | -0.0020 | 0.0001 | -0.0003 | -0.0001 |
| | 308 | 0.0012 | 0.0019 | -0.0049 | 0.0003 | 0.0000 | 0.0000 |
| | 309 | -0.0181 | 0.0006 | -0.0003 | 0.0000 | 0.0005 | 0.0001 |
| | 310 | -0.0181 | 0.0002 | -0.0003 | 0.0000 | 0.0005 | 0.0001 |
| | 311 | 0.0245 | 0.0010 | -0.0003 | 0.0000 | -0.0008 | -0.0002 |
| | 312 | 0.0245 | 0.0006 | -0.0003 | 0.0000 | -0.0008 | -0.0002 |
| | 313 | -0.0032 | 0.0012 | -0.0003 | 0.0000 | 0.0001 | 0.0000 |
| | 314 | 0.0096 | 0.0013 | -0.0003 | 0.0000 | -0.0003 | -0.0001 |
| | 315 | -0.0032 | 0.0000 | -0.0003 | 0.0000 | 0.0001 | 0.0000 |
| | 316 | 0.0095 | 0.0001 | -0.0003 | 0.0000 | -0.0003 | -0.0001 |
| 38 | 301 | -0.0001 | -0.0055 | 0.0011 | 0.0000 | 0.0005 | -0.0002 |
| | 302 | 0.0002 | 0.0007 | -0.0008 | 0.0000 | 0.0002 | 0.0000 |
| | 303 | -0.0002 | -0.0111 | 0.0873 | -0.0009 | 0.0000 | 0.0000 |
| | 304 | -0.0002 | -0.0065 | 0.0471 | -0.0005 | -0.0001 | 0.0000 |
| | 305 | 0.0004 | 0.0063 | -0.0420 | 0.0004 | 0.0003 | 0.0000 |
| | 306 | 0.0001 | 0.0099 | -0.0811 | 0.0009 | -0.0001 | 0.0000 |
| | 307 | 0.0003 | 0.0033 | -0.0176 | 0.0002 | 0.0003 | 0.0000 |
| | 308 | 0.0001 | 0.0074 | -0.0574 | 0.0006 | 0.0000 | 0.0000 |
| | 309 | -0.0171 | 0.0006 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0171 | 0.0002 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0173 | 0.0009 | -0.0007 | 0.0000 | 0.0002 | 0.0000 |
| | 312 | 0.0173 | 0.0006 | -0.0006 | 0.0000 | 0.0002 | 0.0000 |
| | 313 | -0.0050 | 0.0010 | -0.0007 | 0.0000 | 0.0001 | 0.0000 |
| | 314 | 0.0053 | 0.0011 | -0.0007 | 0.0000 | 0.0002 | 0.0000 |
| | 315 | -0.0050 | 0.0000 | -0.0005 | 0.0000 | 0.0001 | 0.0000 |
| | 316 | 0.0053 | 0.0001 | -0.0006 | 0.0000 | 0.0002 | 0.0000 |
| 39 | 301 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0043 | -0.0006 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0016 | -0.0005 | 0.0001 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0009 | -0.0003 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0008 | 0.0002 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | -0.0015 | 0.0004 | -0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0001 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0011 | 0.0003 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0000 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0000 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 40 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0003 | -0.0004 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0004 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0008 | 0.0015 | 0.0002 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0006 | 0.0011 | 0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0012 | 0.0021 | 0.0003 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0012 | -0.0023 | -0.0003 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0007 | -0.0012 | -0.0002 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 41 | 301 | -0.0001 | -0.0038 | -0.0011 | 0.0000 | 0.0000 | 0.0005 |
| | 302 | 0.0006 | 0.0006 | 0.0006 | 0.0000 | 0.0000 | -0.0001 |
| | 303 | 0.0010 | 0.0027 | 0.0136 | -0.0001 | 0.0002 | -0.0001 |
| | 304 | 0.0005 | 0.0057 | 0.0440 | -0.0005 | 0.0009 | 0.0000 |
| | 305 | 0.0012 | 0.0050 | 0.0322 | -0.0003 | 0.0006 | -0.0001 |
| | 306 | 0.0003 | 0.0075 | 0.0621 | -0.0007 | 0.0013 | 0.0001 |
| | 307 | -0.0007 | -0.0086 | -0.0670 | 0.0007 | -0.0014 | 0.0000 |
| | 308 | -0.0008 | -0.0051 | -0.0361 | 0.0004 | -0.0007 | 0.0001 |
| | 309 | -0.0166 | 0.0004 | 0.0005 | 0.0000 | 0.0000 | -0.0001 |
| | 310 | -0.0166 | 0.0006 | 0.0005 | 0.0000 | 0.0000 | -0.0001 |
| | 311 | 0.0175 | 0.0004 | 0.0005 | 0.0000 | 0.0000 | -0.0001 |
| | 312 | 0.0175 | 0.0006 | 0.0005 | 0.0000 | 0.0000 | -0.0001 |
| | 313 | -0.0046 | 0.0001 | 0.0004 | 0.0000 | 0.0000 | -0.0001 |
| | 314 | 0.0055 | 0.0001 | 0.0004 | 0.0000 | 0.0000 | -0.0001 |
| | 315 | -0.0046 | 0.0009 | 0.0006 | 0.0000 | 0.0000 | -0.0001 |
| | 316 | 0.0055 | 0.0009 | 0.0006 | 0.0000 | 0.0000 | -0.0001 |
| 42 | 301 | -0.0133 | -0.0028 | -0.0012 | 0.0000 | -0.0002 | 0.0000 |
| | 302 | 0.0161 | 0.0007 | 0.0003 | 0.0000 | 0.0007 | -0.0003 |
| | 303 | 0.0231 | 0.0016 | 0.0015 | -0.0001 | 0.0010 | -0.0004 |
| | 304 | -0.0062 | 0.0015 | 0.0037 | -0.0002 | -0.0002 | 0.0001 |
| | 305 | 0.0195 | 0.0021 | 0.0030 | -0.0002 | 0.0009 | -0.0004 |
| | 306 | -0.0227 | 0.0015 | 0.0050 | -0.0003 | -0.0008 | 0.0003 |
| | 307 | 0.0139 | -0.0021 | -0.0056 | 0.0003 | 0.0005 | -0.0002 |
| | 308 | -0.0054 | -0.0017 | -0.0032 | 0.0002 | -0.0003 | 0.0001 |
| | 309 | -0.0012 | 0.0004 | 0.0002 | 0.0000 | 0.0002 | -0.0002 |
| | 310 | -0.0011 | 0.0007 | 0.0002 | 0.0000 | 0.0002 | -0.0002 |
| | 311 | 0.0268 | 0.0004 | 0.0002 | 0.0000 | 0.0009 | -0.0002 |
| | 312 | 0.0269 | 0.0007 | 0.0002 | 0.0000 | 0.0009 | -0.0002 |
| | 313 | 0.0086 | 0.0001 | 0.0002 | 0.0000 | 0.0004 | -0.0002 |
| | 314 | 0.0169 | 0.0001 | 0.0002 | 0.0000 | 0.0006 | -0.0002 |
| | 315 | 0.0088 | 0.0011 | 0.0002 | 0.0000 | 0.0004 | -0.0002 |
| | 316 | 0.0171 | 0.0011 | 0.0002 | 0.0000 | 0.0006 | -0.0002 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 43 | 301 | -0.0147 | -0.0006 | -0.0001 | -0.0001 | -0.0005 | 0.0009 |
| | 302 | -0.0100 | -0.0003 | -0.0008 | -0.0001 | -0.0002 | 0.0006 |
| | 303 | -0.0160 | -0.0005 | -0.0014 | -0.0001 | -0.0003 | 0.0010 |
| | 304 | -0.0014 | -0.0003 | -0.0003 | 0.0000 | -0.0001 | 0.0001 |
| | 305 | -0.0159 | -0.0006 | -0.0015 | -0.0001 | -0.0003 | 0.0010 |
| | 306 | 0.0066 | -0.0003 | 0.0003 | 0.0000 | 0.0000 | -0.0004 |
| | 307 | -0.0006 | 0.0005 | 0.0003 | 0.0000 | 0.0001 | 0.0000 |
| | 308 | 0.0076 | 0.0004 | 0.0009 | 0.0001 | 0.0002 | -0.0005 |
| | 309 | -0.0015 | -0.0002 | -0.0004 | 0.0000 | -0.0002 | 0.0003 |
| | 310 | -0.0015 | -0.0002 | -0.0009 | 0.0000 | -0.0002 | 0.0003 |
| | 311 | -0.0145 | -0.0002 | -0.0004 | 0.0000 | -0.0001 | 0.0007 |
| | 312 | -0.0145 | -0.0002 | -0.0009 | 0.0000 | -0.0001 | 0.0007 |
| | 313 | -0.0061 | -0.0002 | 0.0001 | 0.0000 | -0.0002 | 0.0004 |
| | 314 | -0.0099 | -0.0002 | 0.0001 | 0.0000 | -0.0001 | 0.0006 |
| | 315 | -0.0061 | -0.0002 | -0.0014 | -0.0001 | -0.0002 | 0.0004 |
| | 316 | -0.0099 | -0.0002 | -0.0014 | -0.0001 | -0.0001 | 0.0006 |
| 44 | 301 | 0.0112 | -0.0153 | -0.0027 | -0.0002 | 0.0000 | -0.0012 |
| | 302 | -0.0065 | -0.0093 | -0.0024 | -0.0001 | 0.0000 | 0.0001 |
| | 303 | -0.0089 | -0.0175 | -0.0042 | -0.0002 | -0.0001 | 0.0001 |
| | 304 | 0.0039 | -0.0046 | -0.0010 | 0.0000 | 0.0000 | -0.0003 |
| | 305 | -0.0070 | -0.0161 | -0.0041 | -0.0002 | -0.0001 | 0.0000 |
| | 306 | 0.0111 | 0.0033 | 0.0009 | 0.0000 | 0.0001 | -0.0005 |
| | 307 | -0.0077 | 0.0046 | 0.0009 | 0.0000 | 0.0000 | 0.0005 |
| | 308 | 0.0012 | 0.0085 | 0.0023 | 0.0001 | 0.0000 | 0.0002 |
| | 309 | 0.0095 | -0.0071 | -0.0016 | -0.0001 | -0.0001 | -0.0002 |
| | 310 | 0.0094 | -0.0078 | -0.0022 | -0.0001 | -0.0001 | -0.0002 |
| | 311 | -0.0198 | -0.0071 | -0.0015 | -0.0001 | 0.0000 | 0.0004 |
| | 312 | -0.0199 | -0.0077 | -0.0022 | -0.0001 | 0.0000 | 0.0004 |
| | 313 | -0.0007 | -0.0064 | -0.0008 | -0.0001 | 0.0000 | 0.0000 |
| | 314 | -0.0094 | -0.0064 | -0.0008 | -0.0001 | 0.0000 | 0.0002 |
| | 315 | -0.0010 | -0.0085 | -0.0030 | -0.0001 | 0.0000 | 0.0000 |
| | 316 | -0.0097 | -0.0085 | -0.0030 | -0.0001 | 0.0000 | 0.0002 |
| 45 | 301 | -0.0011 | -0.0022 | -0.0013 | -0.0001 | 0.0000 | 0.0001 |
| | 302 | -0.0009 | -0.0013 | -0.0010 | -0.0001 | 0.0000 | 0.0001 |
| | 303 | -0.0014 | -0.0018 | -0.0012 | 0.0000 | 0.0000 | 0.0001 |
| | 304 | -0.0001 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 305 | -0.0015 | -0.0021 | -0.0018 | -0.0001 | 0.0000 | 0.0001 |
| | 306 | 0.0006 | 0.0009 | 0.0006 | 0.0001 | 0.0000 | -0.0001 |
| | 307 | -0.0001 | -0.0003 | -0.0004 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0006 | 0.0011 | 0.0014 | 0.0001 | 0.0000 | -0.0001 |
| | 309 | 0.0276 | -0.0011 | -0.0003 | -0.0001 | 0.0000 | -0.0003 |
| | 310 | 0.0276 | -0.0011 | -0.0014 | -0.0001 | 0.0000 | -0.0003 |
| | 311 | -0.0291 | -0.0010 | -0.0003 | -0.0001 | 0.0000 | 0.0004 |
| | 312 | -0.0291 | -0.0010 | -0.0013 | -0.0001 | 0.0000 | 0.0004 |
| | 313 | 0.0077 | -0.0010 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0092 | -0.0010 | 0.0009 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0077 | -0.0010 | -0.0025 | -0.0001 | 0.0000 | 0.0000 |
| | 316 | -0.0092 | -0.0010 | -0.0025 | -0.0001 | 0.0000 | 0.0002 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 46 | 301 | 0.0014 | -0.1451 | 0.0001 | 0.0000 | 0.0000 | -0.0001 |
| | 302 | 0.0007 | -0.0857 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0006 | -0.0692 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | -0.0002 | 0.0295 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0012 | -0.1388 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | -0.0005 | 0.0592 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0005 | -0.0720 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0019 | 0.0792 | 0.0005 | 0.0000 | 0.0000 | -0.0017 |
| | 309 | 0.0287 | -0.0686 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | 0.0287 | -0.0687 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | -0.0275 | -0.0685 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | -0.0275 | -0.0686 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | 0.0090 | -0.0685 | 0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0078 | -0.0684 | 0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0090 | -0.0687 | -0.0016 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | -0.0078 | -0.0687 | -0.0016 | 0.0000 | 0.0000 | 0.0000 |
| 47 | 301 | -0.0012 | -0.0021 | 0.0013 | 0.0001 | 0.0000 | 0.0001 |
| | 302 | -0.0010 | -0.0012 | 0.0011 | 0.0001 | 0.0000 | 0.0001 |
| | 303 | 0.0000 | -0.0003 | 0.0003 | 0.0001 | 0.0000 | 0.0000 |
| | 304 | 0.0008 | 0.0008 | -0.0008 | 0.0000 | 0.0000 | -0.0001 |
| | 305 | -0.0016 | -0.0020 | 0.0019 | 0.0001 | 0.0000 | 0.0001 |
| | 306 | 0.0007 | 0.0008 | -0.0006 | -0.0001 | 0.0000 | -0.0001 |
| | 307 | -0.0015 | -0.0017 | 0.0014 | 0.0000 | 0.0000 | 0.0001 |
| | 308 | -0.0002 | 0.0002 | -0.0005 | -0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0255 | -0.0010 | 0.0014 | 0.0001 | 0.0000 | -0.0003 |
| | 310 | 0.0255 | -0.0010 | 0.0004 | 0.0001 | 0.0000 | -0.0003 |
| | 311 | -0.0271 | -0.0010 | 0.0014 | 0.0001 | 0.0000 | 0.0004 |
| | 312 | -0.0271 | -0.0010 | 0.0004 | 0.0001 | 0.0000 | 0.0004 |
| | 313 | 0.0071 | -0.0010 | 0.0025 | 0.0001 | 0.0000 | 0.0000 |
| | 314 | -0.0086 | -0.0010 | 0.0025 | 0.0001 | 0.0000 | 0.0002 |
| | 315 | 0.0071 | -0.0010 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | -0.0086 | -0.0010 | -0.0008 | 0.0000 | 0.0000 | 0.0002 |
| 48 | 301 | 0.0092 | -0.0158 | 0.0028 | 0.0002 | 0.0000 | -0.0011 |
| | 302 | -0.0062 | -0.0098 | 0.0025 | 0.0001 | 0.0001 | 0.0001 |
| | 303 | -0.0069 | 0.0043 | -0.0008 | 0.0000 | 0.0000 | 0.0005 |
| | 304 | 0.0012 | 0.0101 | -0.0024 | -0.0001 | 0.0000 | 0.0001 |
| | 305 | -0.0069 | -0.0168 | 0.0042 | 0.0002 | 0.0001 | 0.0000 |
| | 306 | 0.0101 | 0.0037 | -0.0010 | -0.0001 | -0.0001 | -0.0005 |
| | 307 | -0.0086 | -0.0181 | 0.0043 | 0.0002 | 0.0001 | 0.0001 |
| | 308 | 0.0035 | -0.0059 | 0.0011 | 0.0001 | 0.0000 | -0.0003 |
| | 309 | 0.0094 | -0.0081 | 0.0023 | 0.0001 | 0.0001 | -0.0002 |
| | 310 | 0.0095 | -0.0075 | 0.0016 | 0.0001 | 0.0001 | -0.0002 |
| | 311 | -0.0194 | -0.0081 | 0.0023 | 0.0001 | 0.0000 | 0.0004 |
| | 312 | -0.0193 | -0.0075 | 0.0016 | 0.0001 | 0.0000 | 0.0004 |
| | 313 | -0.0008 | -0.0088 | 0.0031 | 0.0001 | 0.0000 | 0.0000 |
| | 314 | -0.0094 | -0.0088 | 0.0031 | 0.0001 | 0.0000 | 0.0002 |
| | 315 | -0.0005 | -0.0068 | 0.0009 | 0.0001 | 0.0000 | 0.0000 |
| | 316 | -0.0091 | -0.0068 | 0.0009 | 0.0001 | 0.0000 | 0.0002 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 49 | 301 | -0.0147 | -0.0006 | 0.0001 | 0.0001 | 0.0005 | 0.0009 |
| | 302 | -0.0099 | -0.0003 | 0.0009 | 0.0001 | 0.0002 | 0.0006 |
| | 303 | -0.0006 | 0.0005 | -0.0003 | 0.0000 | -0.0001 | 0.0000 |
| | 304 | 0.0075 | 0.0005 | -0.0008 | -0.0001 | -0.0002 | -0.0005 |
| | 305 | -0.0158 | -0.0006 | 0.0015 | 0.0001 | 0.0003 | 0.0010 |
| | 306 | 0.0066 | -0.0002 | -0.0003 | 0.0000 | 0.0000 | -0.0004 |
| | 307 | -0.0159 | -0.0005 | 0.0014 | 0.0001 | 0.0003 | 0.0010 |
| | 308 | -0.0014 | -0.0004 | 0.0003 | 0.0000 | 0.0001 | 0.0001 |
| | 309 | -0.0013 | -0.0002 | 0.0009 | 0.0000 | 0.0002 | 0.0003 |
| | 310 | -0.0013 | -0.0002 | 0.0005 | 0.0000 | 0.0002 | 0.0003 |
| | 311 | -0.0146 | -0.0002 | 0.0009 | 0.0000 | 0.0001 | 0.0007 |
| | 312 | -0.0146 | -0.0002 | 0.0005 | 0.0000 | 0.0001 | 0.0007 |
| | 313 | -0.0060 | -0.0002 | 0.0015 | 0.0001 | 0.0002 | 0.0004 |
| | 314 | -0.0100 | -0.0002 | 0.0015 | 0.0001 | 0.0001 | 0.0006 |
| | 315 | -0.0060 | -0.0002 | -0.0001 | 0.0000 | 0.0002 | 0.0004 |
| | 316 | -0.0099 | -0.0002 | -0.0001 | 0.0000 | 0.0001 | 0.0006 |
| 50 | 301 | -0.0129 | -0.0027 | 0.0012 | 0.0000 | 0.0002 | 0.0000 |
| | 302 | 0.0158 | 0.0007 | -0.0003 | 0.0000 | -0.0007 | -0.0003 |
| | 303 | 0.0136 | -0.0021 | 0.0056 | -0.0003 | -0.0004 | -0.0002 |
| | 304 | -0.0052 | -0.0017 | 0.0032 | -0.0002 | 0.0003 | 0.0001 |
| | 305 | 0.0192 | 0.0021 | -0.0030 | 0.0002 | -0.0009 | -0.0004 |
| | 306 | -0.0223 | 0.0014 | -0.0050 | 0.0003 | 0.0008 | 0.0003 |
| | 307 | 0.0228 | 0.0016 | -0.0016 | 0.0001 | -0.0010 | -0.0004 |
| | 308 | -0.0061 | 0.0015 | -0.0038 | 0.0002 | 0.0002 | 0.0001 |
| | 309 | -0.0013 | 0.0007 | -0.0002 | 0.0000 | -0.0002 | -0.0002 |
| | 310 | -0.0013 | 0.0004 | -0.0002 | 0.0000 | -0.0002 | -0.0002 |
| | 311 | 0.0267 | 0.0007 | -0.0002 | 0.0000 | -0.0009 | -0.0002 |
| | 312 | 0.0266 | 0.0004 | -0.0002 | 0.0000 | -0.0009 | -0.0002 |
| | 313 | 0.0086 | 0.0011 | -0.0002 | 0.0000 | -0.0004 | -0.0002 |
| | 314 | 0.0169 | 0.0011 | -0.0002 | 0.0000 | -0.0006 | -0.0002 |
| | 315 | 0.0084 | 0.0001 | -0.0002 | 0.0000 | -0.0004 | -0.0002 |
| | 316 | 0.0168 | 0.0001 | -0.0002 | 0.0000 | -0.0006 | -0.0002 |
| 51 | 301 | -0.0002 | -0.0037 | 0.0011 | 0.0000 | 0.0000 | 0.0005 |
| | 302 | 0.0005 | 0.0007 | -0.0007 | 0.0000 | 0.0000 | -0.0001 |
| | 303 | -0.0007 | -0.0085 | 0.0669 | -0.0007 | 0.0014 | 0.0000 |
| | 304 | -0.0007 | -0.0051 | 0.0362 | -0.0004 | 0.0007 | 0.0001 |
| | 305 | 0.0011 | 0.0050 | -0.0323 | 0.0003 | -0.0006 | -0.0001 |
| | 306 | 0.0003 | 0.0075 | -0.0621 | 0.0007 | -0.0013 | 0.0001 |
| | 307 | 0.0009 | 0.0027 | -0.0136 | 0.0001 | -0.0002 | -0.0001 |
| | 308 | 0.0005 | 0.0057 | -0.0440 | 0.0005 | -0.0009 | 0.0000 |
| | 309 | -0.0167 | 0.0007 | -0.0006 | 0.0000 | 0.0000 | -0.0001 |
| | 310 | -0.0167 | 0.0004 | -0.0005 | 0.0000 | 0.0000 | -0.0001 |
| | 311 | 0.0175 | 0.0006 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0175 | 0.0004 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0047 | 0.0010 | -0.0006 | 0.0000 | 0.0000 | -0.0001 |
| | 314 | 0.0055 | 0.0010 | -0.0006 | 0.0000 | 0.0000 | -0.0001 |
| | 315 | -0.0047 | 0.0001 | -0.0005 | 0.0000 | 0.0000 | -0.0001 |
| | 316 | 0.0055 | 0.0001 | -0.0005 | 0.0000 | 0.0000 | -0.0001 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 52 | 301 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0003 | -0.0004 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0012 | 0.0023 | -0.0003 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0007 | 0.0012 | -0.0002 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0006 | -0.0011 | 0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | -0.0012 | -0.0021 | 0.0003 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0004 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0008 | -0.0015 | 0.0002 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 53 | 301 | -0.0147 | 0.1185 | 0.0179 | -0.0001 | 0.0001 | 0.0008 |
| | 302 | -0.0100 | 0.0847 | 0.0110 | -0.0001 | 0.0000 | 0.0006 |
| | 303 | -0.0160 | 0.1347 | 0.0182 | -0.0001 | 0.0000 | 0.0009 |
| | 304 | -0.0014 | 0.0108 | 0.0020 | 0.0000 | 0.0000 | 0.0001 |
| | 305 | -0.0159 | 0.1336 | 0.0175 | -0.0001 | 0.0000 | 0.0009 |
| | 306 | 0.0066 | -0.0581 | -0.0073 | 0.0000 | 0.0001 | -0.0004 |
| | 307 | -0.0006 | 0.0070 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0076 | -0.0630 | -0.0080 | 0.0001 | 0.0000 | -0.0004 |
| | 309 | -0.0015 | 0.0331 | 0.0138 | 0.0000 | -0.0001 | 0.0002 |
| | 310 | -0.0015 | 0.0331 | 0.0130 | 0.0000 | -0.0001 | 0.0002 |
| | 311 | -0.0145 | 0.1024 | 0.0046 | 0.0000 | 0.0000 | 0.0007 |
| | 312 | -0.0145 | 0.1024 | 0.0038 | 0.0000 | 0.0000 | 0.0007 |
| | 313 | -0.0061 | 0.0574 | 0.0115 | 0.0000 | 0.0000 | 0.0004 |
| | 314 | -0.0099 | 0.0781 | 0.0088 | 0.0000 | 0.0000 | 0.0005 |
| | 315 | -0.0061 | 0.0574 | 0.0089 | -0.0001 | 0.0000 | 0.0004 |
| | 316 | -0.0100 | 0.0781 | 0.0061 | -0.0001 | 0.0000 | 0.0005 |
| 54 | 301 | -0.0147 | 0.1185 | -0.0198 | 0.0001 | 0.0000 | 0.0008 |
| | 302 | -0.0100 | 0.0844 | -0.0123 | 0.0001 | 0.0000 | 0.0006 |
| | 303 | -0.0006 | 0.0070 | -0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0075 | -0.0629 | 0.0096 | -0.0001 | 0.0000 | -0.0004 |
| | 305 | -0.0158 | 0.1332 | -0.0195 | 0.0001 | 0.0001 | 0.0009 |
| | 306 | 0.0066 | -0.0579 | 0.0081 | 0.0000 | -0.0001 | -0.0004 |
| | 307 | -0.0159 | 0.1343 | -0.0201 | 0.0001 | 0.0001 | 0.0009 |
| | 308 | -0.0014 | 0.0108 | -0.0026 | 0.0000 | 0.0000 | 0.0001 |
| | 309 | -0.0013 | 0.0320 | -0.0135 | 0.0000 | 0.0001 | 0.0002 |
| | 310 | -0.0013 | 0.0320 | -0.0142 | 0.0000 | 0.0001 | 0.0002 |
| | 311 | -0.0146 | 0.1031 | -0.0055 | 0.0000 | 0.0000 | 0.0007 |
| | 312 | -0.0146 | 0.1031 | -0.0062 | 0.0000 | 0.0000 | 0.0007 |
| | 313 | -0.0060 | 0.0569 | -0.0098 | 0.0001 | 0.0000 | 0.0004 |
| | 314 | -0.0100 | 0.0782 | -0.0074 | 0.0001 | 0.0000 | 0.0005 |
| | 315 | -0.0060 | 0.0569 | -0.0123 | 0.0000 | 0.0000 | 0.0004 |
| | 316 | -0.0099 | 0.0781 | -0.0099 | 0.0000 | 0.0000 | 0.0005 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 55 | 301 | 0.0147 | 0.1182 | 0.0181 | -0.0001 | -0.0001 | -0.0008 |
| | 302 | 0.0100 | 0.0845 | 0.0111 | -0.0001 | 0.0000 | -0.0006 |
| | 303 | 0.0159 | 0.1344 | 0.0184 | -0.0001 | 0.0001 | -0.0009 |
| | 304 | 0.0014 | 0.0108 | 0.0020 | 0.0000 | 0.0000 | -0.0001 |
| | 305 | 0.0159 | 0.1333 | 0.0177 | -0.0001 | 0.0000 | -0.0009 |
| | 306 | -0.0066 | -0.0580 | -0.0073 | 0.0000 | -0.0001 | 0.0004 |
| | 307 | 0.0006 | 0.0070 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | -0.0076 | -0.0630 | -0.0087 | 0.0001 | 0.0000 | 0.0004 |
| | 309 | 0.0145 | 0.1024 | 0.0047 | 0.0000 | 0.0000 | -0.0007 |
| | 310 | 0.0145 | 0.1024 | 0.0039 | 0.0000 | 0.0000 | -0.0007 |
| | 311 | 0.0015 | 0.0328 | 0.0139 | 0.0000 | 0.0001 | -0.0002 |
| | 312 | 0.0015 | 0.0328 | 0.0131 | 0.0000 | 0.0001 | -0.0002 |
| | 313 | 0.0099 | 0.0780 | 0.0089 | 0.0000 | 0.0000 | -0.0005 |
| | 314 | 0.0060 | 0.0572 | 0.0116 | 0.0000 | 0.0000 | -0.0004 |
| | 315 | 0.0099 | 0.0780 | 0.0062 | -0.0001 | 0.0000 | -0.0005 |
| | 316 | 0.0061 | 0.0572 | 0.0089 | -0.0001 | 0.0000 | -0.0004 |
| 56 | 301 | 0.0147 | 0.1185 | -0.0198 | 0.0001 | 0.0000 | -0.0008 |
| | 302 | 0.0100 | 0.0844 | -0.0123 | 0.0001 | 0.0000 | -0.0006 |
| | 303 | 0.0006 | 0.0070 | -0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | -0.0075 | -0.0629 | 0.0096 | -0.0001 | 0.0000 | 0.0004 |
| | 305 | 0.0158 | 0.1332 | -0.0195 | 0.0001 | -0.0001 | -0.0009 |
| | 306 | -0.0066 | -0.0579 | 0.0081 | 0.0000 | 0.0001 | 0.0004 |
| | 307 | 0.0159 | 0.1343 | -0.0202 | 0.0001 | -0.0001 | -0.0009 |
| | 308 | 0.0014 | 0.0108 | -0.0021 | 0.0000 | 0.0000 | -0.0001 |
| | 309 | 0.0146 | 0.1031 | -0.0055 | 0.0000 | 0.0000 | -0.0007 |
| | 310 | 0.0146 | 0.1031 | -0.0062 | 0.0000 | 0.0000 | -0.0007 |
| | 311 | 0.0013 | 0.0320 | -0.0135 | 0.0000 | -0.0001 | -0.0002 |
| | 312 | 0.0013 | 0.0320 | -0.0142 | 0.0000 | -0.0001 | -0.0002 |
| | 313 | 0.0100 | 0.0782 | -0.0074 | 0.0001 | 0.0000 | -0.0005 |
| | 314 | 0.0060 | 0.0569 | -0.0098 | 0.0001 | 0.0000 | -0.0004 |
| | 315 | 0.0099 | 0.0781 | -0.0099 | 0.0000 | 0.0000 | -0.0005 |
| | 316 | 0.0060 | 0.0569 | -0.0123 | 0.0000 | 0.0000 | -0.0004 |
| 57 | 301 | -0.0011 | -0.1729 | 0.0022 | -0.0001 | 0.0000 | -0.0008 |
| | 302 | -0.0009 | -0.1017 | 0.0015 | -0.0001 | 0.0000 | -0.0004 |
| | 303 | -0.0014 | -0.1654 | 0.0006 | 0.0000 | 0.0000 | -0.0007 |
| | 304 | -0.0001 | -0.0140 | -0.0009 | 0.0001 | 0.0000 | -0.0001 |
| | 305 | -0.0015 | -0.1653 | 0.0024 | -0.0001 | 0.0000 | -0.0007 |
| | 306 | 0.0006 | 0.0700 | -0.0009 | 0.0001 | 0.0000 | 0.0003 |
| | 307 | -0.0001 | -0.0005 | 0.0018 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0006 | 0.0841 | 0.0017 | 0.0001 | 0.0000 | 0.0004 |
| | 309 | 0.0276 | -0.1869 | 0.0004 | -0.0001 | 0.0000 | -0.0007 |
| | 310 | 0.0276 | -0.1868 | -0.0006 | -0.0001 | 0.0000 | -0.0007 |
| | 311 | -0.0291 | 0.0242 | 0.0029 | -0.0001 | 0.0000 | 0.0000 |
| | 312 | -0.0291 | 0.0242 | 0.0019 | -0.0001 | 0.0000 | 0.0000 |
| | 313 | 0.0077 | -0.1129 | 0.0024 | 0.0000 | 0.0000 | -0.0005 |
| | 314 | -0.0092 | -0.0500 | 0.0032 | 0.0000 | 0.0000 | -0.0003 |
| | 315 | 0.0077 | -0.1127 | -0.0008 | -0.0001 | 0.0000 | -0.0005 |
| | 316 | -0.0092 | -0.0498 | -0.0001 | -0.0001 | 0.0000 | -0.0003 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 58 | 301 | -0.0012 | -0.1734 | -0.0017 | 0.0001 | 0.0000 | -0.0008 |
| | 302 | -0.0010 | -0.1015 | -0.0012 | 0.0001 | 0.0000 | -0.0004 |
| | 303 | 0.0000 | -0.0007 | -0.0018 | 0.0001 | 0.0000 | 0.0000 |
| | 304 | 0.0008 | 0.0833 | -0.0002 | 0.0000 | 0.0000 | 0.0004 |
| | 305 | -0.0016 | -0.1650 | -0.0020 | 0.0001 | 0.0000 | -0.0007 |
| | 306 | 0.0007 | 0.0699 | 0.0008 | -0.0001 | 0.0000 | 0.0003 |
| | 307 | -0.0015 | -0.1650 | -0.0001 | 0.0000 | 0.0000 | -0.0007 |
| | 308 | -0.0002 | -0.0133 | -0.0003 | -0.0001 | 0.0000 | -0.0001 |
| | 309 | 0.0255 | -0.1790 | -0.0009 | 0.0001 | 0.0000 | -0.0007 |
| | 310 | 0.0255 | -0.1791 | -0.0019 | 0.0001 | 0.0000 | -0.0007 |
| | 311 | -0.0271 | 0.0167 | -0.0001 | 0.0001 | 0.0000 | 0.0000 |
| | 312 | -0.0271 | 0.0166 | -0.0011 | 0.0001 | 0.0000 | 0.0000 |
| | 313 | 0.0071 | -0.1103 | 0.0006 | 0.0001 | 0.0000 | -0.0005 |
| | 314 | -0.0086 | -0.0520 | 0.0008 | 0.0001 | 0.0000 | -0.0003 |
| | 315 | 0.0071 | -0.1104 | -0.0027 | 0.0000 | 0.0000 | -0.0005 |
| | 316 | -0.0086 | -0.0521 | -0.0025 | 0.0000 | 0.0000 | -0.0003 |
| 59 | 301 | 0.0014 | -0.1719 | 0.0023 | -0.0001 | 0.0000 | 0.0008 |
| | 302 | 0.0011 | -0.1010 | 0.0015 | -0.0001 | 0.0000 | 0.0004 |
| | 303 | 0.0017 | -0.1644 | 0.0007 | 0.0000 | 0.0000 | 0.0007 |
| | 304 | 0.0001 | -0.0139 | -0.0009 | 0.0001 | 0.0000 | 0.0001 |
| | 305 | 0.0018 | -0.1643 | 0.0025 | -0.0001 | 0.0000 | 0.0007 |
| | 306 | -0.0008 | 0.0696 | -0.0010 | 0.0001 | 0.0000 | -0.0003 |
| | 307 | 0.0001 | -0.0005 | 0.0018 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | -0.0005 | 0.0842 | -0.0008 | 0.0000 | 0.0000 | -0.0004 |
| | 309 | 0.0292 | 0.0245 | 0.0029 | -0.0001 | 0.0000 | 0.0000 |
| | 310 | 0.0292 | 0.0245 | 0.0019 | -0.0001 | 0.0000 | 0.0000 |
| | 311 | -0.0274 | -0.1862 | 0.0005 | -0.0001 | 0.0000 | 0.0007 |
| | 312 | -0.0274 | -0.1861 | -0.0005 | -0.0001 | 0.0000 | 0.0007 |
| | 313 | 0.0093 | -0.0495 | 0.0032 | 0.0000 | 0.0000 | 0.0003 |
| | 314 | -0.0076 | -0.1123 | 0.0025 | 0.0000 | 0.0000 | 0.0005 |
| | 315 | 0.0093 | -0.0494 | -0.0001 | -0.0001 | 0.0000 | 0.0003 |
| | 316 | -0.0075 | -0.1122 | -0.0008 | -0.0001 | 0.0000 | 0.0005 |
| 60 | 301 | 0.0012 | -0.1732 | -0.0016 | 0.0001 | 0.0000 | 0.0008 |
| | 302 | 0.0010 | -0.1014 | -0.0011 | 0.0001 | 0.0000 | 0.0004 |
| | 303 | 0.0001 | -0.0005 | -0.0017 | 0.0001 | 0.0000 | 0.0000 |
| | 304 | -0.0008 | 0.0833 | -0.0001 | 0.0000 | 0.0000 | -0.0004 |
| | 305 | 0.0016 | -0.1648 | -0.0019 | 0.0001 | 0.0000 | 0.0007 |
| | 306 | -0.0007 | 0.0698 | 0.0007 | -0.0001 | 0.0000 | -0.0003 |
| | 307 | 0.0015 | -0.1650 | -0.0002 | 0.0000 | 0.0000 | 0.0007 |
| | 308 | 0.0004 | -0.0129 | 0.0016 | -0.0001 | 0.0000 | 0.0001 |
| | 309 | 0.0271 | 0.0168 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 310 | 0.0271 | 0.0168 | -0.0010 | 0.0001 | 0.0000 | 0.0000 |
| | 311 | -0.0255 | -0.1790 | -0.0008 | 0.0001 | 0.0000 | 0.0007 |
| | 312 | -0.0255 | -0.1790 | -0.0018 | 0.0001 | 0.0000 | 0.0007 |
| | 313 | 0.0087 | -0.0519 | 0.0009 | 0.0001 | 0.0000 | 0.0003 |
| | 314 | -0.0070 | -0.1102 | 0.0006 | 0.0001 | 0.0000 | 0.0005 |
| | 315 | 0.0086 | -0.0520 | -0.0025 | 0.0000 | 0.0000 | 0.0003 |
| | 316 | -0.0071 | -0.1103 | -0.0027 | 0.0000 | 0.0000 | 0.0005 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 61 | 301 | 0.0018 | -0.4874 | 0.0008 | 0.0000 | 0.0000 | -0.0013 |
| | 302 | 0.0010 | -0.2655 | 0.0005 | 0.0000 | 0.0000 | -0.0007 |
| | 303 | 0.0008 | -0.2160 | 0.0006 | 0.0000 | 0.0000 | -0.0006 |
| | 304 | -0.0003 | 0.0891 | -0.0001 | 0.0000 | 0.0000 | 0.0002 |
| | 305 | 0.0016 | -0.4301 | 0.0007 | 0.0000 | 0.0000 | -0.0012 |
| | 306 | -0.0006 | 0.1807 | -0.0003 | 0.0000 | 0.0000 | 0.0005 |
| | 307 | 0.0007 | -0.2109 | 0.0002 | 0.0000 | 0.0000 | -0.0006 |
| | 308 | 0.0021 | -0.4043 | 0.0007 | 0.0000 | 0.0000 | -0.0015 |
| | 309 | 0.0288 | -0.2132 | -0.0001 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0288 | -0.2133 | -0.0011 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | -0.0272 | -0.2115 | 0.0018 | 0.0000 | 0.0000 | -0.0006 |
| | 312 | -0.0272 | -0.2116 | 0.0008 | 0.0000 | 0.0000 | -0.0006 |
| | 313 | 0.0091 | -0.2125 | 0.0017 | 0.0000 | 0.0000 | -0.0006 |
| | 314 | -0.0076 | -0.2120 | 0.0023 | 0.0000 | 0.0000 | -0.0006 |
| | 315 | 0.0091 | -0.2128 | -0.0016 | 0.0000 | 0.0000 | -0.0006 |
| | 316 | -0.0076 | -0.2123 | -0.0010 | 0.0000 | 0.0000 | -0.0006 |
| 62 | 301 | -0.0016 | -0.4874 | 0.0009 | 0.0000 | 0.0000 | 0.0013 |
| | 302 | -0.0009 | -0.2655 | 0.0005 | 0.0000 | 0.0000 | 0.0007 |
| | 303 | -0.0006 | -0.2160 | 0.0007 | 0.0000 | 0.0000 | 0.0006 |
| | 304 | 0.0003 | 0.0891 | -0.0001 | 0.0000 | 0.0000 | -0.0002 |
| | 305 | -0.0014 | -0.4301 | 0.0008 | 0.0000 | 0.0000 | 0.0012 |
| | 306 | 0.0005 | 0.1807 | -0.0003 | 0.0000 | 0.0000 | -0.0005 |
| | 307 | -0.0008 | -0.2179 | 0.0002 | 0.0000 | 0.0000 | 0.0006 |
| | 308 | 0.0012 | 0.0887 | -0.0002 | 0.0000 | 0.0000 | -0.0002 |
| | 309 | 0.0273 | -0.2115 | 0.0019 | 0.0000 | 0.0000 | 0.0006 |
| | 310 | 0.0273 | -0.2116 | 0.0009 | 0.0000 | 0.0000 | 0.0006 |
| | 311 | -0.0287 | -0.2132 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | -0.0287 | -0.2133 | -0.0010 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0077 | -0.2120 | 0.0023 | 0.0000 | 0.0000 | 0.0006 |
| | 314 | -0.0090 | -0.2126 | 0.0018 | 0.0000 | 0.0000 | 0.0006 |
| | 315 | 0.0076 | -0.2123 | -0.0009 | 0.0000 | 0.0000 | 0.0006 |
| | 316 | -0.0091 | -0.2128 | -0.0015 | 0.0000 | 0.0000 | 0.0006 |
| 63 | 301 | -0.0004 | -0.0407 | 0.0025 | 0.0000 | 0.0000 | -0.0013 |
| | 302 | 0.0000 | -0.0243 | 0.0021 | 0.0000 | 0.0000 | -0.0009 |
| | 303 | 0.0002 | 0.0094 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0001 | 0.0244 | -0.0019 | 0.0000 | 0.0000 | 0.0007 |
| | 305 | 0.0000 | -0.0399 | 0.0036 | 0.0000 | 0.0000 | -0.0014 |
| | 306 | -0.0002 | 0.0126 | -0.0009 | 0.0000 | 0.0000 | 0.0006 |
| | 307 | 0.0000 | -0.0469 | 0.0035 | 0.0000 | -0.0001 | -0.0014 |
| | 308 | -0.0001 | -0.0104 | 0.0007 | 0.0000 | 0.0000 | -0.0001 |
| | 309 | 0.0191 | -0.0195 | 0.0021 | 0.0000 | 0.0000 | -0.0007 |
| | 310 | 0.0191 | -0.0193 | 0.0013 | 0.0000 | 0.0000 | -0.0007 |
| | 311 | -0.0191 | -0.0196 | 0.0021 | 0.0000 | -0.0001 | -0.0007 |
| | 312 | -0.0191 | -0.0193 | 0.0013 | 0.0000 | -0.0001 | -0.0007 |
| | 313 | 0.0057 | -0.0199 | 0.0030 | 0.0000 | 0.0000 | -0.0007 |
| | 314 | -0.0057 | -0.0199 | 0.0030 | 0.0000 | 0.0000 | -0.0007 |
| | 315 | 0.0057 | -0.0189 | 0.0004 | 0.0000 | 0.0000 | -0.0007 |
| | 316 | -0.0057 | -0.0190 | 0.0004 | 0.0000 | 0.0000 | -0.0007 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 64 | 301 | -0.0001 | -0.0506 | 0.0025 | 0.0000 | -0.0001 | -0.0007 |
| | 302 | 0.0000 | -0.0309 | 0.0024 | 0.0000 | 0.0000 | -0.0001 |
| | 303 | 0.0001 | 0.0078 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0287 | -0.0024 | 0.0000 | 0.0000 | 0.0001 |
| | 305 | 0.0000 | -0.0510 | 0.0040 | 0.0000 | 0.0000 | -0.0002 |
| | 306 | -0.0001 | 0.0162 | -0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | -0.0555 | 0.0042 | 0.0000 | 0.0000 | -0.0002 |
| | 308 | -0.0001 | -0.0106 | 0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0191 | -0.0251 | 0.0027 | 0.0000 | 0.0000 | -0.0005 |
| | 310 | 0.0191 | -0.0248 | 0.0018 | 0.0000 | 0.0000 | -0.0005 |
| | 311 | -0.0191 | -0.0247 | 0.0019 | 0.0000 | 0.0000 | 0.0003 |
| | 312 | -0.0191 | -0.0244 | 0.0011 | 0.0000 | 0.0000 | 0.0003 |
| | 313 | 0.0057 | -0.0254 | 0.0034 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | -0.0057 | -0.0253 | 0.0032 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0057 | -0.0242 | 0.0006 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | -0.0057 | -0.0241 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
| 65 | 301 | 0.0001 | -0.0506 | 0.0024 | 0.0000 | 0.0001 | 0.0007 |
| | 302 | 0.0000 | -0.0309 | 0.0023 | 0.0000 | 0.0000 | 0.0001 |
| | 303 | 0.0000 | 0.0078 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0287 | -0.0024 | 0.0000 | 0.0000 | -0.0001 |
| | 305 | 0.0000 | -0.0510 | 0.0040 | 0.0000 | 0.0000 | 0.0002 |
| | 306 | 0.0000 | 0.0162 | -0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | -0.0554 | 0.0042 | 0.0000 | 0.0000 | 0.0002 |
| | 308 | 0.0000 | -0.0162 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0191 | -0.0247 | 0.0019 | 0.0000 | 0.0000 | -0.0003 |
| | 310 | 0.0191 | -0.0244 | 0.0011 | 0.0000 | 0.0000 | -0.0003 |
| | 311 | -0.0191 | -0.0251 | 0.0027 | 0.0000 | 0.0000 | 0.0005 |
| | 312 | -0.0191 | -0.0248 | 0.0018 | 0.0000 | 0.0000 | 0.0005 |
| | 313 | 0.0057 | -0.0253 | 0.0032 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0057 | -0.0254 | 0.0034 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0057 | -0.0241 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | -0.0057 | -0.0242 | 0.0006 | 0.0000 | 0.0000 | 0.0002 |
| 66 | 301 | 0.0004 | -0.0407 | 0.0025 | 0.0000 | 0.0000 | 0.0013 |
| | 302 | 0.0000 | -0.0243 | 0.0021 | 0.0000 | 0.0000 | 0.0009 |
| | 303 | -0.0002 | 0.0094 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | -0.0001 | 0.0244 | -0.0019 | 0.0000 | 0.0000 | -0.0007 |
| | 305 | 0.0001 | -0.0399 | 0.0036 | 0.0000 | 0.0000 | 0.0014 |
| | 306 | 0.0002 | 0.0126 | -0.0009 | 0.0000 | 0.0000 | -0.0006 |
| | 307 | 0.0000 | -0.0468 | 0.0035 | 0.0000 | 0.0001 | 0.0014 |
| | 308 | 0.0001 | -0.0160 | 0.0006 | 0.0000 | 0.0000 | 0.0001 |
| | 309 | 0.0191 | -0.0196 | 0.0021 | 0.0000 | 0.0001 | 0.0007 |
| | 310 | 0.0191 | -0.0193 | 0.0013 | 0.0000 | 0.0001 | 0.0007 |
| | 311 | -0.0191 | -0.0195 | 0.0021 | 0.0000 | 0.0000 | 0.0007 |
| | 312 | -0.0191 | -0.0193 | 0.0013 | 0.0000 | 0.0000 | 0.0007 |
| | 313 | 0.0057 | -0.0199 | 0.0030 | 0.0000 | 0.0000 | 0.0007 |
| | 314 | -0.0057 | -0.0199 | 0.0030 | 0.0000 | 0.0000 | 0.0007 |
| | 315 | 0.0057 | -0.0190 | 0.0004 | 0.0000 | 0.0000 | 0.0007 |
| | 316 | -0.0057 | -0.0189 | 0.0004 | 0.0000 | 0.0000 | 0.0007 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 67 | 301 | -0.0006 | -0.0440 | -0.0024 | 0.0001 | 0.0000 | -0.0013 |
| | 302 | -0.0001 | -0.0258 | -0.0020 | 0.0000 | 0.0000 | -0.0009 |
| | 303 | -0.0002 | -0.0511 | -0.0033 | 0.0001 | 0.0000 | -0.0014 |
| | 304 | -0.0001 | -0.0117 | -0.0007 | 0.0000 | 0.0000 | -0.0001 |
| | 305 | -0.0003 | -0.0424 | -0.0034 | 0.0001 | 0.0000 | -0.0014 |
| | 306 | -0.0001 | 0.0135 | 0.0009 | 0.0000 | 0.0000 | 0.0006 |
| | 307 | 0.0002 | 0.0111 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0001 | 0.0265 | 0.0019 | 0.0000 | 0.0000 | 0.0007 |
| | 309 | 0.0212 | -0.0206 | -0.0012 | 0.0000 | 0.0000 | -0.0007 |
| | 310 | 0.0212 | -0.0207 | -0.0020 | 0.0000 | 0.0000 | -0.0007 |
| | 311 | -0.0214 | -0.0206 | -0.0012 | 0.0000 | 0.0001 | -0.0007 |
| | 312 | -0.0214 | -0.0208 | -0.0020 | 0.0000 | 0.0001 | -0.0007 |
| | 313 | 0.0063 | -0.0204 | -0.0002 | 0.0000 | 0.0000 | -0.0007 |
| | 314 | -0.0065 | -0.0204 | -0.0002 | 0.0000 | 0.0000 | -0.0007 |
| | 315 | 0.0063 | -0.0209 | -0.0030 | 0.0000 | 0.0000 | -0.0007 |
| | 316 | -0.0064 | -0.0209 | -0.0030 | 0.0000 | 0.0000 | -0.0007 |
| 68 | 301 | -0.0003 | -0.0556 | -0.0022 | 0.0001 | 0.0001 | -0.0006 |
| | 302 | -0.0001 | -0.0336 | -0.0021 | 0.0000 | 0.0000 | -0.0001 |
| | 303 | -0.0002 | -0.0610 | -0.0039 | 0.0001 | 0.0000 | -0.0002 |
| | 304 | -0.0001 | -0.0118 | -0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0002 | -0.0552 | -0.0037 | 0.0001 | 0.0000 | -0.0002 |
| | 306 | 0.0000 | 0.0179 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0001 | 0.0089 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0001 | 0.0314 | 0.0022 | 0.0000 | 0.0000 | 0.0001 |
| | 309 | 0.0212 | -0.0270 | -0.0016 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0212 | -0.0272 | -0.0024 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | -0.0214 | -0.0266 | -0.0009 | 0.0000 | 0.0000 | 0.0004 |
| | 312 | -0.0214 | -0.0268 | -0.0018 | 0.0000 | 0.0000 | 0.0004 |
| | 313 | 0.0062 | -0.0266 | -0.0003 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | -0.0065 | -0.0265 | -0.0001 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | 0.0063 | -0.0273 | -0.0033 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | -0.0064 | -0.0272 | -0.0031 | 0.0000 | 0.0000 | 0.0001 |
| 69 | 301 | -0.0002 | -0.0557 | -0.0022 | 0.0001 | -0.0001 | 0.0006 |
| | 302 | -0.0001 | -0.0336 | -0.0021 | 0.0000 | 0.0000 | 0.0001 |
| | 303 | -0.0002 | -0.0609 | -0.0039 | 0.0001 | 0.0000 | 0.0002 |
| | 304 | 0.0000 | -0.0118 | -0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0002 | -0.0551 | -0.0037 | 0.0001 | 0.0000 | 0.0002 |
| | 306 | 0.0001 | 0.0179 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0090 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0246 | 0.0025 | 0.0000 | 0.0000 | -0.0001 |
| | 309 | 0.0212 | -0.0266 | -0.0010 | 0.0000 | 0.0000 | -0.0004 |
| | 310 | 0.0212 | -0.0268 | -0.0018 | 0.0000 | 0.0000 | -0.0004 |
| | 311 | -0.0214 | -0.0269 | -0.0016 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | -0.0214 | -0.0271 | -0.0025 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0062 | -0.0264 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0065 | -0.0265 | -0.0003 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0063 | -0.0272 | -0.0031 | 0.0000 | 0.0000 | -0.0001 |
| | 316 | -0.0065 | -0.0273 | -0.0033 | 0.0000 | 0.0000 | 0.0002 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 70 | 301 | 0.0001 | -0.0440 | -0.0024 | 0.0001 | 0.0000 | 0.0013 |
| | 302 | -0.0001 | -0.0258 | -0.0020 | 0.0000 | 0.0000 | 0.0009 |
| | 303 | -0.0002 | -0.0511 | -0.0033 | 0.0001 | 0.0000 | 0.0014 |
| | 304 | 0.0001 | -0.0117 | -0.0007 | 0.0000 | 0.0000 | 0.0001 |
| | 305 | -0.0002 | -0.0423 | -0.0034 | 0.0001 | 0.0000 | 0.0014 |
| | 306 | 0.0002 | 0.0135 | 0.0009 | 0.0000 | 0.0000 | -0.0006 |
| | 307 | -0.0001 | 0.0111 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0198 | 0.0021 | 0.0000 | 0.0000 | -0.0007 |
| | 309 | 0.0212 | -0.0206 | -0.0012 | 0.0000 | -0.0001 | 0.0007 |
| | 310 | 0.0212 | -0.0208 | -0.0020 | 0.0000 | -0.0001 | 0.0007 |
| | 311 | -0.0215 | -0.0206 | -0.0012 | 0.0000 | 0.0000 | 0.0007 |
| | 312 | -0.0214 | -0.0207 | -0.0020 | 0.0000 | 0.0000 | 0.0007 |
| | 313 | 0.0062 | -0.0204 | -0.0002 | 0.0000 | 0.0000 | 0.0007 |
| | 314 | -0.0065 | -0.0204 | -0.0002 | 0.0000 | 0.0000 | 0.0007 |
| | 315 | 0.0063 | -0.0209 | -0.0030 | 0.0000 | 0.0000 | 0.0007 |
| | 316 | -0.0065 | -0.0209 | -0.0030 | 0.0000 | 0.0000 | 0.0007 |
| 71 | 301 | 0.0004 | 0.1697 | -0.0229 | 0.0000 | 0.0002 | 0.0011 |
| | 302 | 0.0000 | 0.1155 | -0.0147 | 0.0000 | 0.0001 | 0.0008 |
| | 303 | -0.0002 | 0.0147 | -0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | -0.0001 | -0.0831 | 0.0106 | 0.0000 | -0.0001 | -0.0006 |
| | 305 | 0.0000 | 0.1838 | -0.0232 | 0.0000 | 0.0002 | 0.0012 |
| | 306 | 0.0002 | -0.0769 | 0.0099 | 0.0000 | -0.0001 | -0.0005 |
| | 307 | 0.0000 | 0.1763 | -0.0227 | 0.0000 | 0.0002 | 0.0012 |
| | 308 | 0.0001 | 0.0060 | -0.0012 | 0.0000 | 0.0000 | 0.0001 |
| | 309 | 0.0191 | 0.0921 | -0.0196 | 0.0000 | 0.0002 | 0.0006 |
| | 310 | 0.0191 | 0.0923 | -0.0201 | 0.0000 | 0.0001 | 0.0006 |
| | 311 | -0.0191 | 0.0925 | -0.0034 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | -0.0191 | 0.0928 | -0.0039 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0057 | 0.0919 | -0.0133 | 0.0000 | 0.0001 | 0.0006 |
| | 314 | -0.0057 | 0.0921 | -0.0085 | 0.0000 | 0.0001 | 0.0006 |
| | 315 | 0.0057 | 0.0928 | -0.0150 | 0.0000 | 0.0001 | 0.0006 |
| | 316 | -0.0057 | 0.0929 | -0.0101 | 0.0000 | 0.0001 | 0.0006 |
| 72 | 301 | -0.0004 | 0.1697 | -0.0229 | 0.0000 | -0.0002 | -0.0011 |
| | 302 | 0.0000 | 0.1155 | -0.0147 | 0.0000 | -0.0001 | -0.0008 |
| | 303 | 0.0002 | 0.0147 | -0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0001 | -0.0831 | 0.0106 | 0.0000 | 0.0001 | 0.0006 |
| | 305 | 0.0000 | 0.1838 | -0.0232 | 0.0000 | -0.0002 | -0.0012 |
| | 306 | -0.0002 | -0.0769 | 0.0099 | 0.0000 | 0.0001 | 0.0005 |
| | 307 | 0.0000 | 0.1763 | -0.0227 | 0.0000 | -0.0002 | -0.0012 |
| | 308 | -0.0001 | 0.0116 | -0.0016 | 0.0000 | 0.0000 | -0.0001 |
| | 309 | 0.0191 | 0.0925 | -0.0034 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0191 | 0.0928 | -0.0039 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | -0.0191 | 0.0921 | -0.0196 | 0.0000 | -0.0002 | -0.0006 |
| | 312 | -0.0191 | 0.0923 | -0.0201 | 0.0000 | -0.0001 | -0.0006 |
| | 313 | 0.0057 | 0.0921 | -0.0085 | 0.0000 | -0.0001 | -0.0006 |
| | 314 | -0.0057 | 0.0919 | -0.0133 | 0.0000 | -0.0001 | -0.0006 |
| | 315 | 0.0057 | 0.0930 | -0.0101 | 0.0000 | -0.0001 | -0.0006 |
| | 316 | -0.0057 | 0.0928 | -0.0150 | 0.0000 | -0.0001 | -0.0006 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 73 | 301 | 0.0001 | 0.1664 | 0.0201 | 0.0001 | -0.0002 | 0.0011 |
| | 302 | -0.0001 | 0.1144 | 0.0129 | 0.0000 | -0.0001 | 0.0008 |
| | 303 | -0.0002 | 0.1725 | 0.0198 | 0.0001 | -0.0002 | 0.0012 |
| | 304 | 0.0001 | 0.0103 | 0.0012 | 0.0000 | 0.0000 | 0.0001 |
| | 305 | -0.0002 | 0.1820 | 0.0204 | 0.0001 | -0.0002 | 0.0012 |
| | 306 | 0.0002 | -0.0761 | -0.0088 | 0.0000 | 0.0001 | -0.0005 |
| | 307 | -0.0001 | 0.0165 | 0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | -0.0879 | -0.0096 | 0.0000 | 0.0001 | -0.0006 |
| | 309 | 0.0212 | 0.0913 | 0.0190 | 0.0000 | -0.0001 | 0.0006 |
| | 310 | 0.0212 | 0.0912 | 0.0185 | 0.0000 | -0.0001 | 0.0006 |
| | 311 | -0.0215 | 0.0918 | 0.0021 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | -0.0215 | 0.0916 | 0.0016 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0062 | 0.0916 | 0.0136 | 0.0000 | -0.0001 | 0.0006 |
| | 314 | -0.0065 | 0.0918 | 0.0086 | 0.0000 | -0.0001 | 0.0006 |
| | 315 | 0.0063 | 0.0912 | 0.0120 | 0.0000 | -0.0001 | 0.0006 |
| | 316 | -0.0065 | 0.0913 | 0.0070 | 0.0000 | -0.0001 | 0.0006 |
| 74 | 301 | -0.0005 | 0.1664 | 0.0203 | 0.0001 | 0.0002 | -0.0011 |
| | 302 | -0.0001 | 0.1144 | 0.0130 | 0.0000 | 0.0001 | -0.0008 |
| | 303 | -0.0002 | 0.1725 | 0.0200 | 0.0001 | 0.0002 | -0.0012 |
| | 304 | -0.0001 | 0.0103 | 0.0013 | 0.0000 | 0.0000 | -0.0001 |
| | 305 | -0.0003 | 0.1820 | 0.0206 | 0.0001 | 0.0002 | -0.0012 |
| | 306 | -0.0001 | -0.0761 | -0.0089 | 0.0000 | -0.0001 | 0.0005 |
| | 307 | 0.0002 | 0.0165 | 0.0017 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0001 | -0.0812 | -0.0093 | 0.0000 | -0.0001 | 0.0006 |
| | 309 | 0.0212 | 0.0918 | 0.0022 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0213 | 0.0916 | 0.0018 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | -0.0214 | 0.0913 | 0.0190 | 0.0000 | 0.0001 | -0.0006 |
| | 312 | -0.0214 | 0.0912 | 0.0186 | 0.0000 | 0.0001 | -0.0006 |
| | 313 | 0.0063 | 0.0918 | 0.0087 | 0.0000 | 0.0001 | -0.0006 |
| | 314 | -0.0065 | 0.0917 | 0.0137 | 0.0000 | 0.0001 | -0.0006 |
| | 315 | 0.0063 | 0.0913 | 0.0071 | 0.0000 | 0.0001 | -0.0006 |
| | 316 | -0.0064 | 0.0912 | 0.0121 | 0.0000 | 0.0001 | -0.0006 |
| 75 | 301 | -0.0011 | -0.0051 | 0.0106 | 0.0000 | -0.0001 | -0.0008 |
| | 302 | 0.0001 | 0.0003 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0003 | -0.0029 | 0.0421 | 0.0000 | -0.0004 | -0.0001 |
| | 304 | 0.0002 | -0.0119 | 0.1436 | 0.0000 | -0.0015 | -0.0001 |
| | 305 | 0.0003 | -0.0081 | 0.1034 | 0.0000 | -0.0011 | -0.0001 |
| | 306 | 0.0002 | -0.0173 | 0.2044 | 0.0000 | -0.0021 | -0.0002 |
| | 307 | -0.0003 | 0.0183 | -0.2190 | 0.0000 | 0.0023 | 0.0002 |
| | 308 | -0.0002 | 0.0095 | -0.1173 | 0.0000 | 0.0012 | 0.0001 |
| | 309 | -0.0127 | 0.0002 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0127 | 0.0003 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0129 | 0.0002 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0129 | 0.0003 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0037 | 0.0000 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0039 | 0.0000 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0037 | 0.0004 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0039 | 0.0004 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 76 | 301 | -0.0002 | -0.0075 | 0.0172 | 0.0000 | 0.0044 | 0.0005 |
| | 302 | 0.0001 | 0.0003 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0001 | -0.0040 | 0.0548 | 0.0000 | 0.0001 | 0.0000 |
| | 304 | 0.0001 | -0.0156 | 0.1876 | 0.0000 | 0.0003 | 0.0000 |
| | 305 | 0.0002 | -0.0107 | 0.1349 | 0.0000 | 0.0002 | 0.0000 |
| | 306 | 0.0000 | -0.0226 | 0.2671 | 0.0000 | 0.0004 | 0.0001 |
| | 307 | -0.0001 | 0.0239 | -0.2861 | 0.0000 | -0.0005 | -0.0001 |
| | 308 | -0.0001 | 0.0125 | -0.1531 | 0.0000 | -0.0003 | 0.0000 |
| | 309 | -0.0127 | 0.0002 | 0.0007 | 0.0000 | -0.0003 | 0.0000 |
| | 310 | -0.0127 | 0.0004 | 0.0007 | 0.0000 | -0.0003 | 0.0000 |
| | 311 | 0.0129 | 0.0001 | 0.0006 | 0.0000 | 0.0003 | 0.0000 |
| | 312 | 0.0129 | 0.0002 | 0.0007 | 0.0000 | 0.0003 | 0.0000 |
| | 313 | -0.0038 | 0.0000 | 0.0006 | 0.0000 | -0.0001 | 0.0000 |
| | 314 | 0.0039 | -0.0001 | 0.0006 | 0.0000 | 0.0001 | 0.0000 |
| | 315 | -0.0038 | 0.0005 | 0.0008 | 0.0000 | -0.0001 | 0.0000 |
| | 316 | 0.0039 | 0.0005 | 0.0008 | 0.0000 | 0.0001 | 0.0000 |
| 77 | 301 | 0.0004 | -0.0075 | 0.0172 | 0.0000 | -0.0044 | -0.0005 |
| | 302 | 0.0000 | 0.0003 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | -0.0040 | 0.0548 | 0.0000 | -0.0001 | 0.0000 |
| | 304 | 0.0000 | -0.0156 | 0.1876 | 0.0000 | -0.0003 | 0.0000 |
| | 305 | 0.0000 | -0.0107 | 0.1349 | 0.0000 | -0.0002 | 0.0000 |
| | 306 | -0.0001 | -0.0226 | 0.2671 | 0.0000 | -0.0004 | -0.0001 |
| | 307 | 0.0001 | 0.0239 | -0.2861 | 0.0000 | 0.0005 | 0.0001 |
| | 308 | 0.0000 | 0.0125 | -0.1531 | 0.0000 | 0.0002 | 0.0000 |
| | 309 | -0.0128 | 0.0001 | 0.0006 | 0.0000 | -0.0003 | 0.0000 |
| | 310 | -0.0128 | 0.0002 | 0.0007 | 0.0000 | -0.0003 | 0.0000 |
| | 311 | 0.0128 | 0.0002 | 0.0007 | 0.0000 | 0.0003 | 0.0000 |
| | 312 | 0.0128 | 0.0004 | 0.0007 | 0.0000 | 0.0003 | 0.0000 |
| | 313 | -0.0038 | -0.0001 | 0.0006 | 0.0000 | -0.0001 | 0.0000 |
| | 314 | 0.0038 | 0.0000 | 0.0006 | 0.0000 | 0.0001 | 0.0000 |
| | 315 | -0.0038 | 0.0005 | 0.0008 | 0.0000 | -0.0001 | 0.0000 |
| | 316 | 0.0038 | 0.0005 | 0.0008 | 0.0000 | 0.0001 | 0.0000 |
| 78 | 301 | 0.0013 | -0.0051 | 0.0106 | 0.0000 | 0.0001 | 0.0008 |
| | 302 | 0.0000 | 0.0003 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0001 | -0.0029 | 0.0421 | 0.0000 | 0.0004 | 0.0001 |
| | 304 | -0.0002 | -0.0119 | 0.1436 | 0.0000 | 0.0015 | 0.0001 |
| | 305 | -0.0002 | -0.0081 | 0.1034 | 0.0000 | 0.0011 | 0.0001 |
| | 306 | -0.0002 | -0.0173 | 0.2044 | 0.0000 | 0.0021 | 0.0002 |
| | 307 | 0.0003 | 0.0183 | -0.2190 | 0.0000 | -0.0023 | -0.0002 |
| | 308 | 0.0002 | 0.0095 | -0.1172 | 0.0000 | -0.0012 | -0.0001 |
| | 309 | -0.0128 | 0.0002 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0128 | 0.0003 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0128 | 0.0002 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0128 | 0.0003 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0038 | 0.0000 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0038 | 0.0000 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0038 | 0.0004 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0038 | 0.0004 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 79 | 301 | 0.0013 | 0.0830 | -0.0032 | 0.0000 | 0.0001 | 0.0006 |
| | 302 | 0.0000 | 0.0020 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0001 | 0.0053 | -0.0110 | 0.0000 | 0.0004 | 0.0001 |
| | 304 | -0.0002 | 0.0075 | -0.0434 | 0.0000 | 0.0014 | 0.0001 |
| | 305 | -0.0002 | 0.0083 | -0.0298 | 0.0000 | 0.0010 | 0.0001 |
| | 306 | -0.0002 | 0.0088 | -0.0628 | 0.0000 | 0.0021 | 0.0002 |
| | 307 | 0.0003 | -0.0108 | 0.0663 | 0.0000 | -0.0022 | -0.0002 |
| | 308 | 0.0002 | -0.0074 | 0.0348 | 0.0000 | -0.0012 | -0.0001 |
| | 309 | -0.0128 | 0.0048 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0128 | 0.0049 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0128 | -0.0016 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0128 | -0.0015 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0038 | 0.0024 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0038 | 0.0005 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0038 | 0.0028 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0038 | 0.0009 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |
| 80 | 301 | -0.0011 | 0.0831 | -0.0032 | 0.0000 | -0.0001 | -0.0006 |
| | 302 | 0.0001 | 0.0021 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0003 | 0.0054 | -0.0110 | 0.0000 | -0.0004 | -0.0001 |
| | 304 | 0.0002 | 0.0075 | -0.0434 | 0.0000 | -0.0014 | -0.0001 |
| | 305 | 0.0003 | 0.0084 | -0.0298 | 0.0000 | -0.0010 | -0.0001 |
| | 306 | 0.0002 | 0.0088 | -0.0628 | 0.0000 | -0.0021 | -0.0002 |
| | 307 | -0.0003 | -0.0108 | 0.0663 | 0.0000 | 0.0022 | 0.0002 |
| | 308 | -0.0002 | -0.0074 | 0.0347 | 0.0000 | 0.0012 | 0.0001 |
| | 309 | -0.0127 | -0.0016 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0127 | -0.0015 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0129 | 0.0048 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0129 | 0.0049 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0037 | 0.0005 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0039 | 0.0024 | 0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0037 | 0.0009 | 0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0039 | 0.0028 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| 81 | 301 | -0.0012 | -0.0051 | -0.0106 | 0.0000 | 0.0001 | -0.0008 |
| | 302 | 0.0001 | 0.0003 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0003 | 0.0183 | 0.2190 | 0.0000 | -0.0023 | 0.0002 |
| | 304 | -0.0002 | 0.0095 | 0.1173 | 0.0000 | -0.0012 | 0.0001 |
| | 305 | 0.0003 | -0.0081 | -0.1034 | 0.0000 | 0.0011 | -0.0001 |
| | 306 | 0.0002 | -0.0173 | -0.2043 | 0.0000 | 0.0021 | -0.0002 |
| | 307 | 0.0002 | -0.0029 | -0.0422 | 0.0000 | 0.0004 | -0.0001 |
| | 308 | 0.0002 | -0.0119 | -0.1436 | 0.0000 | 0.0015 | -0.0001 |
| | 309 | -0.0128 | 0.0003 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0128 | 0.0002 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0130 | 0.0003 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0130 | 0.0002 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0038 | 0.0004 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0039 | 0.0005 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0038 | 0.0000 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0039 | 0.0000 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 82 | 301 | -0.0003 | -0.0075 | -0.0172 | 0.0000 | -0.0044 | 0.0005 |
| | 302 | 0.0000 | 0.0003 | -0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0001 | 0.0239 | 0.2861 | 0.0000 | 0.0005 | -0.0001 |
| | 304 | -0.0001 | 0.0125 | 0.1532 | 0.0000 | 0.0003 | 0.0000 |
| | 305 | 0.0001 | -0.0107 | -0.1350 | 0.0000 | -0.0002 | 0.0000 |
| | 306 | 0.0001 | -0.0226 | -0.2671 | 0.0000 | -0.0004 | 0.0001 |
| | 307 | 0.0000 | -0.0039 | -0.0548 | 0.0000 | -0.0001 | 0.0000 |
| | 308 | 0.0001 | -0.0156 | -0.1876 | 0.0000 | -0.0003 | 0.0000 |
| | 309 | -0.0129 | 0.0004 | -0.0007 | 0.0000 | 0.0003 | 0.0000 |
| | 310 | -0.0129 | 0.0003 | -0.0007 | 0.0000 | 0.0003 | 0.0000 |
| | 311 | 0.0129 | 0.0002 | -0.0007 | 0.0000 | -0.0003 | 0.0000 |
| | 312 | 0.0129 | 0.0001 | -0.0006 | 0.0000 | -0.0003 | 0.0000 |
| | 313 | -0.0038 | 0.0005 | -0.0008 | 0.0000 | 0.0001 | 0.0000 |
| | 314 | 0.0039 | 0.0005 | -0.0008 | 0.0000 | -0.0001 | 0.0000 |
| | 315 | -0.0038 | 0.0000 | -0.0006 | 0.0000 | 0.0001 | 0.0000 |
| | 316 | 0.0039 | 0.0000 | -0.0006 | 0.0000 | -0.0001 | 0.0000 |
| 83 | 301 | 0.0003 | -0.0075 | -0.0172 | 0.0000 | 0.0044 | -0.0005 |
| | 302 | 0.0000 | 0.0003 | -0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0001 | 0.0239 | 0.2861 | 0.0000 | -0.0005 | 0.0001 |
| | 304 | 0.0001 | 0.0125 | 0.1532 | 0.0000 | -0.0003 | 0.0000 |
| | 305 | -0.0001 | -0.0107 | -0.1350 | 0.0000 | 0.0002 | 0.0000 |
| | 306 | -0.0001 | -0.0226 | -0.2671 | 0.0000 | 0.0004 | -0.0001 |
| | 307 | 0.0000 | -0.0039 | -0.0548 | 0.0000 | 0.0001 | 0.0000 |
| | 308 | -0.0001 | -0.0156 | -0.1877 | 0.0000 | 0.0003 | 0.0000 |
| | 309 | -0.0129 | 0.0002 | -0.0007 | 0.0000 | 0.0003 | 0.0000 |
| | 310 | -0.0129 | 0.0001 | -0.0006 | 0.0000 | 0.0003 | 0.0000 |
| | 311 | 0.0129 | 0.0004 | -0.0007 | 0.0000 | -0.0003 | 0.0000 |
| | 312 | 0.0129 | 0.0002 | -0.0007 | 0.0000 | -0.0003 | 0.0000 |
| | 313 | -0.0039 | 0.0005 | -0.0008 | 0.0000 | 0.0001 | 0.0000 |
| | 314 | 0.0038 | 0.0005 | -0.0008 | 0.0000 | -0.0001 | 0.0000 |
| | 315 | -0.0039 | 0.0000 | -0.0006 | 0.0000 | 0.0001 | 0.0000 |
| | 316 | 0.0038 | 0.0000 | -0.0006 | 0.0000 | -0.0001 | 0.0000 |
| 84 | 301 | 0.0012 | -0.0051 | -0.0106 | 0.0000 | -0.0001 | 0.0008 |
| | 302 | -0.0001 | 0.0003 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0003 | 0.0183 | 0.2190 | 0.0000 | 0.0023 | -0.0002 |
| | 304 | 0.0002 | 0.0095 | 0.1173 | 0.0000 | 0.0012 | -0.0001 |
| | 305 | -0.0003 | -0.0081 | -0.1034 | 0.0000 | -0.0011 | 0.0001 |
| | 306 | -0.0002 | -0.0173 | -0.2043 | 0.0000 | -0.0021 | 0.0002 |
| | 307 | -0.0002 | -0.0029 | -0.0422 | 0.0000 | -0.0004 | 0.0001 |
| | 308 | -0.0002 | -0.0119 | -0.1437 | 0.0000 | -0.0015 | 0.0001 |
| | 309 | -0.0130 | 0.0003 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0130 | 0.0002 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0128 | 0.0003 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0128 | 0.0002 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0039 | 0.0005 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0038 | 0.0004 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0039 | 0.0000 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0038 | 0.0000 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 85 | 301 | 0.0012 | 0.0831 | 0.0032 | 0.0000 | -0.0001 | 0.0006 |
| | 302 | -0.0001 | 0.0021 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0003 | -0.0108 | -0.0664 | 0.0000 | 0.0022 | -0.0002 |
| | 304 | 0.0002 | -0.0074 | -0.0347 | 0.0000 | 0.0012 | -0.0001 |
| | 305 | -0.0003 | 0.0084 | 0.0297 | 0.0000 | -0.0010 | 0.0001 |
| | 306 | -0.0002 | 0.0089 | 0.0628 | 0.0000 | -0.0021 | 0.0002 |
| | 307 | -0.0002 | 0.0053 | 0.0109 | 0.0000 | -0.0004 | 0.0001 |
| | 308 | -0.0002 | 0.0075 | 0.0433 | 0.0000 | -0.0014 | 0.0001 |
| | 309 | -0.0130 | 0.0049 | -0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0130 | 0.0048 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0128 | -0.0015 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0128 | -0.0016 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0039 | 0.0028 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0038 | 0.0009 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0039 | 0.0024 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0038 | 0.0005 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| 86 | 301 | -0.0012 | 0.0831 | 0.0032 | 0.0000 | 0.0001 | -0.0006 |
| | 302 | 0.0001 | 0.0021 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0003 | -0.0108 | -0.0664 | 0.0000 | -0.0022 | 0.0002 |
| | 304 | -0.0002 | -0.0074 | -0.0347 | 0.0000 | -0.0012 | 0.0001 |
| | 305 | 0.0003 | 0.0084 | 0.0297 | 0.0000 | 0.0010 | -0.0001 |
| | 306 | 0.0002 | 0.0089 | 0.0628 | 0.0000 | 0.0021 | -0.0002 |
| | 307 | 0.0002 | 0.0053 | 0.0109 | 0.0000 | 0.0004 | -0.0001 |
| | 308 | 0.0002 | 0.0075 | 0.0434 | 0.0000 | 0.0014 | -0.0001 |
| | 309 | -0.0128 | -0.0015 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0128 | -0.0016 | -0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0130 | 0.0049 | -0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0130 | 0.0048 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0038 | 0.0009 | -0.0007 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0039 | 0.0028 | -0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0038 | 0.0005 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0039 | 0.0024 | -0.0006 | 0.0000 | 0.0000 | 0.0000 |
| 87 | 301 | -0.0001 | 0.0413 | -0.0014 | 0.0000 | 0.0000 | 0.0003 |
| | 302 | 0.0006 | -0.0096 | 0.0024 | 0.0000 | 0.0000 | -0.0001 |
| | 303 | 0.0010 | -0.0134 | -0.0156 | -0.0001 | 0.0002 | -0.0001 |
| | 304 | 0.0005 | 0.0052 | -0.0680 | -0.0005 | 0.0009 | 0.0000 |
| | 305 | 0.0012 | -0.0107 | -0.0451 | -0.0003 | 0.0006 | -0.0001 |
| | 306 | 0.0003 | 0.0157 | -0.0993 | -0.0007 | 0.0012 | 0.0001 |
| | 307 | -0.0007 | -0.0106 | 0.1048 | 0.0007 | -0.0013 | 0.0000 |
| | 308 | -0.0008 | 0.0020 | 0.0541 | 0.0004 | -0.0007 | 0.0001 |
| | 309 | -0.0166 | -0.0095 | 0.0025 | 0.0000 | 0.0000 | -0.0001 |
| | 310 | -0.0166 | -0.0092 | 0.0025 | 0.0000 | 0.0000 | -0.0001 |
| | 311 | 0.0175 | -0.0062 | 0.0013 | 0.0000 | 0.0000 | -0.0001 |
| | 312 | 0.0175 | -0.0060 | 0.0014 | 0.0000 | 0.0000 | -0.0001 |
| | 313 | -0.0046 | -0.0086 | 0.0020 | 0.0000 | 0.0000 | -0.0001 |
| | 314 | 0.0055 | -0.0076 | 0.0017 | 0.0000 | 0.0000 | -0.0001 |
| | 315 | -0.0046 | -0.0078 | 0.0022 | 0.0000 | 0.0000 | -0.0001 |
| | 316 | 0.0055 | -0.0068 | 0.0018 | 0.0000 | 0.0000 | -0.0001 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 88 | 301 | -0.0002 | 0.0410 | 0.0013 | 0.0000 | 0.0000 | 0.0003 |
| | 302 | 0.0005 | -0.0095 | -0.0024 | 0.0000 | 0.0000 | -0.0001 |
| | 303 | -0.0007 | -0.0104 | -0.1048 | -0.0007 | 0.0013 | 0.0000 |
| | 304 | -0.0007 | 0.0019 | -0.0540 | -0.0004 | 0.0007 | 0.0001 |
| | 305 | 0.0011 | -0.0106 | 0.0451 | 0.0003 | -0.0006 | -0.0001 |
| | 306 | 0.0003 | 0.0155 | 0.0993 | 0.0007 | -0.0012 | 0.0001 |
| | 307 | 0.0009 | -0.0133 | 0.0156 | 0.0001 | -0.0002 | -0.0001 |
| | 308 | 0.0005 | 0.0052 | 0.0679 | 0.0005 | -0.0009 | 0.0000 |
| | 309 | -0.0167 | -0.0092 | -0.0025 | 0.0000 | 0.0000 | -0.0001 |
| | 310 | -0.0167 | -0.0094 | -0.0025 | 0.0000 | 0.0000 | -0.0001 |
| | 311 | 0.0175 | -0.0058 | -0.0014 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0175 | -0.0060 | -0.0013 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0047 | -0.0077 | -0.0022 | 0.0000 | 0.0000 | -0.0001 |
| | 314 | 0.0055 | -0.0067 | -0.0018 | 0.0000 | 0.0000 | -0.0001 |
| | 315 | -0.0047 | -0.0085 | -0.0020 | 0.0000 | 0.0000 | -0.0001 |
| | 316 | 0.0055 | -0.0075 | -0.0017 | 0.0000 | 0.0000 | -0.0001 |
| 89 | 301 | 0.0003 | 0.0413 | -0.0014 | 0.0000 | 0.0000 | -0.0003 |
| | 302 | -0.0004 | -0.0097 | 0.0024 | 0.0000 | 0.0000 | 0.0001 |
| | 303 | -0.0008 | -0.0134 | -0.0156 | -0.0001 | -0.0002 | 0.0001 |
| | 304 | -0.0005 | 0.0052 | -0.0680 | -0.0005 | -0.0009 | 0.0000 |
| | 305 | -0.0010 | -0.0107 | -0.0451 | -0.0003 | -0.0006 | 0.0001 |
| | 306 | -0.0004 | 0.0157 | -0.0993 | -0.0007 | -0.0012 | -0.0001 |
| | 307 | 0.0007 | -0.0106 | 0.1048 | 0.0007 | 0.0013 | 0.0000 |
| | 308 | 0.0007 | 0.0019 | 0.0540 | 0.0004 | 0.0007 | -0.0001 |
| | 309 | -0.0174 | -0.0062 | 0.0013 | 0.0000 | 0.0000 | 0.0001 |
| | 310 | -0.0174 | -0.0060 | 0.0014 | 0.0000 | 0.0000 | 0.0001 |
| | 311 | 0.0167 | -0.0095 | 0.0025 | 0.0000 | 0.0000 | 0.0001 |
| | 312 | 0.0167 | -0.0092 | 0.0025 | 0.0000 | 0.0000 | 0.0001 |
| | 313 | -0.0054 | -0.0076 | 0.0017 | 0.0000 | 0.0000 | 0.0001 |
| | 314 | 0.0048 | -0.0086 | 0.0020 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | -0.0054 | -0.0069 | 0.0018 | 0.0000 | 0.0000 | 0.0001 |
| | 316 | 0.0047 | -0.0078 | 0.0022 | 0.0000 | 0.0000 | 0.0001 |
| 90 | 301 | 0.0002 | 0.0410 | 0.0013 | 0.0000 | 0.0000 | -0.0003 |
| | 302 | -0.0005 | -0.0095 | -0.0024 | 0.0000 | 0.0000 | 0.0001 |
| | 303 | 0.0007 | -0.0104 | -0.1048 | -0.0007 | -0.0013 | 0.0000 |
| | 304 | 0.0007 | 0.0019 | -0.0540 | -0.0004 | -0.0007 | -0.0001 |
| | 305 | -0.0011 | -0.0106 | 0.0451 | 0.0003 | 0.0006 | 0.0001 |
| | 306 | -0.0003 | 0.0155 | 0.0993 | 0.0007 | 0.0012 | -0.0001 |
| | 307 | -0.0009 | -0.0133 | 0.0156 | 0.0001 | 0.0002 | 0.0001 |
| | 308 | -0.0005 | 0.0051 | 0.0680 | 0.0005 | 0.0009 | 0.0000 |
| | 309 | -0.0175 | -0.0058 | -0.0014 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0175 | -0.0060 | -0.0013 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0167 | -0.0092 | -0.0025 | 0.0000 | 0.0000 | 0.0001 |
| | 312 | 0.0167 | -0.0094 | -0.0025 | 0.0000 | 0.0000 | 0.0001 |
| | 313 | -0.0055 | -0.0067 | -0.0018 | 0.0000 | 0.0000 | 0.0001 |
| | 314 | 0.0047 | -0.0077 | -0.0022 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | -0.0055 | -0.0075 | -0.0017 | 0.0000 | 0.0000 | 0.0001 |
| | 316 | 0.0047 | -0.0085 | -0.0020 | 0.0000 | 0.0000 | 0.0001 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 91 | 301 | -0.0002 | -0.3556 | -0.0181 | 0.0001 | 0.0000 | 0.0002 |
| | 302 | -0.0001 | -0.2160 | -0.0118 | 0.0000 | 0.0000 | 0.0001 |
| | 303 | -0.0002 | -0.3527 | -0.0194 | 0.0001 | 0.0000 | 0.0002 |
| | 304 | 0.0000 | -0.0408 | -0.0024 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | -0.0002 | -0.3471 | -0.0191 | 0.0001 | 0.0000 | 0.0002 |
| | 306 | 0.0001 | 0.1347 | 0.0072 | 0.0000 | 0.0000 | -0.0001 |
| | 307 | 0.0000 | 0.0025 | 0.0004 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0001 | 0.1685 | 0.0097 | 0.0000 | 0.0000 | -0.0001 |
| | 309 | 0.0212 | -0.1727 | -0.0094 | 0.0000 | 0.0000 | 0.0001 |
| | 310 | 0.0212 | -0.1731 | -0.0102 | 0.0000 | 0.0000 | 0.0001 |
| | 311 | -0.0214 | -0.1726 | -0.0087 | 0.0000 | 0.0000 | 0.0001 |
| | 312 | -0.0214 | -0.1730 | -0.0095 | 0.0000 | 0.0000 | 0.0001 |
| | 313 | 0.0062 | -0.1722 | -0.0082 | 0.0000 | 0.0000 | 0.0001 |
| | 314 | -0.0065 | -0.1722 | -0.0080 | 0.0000 | 0.0000 | 0.0001 |
| | 315 | 0.0063 | -0.1735 | -0.0109 | 0.0000 | 0.0000 | 0.0001 |
| | 316 | -0.0064 | -0.1734 | -0.0107 | 0.0000 | 0.0000 | 0.0001 |
| 93 | 301 | 0.0001 | -0.4674 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0001 | -0.2871 | 0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0001 | -0.2335 | -0.0005 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0965 | -0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0001 | -0.4649 | 0.0008 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.1956 | -0.0003 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | -0.2361 | 0.0014 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0013 | 0.1203 | 0.0005 | 0.0000 | 0.0000 | 0.0001 |
| | 309 | 0.0283 | -0.2296 | 0.0010 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | 0.0283 | -0.2297 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | -0.0282 | -0.2296 | 0.0009 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | -0.0282 | -0.2297 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | 0.0085 | -0.2295 | 0.0022 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0084 | -0.2295 | 0.0022 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0085 | -0.2298 | -0.0013 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | -0.0084 | -0.2298 | -0.0013 | 0.0000 | 0.0000 | 0.0000 |
| 95 | 301 | 0.0000 | -0.3496 | 0.0187 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | -0.2130 | 0.0122 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0014 | -0.0003 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.1689 | -0.0098 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | -0.3422 | 0.0197 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.1327 | -0.0075 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | -0.3465 | 0.0199 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | -0.0422 | 0.0024 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0191 | -0.1706 | 0.0102 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | 0.0191 | -0.1701 | 0.0094 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | -0.0191 | -0.1706 | 0.0102 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | -0.0191 | -0.1701 | 0.0094 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | 0.0057 | -0.1712 | 0.0111 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | -0.0057 | -0.1712 | 0.0111 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0057 | -0.1695 | 0.0084 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | -0.0057 | -0.1695 | 0.0084 | 0.0000 | 0.0000 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 96 | 301 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0002 |
| 97 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| 98 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 99 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0005 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0005 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0005 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0005 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| 100 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| 101 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0005 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0005 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0005 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0005 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 102 | 301 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0002 |
| 103 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 |
| 104 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0005 | -0.0008 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0005 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0003 | -0.0008 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0003 | -0.0008 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0004 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0002 | 0.0004 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0007 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0007 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0001 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0001 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0005 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0003 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 | -0.0005 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0002 | -0.0003 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 105 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0005 | -0.0008 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0002 | -0.0005 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0002 | 0.0004 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0003 | -0.0008 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0004 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0003 | -0.0008 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0007 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0007 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 | -0.0005 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0002 | -0.0003 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0005 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0003 |
| 106 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0002 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | -0.0002 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | -0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0001 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0004 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0004 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | -0.0003 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| 107 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0002 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | -0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0001 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | -0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | -0.0002 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 | -0.0006 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0006 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0004 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0004 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | -0.0003 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 108 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0002 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0002 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | -0.0001 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0004 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0004 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0003 |
| 109 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0002 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | -0.0001 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0002 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0004 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0004 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 | 0.0006 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0006 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0003 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 |
| 110 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0005 | 0.0008 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0005 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0003 | 0.0008 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0003 | 0.0008 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0004 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0002 | -0.0004 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0001 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0001 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0007 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0007 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0003 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0005 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0002 | 0.0003 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 | 0.0005 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 111 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0005 | 0.0008 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0002 | 0.0005 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0002 | -0.0004 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0003 | 0.0008 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0004 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0003 | 0.0008 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0001 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0001 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0007 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0007 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0002 | 0.0003 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 | 0.0005 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0003 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0005 |
| 112 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | -0.0002 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0002 | -0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0002 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | -0.0002 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | -0.0002 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| 113 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0002 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0002 | -0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0002 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0002 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0002 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0002 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0002 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 114 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0004 | -0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 | -0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | -0.0002 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | -0.0002 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0001 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0001 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 115 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 | -0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0002 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0002 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0002 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0002 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 116 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0004 | 0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 | 0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0002 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0002 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 117 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 | 0.0001 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0002 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | -0.0002 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0002 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0002 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| 118 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0002 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0000 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0002 | 0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | -0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0002 | 0.0000 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0002 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | 0.0002 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0002 |
| 119 | 301 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0002 | 0.0001 |
| | 302 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0001 |
| | 303 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | -0.0002 | 0.0000 |
| | 304 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 305 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | -0.0002 | 0.0002 |
| | 306 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0003 | -0.0001 |
| | 307 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0002 |
| | 308 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0000 |
| | 309 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0002 |
| | 312 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0003 | 0.0002 |
| | 313 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 314 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0002 |
| | 315 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 316 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | -0.0002 | 0.0002 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 128 | 301 | 0.0092 | -0.0014 | 0.0001 | 0.0000 | -0.0002 | -0.0001 |
| | 302 | -0.0088 | 0.0006 | -0.0001 | 0.0000 | 0.0002 | -0.0001 |
| | 303 | -0.0083 | 0.0008 | 0.0008 | -0.0002 | 0.0002 | 0.0000 |
| | 304 | 0.0025 | -0.0001 | 0.0005 | -0.0001 | -0.0001 | 0.0001 |
| | 305 | -0.0103 | 0.0006 | -0.0005 | 0.0001 | 0.0002 | -0.0002 |
| | 306 | 0.0131 | -0.0011 | -0.0006 | 0.0001 | -0.0003 | 0.0001 |
| | 307 | -0.0125 | 0.0009 | -0.0003 | 0.0001 | 0.0003 | -0.0002 |
| | 308 | 0.0038 | -0.0004 | -0.0005 | 0.0001 | -0.0001 | 0.0000 |
| | 309 | -0.0143 | 0.0006 | -0.0001 | 0.0000 | 0.0003 | -0.0002 |
| | 310 | -0.0142 | 0.0003 | -0.0001 | 0.0000 | 0.0003 | -0.0002 |
| | 311 | 0.0002 | 0.0006 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0002 | 0.0003 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0092 | 0.0009 | 0.0000 | 0.0000 | 0.0002 | -0.0002 |
| | 314 | -0.0049 | 0.0010 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 315 | -0.0091 | 0.0000 | -0.0001 | 0.0000 | 0.0002 | -0.0002 |
| | 316 | -0.0048 | 0.0000 | -0.0001 | 0.0000 | 0.0001 | -0.0001 |
| 129 | 301 | -0.0145 | -0.0023 | 0.0001 | 0.0000 | 0.0003 | -0.0001 |
| | 302 | -0.0021 | 0.0006 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | 0.0005 | 0.0010 | 0.0010 | -0.0002 | 0.0000 | 0.0000 |
| | 304 | 0.0019 | 0.0000 | 0.0006 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | -0.0036 | 0.0005 | -0.0007 | 0.0001 | 0.0001 | -0.0001 |
| | 306 | 0.0008 | -0.0014 | -0.0008 | 0.0002 | 0.0000 | 0.0000 |
| | 307 | -0.0034 | 0.0009 | -0.0004 | 0.0001 | 0.0001 | -0.0001 |
| | 308 | -0.0007 | -0.0005 | -0.0007 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | -0.0132 | 0.0008 | -0.0001 | 0.0000 | 0.0003 | -0.0002 |
| | 310 | -0.0132 | 0.0005 | -0.0001 | 0.0000 | 0.0003 | -0.0002 |
| | 311 | 0.0099 | 0.0005 | -0.0001 | 0.0000 | -0.0002 | 0.0002 |
| | 312 | 0.0099 | 0.0001 | -0.0002 | 0.0000 | -0.0002 | 0.0002 |
| | 313 | -0.0051 | 0.0011 | 0.0000 | 0.0000 | 0.0001 | -0.0001 |
| | 314 | 0.0018 | 0.0010 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0051 | 0.0000 | -0.0002 | 0.0000 | 0.0001 | -0.0001 |
| | 316 | 0.0018 | -0.0001 | -0.0002 | 0.0000 | 0.0000 | 0.0000 |
| 130 | 301 | 0.0145 | -0.0024 | 0.0001 | 0.0000 | -0.0003 | 0.0001 |
| | 302 | 0.0021 | 0.0006 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0005 | 0.0010 | 0.0010 | -0.0002 | 0.0000 | 0.0000 |
| | 304 | -0.0019 | 0.0000 | 0.0006 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | 0.0036 | 0.0005 | -0.0007 | 0.0001 | -0.0001 | 0.0001 |
| | 306 | -0.0008 | -0.0014 | -0.0008 | 0.0002 | 0.0000 | 0.0000 |
| | 307 | 0.0034 | 0.0009 | -0.0004 | 0.0001 | -0.0001 | 0.0001 |
| | 308 | 0.0007 | -0.0005 | -0.0007 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | -0.0099 | 0.0005 | -0.0001 | 0.0000 | 0.0002 | -0.0002 |
| | 310 | -0.0099 | 0.0001 | -0.0002 | 0.0000 | 0.0002 | -0.0002 |
| | 311 | 0.0132 | 0.0008 | -0.0001 | 0.0000 | -0.0003 | 0.0002 |
| | 312 | 0.0132 | 0.0005 | -0.0001 | 0.0000 | -0.0003 | 0.0002 |
| | 313 | -0.0018 | 0.0010 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0051 | 0.0011 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 315 | -0.0018 | -0.0001 | -0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0051 | 0.0000 | -0.0002 | 0.0000 | -0.0001 | 0.0001 |

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 131 | 301 | -0.0092 | -0.0014 | 0.0001 | 0.0000 | 0.0002 | 0.0001 |
| | 302 | 0.0088 | 0.0006 | -0.0001 | 0.0000 | -0.0002 | 0.0001 |
| | 303 | 0.0083 | 0.0008 | 0.0008 | -0.0002 | -0.0002 | 0.0000 |
| | 304 | -0.0025 | -0.0001 | 0.0005 | -0.0001 | 0.0001 | -0.0001 |
| | 305 | 0.0103 | 0.0006 | -0.0005 | 0.0001 | -0.0002 | 0.0002 |
| | 306 | -0.0131 | -0.0011 | -0.0006 | 0.0001 | 0.0003 | -0.0001 |
| | 307 | 0.0125 | 0.0009 | -0.0003 | 0.0001 | -0.0003 | 0.0002 |
| | 308 | -0.0039 | -0.0004 | -0.0006 | 0.0001 | 0.0001 | 0.0000 |
| | 309 | -0.0002 | 0.0006 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0002 | 0.0003 | -0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0143 | 0.0006 | -0.0001 | 0.0000 | -0.0003 | 0.0002 |
| | 312 | 0.0142 | 0.0003 | -0.0001 | 0.0000 | -0.0003 | 0.0002 |
| | 313 | 0.0049 | 0.0009 | 0.0000 | 0.0000 | -0.0001 | 0.0001 |
| | 314 | 0.0092 | 0.0009 | 0.0000 | 0.0000 | -0.0002 | 0.0002 |
| | 315 | 0.0048 | 0.0000 | -0.0001 | 0.0000 | -0.0001 | 0.0001 |
| | 316 | 0.0091 | 0.0000 | -0.0001 | 0.0000 | -0.0002 | 0.0002 |
| 132 | 301 | 0.0096 | -0.0014 | -0.0001 | 0.0000 | 0.0002 | -0.0001 |
| | 302 | -0.0088 | 0.0006 | 0.0001 | 0.0000 | -0.0002 | -0.0001 |
| | 303 | -0.0126 | 0.0008 | 0.0003 | -0.0001 | -0.0003 | -0.0002 |
| | 304 | 0.0039 | -0.0004 | 0.0005 | -0.0001 | 0.0001 | 0.0000 |
| | 305 | -0.0104 | 0.0006 | 0.0005 | -0.0001 | -0.0002 | -0.0002 |
| | 306 | 0.0133 | -0.0011 | 0.0007 | -0.0001 | 0.0003 | 0.0001 |
| | 307 | -0.0085 | 0.0007 | -0.0008 | 0.0002 | -0.0002 | 0.0000 |
| | 308 | 0.0025 | -0.0001 | -0.0005 | 0.0001 | 0.0001 | 0.0001 |
| | 309 | -0.0144 | 0.0003 | 0.0001 | 0.0000 | -0.0003 | -0.0002 |
| | 310 | -0.0144 | 0.0006 | 0.0000 | 0.0000 | -0.0003 | -0.0002 |
| | 311 | 0.0002 | 0.0003 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 312 | 0.0002 | 0.0006 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 313 | -0.0092 | 0.0000 | 0.0001 | 0.0000 | -0.0002 | -0.0001 |
| | 314 | -0.0048 | 0.0000 | 0.0001 | 0.0000 | -0.0001 | -0.0001 |
| | 315 | -0.0093 | 0.0009 | 0.0000 | 0.0000 | -0.0002 | -0.0001 |
| | 316 | -0.0050 | 0.0009 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| 133 | 301 | -0.0153 | -0.0024 | -0.0001 | 0.0000 | -0.0004 | -0.0001 |
| | 302 | -0.0022 | 0.0006 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 303 | -0.0036 | 0.0009 | 0.0004 | -0.0001 | -0.0001 | -0.0001 |
| | 304 | -0.0008 | -0.0005 | 0.0007 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | -0.0038 | 0.0004 | 0.0007 | -0.0001 | -0.0001 | -0.0001 |
| | 306 | 0.0008 | -0.0014 | 0.0008 | -0.0002 | 0.0000 | 0.0000 |
| | 307 | 0.0006 | 0.0009 | -0.0010 | 0.0002 | 0.0000 | 0.0000 |
| | 308 | 0.0020 | 0.0000 | -0.0006 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | -0.0132 | 0.0005 | 0.0001 | 0.0000 | -0.0003 | -0.0002 |
| | 310 | -0.0132 | 0.0008 | 0.0001 | 0.0000 | -0.0003 | -0.0002 |
| | 311 | 0.0097 | 0.0001 | 0.0002 | 0.0000 | 0.0002 | 0.0001 |
| | 312 | 0.0097 | 0.0004 | 0.0001 | 0.0000 | 0.0002 | 0.0001 |
| | 313 | -0.0051 | 0.0000 | 0.0002 | 0.0000 | -0.0001 | -0.0001 |
| | 314 | 0.0017 | -0.0001 | 0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 315 | -0.0051 | 0.0011 | 0.0000 | 0.0000 | -0.0001 | -0.0001 |
| | 316 | 0.0017 | 0.0010 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |



JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

| JOINT | LOAD | X-TRANS | Y-TRANS | Z-TRANS | X-ROTAN | Y-ROTAN | Z-ROTAN |
|---|---|---|---|---|---|---|---|
| 134 | 301 | 0.0155 | -0.0024 | -0.0001 | 0.0000 | 0.0004 | 0.0001 |
| | 302 | 0.0023 | 0.0006 | 0.0001 | 0.0000 | 0.0001 | 0.0000 |
| | 303 | 0.0037 | 0.0008 | 0.0004 | -0.0001 | 0.0001 | 0.0001 |
| | 304 | 0.0008 | -0.0005 | 0.0007 | -0.0001 | 0.0000 | 0.0000 |
| | 305 | 0.0039 | 0.0004 | 0.0007 | -0.0001 | 0.0001 | 0.0001 |
| | 306 | -0.0008 | -0.0013 | 0.0008 | -0.0002 | 0.0000 | 0.0000 |
| | 307 | -0.0006 | 0.0009 | -0.0010 | 0.0002 | 0.0000 | 0.0000 |
| | 308 | -0.0021 | 0.0000 | -0.0006 | 0.0001 | 0.0000 | 0.0000 |
| | 309 | -0.0096 | 0.0001 | 0.0002 | 0.0000 | -0.0002 | -0.0001 |
| | 310 | -0.0096 | 0.0004 | 0.0001 | 0.0000 | -0.0002 | -0.0001 |
| | 311 | 0.0132 | 0.0005 | 0.0001 | 0.0000 | 0.0003 | 0.0002 |
| | 312 | 0.0132 | 0.0008 | 0.0001 | 0.0000 | 0.0003 | 0.0002 |
| | 313 | -0.0016 | -0.0001 | 0.0002 | 0.0000 | 0.0000 | 0.0000 |
| | 314 | 0.0052 | 0.0000 | 0.0002 | 0.0000 | 0.0001 | 0.0001 |
| | 315 | -0.0016 | 0.0010 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 316 | 0.0052 | 0.0011 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| 135 | 301 | -0.0095 | -0.0014 | -0.0001 | 0.0000 | -0.0002 | 0.0001 |
| | 302 | 0.0089 | 0.0006 | 0.0001 | 0.0000 | 0.0002 | 0.0001 |
| | 303 | 0.0127 | 0.0008 | 0.0003 | -0.0001 | 0.0003 | 0.0002 |
| | 304 | -0.0039 | -0.0004 | 0.0005 | -0.0001 | -0.0001 | 0.0000 |
| | 305 | 0.0105 | 0.0006 | 0.0005 | -0.0001 | 0.0002 | 0.0002 |
| | 306 | -0.0133 | -0.0011 | 0.0007 | -0.0001 | -0.0003 | -0.0001 |
| | 307 | 0.0085 | 0.0007 | -0.0008 | 0.0002 | 0.0002 | 0.0000 |
| | 308 | -0.0026 | -0.0001 | -0.0004 | 0.0001 | -0.0001 | -0.0001 |
| | 309 | -0.0002 | 0.0003 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | 310 | -0.0001 | 0.0006 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| | 311 | 0.0144 | 0.0003 | 0.0001 | 0.0000 | 0.0003 | 0.0002 |
| | 312 | 0.0144 | 0.0006 | 0.0000 | 0.0000 | 0.0003 | 0.0002 |
| | 313 | 0.0049 | 0.0000 | 0.0001 | 0.0000 | 0.0001 | 0.0001 |
| | 314 | 0.0093 | 0.0000 | 0.0001 | 0.0000 | 0.0002 | 0.0002 |
| | 315 | 0.0050 | 0.0009 | 0.0000 | 0.0000 | 0.0001 | 0.0001 |
| | 316 | 0.0094 | 0.0009 | 0.0000 | 0.0000 | 0.0002 | 0.0002 |

************** END OF LATEST ANALYSIS RESULT **************

```
591. LOAD LIST 201 TO 236
592. PARAMETER 1
593. CODE BS5950
594. LZ 9.3 MEMB 1 TO 3 10 TO 15 22 TO 27 34 TO 39 46 TO 48 95 TO 102 562 564 566 -
595. 567 572 TO 575
596. LZ 6.65 MEMB 6 7 18 19 30 31 42 43
597. LZ 5.6 MEMB 4 5 8 9 16 17 20 21 28 29 32 33 40 41 44 45 52 TO 59
598. LZ 6 MEMB 208 218 228 236 238 240
```

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599. UNL 1.5 MEMB 201 TO 218 220 TO 236 238 240 241 301 TO 320

600. KY 0.65 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

601. KZ 0.65 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

602. MAIN 3 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

603. PY 345000 ALL

604. SBLT 1 ALL

605. SBLT 0 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

606. TRACK 2 ALL

607. CHECK CODE ALL
```

STEEL DESIGN

STAAD.Pro CODE CHECKING - (BSI )

***********************

PROGRAM CODE REVISION V2.9_5950-1_2000



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

     1    TAPERED       PASS     BS-4.7   (C)       0.262       214
                       93.79 C          3.69        -37.46       2.70
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     1 UNIT - kN,m  SECTION CLASS  3  |
| MCZ=  275.6   MCY=   56.6   PC=  1421.2   PT=  2063.1    MB=  210.2 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
|  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|-----------------------------------------------------------------------|
     2    TAPERED       PASS     BS-4.8.2.2         0.154       211
                       80.47 T          0.25         36.49       1.02
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     2 UNIT - kN,m  SECTION CLASS  3  |
| MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
|  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|-----------------------------------------------------------------------|
     3    TAPERED       PASS     BS-4.8.2.2         0.153       211
                       79.54 T         -0.24        -36.49       0.00
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     3 UNIT - kN,m  SECTION CLASS  3  |
| MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
|  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|-----------------------------------------------------------------------|
     4    TAPERED       PASS     BS-4.7   (C)       0.073       201
                       19.01 C          0.74         16.15       0.00
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     4 UNIT - kN,m  SECTION CLASS  3  |
| MCZ=  324.8   MCY=   56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
|  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|-----------------------------------------------------------------------|
     5    TAPERED       PASS     BS-4.7   (C)       0.095       201
                       16.42 C          0.26         26.30       2.18
```

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     5 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1642.4   PT=  2166.6    MB=  315.5 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      6    TAPERED       PASS     BS-4.7   (C)       0.196       207
                       32.01 C         -0.10          56.14        3.19
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     6 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1301.0   PT=  2166.6    MB=  244.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      7    TAPERED       PASS     BS-4.7   (C)       0.197       207
                       32.05 C          0.10         -56.14        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     7 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1207.8   PT=  2166.6    MB=  228.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      8    TAPERED       PASS     BS-4.7   (C)       0.127       207
                       13.00 C         -0.14          37.91        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     8 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1488.2   PT=  2166.6    MB=  280.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      9    TAPERED       PASS     BS-4.7   (C)       0.077       213
                       20.27 C         -0.85          16.67        0.00
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     9 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     10    TAPERED       PASS     BS-4.8.2.2         0.153       211
                       79.43 T          0.24         -36.54        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    10 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     11    TAPERED       PASS     BS-4.8.2.2         0.154       211
                       80.34 T         -0.24          36.54        1.02
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    11 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     12    TAPERED       PASS     BS-4.7   (C)       0.262       202
                       93.71 C         -3.69         -37.46        2.70
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    12 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1421.2   PT=  2063.1    MB=  210.2 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     13    TAPERED       PASS     BS-4.7   (C)       0.259       214
                      124.52 C          0.14         -48.35        2.70
```

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    13 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1421.2   PT=  2063.1    MB=  202.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     14    TAPERED       PASS     BS-4.8.2.2         0.199       211
                      107.05 T         -0.07          48.26        1.02
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    14 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     15    TAPERED       PASS     BS-4.8.2.2         0.198       211
                      105.62 T          0.06         -48.26        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    15 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     16    TAPERED       PASS     BS-4.7   (C)       0.078       201
                       20.96 C         -0.29          20.21        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    16 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     17    TAPERED       PASS     BS-4.7   (C)       0.110       201
                       17.87 C         -0.29          30.95        2.18
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    17 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1642.4   PT=  2166.6    MB=  315.5 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     18    TAPERED       PASS     BS-4.7   (C)       0.144       207
                       23.92 C         -0.08          41.01        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    18 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1301.0   PT=  2166.6    MB=  244.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     19    TAPERED       PASS     BS-4.7   (C)       0.143       207
                       23.92 C          0.03         -40.65        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    19 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1207.8   PT=  2166.6    MB=  228.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     20    TAPERED       PASS     BS-4.7   (C)       0.122       207
                       16.57 C         -0.09          35.84        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    20 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1488.2   PT=  2166.6    MB=  280.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
     21    TAPERED       PASS     BS-4.7   (C)       0.081       213
                       20.97 C          0.34          20.75        0.00
```

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    21 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     22    TAPERED       PASS     BS-4.8.2.2         0.198       211
                      105.49 T         -0.06          -48.31        0.00
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    22 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     23    TAPERED       PASS     BS-4.8.2.2         0.199       211
                      106.90 T          0.06           48.31        1.02
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    23 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     24    TAPERED       PASS     BS-4.7   (C)       0.259       202
                      124.29 C         -0.14          -48.34        2.70
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    24 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=   56.6   PC=  1421.2   PT=  2063.1    MB=  210.2 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     25    TAPERED       PASS     BS-4.7   (C)       0.259       214
                      124.51 C         -0.14          -48.35        2.70
```





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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    25 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=   56.6   PC=  1421.2   PT=  2063.1    MB=  202.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     26    TAPERED       PASS     BS-4.8.2.2         0.199       211
                      107.06 T          0.07           48.26        1.02
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    26 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     27    TAPERED       PASS     BS-4.8.2.2         0.198       211
                      105.63 T         -0.06          -48.26        0.00
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    27 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     28    TAPERED       PASS     BS-4.7   (C)       0.078       201
                       20.43 C          0.28           20.25        0.00
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    28 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     29    TAPERED       PASS     BS-4.7   (C)       0.109       201
                       17.33 C          0.26           30.94        2.18
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    29 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1642.4   PT=  2166.6    MB=  315.5 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     30    TAPERED       PASS     BS-4.7   (C)       0.143       207
                       24.16 C          0.06          40.96        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    30 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1301.0   PT=  2166.6    MB=  244.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     31    TAPERED       PASS     BS-4.7   (C)       0.143       207
                       24.16 C         -0.03         -40.66        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    31 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1207.8   PT=  2166.6    MB=  228.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     32    TAPERED       PASS     BS-4.7   (C)       0.122       207
                       16.73 C          0.09          35.81        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    32 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1488.2   PT=  2166.6    MB=  280.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     33    TAPERED       PASS     BS-4.7   (C)       0.081       213
                       21.00 C         -0.34          20.74        0.00
```





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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    33 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     34    TAPERED       PASS     BS-4.8.2.2         0.198       211
                      105.49 T          0.06         -48.31        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    34 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     35    TAPERED       PASS     BS-4.8.2.2         0.199       211
                      106.90 T         -0.06          48.31        1.02
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    35 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     36    TAPERED       PASS     BS-4.7   (C)       0.259       202
                      124.32 C          0.14         -48.34        2.70
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    36 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1421.2   PT=  2063.1    MB=  210.2 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     37    TAPERED       PASS     BS-4.7   (C)       0.262       214
                       93.79 C         -3.69         -37.46        2.70
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    37 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1421.2   PT=  2063.1    MB=  202.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      38    TAPERED       PASS     BS-4.8.2.2         0.154       211
                        80.47 T         -0.25          36.49        1.02
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    38 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      39    TAPERED       PASS     BS-4.8.2.2         0.153       211
                        79.54 T          0.24         -36.49        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    39 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      40    TAPERED       PASS     BS-4.7   (C)       0.073       201
                        19.00 C         -0.75          16.15        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    40 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      41    TAPERED       PASS     BS-4.7   (C)       0.095       201
                        16.41 C         -0.28          26.30        2.18
```



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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    41 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1642.4   PT=  2166.6    MB=  315.5 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      42    TAPERED       PASS     BS-4.7   (C)       0.196       207
                        32.02 C          0.10          56.14        3.19
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    42 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1301.0   PT=  2166.6    MB=  244.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      43    TAPERED       PASS     BS-4.7   (C)       0.197       207
                        32.05 C         -0.10         -56.14        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    43 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1207.8   PT=  2166.6    MB=  228.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      44    TAPERED       PASS     BS-4.7   (C)       0.127       207
                        13.00 C          0.14          37.91        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    44 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1488.2   PT=  2166.6    MB=  280.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
      45    TAPERED       PASS     BS-4.7   (C)       0.077       213
                        20.23 C          0.85          16.66        0.00
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE        RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    45 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1902.8   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     46    TAPERED       PASS     BS-4.8.2.2         0.153       211
                       79.43 T         -0.24         -36.54        0.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    46 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     47    TAPERED       PASS     BS-4.8.2.2         0.154       211
                       80.34 T          0.24          36.54        1.02
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    47 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     48    TAPERED       PASS     BS-4.7   (C)       0.262       202
                       93.71 C          3.69         -37.46        2.70
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    48 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=   56.6   PC=  1421.2   PT=  2063.1    MB=  210.2 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     52    TAPERED       PASS     BS-4.7   (C)       0.095       213
                       17.27 C         -0.29         -26.11        0.00
```



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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE        RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    52 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1763.3   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     53    TAPERED       PASS     BS-4.7   (C)       0.108       213
                       17.41 C          0.32         -30.33        0.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    53 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1763.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     54    TAPERED       PASS     BS-4.7   (C)       0.108       213
                       17.44 C         -0.32         -30.31        0.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    54 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1763.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     55    TAPERED       PASS     BS-4.7   (C)       0.095       213
                       17.23 C          0.29         -26.10        0.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    55 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1763.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
     56    TAPERED       PASS     BS-4.7   (C)       0.136       207
                       11.83 C          0.15         -41.04        2.52
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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    56 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1532.6   PT=  2166.6    MB=  290.0 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     57    TAPERED       PASS     BS-4.7   (C)       0.132       207
                       16.99 C          0.10         -38.96        2.52
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    57 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1532.6   PT=  2166.6    MB=  290.0 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     58    TAPERED       PASS     BS-4.7   (C)       0.131       207
                       16.48 C         -0.08         -39.02        2.52
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    58 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1532.6   PT=  2166.6    MB=  290.0 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     59    TAPERED       PASS     BS-4.7   (C)       0.135       207
                       11.83 C         -0.13         -41.04        2.52
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    59 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1532.6   PT=  2166.6    MB=  290.0 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     95    TAPERED       PASS     BS-4.7   (C)       0.302       214
                       68.91 C          6.87          37.46        0.00
```



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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    95 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=   56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     96    TAPERED       PASS     BS-4.7   (C)       0.241       215
                       89.45 C         -0.23          49.36        0.00
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    96 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=   56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     97    TAPERED       PASS     BS-4.7   (C)       0.241       215
                       89.44 C          0.23          49.36        0.00
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    97 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=   56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     98    TAPERED       PASS     BS-4.7   (C)       0.302       214
                       68.91 C         -6.87          37.46        0.00
 |---------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    98 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=   56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |---------------------------------------------------------------------|
     99    TAPERED       PASS     BS-4.7   (C)       0.302       202
                       68.82 C         -6.87          37.46        0.00
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    99 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    100    TAPERED       PASS     BS-4.7   (C)       0.241       203
                       89.27 C          0.23          49.36        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   100 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    101    TAPERED       PASS     BS-4.7   (C)       0.241       203
                       89.30 C         -0.23          49.36        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   101 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    102    TAPERED       PASS     BS-4.7   (C)       0.302       202
                       68.82 C          6.87          37.46        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   102 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  275.6   MCY=    56.6   PC=  1426.5   PT=  2063.1    MB=  243.9 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  409 kN : qw       = 207 N/mm2  |
 |  d   = 330 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    201    TAPERED       PASS     BS-4.7   (C)       0.092       207
                        0.55 C         -1.75          -7.99        1.50
```



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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   201 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  1266.8   PT=     0.0    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    202    TAPERED       PASS     BS-4.7   (C)       0.307       201
                       43.82 C         -0.13         -28.13        3.79
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   202 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    203    TAPERED       PASS     BS-4.7   (C)       0.293       201
                       37.62 C          0.05         -31.67        3.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   203 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   282.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    204    TAPERED       PASS     BS-4.7   (C)       0.307       201
                       43.96 C         -0.13         -28.12        2.71
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   204 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    205    TAPERED       PASS     BS-4.7   (C)       0.092       207
                        0.55 C          1.75           7.99        0.00
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   205 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  1267.4   PT=     0.0    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    206    TAPERED       PASS     BS-4.7   (C)       0.083       207
                        0.17 C          0.55         -11.69        1.80
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   206 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  1174.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    207    TAPERED       PASS     BS-4.3.6           0.206       218
                        0.28 T          0.01           8.91        3.25
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   207 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  5224.1   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    208    TAPERED       PASS     BS-4.3.6           0.191       218
                        0.13 T         -0.09           7.74        2.80
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   208 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   839.5   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    209    TAPERED       PASS     BS-4.3.6           0.206       218
                        0.28 T          0.01           8.91        3.25
```



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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   209 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  3734.2   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    210    TAPERED       PASS     BS-4.7   (C)       0.083       207
                        0.18 C         -0.56          11.69        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   210 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  1174.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    211    TAPERED       PASS     BS-4.3.6           0.215       201
                        0.01 T          0.02          35.20        3.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   211 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    212    TAPERED       PASS     BS-4.3.6           0.254       218
                        1.66 T         -0.01          11.36        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   212 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d    = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    213    TAPERED       PASS     BS-4.3.6           0.228       218
                        0.54 C          0.00          10.55        3.00
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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   213 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   282.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    214    TAPERED       PASS     BS-4.3.6           0.253       218
                        2.17 T          0.00          11.38        6.50
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   214 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    215    TAPERED       PASS     BS-4.3.6           0.215       201
                        0.01 T         -0.03          35.20        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   215 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    216    TAPERED       PASS     BS-4.3.6           0.275       207
                        3.38 T          0.00          44.73        3.20
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   216 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   719.1   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    217    TAPERED       PASS     BS-4.3.6           0.303       218
                        1.37 T          0.00          23.80        0.00
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   217 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    218    TAPERED       PASS     BS-4.3.6           0.240       212
                        0.44 C          0.00           8.55        3.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   218 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    220    TAPERED       PASS     BS-4.3.6           0.275       207
                        3.39 T          0.00          44.73        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   220 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   719.1   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    221    TAPERED       PASS     BS-4.3.6           0.215       207
                        0.01 T         -0.03          35.22        3.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   221 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
    222    TAPERED       PASS     BS-4.3.6           0.253       206
                        2.35 T          0.01          11.33        0.00
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   222 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    223    TAPERED       PASS     BS-4.3.6           0.228       206
                        0.15 C          0.00          10.55        3.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   223 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   282.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    224    TAPERED       PASS     BS-4.3.6           0.253       206
                        2.35 T         -0.01          11.33        6.50
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   224 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    225    TAPERED       PASS     BS-4.3.6           0.215       207
                        0.01 T          0.03          35.22        0.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   225 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    226    TAPERED       PASS     BS-4.7   (C)       0.085       207
                        0.23 C         -0.62         -11.68        1.80
```



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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   226 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  1174.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    227    TAPERED       PASS     BS-4.3.6           0.207       206
                        0.34 T         -0.02           8.90        3.25
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   227 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  5224.1   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    228    TAPERED       PASS     BS-4.3.6           0.191       206
                        0.17 T          0.10           7.73        3.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   228 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    229    TAPERED       PASS     BS-4.3.6           0.206       206
                        0.34 T         -0.02           8.90        3.25
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   229 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=    38.8   PC=  3457.6   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    230    TAPERED       PASS     BS-4.7   (C)       0.085       207
                        0.23 C          0.62          11.68        0.00
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   230 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=  1174.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    231    TAPERED       PASS     BS-4.7   (C)       0.090       207
                        0.60 C          1.64          -7.97        1.50
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   231 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=  1267.4   PT=     0.0    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    232    TAPERED       PASS     BS-4.7   (C)       0.308       213
                       44.28 C          0.12         -28.12        3.79
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   232 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    233    TAPERED       PASS     BS-4.7   (C)       0.290       213
                       36.44 C         -0.04         -31.67        3.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   233 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   282.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    234    TAPERED       PASS     BS-4.7   (C)       0.308       213
                       44.27 C          0.12         -28.12        2.71
```





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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   234 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    235    TAPERED       PASS     BS-4.7   (C)       0.090       207
                        0.60 C         -1.64           7.97        0.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   235 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=  1267.4   PT=     0.0    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    236    TAPERED       PASS     BS-4.3.6           0.193       218
                        0.15 T          0.08           7.86        0.27
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   236 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   719.1   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    238    TAPERED       PASS     BS-4.3.6           0.240       212
                        0.44 C          0.00           8.55        0.00
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   238 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    240    TAPERED       PASS     BS-4.3.6           0.191       206
                        0.17 T         -0.10           7.73        0.00
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   240 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   777.3   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    241    TAPERED       PASS     BS-4.3.6           0.295       212
                        0.42 C          0.00            9.80        6.50
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   241 UNIT - kN,m  SECTION CLASS  2  |
 | MCZ=  170.4   MCY=   38.8   PC=   248.4   PT=  1614.6    MB=  190.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  347 kN : qw       = 207 N/mm2  |
 |  d   = 280 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    301    TAPERED       PASS     BS-4.7   (C)       0.063       214
                        0.00           -3.87           6.28        1.30
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   301 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.0   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    302    TAPERED       PASS     BS-4.8.2.2         0.237       211
                        2.46 T        -10.34         -34.31        3.25
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   302 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC= 10382.0   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    303    TAPERED       PASS     BS-4.8.2.2         0.187       214
                        3.17 T         -5.56          33.45        3.00
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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   303 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC= 46181.7   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    304    TAPERED       PASS     BS-4.8.2.2         0.237       211
                        2.46 T        -10.34         -34.31        3.25
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   304 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=205426.3   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    305    TAPERED       PASS     BS-4.7   (C)       0.063       214
                        0.00            3.87          -6.28        0.00
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   305 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.1   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    306    TAPERED       PASS     BS-4.7   (C)       0.063       202
                        0.00           -3.87          -6.28        1.30
 |--------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   306 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.1   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |--------------------------------------------------------------------|
    307    TAPERED       PASS     BS-4.8.2.2         0.237       211
                        2.45 T        -10.34          34.31        3.25
```

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   307 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC= 10382.5   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    308    TAPERED       PASS     BS-4.8.2.2         0.187       202
                        3.16 T         -5.56         -33.45        3.00
 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   308 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC= 46183.5   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    309    TAPERED       PASS     BS-4.8.2.2         0.237       211
                        2.45 T        -10.34          34.31        3.25
 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   309 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=205434.5   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    310    TAPERED       PASS     BS-4.7   (C)       0.063       202
                        0.00           3.87           6.28         0.00
 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   310 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.1   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    311    TAPERED       PASS     BS-4.7   (C)       0.048       214
                        0.00          -3.28           3.93         1.30
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   311 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.1   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    312    TAPERED       PASS     BS-4.7   (C)       0.168       214
                        3.35 C         -7.86          22.30        3.25
 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   312 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  1045.9   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    313    TAPERED       PASS     BS-4.7   (C)       0.135       214
                        3.53 C         -5.18          20.92        3.00
 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   313 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  1154.2   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    314    TAPERED       PASS     BS-4.7   (C)       0.168       214
                        3.35 C         -7.86          22.31        3.25
 |-------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   314 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  257.7   MCY=  100.2   PC=  1045.9   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-------------------------------------------------------------------|
    315    TAPERED       PASS     BS-4.7   (C)       0.048       214
                        0.00           3.28          -3.93         0.00
```

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   315 UNIT - kN,m  SECTION CLASS  3   |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.1   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     316    TAPERED       PASS     BS-4.7   (C)       0.048       202
                         0.00           -3.28          -3.93        1.30
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   316 UNIT - kN,m  SECTION CLASS  3   |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.1   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     317    TAPERED       PASS     BS-4.7   (C)       0.168       202
                         3.33 C         -7.86         -22.31        3.25
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   317 UNIT - kN,m  SECTION CLASS  3   |
 | MCZ=  257.7   MCY=  100.2   PC=  1045.9   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     318    TAPERED       PASS     BS-4.7   (C)       0.135       202
                         3.52 C         -5.19         -20.92        3.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   318 UNIT - kN,m  SECTION CLASS  3   |
 | MCZ=  257.7   MCY=  100.2   PC=  1154.2   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     319    TAPERED       PASS     BS-4.7   (C)       0.168       202
                         3.33 C         -7.86         -22.31        3.25
```



```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   319 UNIT - kN,m  SECTION CLASS  3   |
 | MCZ=  257.7   MCY=  100.2   PC=  1045.9   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     320    TAPERED       PASS     BS-4.7   (C)       0.048       202
                         0.00            3.28           3.93        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   320 UNIT - kN,m  SECTION CLASS  3   |
 | MCZ=  257.7   MCY=  100.2   PC=  2334.1   PT=  2537.8    MB=  257.7 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  280 kN : qw       = 207 N/mm2  |
 |  d   = 226 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
```

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    501 ST UA70X70X6     PASS     BS-4.6   (T)       0.005       207
                       1.30 T          0.00           0.00        3.50
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    349.70
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
     Moment of inertia        :      15.391          60.246
     Plastic modulus          :      10.133          21.909
     Elastic modulus          :       5.629          12.171
     Effective modulus        :       5.629          12.171
     Shear Area               :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER

                                  z-z  axis       y-y  axis
     Tension Capacity         :       282.9           282.9
     Shear Capacity           :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)

     LTB Moment Capacity (kNm) and LTB Length (m):     2.69,    3.497
     LTB Coefficients & Associated Moments (kNm):
     mLT =    1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =     0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.005   207     1.3       -       -       -       -
   Torsion and deflections have not been considered in the design.
```





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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    502 ST UA70X70X6     PASS     BS-4.6   (T)       0.001       207
                       0.36 T          0.00           0.00        3.64
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    363.58
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
     Moment of inertia        :      15.391          60.246
     Plastic modulus          :      10.133          21.909
     Elastic modulus          :       5.629          12.171
     Effective modulus        :       5.629          12.171
     Shear Area               :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.001

                                  z-z  axis       y-y  axis
     Compression Capacity     :        46.3            97.2
     Tension Capacity         :       282.9           282.9
     Shear Capacity           :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                               v-v axis      a-a axis       b-b axis
     Slenderness              :  172.501       111.179        111.179
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.363         2.363          2.363

     LTB Moment Capacity (kNm) and LTB Length (m):     2.63,    3.636
     LTB Coefficients & Associated Moments (kNm):
     mLT =    1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =     0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.001   207     0.4       -       -       -       -
  BS-4.7   (C)   0.001   218     0.0       -       -       -       -
```

Torsion and deflections have not been considered in the design.



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    503 ST UA70X70X6     PASS     BS-4.7   (C)       0.001       220
                        0.02 C          0.00          0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    437.92
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                   z-z  axis       y-y  axis
     Moment of inertia       :       15.391          60.246
     Plastic modulus         :       10.133          21.909
     Elastic modulus         :        5.629          12.171
     Effective modulus       :        5.629          12.171
     Shear Area              :        2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class           :   SLENDER
     Squash Load             :    282.90
     Axial force/Squash load  :     0.000

                                   z-z  axis       y-y  axis
     Compression Capacity    :         33.1           71.9
     Tension Capacity        :        282.9          282.9
     Shear Capacity          :         52.2           52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                                v-v axis      a-a axis       b-b axis
     Slenderness             :  207.770       133.911        133.911
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length        :    2.846         2.846          2.846

     LTB Moment Capacity (kNm) and LTB Length (m):     2.37,    4.379
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   Torsion and deflections have not been considered in the design.
```

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

     504 ST UA70X70X6     PASS     BS-4.7   (C)       0.080       207
                        2.49 C          0.00           0.00        0.00
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    451.85
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
      Moment of inertia       :      15.391          60.246
      Plastic modulus         :      10.133          21.909
      Elastic modulus         :       5.629          12.171
      Effective modulus       :       5.629          12.171
      Shear Area              :       2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class           :   SLENDER
      Squash Load             :    282.90
      Axial force/Squash load :     0.009

                                 z-z  axis       y-y  axis
      Compression Capacity    :        31.2            68.2
      Tension Capacity        :       282.9           282.9
      Shear Capacity          :        52.2            52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
      Slenderness             :  214.382       138.172        138.172
      Radius of gyration (cm) :    1.370         2.126          2.126
      Effective Length        :    2.937         2.937          2.937

      LTB Moment Capacity (kNm) and LTB Length (m):     2.33,    4.519
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.001   212     0.4       -       -       -       -
   BS-4.7   (C)   0.080   207     2.5       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY            MZ       LOCATION
=======================================================================

    505 ST UA70X70X6     PASS     BS-4.7   (C)       0.083       207
                        2.40 C          0.00          0.00        4.71
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    471.31
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                              z-z  axis       y-y  axis
     Moment of inertia       :      15.391         60.246
     Plastic modulus         :      10.133         21.909
     Elastic modulus         :       5.629         12.171
     Effective modulus       :       5.629         12.171
     Shear Area              :       2.520          2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class           :   SLENDER
     Squash Load             :    282.90
     Axial force/Squash load :     0.008

                              z-z  axis       y-y  axis
     Compression Capacity    :        28.9          63.5
     Tension Capacity        :       282.9         282.9
     Shear Capacity          :        52.2          52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                           v-v axis      a-a axis       b-b axis
     Slenderness           :  223.614       144.122        144.122
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length      :    3.064         3.064          3.064

     LTB Moment Capacity (kNm) and LTB Length (m):     2.27,    4.713
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.001   212     0.4       -       -       -       -
  BS-4.7   (C)   0.083   207     2.4       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

   506 ST UA70X70X6     PASS     BS-4.7   (C)       0.001       207
                       0.02 C          0.00           0.00        4.58
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    457.97
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia       :      15.391          60.246
     Plastic modulus         :      10.133          21.909
     Elastic modulus         :       5.629          12.171
     Effective modulus       :       5.629          12.171
     Shear Area              :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class           :  SLENDER
     Squash Load             :   282.90
     Axial force/Squash load :     0.000

                                 z-z  axis       y-y  axis
     Compression Capacity    :        30.4            66.7
     Tension Capacity        :       282.9           282.9
     Shear Capacity          :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness             : 217.283       140.042        140.042
     Radius of gyration (cm) :   1.370         2.126          2.126
     Effective Length        :   2.977         2.977          2.977

     LTB Moment Capacity (kNm) and LTB Length (m):     2.31,    4.580
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   Torsion and deflections have not been considered in the design.
```





```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

   507 ST UA70X70X6     PASS     BS-4.6   (T)       0.001       207
                       0.42 T          0.00           0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    329.16
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia       :      15.391          60.246
     Plastic modulus         :      10.133          21.909
     Elastic modulus         :       5.629          12.171
     Effective modulus       :       5.629          12.171
     Shear Area              :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class           :  SLENDER
     Squash Load             :   282.90
     Axial force/Squash load :     0.001

                                 z-z  axis       y-y  axis
     Compression Capacity    :        55.2           112.8
     Tension Capacity        :       282.9           282.9
     Shear Capacity          :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness             : 156.172       100.655        100.655
     Radius of gyration (cm) :   1.370         2.126          2.126
     Effective Length        :   2.140         2.140          2.140

     LTB Moment Capacity (kNm) and LTB Length (m):     2.77,    3.292
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.001   207     0.4      -       -       -       -
   Torsion and deflections have not been considered in the design.
```

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    508 ST UA70X70X6     PASS     BS-4.6   (T)       0.004       207
                       1.26 T          0.00           0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    313.76
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia        :      15.391          60.246
     Plastic modulus          :      10.133          21.909
     Elastic modulus          :       5.629          12.171
     Effective modulus        :       5.629          12.171
     Shear Area               :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER

                                 z-z  axis       y-y  axis
     Tension Capacity         :       282.9           282.9
     Shear Capacity           :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)

     LTB Moment Capacity (kNm) and LTB Length (m):     2.83,    3.138
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.004   207     1.3       -       -       -       -
   Torsion and deflections have not been considered in the design.
                                  _________________________
```





```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    509 ST UA70X70X6     PASS     BS-4.6   (T)       0.001       207
                       0.36 T          0.00           0.00        3.64
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    363.58
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia        :      15.391          60.246
     Plastic modulus          :      10.133          21.909
     Elastic modulus          :       5.629          12.171
     Effective modulus        :       5.629          12.171
     Shear Area               :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.001

                                 z-z  axis       y-y  axis
     Compression Capacity     :        46.3            97.2
     Tension Capacity         :       282.9           282.9
     Shear Capacity           :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                             v-v axis      a-a axis       b-b axis
     Slenderness            :  172.501       111.179        111.179
     Radius of gyration (cm)  :   1.370         2.126          2.126
     Effective Length       :   2.363         2.363          2.363

     LTB Moment Capacity (kNm) and LTB Length (m):     2.63,    3.636
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.001   207     0.4       -       -       -       -
  BS-4.7   (C)   0.001   218     0.1       -       -       -       -
```

Torsion and deflections have not been considered in the design.



```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

     510 ST UA70X70X6     PASS     BS-4.6   (T)       0.005       207
                        1.30 T          0.00          0.00        3.50
 =======================================================================
   MATERIAL DATA
      Grade of steel          =  S 275
      Modulus of elasticity   =  205 kN/mm2
      Design Strength  (py)   =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    349.70
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia       :      15.391          60.246
      Plastic modulus         :      10.133          21.909
      Elastic modulus         :       5.629          12.171
      Effective modulus       :       5.629          12.171
      Shear Area              :       2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class           :   SLENDER

                                  z-z  axis       y-y  axis
      Tension Capacity        :       282.9          282.9
      Shear Capacity          :        52.2           52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)

      LTB Moment Capacity (kNm) and LTB Length (m):     2.69,    3.497
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.005   207     1.3       -       -       -       -
    Torsion and deflections have not been considered in the design.
```

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

     511 ST UA70X70X6     PASS     BS-4.7   (C)       0.080       207
                         2.49 C          0.00           0.00        0.00
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    451.85
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia        :       15.391          60.246
      Plastic modulus          :       10.133          21.909
      Elastic modulus          :        5.629          12.171
      Effective modulus        :        5.629          12.171
      Shear Area               :        2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class            :   SLENDER
      Squash Load              :    282.90
      Axial force/Squash load  :     0.009

                                  z-z  axis       y-y  axis
      Compression Capacity     :         31.2           68.2
      Tension Capacity         :        282.9          282.9
      Shear Capacity           :         52.2           52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                               v-v axis      a-a axis       b-b axis
      Slenderness              :  214.382       138.172        138.172
      Radius of gyration (cm)  :    1.370         2.126          2.126
      Effective Length         :    2.937         2.937          2.937

      LTB Moment Capacity (kNm) and LTB Length (m):     2.33,    4.519
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE         RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.001   212     0.4       -       -       -       -
   BS-4.7   (C)   0.080   207     2.5       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX           MY             MZ       LOCATION
 =======================================================================

    512 ST UA70X70X6     PASS     BS-4.7   (C)       0.001       222
                        0.02 C         0.00           0.00        0.00
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    437.92
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia       :       15.391          60.246
      Plastic modulus         :       10.133          21.909
      Elastic modulus         :        5.629          12.171
      Effective modulus       :        5.629          12.171
      Shear Area              :        2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class           :  SLENDER
      Squash Load             :   282.90
      Axial force/Squash load :     0.000

                                  z-z  axis       y-y  axis
      Compression Capacity    :         33.1            71.9
      Tension Capacity        :        282.9           282.9
      Shear Capacity          :         52.2            52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                               v-v axis      a-a axis       b-b axis
      Slenderness             :  207.770       133.911        133.911
      Radius of gyration (cm)  :    1.370         2.126          2.126
      Effective Length        :    2.846         2.846          2.846

      LTB Moment Capacity (kNm) and LTB Length (m):     2.37,    4.379
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
    Torsion and deflections have not been considered in the design.
```



```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX           MY             MZ       LOCATION
 =======================================================================

    513 ST UA70X70X6     PASS     BS-4.7   (C)       0.001       207
                        0.02 C         0.00           0.00        4.58
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    457.97
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia       :       15.391          60.246
      Plastic modulus         :       10.133          21.909
      Elastic modulus         :        5.629          12.171
      Effective modulus       :        5.629          12.171
      Shear Area              :        2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class           :  SLENDER
      Squash Load             :   282.90
      Axial force/Squash load :     0.000

                                  z-z  axis       y-y  axis
      Compression Capacity    :         30.4            66.7
      Tension Capacity        :        282.9           282.9
      Shear Capacity          :         52.2            52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                               v-v axis      a-a axis       b-b axis
      Slenderness             :  217.283       140.042        140.042
      Radius of gyration (cm)  :    1.370         2.126          2.126
      Effective Length        :    2.977         2.977          2.977

      LTB Moment Capacity (kNm) and LTB Length (m):     2.31,    4.580
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
    Torsion and deflections have not been considered in the design.
```

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

    514 ST UA70X70X6     PASS     BS-4.7   (C)       0.083       207
                        2.39 C          0.00           0.00        4.71
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    471.31
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia        :      15.391          60.246
      Plastic modulus          :      10.133          21.909
      Elastic modulus          :       5.629          12.171
      Effective modulus        :       5.629          12.171
      Shear Area               :       2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class            :   SLENDER
      Squash Load              :    282.90
      Axial force/Squash load  :     0.008

                                  z-z  axis       y-y  axis
      Compression Capacity     :        28.9            63.5
      Tension Capacity         :       282.9           282.9
      Shear Capacity           :        52.2            52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                               v-v axis      a-a axis       b-b axis
      Slenderness              :  223.614       144.122        144.122
      Radius of gyration (cm)  :    1.370         2.126          2.126
      Effective Length         :    3.064         3.064          3.064

      LTB Moment Capacity (kNm) and LTB Length (m):     2.27,    4.713
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.001   212     0.4       -       -       -       -
   BS-4.7   (C)   0.083   207     2.4       -       -       -       -
```



Torsion and deflections have not been considered in the design.

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

    515 ST UA70X70X6     PASS     BS-4.6   (T)       0.001       207
                         0.42 T          0.00           0.00        0.00
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    329.16
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia       :      15.391          60.246
      Plastic modulus         :      10.133          21.909
      Elastic modulus         :       5.629          12.171
      Effective modulus       :       5.629          12.171
      Shear Area              :       2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class           :   SLENDER
      Squash Load             :   282.90
      Axial force/Squash load :     0.001

                                  z-z  axis       y-y  axis
      Compression Capacity    :        55.2          112.8
      Tension Capacity        :       282.9          282.9
      Shear Capacity          :        52.2           52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                               v-v axis      a-a axis       b-b axis
      Slenderness             : 156.172       100.655        100.655
      Radius of gyration (cm) :   1.370         2.126          2.126
      Effective Length        :   2.140         2.140          2.140

      LTB Moment Capacity (kNm) and LTB Length (m):     2.77,    3.292
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE         RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.001   207     0.4       -       -       -       -
    Torsion and deflections have not been considered in the design.
```



```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

    516 ST UA70X70X6     PASS     BS-4.6   (T)       0.004       207
                         1.26 T          0.00           0.00        3.14
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    313.76
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia       :      15.391          60.246
      Plastic modulus         :      10.133          21.909
      Elastic modulus         :       5.629          12.171
      Effective modulus       :       5.629          12.171
      Shear Area              :       2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class           :   SLENDER

                                  z-z  axis       y-y  axis
      Tension Capacity        :       282.9          282.9
      Shear Capacity          :        52.2           52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)

      LTB Moment Capacity (kNm) and LTB Length (m):     2.83,    3.138
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE         RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.004   207     1.3       -       -       -       -
    Torsion and deflections have not been considered in the design.
```

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

    522 ST UA70X70X6     PASS     BS-4.7   (C)       0.313       221
                        9.86 C          0.00           0.00        0.00
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    449.65
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia        :      15.391          60.246
      Plastic modulus          :      10.133          21.909
      Elastic modulus          :       5.629          12.171
      Effective modulus        :       5.629          12.171
      Shear Area               :       2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class            :   SLENDER
      Squash Load              :    282.90
      Axial force/Squash load  :     0.035

                                  z-z  axis       y-y  axis
      Compression Capacity     :        31.5           68.8
      Tension Capacity         :       282.9          282.9
      Shear Capacity           :        52.2           52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
      Slenderness            :  213.340       137.500        137.500
      Radius of gyration (cm)  :    1.370         2.126          2.126
      Effective Length         :    2.923         2.923          2.923

      LTB Moment Capacity (kNm) and LTB Length (m):     2.33,    4.497
      LTB Coefficients & Associated Moments (kNm):
      mLT =    1.00  :   mx =    0.00   :  my =    0.00   :  myx =    0.00
      Mlt =     0.00 :   Mx =     0.00  :  My =    0.00   :   My =    0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.009   228     2.7       -       -       -       -
   BS-4.7   (C)   0.313   221     9.9       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

     523 ST UA70X70X6     PASS     BS-4.7   (C)       0.224       201
                        9.73 C          0.00           0.00        0.00
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    376.60
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia        :      15.391          60.246
      Plastic modulus          :      10.133          21.909
      Elastic modulus          :       5.629          12.171
      Effective modulus        :       5.629          12.171
      Shear Area               :       2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class            :   SLENDER
      Squash Load              :    282.90
      Axial force/Squash load  :     0.034

                                  z-z  axis       y-y  axis
      Compression Capacity     :        43.5           92.0
      Tension Capacity         :       282.9          282.9
      Shear Capacity           :        52.2           52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
      Slenderness              :  178.677       115.160        115.160
      Radius of gyration (cm)  :    1.370         2.126          2.126
      Effective Length         :    2.448         2.448          2.448

      LTB Moment Capacity (kNm) and LTB Length (m):     2.58,    3.766
      LTB Coefficients & Associated Moments (kNm):
      mLT =    1.00  :   mx =    0.00   :  my =    0.00   :  myx =    0.00
      Mlt =     0.00 :   Mx =     0.00  :  My =    0.00   :   My =    0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.009   218     2.6       -       -       -       -
   BS-4.7   (C)   0.224   201     9.7       -       -       -       -
```



Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    524 ST UA70X70X6     PASS     BS-4.7   (C)       0.274       219
                       9.69 C          0.00           0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    422.12
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                               z-z  axis       y-y  axis
     Moment of inertia        :       15.391          60.246
     Plastic modulus          :       10.133          21.909
     Elastic modulus          :        5.629          12.171
     Effective modulus        :        5.629          12.171
     Shear Area               :        2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.034

                               z-z  axis       y-y  axis
     Compression Capacity     :         35.4           76.5
     Tension Capacity         :        282.9          282.9
     Shear Capacity           :         52.2           52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                            v-v axis      a-a axis       b-b axis
     Slenderness              :  200.278       129.082        129.082
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.744         2.744          2.744

     LTB Moment Capacity (kNm) and LTB Length (m):     2.42,    4.221
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.009   218     2.5       -       -       -       -
  BS-4.7   (C)   0.274   219     9.7       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY            MZ       LOCATION
=======================================================================

    525 ST UA70X70X6     PASS     BS-4.7   (C)       0.251       219
                        9.48 C          0.00          0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =     407.22
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                z-z  axis       y-y  axis
     Moment of inertia        :      15.391          60.246
     Plastic modulus          :      10.133          21.909
     Elastic modulus          :       5.629          12.171
     Effective modulus        :       5.629          12.171
     Shear Area               :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.034

                                z-z  axis       y-y  axis
     Compression Capacity     :        37.7           81.1
     Tension Capacity         :       282.9          282.9
     Shear Capacity           :        52.2           52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                             v-v axis      a-a axis       b-b axis
     Slenderness             :  193.205       124.523        124.523
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.647         2.647          2.647

     LTB Moment Capacity (kNm) and LTB Length (m):     2.47,    4.072
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.009   218     2.5       -       -       -       -
  BS-4.7   (C)   0.251   219     9.5       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY            MZ       LOCATION
=======================================================================

    530 ST UA70X70X6     PASS     BS-4.7   (C)       0.137       207
                        4.51 C          0.00          0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    437.92
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia        :       15.391          60.246
     Plastic modulus          :       10.133          21.909
     Elastic modulus          :        5.629          12.171
     Effective modulus        :        5.629          12.171
     Shear Area               :        2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.016

                                 z-z  axis       y-y  axis
     Compression Capacity     :         33.1          71.9
     Tension Capacity         :        282.9         282.9
     Shear Capacity           :         52.2          52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness              :  207.770       133.911        133.911
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.846         2.846          2.846

     LTB Moment Capacity (kNm) and LTB Length (m):     2.37,    4.379
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.004   212     1.1       -       -       -       -
  BS-4.7   (C)   0.137   207     4.5       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX           MY             MZ       LOCATION
=======================================================================

    531 ST UA70X70X6     PASS     BS-4.7   (C)       0.144       207
                        4.37 C          0.00          0.00        4.58
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    457.97
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia       :      15.391          60.246
     Plastic modulus         :      10.133          21.909
     Elastic modulus         :       5.629          12.171
     Effective modulus       :       5.629          12.171
     Shear Area              :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class           :   SLENDER
     Squash Load             :    282.90
     Axial force/Squash load :     0.015

                                 z-z  axis       y-y  axis
     Compression Capacity    :        30.4            66.7
     Tension Capacity        :       282.9           282.9
     Shear Capacity          :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness             :  217.283       140.042        140.042
     Radius of gyration (cm) :    1.370         2.126          2.126
     Effective Length        :    2.977         2.977          2.977

     LTB Moment Capacity (kNm) and LTB Length (m):     2.31,    4.580
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.004   212     1.1       -       -       -       -
  BS-4.7   (C)   0.144   207     4.4       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    532 ST UA70X70X6     PASS     BS-4.7   (C)       0.136       207
                       4.48 C          0.00           0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    437.92
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia        :      15.391          60.246
     Plastic modulus          :      10.133          21.909
     Elastic modulus          :       5.629          12.171
     Effective modulus        :       5.629          12.171
     Shear Area               :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.016

                                 z-z  axis       y-y  axis
     Compression Capacity     :        33.1            71.9
     Tension Capacity         :       282.9           282.9
     Shear Capacity           :        52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness              :  207.770       133.911        133.911
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.846         2.846          2.846

     LTB Moment Capacity (kNm) and LTB Length (m):     2.37,    4.379
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.004   212     1.1       -       -       -       -
  BS-4.7   (C)   0.136   207     4.5       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX           MY              MZ       LOCATION
=======================================================================

    533 ST UA70X70X6     PASS     BS-4.7   (C)       0.142       207
                       4.32 C          0.00           0.00        4.58
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    457.97
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia        :      15.391          60.246
     Plastic modulus          :      10.133          21.909
     Elastic modulus          :       5.629          12.171
     Effective modulus        :       5.629          12.171
     Shear Area               :       2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.015

                                 z-z  axis       y-y  axis
     Compression Capacity     :        30.4           66.7
     Tension Capacity         :       282.9          282.9
     Shear Capacity           :        52.2           52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness              :  217.283       140.042        140.042
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.977         2.977          2.977

     LTB Moment Capacity (kNm) and LTB Length (m):     2.31,    4.580
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.004   218     1.1       -       -       -       -
  BS-4.7   (C)   0.142   207     4.3       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY            MZ       LOCATION
=======================================================================

    538 ST UA70X70X6     PASS     BS-4.7   (C)       0.258       213
                       10.04 C          0.00          0.00        4.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    400.28
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia        :       15.391          60.246
     Plastic modulus          :       10.133          21.909
     Elastic modulus          :        5.629          12.171
     Effective modulus        :        5.629          12.171
     Shear Area               :        2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.035

                                 z-z  axis       y-y  axis
     Compression Capacity     :         38.9            83.4
     Tension Capacity         :        282.9           282.9
     Shear Capacity           :         52.2            52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness              :  189.914       122.402        122.402
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.602         2.602          2.602

     LTB Moment Capacity (kNm) and LTB Length (m):     2.50,    4.003
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.010   206     2.9       -       -       -       -
  BS-4.7   (C)   0.258   213    10.0       -       -       -       -
```





Torsion and deflections have not been considered in the design.

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

     539 ST UA70X70X6     PASS     BS-4.7   (C)       0.268       222
                        10.11 C          0.00           0.00        4.07
 =======================================================================
   MATERIAL DATA
      Grade of steel           =  S 275
      Modulus of elasticity    =  205 kN/mm2
      Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
      Member Length =    407.37
      Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                  z-z  axis       y-y  axis
      Moment of inertia        :       15.391          60.246
      Plastic modulus          :       10.133          21.909
      Elastic modulus          :        5.629          12.171
      Effective modulus        :        5.629          12.171
      Shear Area               :        2.520           2.520

   DESIGN DATA (units - kN,m)   BS5950-1/2000
      Section Class            :   SLENDER
      Squash Load              :    282.90
      Axial force/Squash load  :     0.036

                                  z-z  axis       y-y  axis
      Compression Capacity     :         37.7           81.1
      Tension Capacity         :        282.9          282.9
      Shear Capacity           :         52.2           52.2

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
      Slenderness            :  193.276       124.569       124.569
      Radius of gyration (cm)  :    1.370         2.126         2.126
      Effective Length         :    2.648         2.648         2.648

      LTB Moment Capacity (kNm) and LTB Length (m):     2.47,    4.074
      LTB Coefficients & Associated Moments (kNm):
      mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
      Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
   BS-4.6   (T)   0.010   206     2.9       -       -       -       -
   BS-4.7   (C)   0.268   222    10.1       -       -       -       -
```



Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

    540 ST UA70X70X6     PASS     BS-4.7   (C)       0.268       220
                       10.11 C          0.00           0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    407.37
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                               z-z  axis       y-y  axis
     Moment of inertia        :       15.391          60.246
     Plastic modulus          :       10.133          21.909
     Elastic modulus          :        5.629          12.171
     Effective modulus        :        5.629          12.171
     Shear Area               :        2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   SLENDER
     Squash Load              :    282.90
     Axial force/Squash load  :     0.036

                               z-z  axis       y-y  axis
     Compression Capacity     :         37.7           81.1
     Tension Capacity         :        282.9          282.9
     Shear Capacity           :         52.2           52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness              :  193.276       124.569        124.569
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length         :    2.648         2.648          2.648

     LTB Moment Capacity (kNm) and LTB Length (m):     2.47,    4.074
     LTB Coefficients & Associated Moments (kNm):
     mLT =   1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE        RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.010   206     2.9       -       -       -       -
  BS-4.7   (C)   0.268   220    10.1       -       -       -       -
```



Torsion and deflections have not been considered in the design.

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX           MY             MZ       LOCATION
=======================================================================

   541 ST UA70X70X6     PASS     BS-4.7   (C)       0.258       213
                       10.04 C         0.00          0.00        0.00
=======================================================================
  MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =    400.28
     Gross Area =    8.20  Net Area =    8.20  Eff. Area =    8.20

                                 z-z  axis       y-y  axis
     Moment of inertia       :       15.391          60.246
     Plastic modulus         :       10.133          21.909
     Elastic modulus         :        5.629          12.171
     Effective modulus       :        5.629          12.171
     Shear Area              :        2.520           2.520

  DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class           :   SLENDER
     Squash Load             :    282.90
     Axial force/Squash load  :     0.035

                                 z-z  axis       y-y  axis
     Compression Capacity    :         38.9           83.4
     Tension Capacity        :        282.9          282.9
     Shear Capacity          :         52.2           52.2

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              v-v axis      a-a axis       b-b axis
     Slenderness             :  189.914       122.402        122.402
     Radius of gyration (cm)  :    1.370         2.126          2.126
     Effective Length        :    2.602         2.602          2.602

     LTB Moment Capacity (kNm) and LTB Length (m):     2.50,    4.003
     LTB Coefficients & Associated Moments (kNm):
     mLT =    1.00  :   mx =   0.00   :  my =   0.00   :  myx =   0.00
     Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX       VY      VZ      MZ      MY
  BS-4.6   (T)   0.010   206     2.9       -       -       -       -
  BS-4.7   (C)   0.258   213    10.0       -       -       -       -
```





```
Torsion and deflections have not been considered in the design.

   542    TAPERED       PASS     BS-4.7   (C)       0.229       222
                       62.99 C        -5.43          6.42        0.50
|---------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB   542 UNIT - kN,m  SECTION CLASS  4  |
| MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
|  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|---------------------------------------------------------------------|
   543    TAPERED       PASS     BS-4.7   (C)       0.241       221
                       66.36 C        -5.72         -6.70        0.50
|---------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB   543 UNIT - kN,m  SECTION CLASS  4  |
| MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
|  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|---------------------------------------------------------------------|
   544    TAPERED       PASS     BS-4.7   (C)       0.068       213
                       85.58 C        -0.01         -2.80        0.50
|---------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB   544 UNIT - kN,m  SECTION CLASS  4  |
| MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
|  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|---------------------------------------------------------------------|
   545    TAPERED       PASS     BS-4.7   (C)       0.072       201
                       90.24 C        -0.01          3.05        0.50
|---------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB   545 UNIT - kN,m  SECTION CLASS  4  |
| MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
|  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|---------------------------------------------------------------------|
   546    TAPERED       PASS     BS-4.7   (C)       0.067       213
                       85.66 C         0.01         -2.71        0.50
|---------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB   546 UNIT - kN,m  SECTION CLASS  4  |
| MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
|  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
|  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
|  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
|---------------------------------------------------------------------|
   547    TAPERED       PASS     BS-4.7   (C)       0.073       201
                       90.22 C         0.01          3.16        0.50
```

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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY              MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   547 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    548    TAPERED       PASS     BS-4.7   (C)       0.229       220
                       62.99 C          5.44           6.42        0.50
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   548 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    549    TAPERED       PASS     BS-4.7   (C)       0.242       219
                       66.37 C          5.77          -6.70        0.50
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   549 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    550    TAPERED       PASS     BS-4.7   (C)       0.285       207
                       86.23 C         -7.66           2.71        0.20
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   550 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    551    TAPERED       PASS     BS-4.7   (C)       0.089       208
                       99.55 C         -0.16           4.48        0.20
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY              MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   551 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    552    TAPERED       PASS     BS-4.7   (C)       0.089       208
                       99.52 C          0.16           4.50        0.20
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   552 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    553    TAPERED       PASS     BS-4.7   (C)       0.285       207
                       86.24 C          7.66           2.71        0.20
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   553 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    554    TAPERED       PASS     BS-4.7   (C)       0.282       207
                       83.98 C         -7.62          -2.59        0.20
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   554 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    555    TAPERED       PASS     BS-4.7   (C)       0.087       208
                       97.28 C         -0.14          -4.46        0.20
```

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY              MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   555 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    556    TAPERED       PASS     BS-4.7   (C)       0.087       208
                        97.28 C          0.14          -4.48        0.20
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   556 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    557    TAPERED       PASS     BS-4.7   (C)       0.281       207
                        83.98 C          7.59          -2.59        0.20
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   557 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    558    TAPERED       PASS     BS-4.7   (C)       0.085       203
                        36.68 C         -0.01         -11.12        0.44
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   558 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    559    TAPERED       PASS     BS-4.7   (C)       0.110       203
                        46.89 C          0.02         -14.36        0.44
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY              MZ       LOCATION
=======================================================================

 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   559 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    560    TAPERED       PASS     BS-4.7   (C)       0.110       203
                        46.85 C         -0.02         -14.36        0.44
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   560 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    561    TAPERED       PASS     BS-4.7   (C)       0.085       203
                        36.68 C          0.01         -11.12        0.44
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   561 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=    35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    562    TAPERED       PASS     BS-4.8.2.2         0.138       207
                        25.04 T          5.03         -12.26        0.66
 |-----------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   562 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------|
    564    TAPERED       PASS     BS-4.8.2.2         0.101       212
                        56.70 T          0.17         -23.46        0.00
```

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 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   564 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     566    TAPERED       PASS     BS-4.8.2.2         0.101       212
                        56.71 T         -0.17         -23.46        0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   566 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     567    TAPERED       PASS     BS-4.8.2.2         0.138       207
                        25.04 T         -5.03         -12.26        0.66
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   567 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     568    TAPERED       PASS     BS-4.7   (C)       0.085       215
                        36.82 C         -0.01         -11.13        0.44
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   568 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     569    TAPERED       PASS     BS-4.7   (C)       0.110       215
                        46.94 C          0.02         -14.35        0.44
```





```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   569 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     570    TAPERED       PASS     BS-4.7   (C)       0.110       215
                        46.93 C         -0.02         -14.35        0.44
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   570 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     571    TAPERED       PASS     BS-4.7   (C)       0.085       215
                        36.81 C          0.01         -11.13        0.44
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   571 UNIT - kN,m  SECTION CLASS  4  |
 | MCZ=  177.6   MCY=   35.1   PC=  1550.7   PT=  1691.9    MB=  177.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  352 kN : qw       = 207 N/mm2  |
 |  d   = 284 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     572    TAPERED       PASS     BS-4.8.2.2         0.136       207
                        25.32 T         -5.02         -11.78        0.66
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB   572 UNIT - kN,m  SECTION CLASS  3  |
 | MCZ=  324.8   MCY=   56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
 |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
 |-----------------------------------------------------------------------|
     573    TAPERED       PASS     BS-4.8.2.2         0.101       212
                        56.65 T         -0.16         -23.37        0.00
```

```
 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

  |-----------------------------------------------------------------------|
  | CALCULATED CAPACITIES FOR MEMB   573 UNIT - kN,m  SECTION CLASS  3  |
  | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
  | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
  |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
  |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
  |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
  |-----------------------------------------------------------------------|
     574    TAPERED       PASS     BS-4.8.2.2         0.101       212
                        56.65 T          0.16         -23.37        0.00
  |-----------------------------------------------------------------------|
  | CALCULATED CAPACITIES FOR MEMB   574 UNIT - kN,m  SECTION CLASS  3  |
  | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
  | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
  |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
  |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
  |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
  |-----------------------------------------------------------------------|
     575    TAPERED       PASS     BS-4.8.2.2         0.136       207
                        25.32 T          5.02         -11.78        0.66
  |-----------------------------------------------------------------------|
  | CALCULATED CAPACITIES FOR MEMB   575 UNIT - kN,m  SECTION CLASS  3  |
  | MCZ=  324.8   MCY=    56.7   PC=  1663.1   PT=  2166.6    MB=  324.8 |
  | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
  |  Shear Buckling check is required : Vb =  471 kN : qw       = 207 N/mm2  |
  |  d   = 380 mm  :   t  =   6 mm         : a  =    0 mm : pyf = 344 N/mm2  |
  |  BS-4.4.3.2     status = PASS         : BS-4.4.3.3 status = PASS        |
  |-----------------------------------------------------------------------|

   ************** END OF TABULATED RESULT OF DESIGN **************

   608. STEEL TAKE OFF ALL
```





```
 STEEL TAKE OFF

    STEEL TAKE-OFF
    --------------

       PROFILE                 LENGTH(METE)      WEIGHT(KN  )

    ISECT: TAPERED        1        21.60              9.822
    ISECT: TAPERED        2        87.01             42.619
    ISECT: TAPERED       95        21.22             10.148
    ISECT: TAPERED      201       164.60             60.086
    ISECT: TAPERED      301        86.40             49.574
    ST  UA70X70X6                 114.13              7.300
    ISECT: TAPERED      542         9.08              3.474
                                               ----------------
                                         TOTAL =        183.023

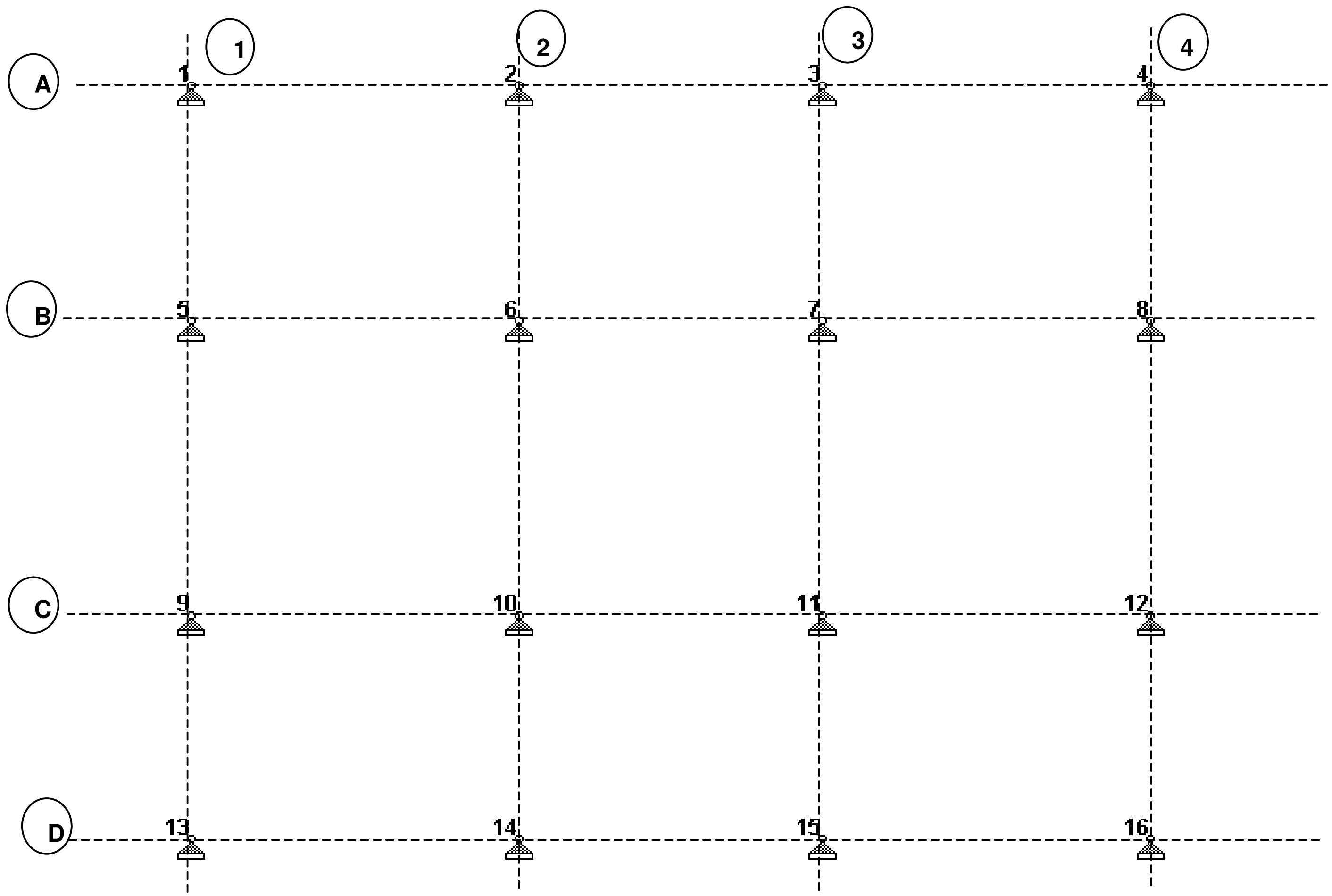
    ************ END OF DATA FROM INTERNAL STORAGE ************

    609. FINISH

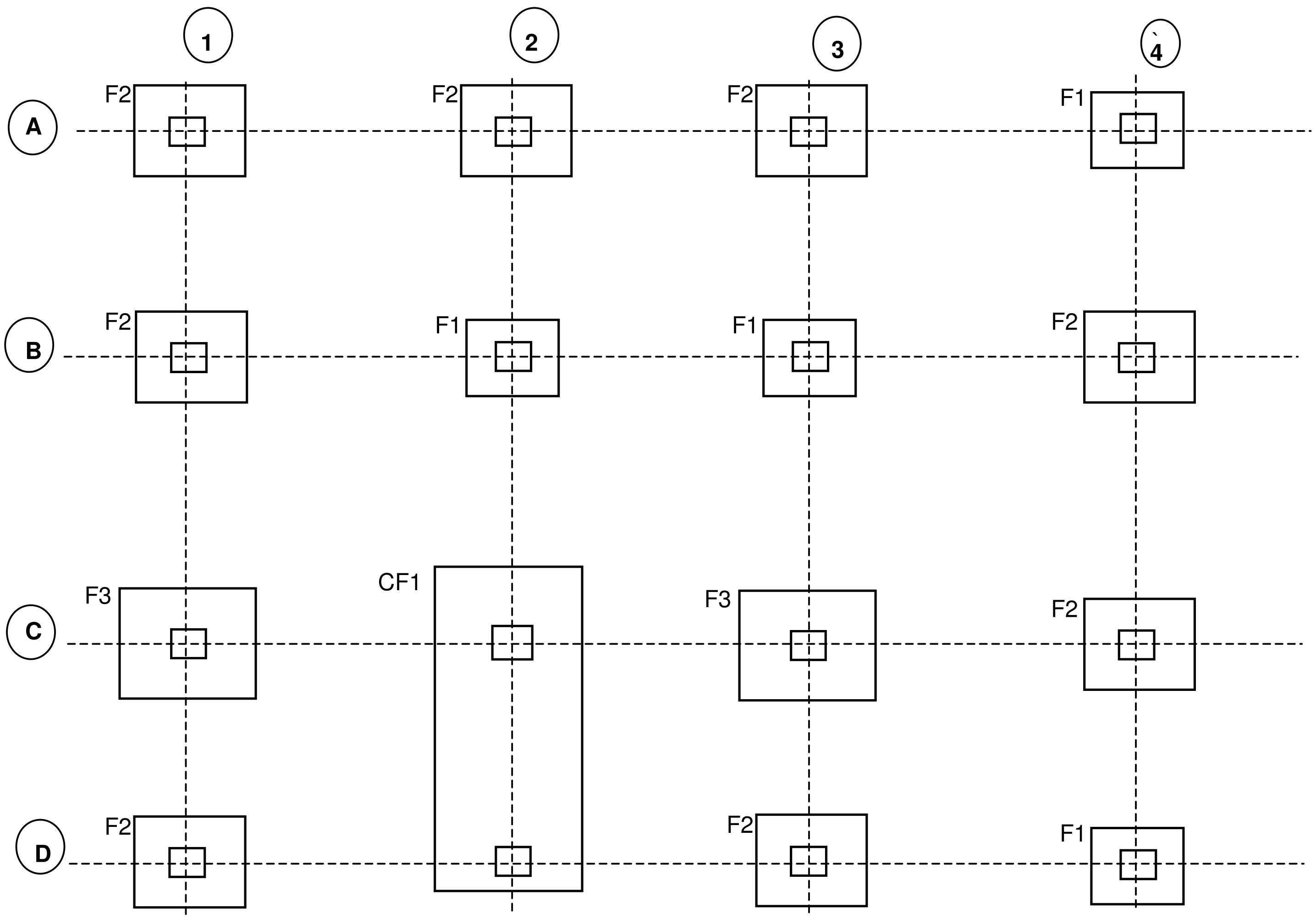
              *********** END OF THE STAAD.Pro RUN ***********

                **** DATE= DEC 25,2007   TIME= 15:43:40 ****
```

# 5 Foundation Design




Grid Marking With Node Details (STAAD) @ Foundation Level




**RC Foundation Layout Plan**

# Footing Design Summary

## (A) Isolated Footing

| Ref. node on STAAD | Location on drawing | Footing ID on Drg. | Length $l_x$ (M) | Width $l_y$ (M) | Depth of footing D (mm) | Bottom reinforcement ( Parallel to ) | | | | Top reinforcement ( Parallel to ) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Length (x dir.) | | Width (y dir.) | | Length (x dir.) | | Width (y dir.) | |
| | | | | | | Bar dia | Nos. | Bar dia | Nos. | Bar dia | Nos. | Bar dia | Nos. |
| | | | | | | Req./Pro. | Req./Pro. | Req./Pro. | Req./Pro. | Req./Pro. | Req./Pro. | Req./Pro. | Req./Pro. |
| 4 , 6 7 , 16 | A-4 , B-2 B-3 , D-4 | F1 | 2.8 | 2.8 | 600 | 12 / 12 | 19 / 21 | 12 / 12 | 19 / 21 | 0 / 10 | 0 / 19 | 0 / 10 | 0 / 19 |
| 1,2,3, 5,8,12 13,15 | A-1,A-2,A-3, B-1,B-4,C-4 D-1, D-3 | F2 | 3.5 | 3.5 | 600 | 16 / 16 | 20 / 24 | 16 / 16 | 20 / 24 | 0 / 10 | 0 / 24 | 0 / 10 | 0 / 24 |
| 9, 11 | C-1, C-3 | F3 | 4.1 | 4.1 | 600 | 16 / 16 | 33 / 34 | 16 / 16 | 33 / 34 | 0 / 10 | 0 / 28 | 0 / 10 | 0 / 28 |

## ( B) Combine Footing

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 10, 14 | C-2, D-2 | CF1 | 8.5 | 3.8 | 600 | 16 / 16 | 58 / 68 | 16 / 16 | 28 / 31 | 12 / 12 | 75 / 85 | 10 / 10 | 34/ 38 |

## Reinforced Concrete Footing Design Based On BS8110

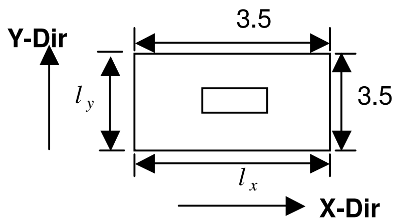
**Footing Location on Drg.** A-2 — **Ref. node on staad** 2

### Material properties

| | | | | |
|---|---|---|---|---|
| Characteristic strength of concrete $f_{cu}$ | 40 $N/mm^2$ | Net Bearing capacity of soil | 150 $kN/m^2$ |
| Characteristic strength of steel $f_y$ | 460 $N/mm^2$ | at | 2.7 m |
| Concrete density | 24 $kN/mm^3$ | | Below G.L |
| Steel density | 78 $kN/mm^3$ | | |

| | | | |
|---|---|---|---|
| Length of column | $c_x$ | 0.55 | m |
| Width of column | $c_y$ | 0.55 | m |
| Length of footing | $l_x$ | 3.5 | m |
| Width of footing | $l_y$ | 3.5 | m |
| Depth of footing | D | 600 | mm |
| Conc. Ten. Cover | $d_t$ | 75 | mm |
| Conc. Comp. Cover | $d_c$ | 75 | mm |
| Tension bar dia.(x- dir) | $\Phi_x$ | 16 | mm |
| Tension bar dia.(y- dir) | $\Phi_y$ | 16 | mm |
| Comp. bar dia.(x- dir) | $\Phi_{cx}$ | 10 | mm |
| Comp. bar dia.(y- dir) | $\Phi_{cy}$ | 10 | mm |
| Effect. Depth(x- dir) | $d_x$ | 517 | mm |
| Effect. Depth(y- dir) | $d_y$ | 501 | mm |
| Area of footing | $A_B$ | 12.25 | $m^2$ |

Self weight of footing 639.45 kN




Reactions

| L/C | Serviceability | | | L/C | Design | | |
|---|---|---|---|---|---|---|---|
| | Fy(kN) | P/A | p | | Fy(kN) | P/A | p |
| 101 | 1031.0 | 136.4 | 136.4 | 201 | 1489.1 | 199.9 | 199.9 |
| 102 | 917.0 | 127.1 | 127.1 | 202 | 1239.7 | 179.5 | 179.5 |
| 103 | 826.4 | 119.7 | 119.7 | 203 | 1126.9 | 170.3 | 170.3 |
| 104 | 804.9 | 117.9 | 117.9 | 204 | 805.7 | 144.1 | 144.1 |
| 105 | 803.3 | 117.8 | 117.8 | 205 | 1204.9 | 176.7 | 176.7 |
| 106 | 776.0 | 115.6 | 115.6 | 206 | 1086.4 | 167.0 | 167.0 |
| 107 | 843.6 | 121.1 | 121.1 | 207 | 765.2 | 140.8 | 140.8 |
| 108 | 826.4 | 119.7 | 119.7 | 208 | 1265.4 | 181.6 | 181.6 |
| 109 | 791.0 | 116.8 | 116.8 | 209 | 1156.9 | 172.7 | 172.7 |
| 110 | 760.6 | 114.3 | 114.3 | 210 | 835.7 | 146.5 | 146.5 |
| 111 | 824.0 | 119.5 | 119.5 | 211 | 1186.4 | 175.2 | 175.2 |
| 112 | 801.9 | 117.7 | 117.7 | 212 | 1064.8 | 165.2 | 165.2 |
| 113 | 794.4 | 117.1 | 117.1 | 213 | 743.6 | 139.0 | 139.0 |
| 114 | 764.9 | 114.6 | 114.6 | 214 | 1236.0 | 179.2 | 179.2 |
| 115 | 768.7 | 115.0 | 115.0 | 215 | 1122.6 | 169.9 | 169.9 |
| 116 | 852.5 | 121.8 | 121.8 | 216 | 801.5 | 143.7 | 143.7 |
| 117 | 753.4 | 113.7 | 113.7 | 217 | 1191.6 | 175.6 | 175.6 |
| 118 | 837.2 | 120.5 | 120.5 | 218 | 1070.9 | 165.7 | 165.7 |
| 119 | 668.2 | 106.8 | 106.8 | 219 | 749.7 | 139.5 | 139.5 |
| 120 | 663.5 | 106.4 | 106.4 | 220 | 1259.7 | 181.1 | 181.1 |

| L/C | Serviceability | | | L/C | Design | | |
|---|---|---|---|---|---|---|---|
| | Fy(kN) | P/A | p | | Fy(kN) | P/A | p |
| 121 | 942.3 | 129.1 | 129.1 | 221 | 1385.3 | 191.4 | 191.4 |
| 122 | 937.7 | 128.7 | 128.7 | 222 | 1236.3 | 179.2 | 179.2 |
| 123 | 688.4 | 108.4 | 108.4 | 223 | 1361.9 | 189.5 | 189.5 |
| 124 | 772.2 | 115.2 | 115.2 | 224 | 1104.9 | 168.5 | 168.5 |
| 125 | 673.1 | 107.2 | 107.2 | 225 | 1097.9 | 167.9 | 167.9 |
| 126 | 756.9 | 114.0 | 114.0 | 226 | 1523.7 | 202.7 | **202.7** |
| 127 | 587.9 | 100.2 | 100.2 | 227 | 1516.7 | 202.1 | 202.1 |
| 128 | 583.2 | 99.8 | 99.8 | 228 | 743.8 | 139.0 | 139.0 |
| 129 | 862.0 | 122.6 | 122.6 | 229 | 869.4 | 149.3 | 149.3 |
| 130 | 857.4 | 122.2 | 122.2 | 230 | 720.4 | 137.1 | 137.1 |
| 131 | 947.4 | 129.5 | 129.5 | 231 | 846.0 | 147.4 | 147.4 |
| 132 | 1012.1 | 134.8 | 134.8 | 232 | 589.0 | 126.4 | 126.4 |
| 133 | 935.9 | 128.6 | 128.6 | 233 | 582.0 | 125.8 | 125.8 |
| 134 | 1000.6 | 133.9 | 133.9 | 234 | 1007.8 | 160.6 | 160.6 |
| 135 | 873.0 | 123.5 | 123.5 | 235 | 1000.8 | 160.0 | 160.0 |
| 136 | 869.4 | 123.2 | 123.2 | 236 | 1030.9 | 162.5 | 162.5 |
| 137 | 1078.6 | 140.3 | **140.3** | | | | |
| 138 | 1075.0 | 140.0 | 140.0 | | | | |
| L/C - Load combination cases as per specification | | | | | P= Fy + self weight of footing | | |

$P_{max,net}$ 140.3 kN/m$^2$ (for SLS)
$P_{max,net}$ 202.7 kN/m$^2$ (for ULS)

Length of cantilever in x- Dir 1.48 m
Length of cantilever in y- Dir 1.48 m
Moment (x - Dir) 221.98 kN-m/m
Moment (y - Dir) 221.98 kN-m/m

About x- dir

K = $M/(b \times d^2 \times f_{cu})$ 0.021 < K' = 0.156
since K < K' compression reinforcement not required

z = d{0.5 + sqrt(0.25 - K/0.9)} ≤ 0.95d
= 0.976 d
Hence, z = 0.95 d

As = M/(0.95 x fy x z)
= 1034 mm$^2$/m

Min area of steel ( 0.13 % of bD ) = 780 mm$^2$/m
Area of bottom reinforcement req 1034 mm$^2$/m
Area of bottom reinforcement pro. 1340 mm$^2$/m No. of bar
T - 16 @ 145 c/c 24

Area of top reinforcement req = 0 mm$^2$/m
Area of top reinforcement pro. = 524 mm$^2$/m No. of bar
T - 10 @ 145 c/c 24

About y- dir

K  $M/(b \times d^2 \times f_{cu})$  0.022 < K' = 0.156

since K < K' compression reinforcement not required

z = d{0.5 + sqrt(0.25 - K/0.9)} ≤ 0.95d
= 0.975 d
Hence, z = 0.95 d

As = M/(0.95 x fy x z)
= 1067 $mm^2/m$

Min area of steel ( 0.13 % of bD ) = 780 $mm^2/m$
Area of bottom reinforcement req 1067 $mm^2/m$
Area of bottom reinforcement pro. 1340 $mm^2/m$ No. of bar
T - 16 @ 145 c/c 24

Area of top reinforcement req = 0 $mm^2/m$
Area of top reinforcement pro. = 524 $mm^2/m$ No. of bar
T - 10 @ 145 c/c 24

Check for shear

About x- dir
Shear force at face = 299.97 kN/m
Shear stress at face = 0.58 $(N/mm^2)$ O.K

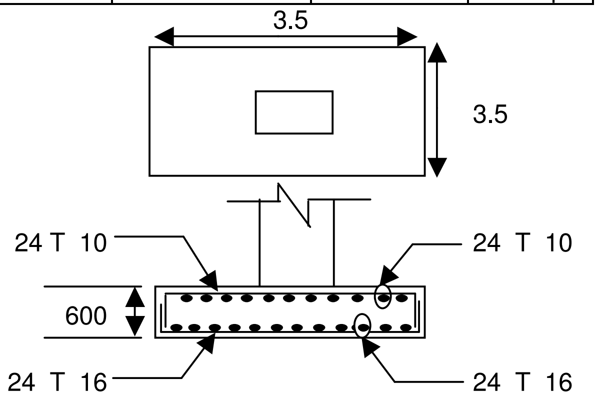
| Distance from support (mm) | av (mm) | V (kN) | v $(N/mm^2)$ | 100As/ (bv.d) | $v_c$ $(N/mm^2)$ | Enhanced $v_c(N/mm^2)$ | Remarks |
|---|---|---|---|---|---|---|---|
| d | 517 | 195.18 | 0.38 | 0.26 | 0.47 | 0.94 | Not reqd |
| 2d | 1034 | 90.40 | 0.17 | 0.26 | 0.47 | 0.47 | Not reqd |

About y- dir
Shear force at face = 299.97 kN/m
Shear stress at face = 0.6 $(N/mm^2)$ O.K

| Distance from support (mm) | av (mm) | V (kN) | v $(N/mm^2)$ | 100As/ (bv.d) | $v_c$ $(N/mm^2)$ | Enhanced $v_c(N/mm^2)$ | Remarks |
|---|---|---|---|---|---|---|---|
| d | 501 | 198.42 | 0.4 | 0.27 | 0.408 | 0.82 | Not reqd |
| 2d | 1002 | 96.88 | 0.19 | 0.27 | 0.408 | 0.41 | Not reqd |

Check for punching shear

P 1489.1 kN
p 121.56 $kN/m^2$
Punching surface perimeter 4600 mm
Net P 1339.95 kN
$\zeta_v$ 0.56 $N/mm^2$
$\zeta_{v,design}$ 5 $N/mm^2$
> ζv .: Safe



## Reinforced Concrete Footing Design Based On BS8110

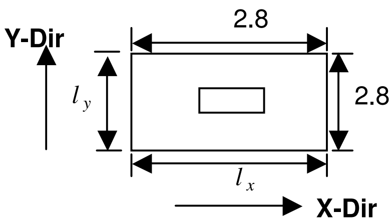
***Footing Location on Drg.*** B-2 **Ref. node on staad** 6

### Material properties

| | | | | | |
|---|---|---|---|---|---|
| Characteristic strength of concrete $f_{cu}$ | 40 | N/mm$^2$ | Net Bearing capacity of soil | 150 | kN/m$^2$ |
| Characteristic strength of steel $f_y$ | 460 | N/mm$^2$ | at | 2.7 | m |
| Concrete density | 24 | kN/mm$^3$ | | Below G.L | |
| Steel density | 78 | kN/mm$^3$ | | | |

| | | | |
|---|---|---|---|
| Length of column | $c_x$ | 0.55 | m |
| Width of column | $c_y$ | 0.55 | m |
| Length of footing | $l_x$ | 2.8 | m |
| Width of footing | $l_y$ | 2.8 | m |
| Depth of footing | D | 600 | mm |
| Conc. Ten. Cover | $d_t$ | 75 | mm |
| Conc. Comp. Cover | $d_c$ | 75 | mm |
| Tension bar dia.(x- dir) | $\Phi_x$ | 12 | mm |
| Tension bar dia.(y- dir) | $\Phi_y$ | 12 | mm |
| Comp. bar dia.(x- dir) | $\Phi_{cx}$ | 10 | mm |
| Comp. bar dia.(y- dir) | $\Phi_{cy}$ | 10 | mm |
| Effect. Depth(x- dir) | $d_x$ | 519 | mm |
| Effect. Depth(y- dir) | $d_y$ | 507 | mm |
| Area of footing | $A_B$ | 7.84 | m$^2$ |

Self weight of footing 409.25 kN




### Reactions

| L/C | Serviceability | | | L/C | Design | | |
|---|---|---|---|---|---|---|---|
| | Fy(kN) | P/A | p | | Fy(kN) | P/A | p |
| 101 | 719.3 | 144.0 | **144.0** | 201 | 1084.0 | 216.6 | **216.6** |
| 102 | 526.8 | 119.4 | 119.4 | 202 | 854.1 | 187.2 | 187.2 |
| 103 | 569.4 | 124.8 | 124.8 | 203 | 457.6 | 136.7 | 136.7 |
| 104 | 326.8 | 93.9 | 93.9 | 204 | 323.8 | 119.6 | 119.6 |
| 105 | 571.6 | 125.1 | 125.1 | 205 | 857.5 | 187.7 | 187.7 |
| 106 | 329.6 | 94.3 | 94.3 | 206 | 461.5 | 137.2 | 137.2 |
| 107 | 574.6 | 125.5 | 125.5 | 207 | 327.8 | 120.1 | 120.1 |
| 108 | 333.4 | 94.7 | 94.7 | 208 | 862.0 | 188.2 | 188.2 |
| 109 | 576.7 | 125.8 | 125.8 | 209 | 466.7 | 137.8 | 137.8 |
| 110 | 335.9 | 95.1 | 95.1 | 210 | 333.0 | 120.8 | 120.8 |
| 111 | 581.5 | 126.4 | 126.4 | 211 | 865.0 | 188.6 | 188.6 |
| 112 | 342.0 | 95.8 | 95.8 | 212 | 470.3 | 138.3 | 138.3 |
| 113 | 581.2 | 126.3 | 126.3 | 213 | 336.6 | 121.2 | 121.2 |
| 114 | 341.6 | 95.8 | 95.8 | 214 | 872.2 | 189.6 | 189.6 |
| 115 | 357.1 | 97.8 | 97.8 | 215 | 478.7 | 139.4 | 139.4 |
| 116 | 320.9 | 93.1 | 93.1 | 216 | 345.0 | 122.3 | 122.3 |
| 117 | 347.8 | 96.6 | 96.6 | 217 | 871.9 | 189.5 | 189.5 |
| 118 | 311.6 | 92.0 | 92.0 | 218 | 478.3 | 139.3 | 139.3 |
| 119 | 395.1 | 102.6 | 102.6 | 219 | 344.6 | 122.3 | 122.3 |
| 120 | 392.2 | 102.2 | 102.2 | 220 | 899.1 | 193.0 | 193.0 |

| L/C | Serviceability | | | L/C | Design | | |
|---|---|---|---|---|---|---|---|
| | Fy(kN) | P/A | p | | Fy(kN) | P/A | p |
| 121 | 276.5 | 87.5 | 87.5 | 221 | 844.7 | 186.1 | 186.1 |
| 122 | 273.7 | 87.1 | 87.1 | 222 | 884.8 | 191.2 | 191.2 |
| 123 | 323.7 | 93.5 | 93.5 | 223 | 830.5 | 184.2 | 184.2 |
| 124 | 287.5 | 88.9 | 88.9 | 224 | 957.5 | 200.4 | 200.4 |
| 125 | 314.4 | 92.3 | 92.3 | 225 | 953.2 | 199.9 | 199.9 |
| 126 | 278.2 | 87.7 | 87.7 | 226 | 776.3 | 177.3 | 177.3 |
| 127 | 361.6 | 98.3 | 98.3 | 227 | 772.1 | 176.8 | 176.8 |
| 128 | 358.8 | 98.0 | 98.0 | 228 | 365.3 | 124.9 | 124.9 |
| 129 | 243.1 | 83.2 | 83.2 | 229 | 311.0 | 118.0 | 118.0 |
| 130 | 240.2 | 82.8 | 82.8 | 230 | 351.1 | 123.1 | 123.1 |
| 131 | 640.5 | 133.9 | 133.9 | 231 | 296.7 | 116.2 | 116.2 |
| 132 | 612.6 | 130.3 | 130.3 | 232 | 423.7 | 132.4 | 132.4 |
| 133 | 633.6 | 133.0 | 133.0 | 233 | 419.5 | 131.8 | 131.8 |
| 134 | 605.6 | 129.4 | 129.4 | 234 | 242.6 | 109.2 | 109.2 |
| 135 | 668.6 | 137.5 | 137.5 | 235 | 238.3 | 108.7 | 108.7 |
| 136 | 666.4 | 137.2 | 137.2 | 236 | 718.8 | 170.0 | 170.0 |
| 137 | 579.7 | 126.1 | 126.1 | | | | |
| 138 | 577.5 | 125.9 | 125.9 | | | | |
| L/C - Load combination cases as per specification | | | | | P= Fy + self weight of footing | | |

$P_{max,net}$ 144.0 kN/m$^2$ (for SLS)
$P_{max,net}$ 216.6 kN/m$^2$ (for ULS)

Length of cantilever in x- Dir 1.13 m
Length of cantilever in y- Dir 1.13 m
Moment (x - Dir) 138.26 kN-m/m
Moment (y - Dir) 138.26 kN-m/m

About x- dir

K = M/(b x d$^2$ x $f_{cu}$) 0.013 < K' = 0.156
since K < K' compression reinforcement not required

z = d{0.5 + sqrt(0.25 - K/0.9)} ≤ 0.95d
= 0.985 d
Hence, z = 0.95 d

As = M/(0.95 x fy x z)
= 642 mm$^2$/m

Min area of steel ( 0.13 % of bD ) = 780 mm$^2$/m
Area of bottom reinforcement req 780 mm$^2$/m
Area of bottom reinforcement pro. 838 mm$^2$/m No. of bar
T - 12 @ 132 c/c 21

Area of top reinforcement req = 0 mm$^2$/m
Area of top reinforcement pro. = 524 mm$^2$/m No. of bar
T - 10 @ 147 c/c 19

About y- dir

K $M/(b \times d^2 \times f_{cu})$ 0.013 < K' = 0.156

since K < K' compression reinforcement not required

z = d{0.5 + sqrt(0.25 - K/0.9)} ≤ 0.95d
= 0.985 d
Hence, z = 0.95 d

As = M/(0.95 x fy x z)
= 657 $mm^2/m$

Min area of steel ( 0.13 % of bD ) = 780 $mm^2/m$
Area of bottom reinforcement req 780 $mm^2/m$
Area of bottom reinforcement pro. 838 $mm^2/m$ No. of bar
T - 12 @ 132 c/c 21

Area of top reinforcement req = 0 $mm^2/m$
Area of top reinforcement pro. = 524 $mm^2/m$ No. of bar
T - 10 @ 147 c/c 19

Check for shear

About x- dir
Shear force at face = 244.71 kN/m
Shear stress at face = 0.47 $(N/mm^2)$ O.K

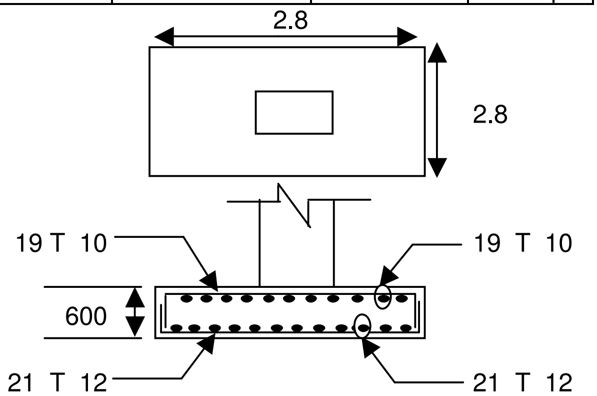
| Distance from support (mm) | av (mm) | V (kN) | v $(N/mm^2)$ | 100As/ (bv.d) | $v_c$ $(N/mm^2)$ | Enhanced $v_c(N/mm^2)$ | Remarks |
|---|---|---|---|---|---|---|---|
| d | 519 | 132.32 | 0.25 | 0.16 | 0.4 | 0.8 | Not reqd |
| 2d | 1038 | 19.92 | 0.04 | 0.16 | 0.4 | 0.4 | Not reqd |

About y- dir
Shear force at face = 244.71 kN/m
Shear stress at face = 0.48 $(N/mm^2)$ O.K

| Distance from support (mm) | av (mm) | V (kN) | v $(N/mm^2)$ | 100As/ (bv.d) | $v_c$ $(N/mm^2)$ | Enhanced $v_c(N/mm^2)$ | Remarks |
|---|---|---|---|---|---|---|---|
| d | 507 | 134.92 | 0.27 | 0.17 | 0.352 | 0.7 | Not reqd |
| 2d | 1014 | 25.12 | 0.05 | 0.17 | 0.352 | 0.35 | Not reqd |

Check for punching shear

P 1084.0 kN
p 138.26 $kN/m^2$
Punching surface perimeter 4600 mm
Net P 913.99 kN
$\zeta_v$ 0.38 $N/mm^2$
$\zeta_{v,design}$ 5 $N/mm^2$
$> \zeta v$ .: Safe



## Reinforced Concrete Footing Design Based On BS8110

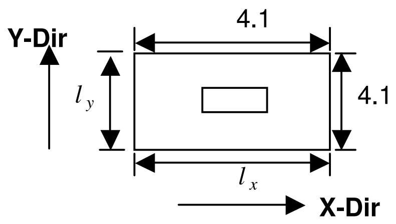
***Footing Location on Drg.*** <u>*C-3*</u> **Ref. node on staad** <u>**11**</u>

### Material properties

| | | | | |
|---|---|---|---|---|
| Characteristic strength of concrete $f_{cu}$ | 40 $N/mm^2$ | Net Bearing capacity of soil | | 150 $kN/m^2$ |
| Characteristic strength of steel $f_y$ | 460 $N/mm^2$ | | at | 2.7 m |
| Concrete density | 24 $kN/mm^3$ | | | Below G.L |
| Steel density | 78 $kN/mm^3$ | | | |

| | | | |
|---|---|---|---|
| Length of column | $c_x$ | 0.55 | m |
| Width of column | $c_y$ | 0.55 | m |
| Length of footing | $l_x$ | 4.1 | m |
| Width of footing | $l_y$ | 4.1 | m |
| Depth of footing | D | 600 | mm |
| Conc. Ten. Cover | $d_t$ | 75 | mm |
| Conc. Comp. Cover | $d_c$ | 75 | mm |
| Tension bar dia.(x- dir) | $\Phi_x$ | 16 | mm |
| Tension bar dia.(y- dir) | $\Phi_y$ | 16 | mm |
| Comp. bar dia.(x- dir) | $\Phi_{cx}$ | 10 | mm |
| Comp. bar dia.(y- dir) | $\Phi_{cy}$ | 10 | mm |
| Effect. Depth(x- dir) | $d_x$ | 517 | mm |
| Effect. Depth(y- dir) | $d_y$ | 501 | mm |
| Area of footing | $A_B$ | 16.81 | $m^2$ |

Self weight of footing 877.48 kN




Reactions

| L/C | Serviceability | | | L/C | Design | | |
|---|---|---|---|---|---|---|---|
| | Fy(kN) | P/A | p | | Fy(kN) | P/A | p |
| 101 | 1624.3 | 148.8 | **148.8** | 201 | 2361.7 | 218.8 | **218.8** |
| 102 | 1405.3 | 135.8 | 135.8 | 202 | 1993.9 | 196.9 | 196.9 |
| 103 | 1329.2 | 131.3 | 131.3 | 203 | 1713.0 | 180.2 | 180.2 |
| 104 | 1223.5 | 125.0 | 125.0 | 204 | 1238.4 | 152.0 | 152.0 |
| 105 | 1287.2 | 128.8 | 128.8 | 205 | 1930.8 | 193.2 | 193.2 |
| 106 | 1171.0 | 121.9 | 121.9 | 206 | 1639.4 | 175.8 | 175.8 |
| 107 | 1325.4 | 131.0 | 131.0 | 207 | 1164.9 | 147.6 | 147.6 |
| 108 | 1218.7 | 124.7 | 124.7 | 208 | 1988.1 | 196.6 | 196.6 |
| 109 | 1254.5 | 126.8 | 126.8 | 209 | 1706.2 | 179.8 | 179.8 |
| 110 | 1130.1 | 119.4 | 119.4 | 210 | 1231.7 | 151.6 | 151.6 |
| 111 | 1270.0 | 127.8 | 127.8 | 211 | 1881.8 | 190.2 | 190.2 |
| 112 | 1149.5 | 120.6 | 120.6 | 212 | 1582.2 | 172.4 | 172.4 |
| 113 | 1241.6 | 126.1 | 126.1 | 213 | 1107.7 | 144.2 | 144.2 |
| 114 | 1114.0 | 118.5 | 118.5 | 214 | 1905.1 | 191.6 | 191.6 |
| 115 | 1105.3 | 118.0 | 118.0 | 215 | 1609.4 | 174.0 | 174.0 |
| 116 | 1175.6 | 122.1 | 122.1 | 216 | 1134.8 | 145.8 | 145.8 |
| 117 | 1197.1 | 123.4 | 123.4 | 217 | 1862.4 | 189.1 | 189.1 |
| 118 | 1267.4 | 127.6 | 127.6 | 218 | 1559.6 | 171.1 | 171.1 |
| 119 | 1057.3 | 115.1 | 115.1 | 219 | 1085.0 | 142.9 | 142.9 |
| 120 | 1085.3 | 116.8 | 116.8 | 220 | 1924.9 | 192.8 | 192.8 |

| L/C | Serviceability | | | L/C | Design | | |
|---|---|---|---|---|---|---|---|
| | Fy(kN) | P/A | p | | Fy(kN) | P/A | p |
| 121 | 1287.3 | 128.8 | 128.8 | 221 | 2030.4 | 199.1 | 199.1 |
| 122 | 1315.4 | 130.5 | 130.5 | 222 | 2065.2 | 201.2 | 201.2 |
| 123 | 986.7 | 110.9 | 110.9 | 223 | 2170.6 | 207.4 | 207.4 |
| 124 | 1056.9 | 115.1 | 115.1 | 224 | 1851.0 | 188.4 | 188.4 |
| 125 | 1078.5 | 116.4 | 116.4 | 225 | 1893.1 | 190.9 | 190.9 |
| 126 | 1148.8 | 120.5 | 120.5 | 226 | 2202.5 | 209.3 | 209.3 |
| 127 | 938.7 | 108.0 | 108.0 | 227 | 2244.5 | 211.8 | 211.8 |
| 128 | 966.7 | 109.7 | 109.7 | 228 | 1051.6 | 140.9 | 140.9 |
| 129 | 1168.7 | 121.7 | 121.7 | 229 | 1157.1 | 147.1 | 147.1 |
| 130 | 1196.8 | 123.4 | 123.4 | 230 | 1191.9 | 149.2 | 149.2 |
| 131 | 1453.3 | 138.7 | 138.7 | 231 | 1297.3 | 155.5 | 155.5 |
| 132 | 1507.6 | 141.9 | 141.9 | 232 | 977.7 | 136.5 | 136.5 |
| 133 | 1522.1 | 142.8 | 142.8 | 233 | 1019.8 | 139.0 | 139.0 |
| 134 | 1576.4 | 146.0 | 146.0 | 234 | 1329.2 | 157.4 | 157.4 |
| 135 | 1417.7 | 136.5 | 136.5 | 235 | 1371.2 | 159.9 | 159.9 |
| 136 | 1439.4 | 137.8 | 137.8 | 236 | 1617.2 | 174.5 | 174.5 |
| 137 | 1590.3 | 146.8 | 146.8 | | | | |
| 138 | 1611.9 | 148.1 | 148.1 | | | | |
| L/C - Load combination cases as per specification | | | | | P= Fy + self weight of footing | | |

$P_{max,net}$ 148.8 kN/m$^2$ (for SLS)
$P_{max,net}$ 218.8 kN/m$^2$ (for ULS)

Length of cantilever in x- Dir 1.78 m
Length of cantilever in y- Dir 1.78 m
Moment (x - Dir) 346.61 kN-m/m
Moment (y - Dir) 346.61 kN-m/m

About x- dir

K = M/(b x d$^2$ x $f_{cu}$) 0.032 < K' = 0.156

since K < K' compression reinforcement not required

z = d{0.5 + sqrt(0.25 - K/0.9)} ≤ 0.95d
= 0.963 d
Hence, z = 0.95 d

As = M/(0.95 x fy x z)
= 1615 mm$^2$/m

Min area of steel ( 0.13 % of bD ) = 780 mm$^2$/m
Area of bottom reinforcement req 1615 mm$^2$/m
Area of bottom reinforcement pro. 1676 mm$^2$/m — No. of bar
T - 16 @ 119 c/c — 34

Area of top reinforcement req = 0 mm$^2$/m
Area of top reinforcement pro. = 524 mm$^2$/m — No. of bar
T - 10 @ 146 c/c — 28

About y- dir

K $M/(b \times d^2 \times f_{cu})$ 0.035 < K' = 0.156

since K < K' compression reinforcement not required

z = d{0.5 + sqrt(0.25 - K/0.9)} ≤ 0.95d
= 0.959 d
Hence, z = 0.95 d

As = M/(0.95 x fy x z)
= 1666 $mm^2/m$

Min area of steel ( 0.13 % of bD ) = 780 $mm^2/m$
Area of bottom reinforcement req 1666 $mm^2/m$
Area of bottom reinforcement pro. 1676 $mm^2/m$ No. of bar
T - 16 @ 119 c/c 34

Area of top reinforcement req = 0 $mm^2/m$
Area of top reinforcement pro. = 524 $mm^2/m$ No. of bar
T - 10 @ 146 c/c 28

Check for shear

About x- dir
Shear force at face = 389.45 kN/m
Shear stress at face = 0.75 ($N/mm^2$) O.K

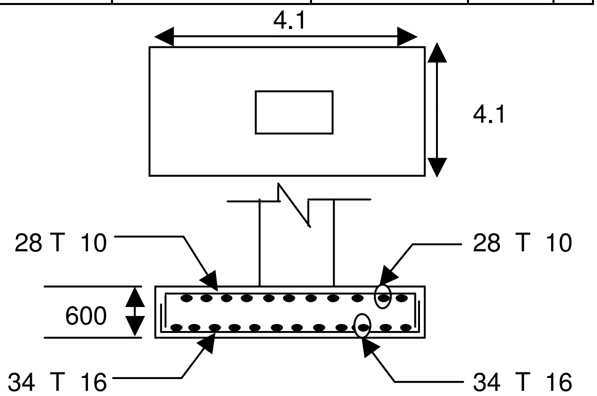
| Distance from support (mm) | av (mm) | V (kN) | v ($N/mm^2$) | 100As/ (bv.d) | $v_c$ ($N/mm^2$) | Enhanced $v_c$($N/mm^2$) | Remarks |
|---|---|---|---|---|---|---|---|
| d | 517 | 276.33 | 0.53 | 0.32 | 0.5 | 1 | Not reqd |
| 2d | 1034 | 163.22 | 0.32 | 0.32 | 0.5 | 0.5 | Not reqd |

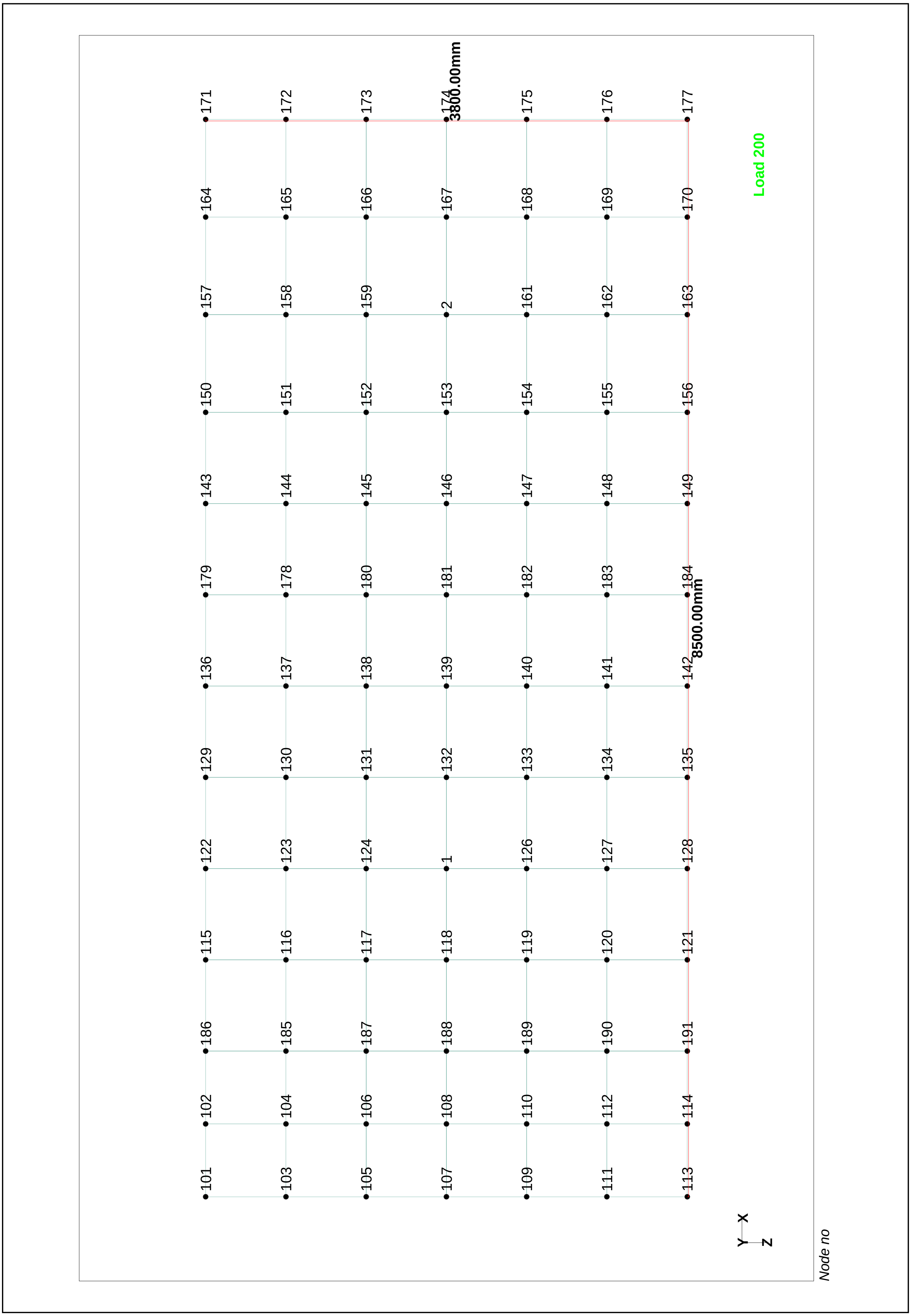
About y- dir
Shear force at face = 389.45 kN/m
Shear stress at face = 0.78 ($N/mm^2$) O.K

| Distance from support (mm) | av (mm) | V (kN) | v ($N/mm^2$) | 100As/ (bv.d) | $v_c$ ($N/mm^2$) | Enhanced $v_c$($N/mm^2$) | Remarks |
|---|---|---|---|---|---|---|---|
| d | 501 | 279.83 | 0.56 | 0.33 | 0.432 | 0.86 | Not reqd |
| 2d | 1002 | 170.22 | 0.34 | 0.33 | 0.432 | 0.43 | Not reqd |

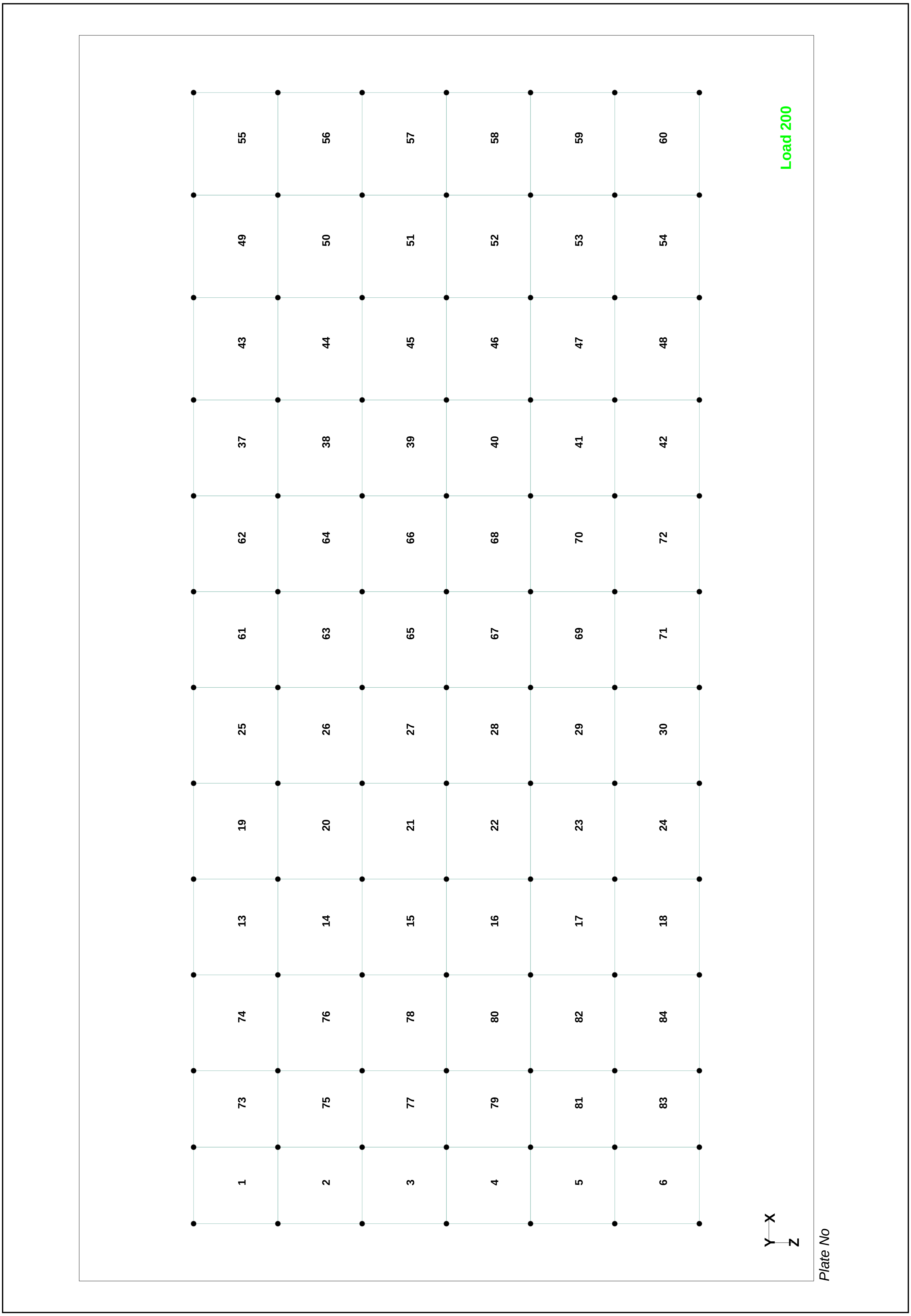
Check for punching shear

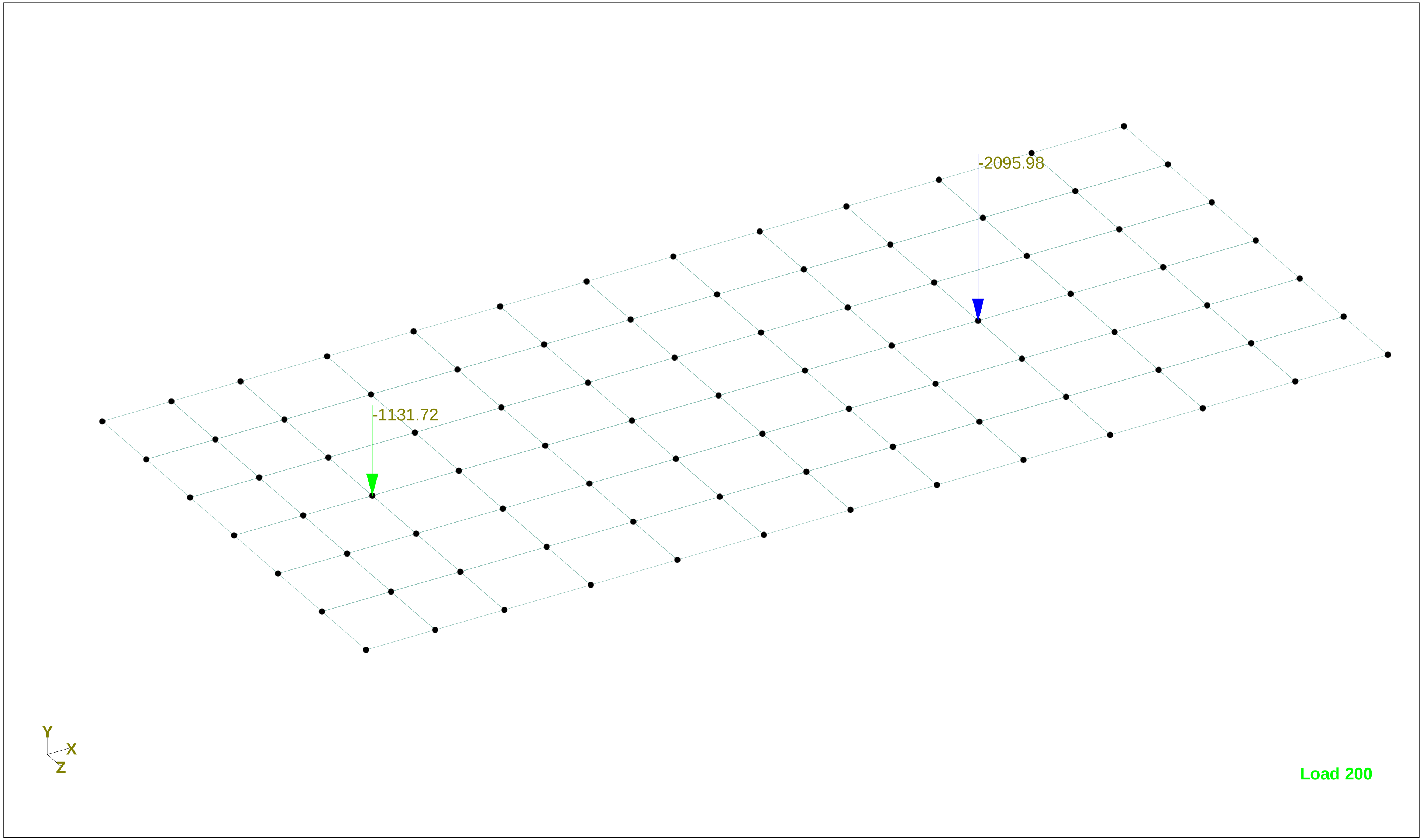
P 2361.7 kN
p 140.49 $kN/m^2$
Punching surface perimeter 4600 mm
Net P 2189.25 kN
$\zeta_v$ 0.92 $N/mm^2$
$\zeta_{v,design}$ 5 $N/mm^2$
> ζv .: Safe



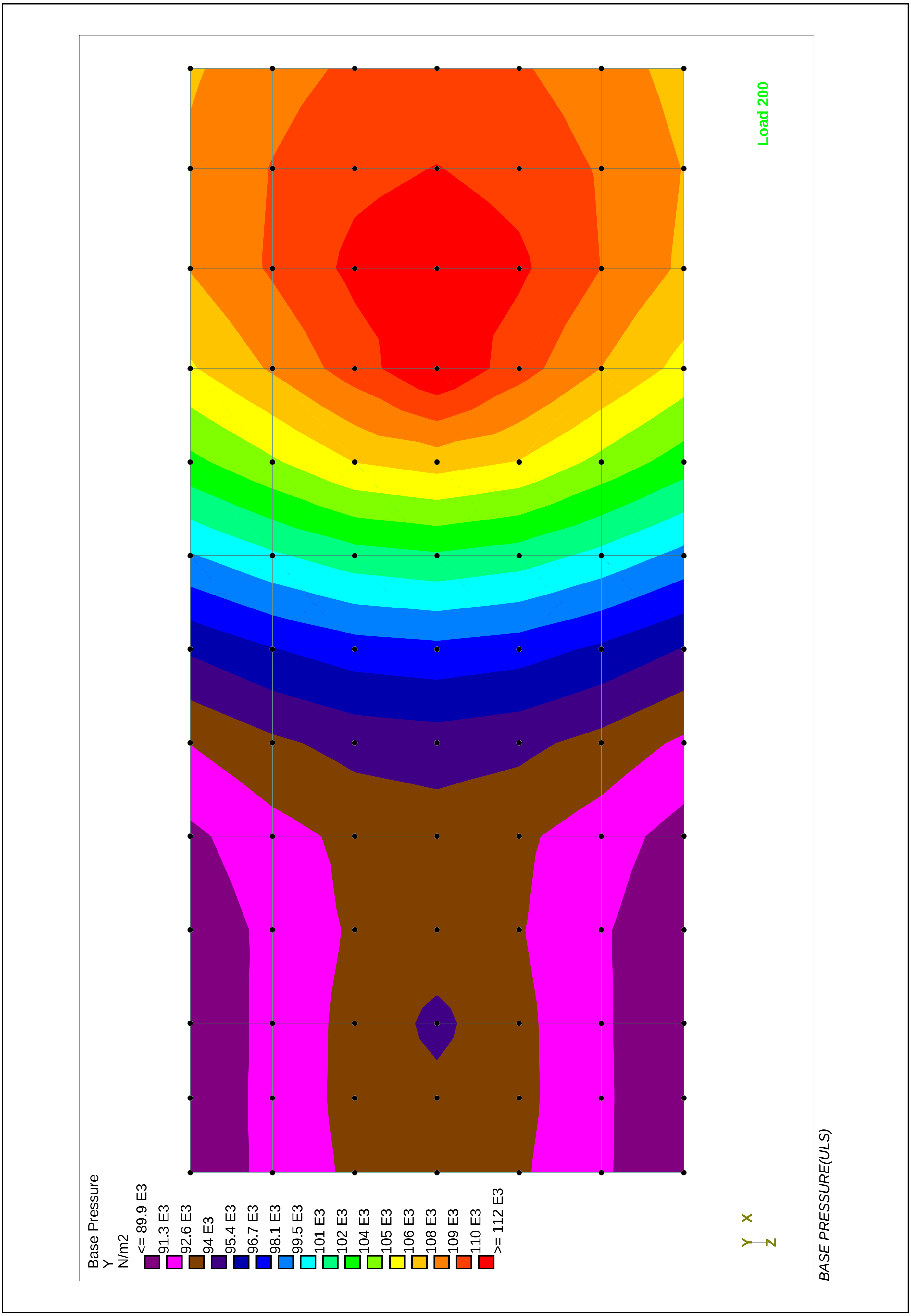


3800.00mm
8500.00mm
Load 200
171 164 157 150 143 179 136 129 122 115 186 102 101
172 165 158 151 144 178 137 130 123 116 185 104 103
173 166 159 152 145 180 138 131 124 117 187 106 105
174 167 2 153 146 181 139 132 1 118 188 108 107
175 168 161 154 147 182 140 133 126 119 189 110 109
176 169 162 155 148 183 141 134 127 120 190 112 111
177 170 163 156 149 184 142 135 128 121 191 114 113
X
Y
Z
Node no

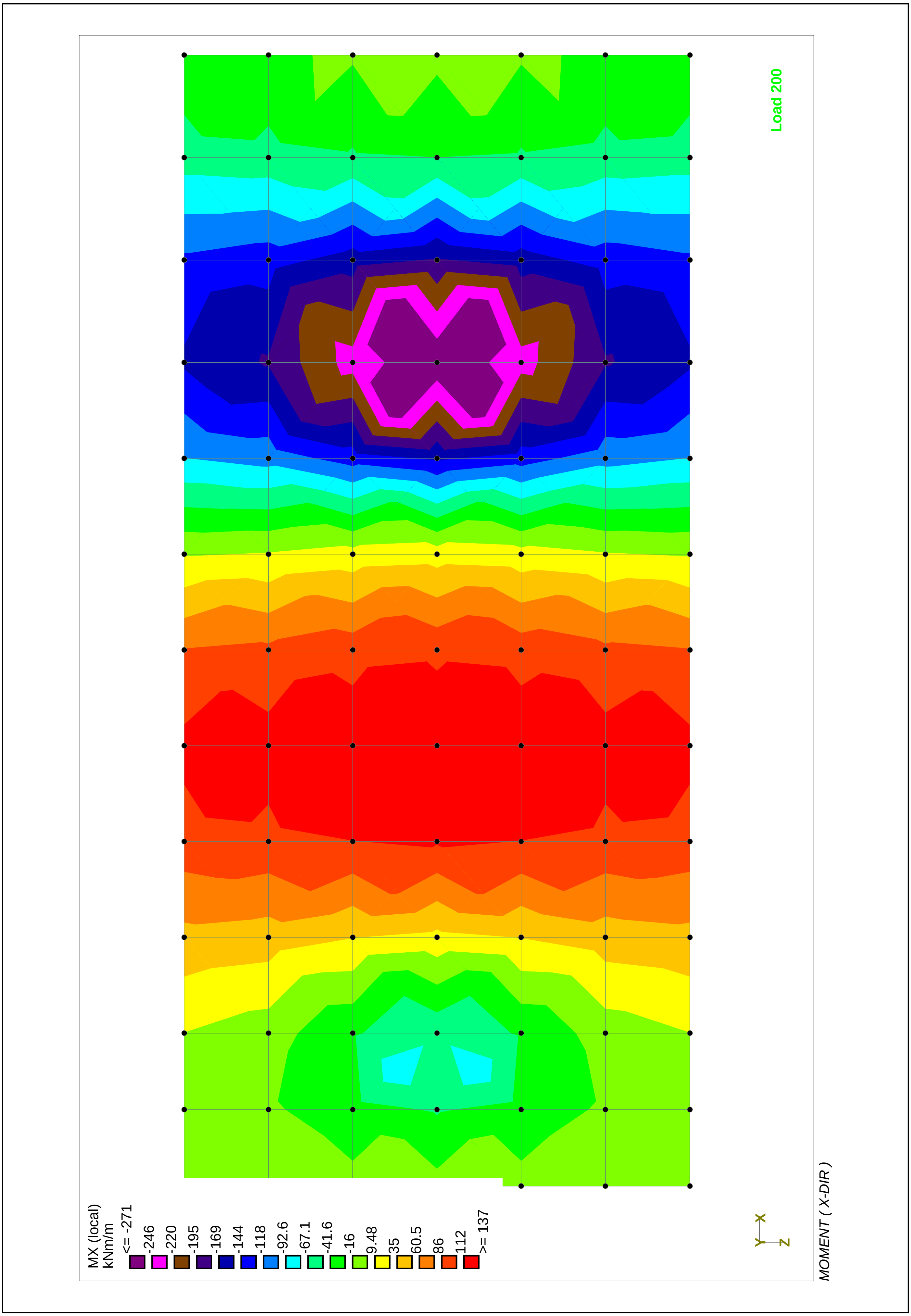

Load 200
Plate No
X
Y
Z

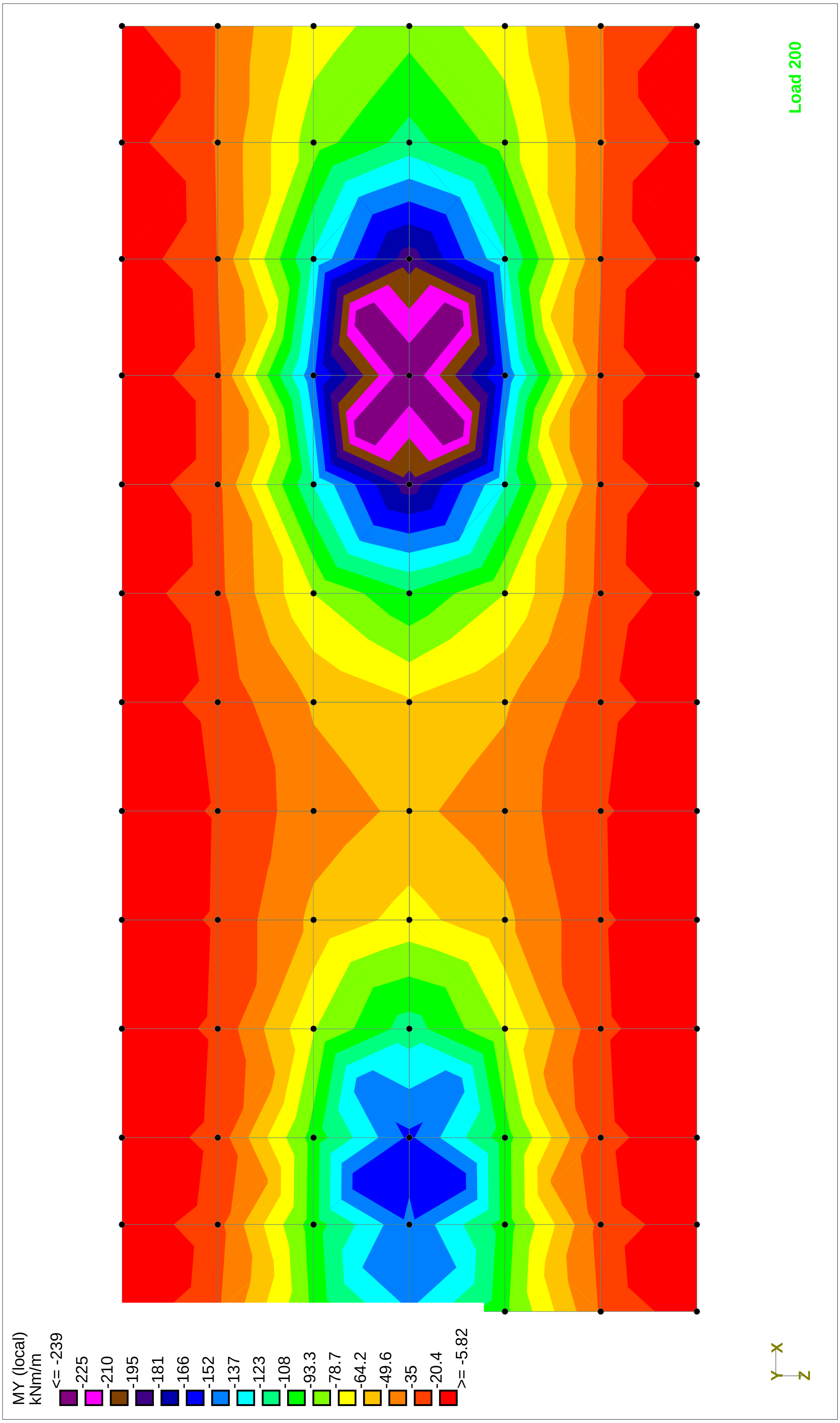



load case 200



Base Pressure
Y
N/m2
<= 89.9 E3
91.3 E3
92.6 E3
94 E3
95.4 E3
96.7 E3
98.1 E3
99.5 E3
101 E3
102 E3
104 E3
105 E3
106 E3
108 E3
109 E3
110 E3
>= 112 E3
Load 200
X
Y
Z
BASE PRESSURE(ULS)


MX (local)
kNm/m
<= -271
-246
-220
-195
-169
-144
-118
-92.6
-67.1
-41.6
-16
9.48
35
60.5
86
112
>= 137
Load 200
Y
X
Z
MOMENT ( X-DIR )


MY (local)
kNm/m
<= -239
-225
-210
-195
-181
-166
-152
-137
-123
-108
-93.3
-78.7
-64.2
-49.6
-35
-20.4
>= -5.82
Load 200
Y
X
Z
MOMENT ( Y-DIR )

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*           Proprietary Program of                 *
*           Research Engineers,  Intl.             *
*           Date=    DEC 29, 2007                  *
*           Time=    15: 8: 6                      *
*                                                  *
*      USER ID: roark                              *
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INPUT FILE: 10 & 14(cf1).STD
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     3. ENGINEER DATE 28-DEC-07
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. **REF NODE ON STAAD 10 & 14
     7. UNIT MMS NEWTON
     8. JOINT COORDINATES
     9. 1 2385 0 1599; 2 6755 0 1599; 101 -205 0 -300; 102 370 0 -300; 103 -205 0 333
    10. 104 370 0 333; 105 -205 0 966; 106 370 0 966; 107 -205 0 1599; 108 370 0 1599
    11. 109 -205 0 2232; 110 370 0 2232; 111 -205 0 2865; 112 370 0 2865
    12. 113 -205 0 3500; 114 370 0 3500; 115 1665 0 -300; 116 1665 0 333
    13. 117 1665 0 966; 118 1665 0 1599; 119 1665 0 2232; 120 1665 0 2865
    14. 121 1665 0 3500; 122 2385 0 -300; 123 2385 0 333; 124 2385 0 966
    15. 126 2385 0 2232; 127 2385 0 2865; 128 2385 0 3500; 129 3105 0 -300
    16. 130 3105 0 333; 131 3105 0 966; 132 3105 0 1599; 133 3105 0 2232
    17. 134 3105 0 2865; 135 3105 0 3500; 136 3825 0 -300; 137 3825 0 333
    18. 138 3825 0 966; 139 3825 0 1599; 140 3825 0 2232; 141 3825 0 2865
    19. 142 3825 0 3500; 143 5265 0 -300; 144 5265 0 333; 145 5265 0 966
    20. 146 5265 0 1599; 147 5265 0 2232; 148 5265 0 2865; 149 5265 0 3500
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    22. 154 5985 0 2232; 155 5985 0 2865; 156 5985 0 3500; 157 6755 0 -300
    23. 158 6755 0 333; 159 6755 0 966; 161 6755 0 2232; 162 6755 0 2865
    24. 163 6755 0 3500; 164 7525 0 -300; 165 7525 0 333; 166 7525 0 966
    25. 167 7525 0 1599; 168 7525 0 2232; 169 7525 0 2865; 170 7525 0 3500
    26. 171 8295 0 -300; 172 8295 0 333; 173 8295 0 966; 174 8295 0 1599
    27. 175 8295 0 2232; 176 8295 0 2865; 177 8295 0 3500; 178 4545 0 333
    28. 179 4545 0 -300; 180 4545 0 966; 181 4545 0 1599; 182 4545 0 2232
    29. 183 4545 0 2865; 184 4545 0 3500; 185 945 0 333; 186 945 0 -300; 187 945 0 966
    30. 188 945 0 1599; 189 945 0 2232; 190 945 0 2865; 191 945 0 3500
    31. ELEMENT INCIDENCES SHELL
    32. 1 103 104 102 101; 2 105 106 104 103; 3 107 108 106 105; 4 109 110 108 107
    33. 5 111 112 110 109; 6 113 114 112 111; 13 116 123 122 115; 14 117 124 123 116
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    48. 77 108 188 187 106; 78 188 118 117 187; 79 110 189 188 108; 80 189 119 118 188
    49. 81 112 190 189 110; 82 190 120 119 189; 83 114 191 190 112; 84 191 121 120 190
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    51. ELEMENT PROPERTY
    52. 1 TO 6 13 TO 30 37 TO 84 THICKNESS 0.6
    53. DEFINE MATERIAL START
    54. ISOTROPIC CONCRETE
    55. E 2.17185E+007
    56. POISSON 0.17
    57. DENSITY 24
    58. ALPHA 1E-005
    59. DAMP 0.05
    60. END DEFINE MATERIAL
    61. CONSTANTS
    62. MATERIAL CONCRETE ALL
    63. SUPPORTS
    64. 1 2 101 TO 124 126 TO 159 161 TO 191 ELASTIC MAT DIRECT Y SUBGRADE 7500
    65. **TRANSFER OF LOAD ON RAFT IS TAKEN FROM OUTPUT OF RCC STRUCTURE  NODE NO 10 &
    66. **
    67. LOAD 200 LC 200
    68. JOINT LOAD
    69. 153 FY -2095.98
    70. 188 FY -1131.72
    71. LOAD 201 LC 201
    72. JOINT LOAD
    73. 153 FY -1773.03
    74. 188 FY -914.86
    75. LOAD 202 LC 202
    76. JOINT LOAD
    77. 153 FY -1508.81
    78. 188 FY -880.24
    79. LOAD 203 LC 203
    80. JOINT LOAD
    81. 153 FY -1092.73
    82. 188 FY -617.98
    83. LOAD 204 LC 204
    84. JOINT LOAD
    85. 153 FY -1716.03
    86. 188 FY -895.14
    87. LOAD 205 LC 205
    88. JOINT LOAD
    89. 153 FY -1442.32
    90. 188 FY -857.23
    91. LOAD 206 LC 206
    92. JOINT LOAD
    93. 153 FY -1026.24
    94. 188 FY -594.97
    95. LOAD 207 LC 207
    96. JOINT LOAD
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 97. 153 FY -1762.57
 98. 188 FY -961.75
 99. LOAD 208 LC 208
100. JOINT LOAD
101. 153 FY -1496.62
102. 188 FY -934.94
103. LOAD 209 LC 209
104. JOINT LOAD
105. 153 FY -1080.53
106. 188 FY -672.68
107. LOAD 210 LC 210
108. JOINT LOAD
109. 153 FY -1668.49
110. 188 FY -918.89
111. LOAD 211 LC 211
112. JOINT LOAD
113. 153 FY -1386.85
114. 188 FY -884.94
115. LOAD 212 LC 212
116. JOINT LOAD
117. 153 FY -970.77
118. 188 FY -622.68
119. LOAD 213 LC 213
120. JOINT LOAD
121. 153 FY -1683.24
122. 188 FY -980.73
123. LOAD 214 LC 214
124. JOINT LOAD
125. 153 FY -1404.06
126. 188 FY -957.09
127. LOAD 215 LC 215
128. JOINT LOAD
129. 153 FY -987.98
130. 188 FY -694.83
131. LOAD 216 LC 216
132. JOINT LOAD
133. 153 FY -1646.76
134. 188 FY -957.03
135. LOAD 217 LC 217
136. JOINT LOAD
137. 153 FY -1361.51
138. 188 FY -929.43
139. LOAD 218 LC 218
140. JOINT LOAD
141. 153 FY -945.42
142. 188 FY -667.18
143. LOAD 219 LC 219
144. JOINT LOAD
145. 153 FY -1785.36
146. 188 FY -1094.3
147. LOAD 220 LC 220
148. JOINT LOAD
149. 153 FY -1889.76
150. 188 FY -922.09
151. LOAD 221 LC 221
152. JOINT LOAD
```



```
153. 153 FY -1735.96
154. 188 FY -1102.8
155. LOAD 222 LC 222
156. JOINT LOAD
157. 153 FY -1840.36
158. 188 FY -930.59
159. LOAD 223 LC 223
160. JOINT LOAD
161. 153 FY -1646.26
162. 188 FY -1298.18
163. LOAD 224 LC 224
164. JOINT LOAD
165. 153 FY -1631.44
166. 188 FY -1300.73
167. LOAD 225 LC 225
168. JOINT LOAD
169. 153 FY -1994.28
170. 188 FY -724.16
171. LOAD 226 LC 226
172. JOINT LOAD
173. 153 FY -1979.46
174. 188 FY -726.71
175. LOAD 227 LC 227
176. JOINT LOAD
177. 153 FY -1002.31
178. 188 FY -730.93
179. LOAD 228 LC 228
180. JOINT LOAD
181. 153 FY -1106.72
182. 188 FY -558.73
183. LOAD 229 LC 229
184. JOINT LOAD
185. 153 FY -952.91
186. 188 FY -739.43
187. LOAD 230 LC 230
188. JOINT LOAD
189. 153 FY -1057.32
190. 188 FY -567.23
191. LOAD 231 LC 231
192. JOINT LOAD
193. 153 FY -863.21
194. 188 FY -934.82
195. LOAD 232 LC 232
196. JOINT LOAD
197. 153 FY -848.39
198. 188 FY -937.37
199. LOAD 233 LC 233
200. JOINT LOAD
201. 153 FY -1211.23
202. 188 FY -360.79
203. LOAD 234 LC 234
204. JOINT LOAD
205. 153 FY -1196.41
206. 188 FY -363.34
207. LOAD 235 LC 235
208. JOINT LOAD
```

```
209. 153 FY -1441.29

210. 188 FY -788.46

211. LOAD 236 LC 236

212. JOINT LOAD

213. 153 FY -1425.2

214. 188 FY -816.36

215. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =    91/    72/    91
  ORIGINAL/FINAL BAND-WIDTH=    82/    13/     84 DOF
  TOTAL PRIMARY LOAD CASES =   37, TOTAL DEGREES OF FREEDOM =    546
  SIZE OF STIFFNESS MATRIX =       46 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     13.0/  22095.6 MB


216. LOAD LIST 200 TO 236

217. START CONCRETE DESIGN
```



```
CONCRETE DESIGN

  218. CODE BS8110

 PROGRAM CODE REVISION V1.5_8110_97/1

  219. UNIT MMS NEWTON

  220. FYSEC 460 ALL

  221. FC 40 ALL

  222. CLEAR 75 ALL

  223. DESIGN ELEMENT ALL
```

ELEMENT DESIGN SUMMARY-BASED ON 16mm BARS

MINIMUM AREAS ARE ACTUAL CODE MIN % REQUIREMENTS.
PRACTICAL LAYOUTS ARE AS FOLLOWS:
FY=460, 6No.16mm BARS AT 150mm C/C = 1206mm2/metre
FY=250, 4No.16mm BARS AT 250mm C/C = 804mm2/metre

| ELEMENT | LONG. REINF (mm2/m) | MOM-X /LOAD (kN-m/m) | TRANS. REINF (mm2/m) | MOM-Y /LOAD (kN-m/m) |
|---|---|---|---|---|
| 1 TOP : | 780. | 0.00 / 0 | 780. | 0.00 / 0 |
| BOTT: | 780. | -7.75 / 224 | 780. | -11.49 / 224 |
| 2 TOP : | 780. | 0.00 / 0 | 780. | 0.00 / 0 |
| BOTT: | 780. | -10.94 / 224 | 780. | -63.92 / 224 |
| 3 TOP : | 780. | 0.00 / 0 | 780. | 0.00 / 0 |
| BOTT: | 780. | -10.04 / 224 | 780. | -155.05 / 224 |
| 4 TOP : | 780. | 0.00 / 0 | 780. | 0.00 / 0 |
| BOTT: | 780. | -10.04 / 224 | 780. | -155.08 / 224 |
| 5 TOP : | 780. | 0.00 / 0 | 780. | 0.00 / 0 |
| BOTT: | 780. | -10.93 / 224 | 780. | -63.97 / 224 |
| 6 TOP : | 780. | 0.00 / 0 | 780. | 0.00 / 0 |
| BOTT: | 780. | -7.72 / 224 | 780. | -11.52 / 224 |
| 13 TOP : | 780. | 83.23 / 223 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -9.02 / 223 |
| 14 TOP : | 780. | 89.78 / 223 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -40.94 / 224 |
| 15 TOP : | 780. | 94.21 / 223 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -95.77 / 224 |
| 16 TOP : | 780. | 94.20 / 223 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -95.79 / 224 |
| 17 TOP : | 780. | 89.76 / 223 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -41.00 / 224 |





| | | | | |
|---|---|---|---|---|
| 18 TOP : | 780. | 83.20 / 223 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -9.05 / 223 |
| 19 TOP : | 780. | 131.43 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -9.96 / 223 |
| 20 TOP : | 780. | 138.64 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -34.95 / 223 |
| 21 TOP : | 780. | 147.17 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -54.90 / 223 |
| 22 TOP : | 780. | 147.15 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -54.92 / 223 |
| 23 TOP : | 780. | 138.59 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -35.01 / 223 |
| 24 TOP : | 780. | 131.33 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -10.00 / 223 |
| 25 TOP : | 780. | 133.71 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -11.11 / 200 |
| 26 TOP : | 780. | 141.19 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -32.97 / 200 |
| 27 TOP : | 780. | 148.36 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -51.69 / 200 |
| 28 TOP : | 780. | 148.34 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -51.71 / 200 |
| 29 TOP : | 780. | 141.13 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -33.04 / 200 |
| 30 TOP : | 780. | 133.59 / 224 | 780. | 0.00 / 0 |
| BOTT: | 780. | 0.00 / 0 | 780. | -11.15 / 200 |
| 37 TOP : | 780. | 0.00 / 0 | 780. | 0.00 / 0 |
| BOTT: | 780. | -156.27 / 225 | 780. | -5.82 / 200 |

```
 38 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    947.    -203.28 /  225      780.     -58.73 /  200

 39 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:   1284.    -275.60 /  225     1150.    -239.18 /  200

 40 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:   1284.    -275.58 /  225     1150.    -239.20 /  200

 41 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    947.    -203.21 /  225      780.     -58.77 /  200

 42 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.    -156.11 /  225      780.      -5.83 /  200

 43 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.    -163.03 /  225      780.      -6.70 /  200

 44 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    971.    -208.38 /  225      780.     -61.38 /  200

 45 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:   1271.    -272.72 /  225     1141.    -237.29 /  200

 46 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:   1271.    -272.70 /  225     1141.    -237.32 /  200

 47 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    971.    -208.30 /  225      780.     -61.43 /  200

 48 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.    -162.86 /  225      780.      -6.71 /  200

 49 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -91.36 /  225      780.     -12.41 /  200

 50 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -81.76 /  225      780.     -61.49 /  200

 51 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -76.73 /  225      780.    -140.45 /  200
```



```
 52 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -76.72 /  225      780.    -140.48 /  200

 53 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -81.72 /  225      780.     -61.56 /  200

 54 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -91.29 /  225      780.     -12.44 /  200

 55 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -22.20 /  225      780.     -15.88 /  200

 56 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -17.09 /  225      780.     -56.36 /  200

 57 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -11.12 /  225      780.     -87.41 /  200

 58 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -11.11 /  225      780.     -87.44 /  200

 59 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -17.07 /  225      780.     -56.44 /  200

 60 TOP :    780.       0.00 /    0      780.       0.00 /    0
    BOTT:    780.     -22.19 /  225      780.     -15.94 /  200

 61 TOP :    780.      88.67 /  224      780.       0.00 /    0
    BOTT:    780.       0.00 /    0      780.     -12.20 /  200

 62 TOP :    780.      24.05 /  232      780.       0.00 /    0
    BOTT:    780.     -73.45 /  225      780.     -10.44 /  200

 63 TOP :    780.      96.36 /  224      780.       0.00 /    0
    BOTT:    780.       0.00 /    0      780.     -46.35 /  200

 64 TOP :    780.      27.43 /  232      780.       0.00 /    0
    BOTT:    780.     -66.56 /  225      780.     -56.23 /  200

 65 TOP :    780.     105.59 /  224      780.       0.00 /    0
    BOTT:    780.       0.00 /    0      780.     -73.94 /  200
```

```
 66 TOP :      780.        29.64 /  232       780.          0.00 /    0
    BOTT:      780.       -62.44 /  225       780.       -139.25 /  200

 67 TOP :      780.       105.58 /  224       780.          0.00 /    0
    BOTT:      780.         0.00 /    0       780.        -73.96 /  200

 68 TOP :      780.        29.63 /  232       780.          0.00 /    0
    BOTT:      780.       -62.43 /  225       780.       -139.27 /  200

 69 TOP :      780.        96.31 /  224       780.          0.00 /    0
    BOTT:      780.         0.00 /    0       780.        -46.42 /  200

 70 TOP :      780.        27.42 /  232       780.          0.00 /    0
    BOTT:      780.       -66.54 /  225       780.        -56.29 /  200

 71 TOP :      780.        88.59 /  224       780.          0.00 /    0
    BOTT:      780.         0.00 /    0       780.        -12.24 /  200

 72 TOP :      780.        24.03 /  232       780.          0.00 /    0
    BOTT:      780.       -73.40 /  225       780.        -10.46 /  200

 73 TOP :      780.         1.04 /  233       780.          0.00 /    0
    BOTT:      780.        -7.14 /  224       780.         -8.23 /  224

 74 TOP :      780.        24.91 /  200       780.          0.00 /    0
    BOTT:      780.         0.00 /    0       780.         -7.07 /  224

 75 TOP :      780.         0.00 /    0       780.          0.00 /    0
    BOTT:      780.       -37.97 /  224       780.        -52.80 /  224

 76 TOP :      780.         6.00 /  225       780.          0.00 /    0
    BOTT:      780.        -8.53 /  232       780.        -47.45 /  224

 77 TOP :      780.         0.00 /    0       780.          0.00 /    0
    BOTT:      780.       -96.27 /  224       893.       -185.68 /  224

 78 TOP :      780.         0.00 /    0       780.          0.00 /    0
    BOTT:      780.       -49.24 /  224       784.       -163.09 /  224

 79 TOP :      780.         0.00 /    0       780.          0.00 /    0
    BOTT:      780.       -96.27 /  224       893.       -185.71 /  224
```





```
 80 TOP :      780.         0.00 /    0       780.          0.00 /    0

    BOTT:      780.       -49.24 /  224       784.       -163.11 /  224

 81 TOP :      780.         0.00 /    0       780.          0.00 /    0

    BOTT:      780.       -37.96 /  224       780.        -52.84 /  224

 82 TOP :      780.         5.97 /  225       780.          0.00 /    0

    BOTT:      780.        -8.52 /  232       780.        -47.50 /  224

 83 TOP :      780.         1.04 /  233       780.          0.00 /    0

    BOTT:      780.        -7.07 /  224       780.         -8.25 /  224

 84 TOP :      780.        24.93 /  200       780.          0.00 /    0

    BOTT:      780.         0.00 /    0       780.         -7.10 /  224

***************************END OF ELEMENT DESIGN**************************

224. END CONCRETE DESIGN

225. FINISH

          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= DEC 29,2007   TIME= 15: 8: 7 ****
```

# 6 Slab Design

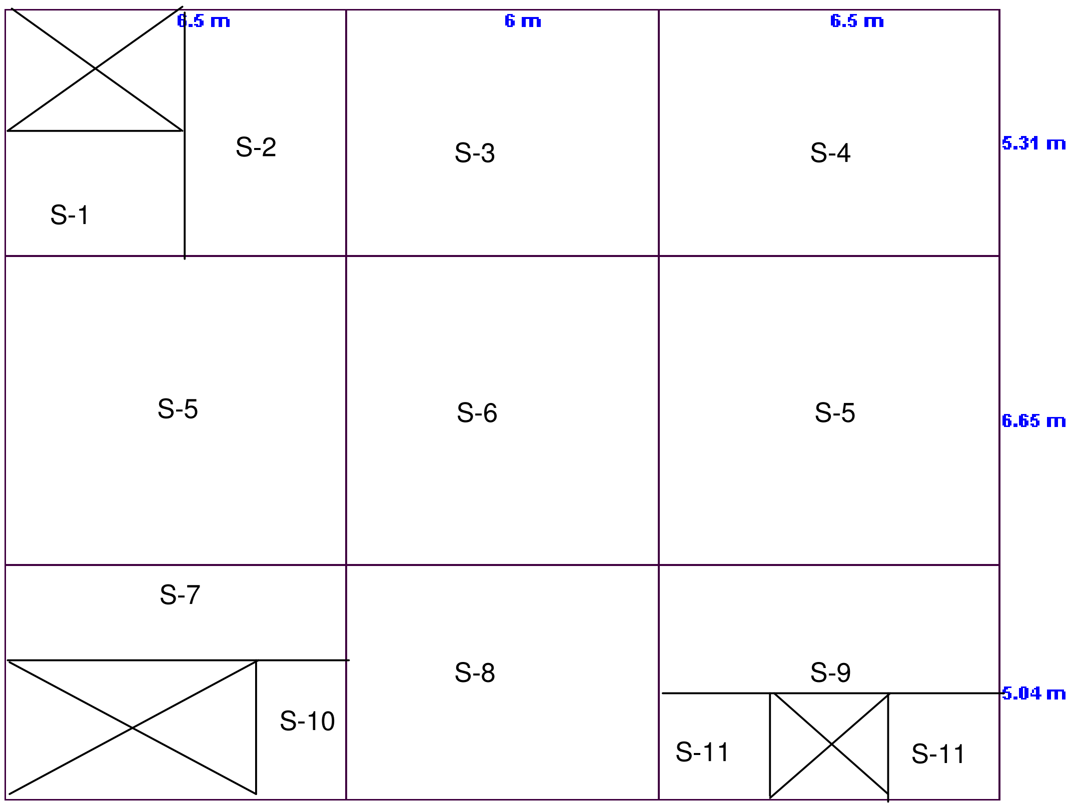
# Slab Design Summary

## (A) Ground Slab Level

| Level | Slab Id no | Length $l_x$ (M) | Width $l_y$ (M) | Thickness of Slab (mm) | Bottom reinforcement | | | | Top reinforcement | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Parallel to Length (x dir.) Bar dia + Spacing (mm)(c/c) | | Parallel to Width (y dir.) Bar dia + Spacing (mm)(c/c) | | Parallel to Length (x dir.) Bar dia + Spacing (mm)(c/c) | | Parallel to Width (y dir.) Bar dia + Spacing (mm)(c/c) | |
| | | | | | Req. | Provided | Req. | Provided | Req. | Provided | Req. | Provided |
| Level - 01 | S1 | 2.49 | 3.33 | 225 | T12 @ 1016 | T12 @ 200 | T12 @ 1120 | T12 @ 200 | T12 @ 761 | T12 @ 150 | T12 @ 1482 | T12 @ 150 |
| Level - 01 | S2 | 3.18 | 5.31 | 225 | T12 @ 523 | T12 @ 200 | T12 @ 908 | T12 @ 200 | T12 @ 392 | T12 @ 150 | T12 @ 686 | T12 @ 150 |
| Level - 01 | S3 | 5.31 | 6.00 | 225 | T12 @ 312 | T12 @ 200 | T12 @ 304 | T12 @ 200 | T12 @ 231 | T12 @ 150 | T12 @ 298 | T12 @ 150 |
| Level - 01 | S4 | 5.31 | 6.50 | 225 | T12 @246 | T12 @ 200 | T12 @ 348 | T12 @ 200 | T12 @ 183 | T12 @ 150 | T12 @ 245 | T12 @ 150 |
| Level - 01 | S5 | 6.50 | 6.65 | 225 | T12 @ 251 | T12 @ 200 | T12 @ 263 | T12 @ 200 | T12 @ 191 | T12 @ 150 | T12 @ 199 | T12 @ 150 |
| Level - 01 | S6 | 6.00 | 6.65 | 225 | T12 @ 327 | T12 @ 200 | T12 @ 360 | T12 @ 200 | T12 @ 247 | T12 @ 150 | T12 @ 270 | T12 @ 150 |
| Level - 01 | S7 | 1.72 | 6.50 | 225 | T12 @ 386 | T12 @ 200 | T12 @ 387 | T12 @ 200 | T12 @ 386 | T12 @ 150 | T12 @ 387 | T12 @ 150 |
| Level - 01 | S8 | 5.04 | 6.00 | 225 | T12 @ 367 | T12 @ 200 | T12 @ 438 | T12 @ 200 | T12 @ 275 | T12 @ 150 | T12 @ 331 | T12 @ 150 |
| Level - 01 | S9 | 3.22 | 6.50 | 225 | T12 @ 322 | T12 @ 200 | T12 @ 387 | T12 @ 200 | T12 @ 322 | T12 @ 150 | T12 @ 387 | T12 @ 150 |
| Level - 01 | S10 | 1.33 | 3.33 | 225 | T12 @ 386 | T12 @ 200 | T12 @ 387 | T12 @ 200 | T12 @ 386 | T12 @ 150 | T12 @ 387 | T12 @ 150 |
| Level - 01 | S11 | 1.83 | 2.94 | 225 | T12 @ 1618 | T12 @ 200 | T12 @ 2076 | T12 @ 200 | T12 @ 1219 | T12 @ 150 | T12 @ 2076 | T12 @ 150 |

## (B) Rc Roof Level

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Roof | RS-1 | 5.31 | 6.50 | 175 | T10 @ 345 | T10 @ 200 | T10 @ 345 | T10 @ 200 | T10 @ 263 | T10 @ 150 | T10 @ 345 | T10 @ 150 |
| Roof | RS-2 | 5.31 | 6.00 | 175 | T10 @ 345 | T10 @ 200 | T10 @ 345 | T10 @ 200 | T10 @ 331 | T10 @ 150 | T10 @ 345 | T10 @ 150 |
| Roof | RS-3 | 6.50 | 6.65 | 175 | T10 @ 345 | T10 @ 200 | T10 @ 345 | T10 @ 200 | T10 @ 273 | T10 @ 150 | T10 @ 283 | T10 @ 150 |
| Roof | RS-4 | 6.00 | 6.65 | 175 | T10 @ 345 | T10 @ 200 | T10 @ 345 | T10 @ 200 | T10 @ 345 | T10 @ 150 | T10 @ 345 | T10 @ 150 |
| Roof | RS-5 | 5.04 | 6.50 | 175 | T10 @ 345 | T10 @ 200 | T10 @ 345 | T10 @ 200 | T10 @ 275 | T10 @ 150 | T10 @ 345 | T10 @ 150 |
| Roof | RS-6 | 5.04 | 6.00 | 175 | T10 @ 345 | T10 @ 200 | T10 @ 345 | T10 @ 200 | T10 @ 340 | T10 @ 150 | T10 @ 345 | T10 @ 150 |




Ground Slab Layout Plan

## Design of Slab Panel - S1

***Material Properties***

| | | |
|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 N/mm$^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 N/mm$^2$ |
| Concrete Density, $\gamma_c$ | = | 24 kN/m$^3$ |

***Data***

| | | |
|---|---|---|
| Thickness of slab | = | 225 mm |
| Clear cover | = | 35 mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **2.485** |
| Long span of the slab, $l_y$ | **3.325** |
| Span ratio, $l_y/l_x$ | **1.34** |

**TWO WAY SLAB**

**Edge Condition** Two Adjacent Edges Discont.

***Loading***

| | | |
|---|---|---|
| Self weight of slab | = | 5.4 kN/m$^2$ |
| Floor finish | = | 1.2 kN/m$^2$ |
| Partitions | = | 0 kN/m$^2$ |
| M&E services | = | 0.5 kN/m$^2$ |
| Total Dead load, Gk | = | **7.1** kN/m$^2$ |
| Live load, Qk | = | **10** kN/m$^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **25.94** kN/m$^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, N/mm$^2$ | 60 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 55 |
| $D_{reqd}$, mm | 96 |
| $D_{prov.}$ mm | 225 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar | Required Reinforcement | | Provided Reinforcement | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\phi$ | As, mm$^2$ | Spacing | Spacing | As, mm$^2$ |
| **Short span at top** | 11.36 | 184 | 0.01 | 174.80 | **12** | 149 | **761** | **150** | **755** |
| **Short span at bottom** | 8.42 | 184 | 0.01 | 174.80 | **12** | 110 | **1026** | **200** | **566** |
| **Long span at top** | 7.21 | 172 | 0.01 | 163.40 | **12** | 101 | **1120** | **150** | **755** |
| **Long span at bottom** | 5.45 | 172 | 0.01 | 163.40 | **12** | 76 | **1482** | **200** | **566** |

**End Condition : Two adjecent edges discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.071 | 11.36 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.053 | 8.42 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.045 | 7.21 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.034 | 5.45 kNm |

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 174.80 mm |
| Required area of the steel, $A_s$ | = | 149 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 760 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 754 $mm^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ |
| Lever arm z | = | 174.8 mm |
| Required area of the steel, $A_s$ | = | 110 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 1027 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 565 $mm^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 101 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 1120 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **754** $mm^2$ |
| Provided % of steel | = | 0.44 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 76 $mm^2$ |
| Diameter of bar | = | **12** mm |

Required spacing of reinf. = **1482** mm
Provided spacing of reinf. = **200** mm
Provided area of the steel = **566** $mm^2$
Provided % of steel = 0.33 % **Provided area of steel is OK**
**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 2.485 m
Actul l/d ratio = 23
Provided % of steel at bottom along short span = 0.410 %
$f_s = (2*f_y$*Req. steel)/(3*Provided steel) = 60 $N/mm^2$
$0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ = 2.00
Required effective depth = 54 mm
Available effective depth = 184 mm **Safe in Deflection**

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| **Location** | **Coeff.** | **S.Force V (kN)** | ***v*** **$N/mm^2$** | **% of steel** | ***$v_c$*, $N/mm^2$** | **Enhanced *$v_c$*, $N/mm^2$** | **$A_{sv}/s_v$** | **Remarks** |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.508** | 32.72 | 0.18 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.334** | 21.52 | 0.12 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.400** | 25.78 | 0.15 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |

## Design of Slab Panel - S2

***Material Properties***

| | | | |
|---|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab | = | 225 | mm |
| Clear cover | = | 35 | mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **3.175** |
| Long span of the slab, $l_y$ | **5.31** |
| Span ratio, $l_y/l_x$ | **1.67** |

**TWO WAY SLAB**

**Edge Condition** Two Adjacent Edges Discont.

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 5.4 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **7.1** | $kN/m^2$ |
| Live load, Qk | = | **10** | $kN/m^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **25.94** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 117 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 70 |
| $D_{reqd}$, mm | 111 |
| $D_{prov.}$ mm | 225 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar $\phi$ | Required Reinforcement As, $mm^2$ | Required Reinforcement Spacing | Provided Reinforcement Spacing | Provided Reinforcement As, $mm^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 22.02 | 184 | 0.02 | 174.80 | **12** | 288 | **392** | **150** | **755** |
| **Short span at bottom** | 16.52 | 184 | 0.02 | 174.80 | **12** | 216 | **523** | **200** | **566** |
| **Long span at top** | 11.77 | 172 | 0.01 | 163.40 | **12** | 165 | **686** | **150** | **755** |
| **Long span at bottom** | 8.90 | 172 | 0.01 | 163.40 | **12** | 125 | **908** | **200** | **566** |

**End Condition : Two adjecent edges discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.084 | 22.02 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.063 | 16.52 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.045 | 11.77 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.034 | 8.90 kNm |

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 174.80 mm |
| Required area of the steel, $A_s$ | = | 288 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 392 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 754 $mm^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ |
| Lever arm z | = | 174.8 mm |
| Required area of the steel, $A_s$ | = | 216 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 523 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 565 $mm^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 165 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 686 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **754** $mm^2$ |
| Provided % of steel | = | 0.44 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 125 $mm^2$ |
| Diameter of bar | = | **12** mm |

| | | | |
|---|---|---|---|
| Required spacing of reinf. | = | **908** mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | **566** $mm^2$ | |
| Provided % of steel | = | 0.33 % | **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | |
|---|---|---|---|
| Length of shorter span | = | 3.175 m | |
| Actul l/d ratio | = | 23 | |
| Provided % of steel at bottom along short span | = | 0.410 % | |
| $f_s = (2*f_y$*Req. steel)/(3*Provided steel) | = | 117 $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | |
| Required effective depth | = | 69 mm | |
| Available effective depth | = | 184 mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.561** | 46.18 | 0.26 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.371** | 30.53 | 0.17 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.400** | 32.94 | 0.19 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |

## Design of Slab Panel - S3

***Material Properties***

Charactesistic strength of concrete, $f_{cu}$ = 40 $N/mm^2$

Charactesistic strength of steel, $f_y$ = 460 $N/mm^2$

Concrete Density, $\gamma_c$ = 24 $kN/m^3$

***Data***

Thickness of slab = 225 mm

Clear cover = 35 mm

Short span of the slab, $l_x$ **5.31**

Long span of the slab, $l_y$ **6**

Span ratio, $l_y/l_x$ **1.13**

**TWO WAY SLAB**

**Edge Condition** One long Edge Distcont.

***Loading***

Self weight of slab = 5.4 $kN/m^2$

Floor finish = 1.2 $kN/m^2$

Partitions = 0 $kN/m^2$

M&E services = 0.5 $kN/m^2$

Total Dead load, Gk = **7.1** $kN/m^2$

Live load, Qk = **10** $kN/m^2$

Design load, N= 1.4*Gk+1.6*Qk = **25.94** $kN/m^2$

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 196 |
| M.F. | 1.66 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 139 |
| $D_{reqd}$, mm | 180 |
| $D_{prov.}$ mm | 225 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar $\phi$ | Required Reinforcement As, $mm^2$ | Required Reinforcement Spacing | Provided Reinforcement Spacing | Provided Reinforcement As, $mm^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 37.38 | 184 | 0.03 | 174.80 | **12** | 489 | **231** | **150** | **755** |
| **Short span at bottom** | 27.65 | 184 | 0.03 | 174.80 | **12** | 362 | **312** | **200** | **566** |
| **Long span at top** | 27.07 | 172 | 0.03 | 163.40 | **12** | 379 | **298** | **150** | **755** |
| **Long span at bottom** | 20.48 | 172 | 0.02 | 163.40 | **12** | 287 | **394** | **200** | **566** |

**End Condition : One long edge discontinous**

Coefficients from BS 8110-1, Table 3.14

**Along short span at top** $\beta_{sx}$ (-) 0.051 37.38 kNm

**Along short span at bottom** $\beta_{sx}$ (+) 0.038 27.65 kNm

**Along long span at top** $\beta_{sy}$ (-) 0.037 27.07 kNm

**Along long span at bottom** $\beta_{sy}$ (+) 0.028 20.48 kNm

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 174.80 mm |
| Required area of the steel, $A_s$ | = | 489 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 231 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 754 $mm^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ |
| Lever arm z | = | 174.8 mm |
| Required area of the steel, $A_s$ | = | 362 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 313 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 565 $mm^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 379 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 298 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **754** $mm^2$ |
| Provided % of steel | = | 0.44 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 287 $mm^2$ |
| Diameter of bar | = | **12** mm |

| | | | | |
|---|---|---|---|---|
| Required spacing of reinf. | = | **395** | mm | |
| Provided spacing of reinf. | = | **200** | mm | |
| Provided area of the steel | = | **566** | $mm^2$ | |
| Provided % of steel | = | 0.33 | % | **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | | |
|---|---|---|---|---|
| Length of shorter span | = | 5.31 | m | |
| Actul l/d ratio | = | 23 | | |
| Provided % of steel at bottom along short span | = | 0.410 | % | |
| $f_s =$ (2*$f_y$*Req. steel)/(3*Provided steel) | = | 196 | $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | | |
| Required effective depth | = | 115 | mm | |
| Available effective depth | = | 184 | mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.412** | 56.75 | 0.31 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.276** | 38.02 | 0.21 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.360** | 49.59 | 0.29 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |

## Design of Slab Panel - S4

***Material Properties***

| | | | |
|---|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab | = | 225 | mm |
| Clear cover | = | 35 | mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **5.31** |
| Long span of the slab, $l_y$ | **6.5** |
| Span ratio, $l_y/l_x$ | **1.22** |

**TWO WAY SLAB**

**Edge Condition**: Two Adjacent Edges Discont.

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 5.4 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **7.1** | $kN/m^2$ |
| Live load, Qk | = | **10** | $kN/m^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **25.94** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 249 |
| M.F. | 1.46 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$, Reqd | 159 |
| $D_{reqd}$, mm | 200 |
| $D_{prov}$, mm | 225 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar $\phi$ | Required Reinforcement As, $mm^2$ | Required Reinforcement Spacing | Provided Reinforcement Spacing | Provided Reinforcement As, $mm^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 47.14 | 184 | 0.04 | 174.80 | **12** | 617 | **183** | **150** | **755** |
| **Short span at bottom** | 35.09 | 184 | 0.03 | 174.80 | **12** | 459 | **246** | **200** | **566** |
| **Long span at top** | 32.92 | 172 | 0.03 | 163.40 | **12** | 461 | **245** | **150** | **755** |
| **Long span at bottom** | 24.87 | 172 | 0.03 | 163.40 | **12** | 348 | **325** | **200** | **566** |

**End Condition : Two adjecent edges discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.064 | 47.14 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.048 | 35.09 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.045 | 32.92 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.034 | 24.87 kNm |

***Reinforcement along the Short Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm | |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.04** $N/mm^2$ | OK |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever arm z | = | 174.80 mm | |
| Required area of the steel, $A_s$ | = | 617 $mm^2$ | |
| Diameter of bar | = | **12** mm | |
| Required spacing of reinf. | = | 183 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | 754 $mm^2$ | |
| Provided % of steel | = | 0.41 % | **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | | |
|---|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ | |
| Lever arm z | = | 174.8 mm | |
| Required area of the steel, $A_s$ | = | 459 $mm^2$ | |
| Diameter of bar | = | **12** mm | |
| Required spacing of reinf. | = | 247 mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | 565 $mm^2$ | |
| Provided % of steel | = | 0.41 % | **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm | |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ | OK |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever Arm z | = | 163.40 mm | |
| Required area of the steel, $A_s$ | = | 461 $mm^2$ | |
| Diameter of bar | = | **12** mm | |
| Required spacing of reinf. | = | 245 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | **754** $mm^2$ | |
| Provided % of steel | = | 0.44 % | **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 348 $mm^2$ |
| Diameter of bar | = | **12** mm |

| | | | |
|---|---|---|---|
| Required spacing of reinf. | = | **325** mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | **566** $mm^2$ | |
| Provided % of steel | = | 0.33 % | **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | |
|---|---|---|---|
| Length of shorter span | = | 5.31 m | |
| Actul l/d ratio | = | 23 | |
| Provided % of steel at bottom along short span | = | 0.410 % | |
| $f_s = (2*f_y$*Req. steel)/(3*Provided steel) | = | 249 $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | |
| Required effective depth | = | 115 mm | |
| Available effective depth | = | 184 mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.477** | 65.73 | 0.36 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.315** | 43.36 | 0.24 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.400** | 55.10 | 0.32 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |

## Design of Slab Panel - S5

***Material Properties***

| | | | |
|---|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 | N/mm$^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 | N/mm$^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | kN/m$^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab | = | 225 | mm |
| Clear cover | = | 35 | mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **6.5** |
| Long span of the slab, $l_y$ | **6.65** |
| Span ratio, $l_y/l_x$ | **1.02** |

**TWO WAY SLAB**

**Edge Condition** One long Edge Distcont.

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 5.4 | kN/m$^2$ |
| Floor finish | = | 1.2 | kN/m$^2$ |
| Partitions | = | 0 | kN/m$^2$ |
| M&E services | = | 0.5 | kN/m$^2$ |
| Total Dead load, Gk | = | **7.1** | kN/m$^2$ |
| Live load, Qk | = | **10** | kN/m$^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **25.94** | kN/m$^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, N/mm$^2$ | 244 |
| M.F. | 1.47 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 192 |
| $D_{reqd}$, mm | 233 |
| $D_{prov.}$ mm | 225 |
| **Unsafe in deflection** | |

| **Location** | **B.M., kN.m** | **$d_{eff}$, mm** | **K = $M/f_{cu}bd^2$** | **Lever Arm z** | **Dia of bar $\phi$** | **Required Reinforcement** As, mm$^2$ | **Required Reinforcement** Spacing | **Provided Reinforcement** Spacing | **Provided Reinforcement** As, mm$^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 45.28 | 184 | 0.04 | 174.80 | **12** | 593 | **191** | **150** | **755** |
| **Short span at bottom** | 34.40 | 184 | 0.03 | 174.80 | **12** | 450 | **251** | **200** | **566** |
| **Long span at top** | 40.56 | 172 | 0.04 | 163.40 | **12** | 568 | **199** | **150** | **755** |
| **Long span at bottom** | 30.69 | 172 | 0.03 | 163.40 | **12** | 430 | **263** | **200** | **566** |

**End Condition : One long edge discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.041 | 45.28 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.031 | 34.40 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.037 | 40.56 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.028 | 30.69 kNm |

*Reinforcement along the Short Span*

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.04** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 174.80 mm |
| Required area of the steel, $A_s$ | = | 593 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 191 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 754 mm$^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.03** N/mm$^2$ |
| Lever arm z | = | 174.8 mm |
| Required area of the steel, $A_s$ | = | 450 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 252 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 565 mm$^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

*Reinforcement along the Long Span*

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.04** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 568 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 199 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **754** mm$^2$ |
| Provided % of steel | = | 0.44 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.03** N/mm$^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 430 mm$^2$ |
| Diameter of bar | = | **12** mm |

Required spacing of reinf. = **264** mm
Provided spacing of reinf. = **200** mm
Provided area of the steel = **566** $mm^2$
Provided % of steel = 0.33 % **Provided area of steel is OK**
**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 6.5 m
Actul l/d ratio = 23
Provided % of steel at bottom along short span = 0.410 %
$f_s = (2*f_y$*Req. steel)/(3*Provided steel) = 244 $N/mm^2$
M.F. for tension reinforcement, = 2.00
Required effective depth = 141 mm
Available effective depth = 184 mm **Safe in Deflection**

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.369** | 62.26 | 0.34 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.247** | 41.63 | 0.23 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.360** | 60.70 | 0.35 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |

## Design of Slab Panel - S6

***Material Properties***

Charactesistic strength of concrete, $f_{cu}$ = 40 N/mm$^2$

Charactesistic strength of steel, $f_y$ = 460 N/mm$^2$

Concrete Density, $\gamma_c$ = 24 kN/m$^3$

***Data***

Thickness of slab = 225 mm

Clear cover = 35 mm

Short span of the slab, $l_x$ **6**

Long span of the slab, $l_y$ **6.65**

Span ratio, $l_y/l_x$ **1.11**

**TWO WAY SLAB**

**Edge Condition** Interior Panel

***Loading***

Self weight of slab = 5.4 kN/m$^2$

Floor finish = 1.2 kN/m$^2$

Partitions = 0 kN/m$^2$

M&E services = 0.5 kN/m$^2$

Total Dead load, Gk = **7.1** kN/m$^2$

Live load, Qk = **10** kN/m$^2$

Design load, N= 1.4*Gk+1.6*Qk = **25.94** kN/m$^2$

| **Deflection check** | |
|---|---|
| $f_s$, N/mm$^2$ | 188 |
| M.F. | 1.97 |
| Span/$d_{eff}$ | **26** |
| $d_{eff}$. Reqd | 118 |
| $D_{reqd}$, mm | 159 |
| $D_{prov}$, mm | 225 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar $\phi$ | Required Reinforcement As, mm$^2$ | Required Reinforcement Spacing | Provided Reinforcement Spacing | Provided Reinforcement As, mm$^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 34.95 | 184 | 0.03 | 174.80 | **12** | 458 | **247** | **150** | **755** |
| **Short span at bottom** | 26.46 | 184 | 0.02 | 174.80 | **12** | 346 | **327** | **200** | **566** |
| **Long span at top** | 29.89 | 172 | 0.03 | 163.40 | **12** | 419 | **270** | **150** | **755** |
| **Long span at bottom** | 22.42 | 172 | 0.02 | 163.40 | **12** | 314 | **360** | **200** | **566** |

**End Condition : Interior panel**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.037 | 34.95 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.028 | 26.46 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.032 | 29.89 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.024 | 22.42 kNm |

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 174.80 mm |
| Required area of the steel, $A_s$ | = | 458 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 247 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 754 mm$^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ |
| Lever arm z | = | 174.8 mm |
| Required area of the steel, $A_s$ | = | 346 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 327 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 565 mm$^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 419 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 270 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **754** mm$^2$ |
| Provided % of steel | = | 0.44 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 314 mm$^2$ |
| Diameter of bar | = | **12** mm |

| | | | | |
|---|---|---|---|---|
| Required spacing of reinf. | = | **361** | mm | |
| Provided spacing of reinf. | = | **200** | mm | |
| Provided area of the steel | = | **566** | $mm^2$ | |
| Provided % of steel | = | 0.33 | % | **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | | |
|---|---|---|---|---|
| Length of shorter span | = | 6 | m | |
| Actul l/d ratio | = | 26 | | |
| Provided % of steel at bottom along short span | = | 0.410 | % | |
| $f_s = (2*f_y$*Req. steel)/(3*Provided steel) | = | 188 | $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | | |
| Required effective depth | = | 115 | mm | |
| Available effective depth | = | 184 | mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.363** | 56.42 | 0.31 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.000** | 0.00 | 0 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.330** | 51.36 | 0.30 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |

**Design of Slab Panal -** S7

***Material Properties***

| | | | |
|---|---|---|---|
| Grade of the concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Grade of the steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab, D | = | 225 | mm |
| Width of slab, b | = | 1000 | mm |
| Bottom cover | = | 35 | mm |
| Short span, $l$ x | = | 1.715 | m |
| Long span, $l$ y | = | 6.5 | m |
| Span ratio, $l$ y/$l$ x | = | 3.79 | Oneway |

**Edge condition** One end continuous

***Loading***

| | | | |
|---|---|---|---|
| Self weight of the slab | = | 5.4 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **7.1** | $kN/m^2$ |
| Total Live load, Qk | = | **10** | $kN/m^2$ |
| Design load on slab, n = 1.4*Gk+1.6*Qk | = | **25.94** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| fs, $N/mm^2$ | 159 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| d eff. $_{reqd}$ | 38 |
| $D_{reqd,}$ mm | 79 |
| D $_{provided}$ ,m | 225 |
| **Safe in deflection** | |

| | | | |
|---|---|---|---|
| (-ve) moment at support, Mt = n * $l^2$/10 | = | **7.63** | kN.m |
| (+ve) moment at span, Mb = n * $l^2$/10 | = | **7.63** | kN.m |
| Minimum Area of steel (0.13% of Gross c/s area) | = | 293 | $mm^2$ |

| Location | K = $M/f_{cu}bd^2$ | $d_{eff,}$ mm | Dia of bar $\phi$ | Required As, $mm^2$ | Required Spacing | Provided Spacing | Provided As, $mm^2$ |
|---|---|---|---|---|---|---|---|
| **Short span at top** | 0.01 | 184 | **12** | 293 | **386** | **150** | **755** |
| **Short span at bottom** | 0.01 | 184 | **12** | 293 | **386** | **200** | **566** |
| **Distribution, 0.13*b*D** | | | **12** | 293 | **387** | **200** | **566** |

***Area of steel at support***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 | mm |
| **At support, Mt / $f_{cu}bd^2$** ( K ) | = | **0.01** | |
| Z | = | 174.8 | mm |
| Required area of the steel, $A_s$ | = | 100 | $mm^2$ |
| Minimum Area of steel | = | 293 | $mm^2$ |
| Diameter of bar | = | **12** | mm |
| Required spacing of reinf. | = | 386 | mm |
| Provided spacing of reinf. | = | **150** | mm |
| Provided area of the steel | = | 754 | $mm^2$ |
| Provided % of steel | = | 0.41 | % **OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

***Area of steel at span***

| | | | | |
|---|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 | mm | |
| **At support, Mb / $f_{cu}bd^2$** | = | **0.01** | | |
| Z | = | 174.80 | | |
| Required area of the steel, $A_s$ | = | 100 | $mm^2$ | |
| Minimum Area of steel | = | 293 | $mm^2$ | |
| Diameter of bar | = | **12** | mm | |
| Required spacing of reinf. | = | 386 | mm | |
| Provided spacing of reinf. | = | **200** | mm | |
| Provided area of the steel | = | 566 | $mm^2$ | |
| Provided % of steel | = | 0.31 | % | **OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

| | | | | |
|---|---|---|---|---|
| Distribution steel, 0.13% of Gross C/A | = | 293 | $mm^2$ | |
| Diameter of bar | = | **12** | mm | |
| Required spacing of reinf. | = | **387** | mm | |
| Provided spacing of reinf. | = | **200** | mm | |
| Provided area of the steel | = | **566** | $mm^2$ | |
| Provided % of steel | = | 0.29 | % | **OK** |

**Provide 12T @ 200 c/c**

***Check for Deflection*** BS8110 -Section 3.4.6.5, Table 3.10

| | | | | |
|---|---|---|---|---|
| Length of shorter span | = | 1.715 | m | |
| Actul l/d ratio | = | 23 | | |
| Provided % of steel at bottom along short span | = | 0.310 | % | |
| $f_s$ = (2*fy*Req. steel)/(3*Provided steel) | = | 54 | $N/mm^2$ | |
| M.F. for tension reinforcement, | = | 2.00 | | |
| Required effective depth | = | 37 | mm | |
| Available effective depth | = | 184 | mm | **Safe in Deflection** |

***Check for shear***

Shear force, Vsx = 22.24355 kN

| **Distance from support, m** | **av (m)** | **Vsx ( kN )** | **v ( N/mm2 )** | **$v_c$ ( N/mm2 )** | **Enhanced $v_c$, N/mm2** | **Asv/sv** | **Remarks** |
|---|---|---|---|---|---|---|---|
| d | 0.184 | 17.47 | 0.09 | 0.67 | 1.34 | 0.00 | No. shear reinf. |
| 2d | 0.368 | 12.70 | 0.07 | 0.67 | 0.67 | 0.00 | No. shear reinf. |
| 3d | 0.552 | 7.92 | 0.04 | 0.67 | 0.67 | 0.00 | No. shear reinf. |

## Design of Slab Panel - S8

***Material Properties***

| | | |
|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 $N/mm^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 $kN/m^3$ |

***Data***

| | | |
|---|---|---|
| Thickness of slab | = | 225 mm |
| Clear cover | = | 35 mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **5.04** |
| Long span of the slab, $l_y$ | **6** |
| Span ratio, $l_y/l_x$ | **1.19** |

**TWO WAY SLAB**

**Edge Condition** One Short Edge Discontinuous

***Loading***

| | | |
|---|---|---|
| Self weight of slab | = | 5.4 $kN/m^2$ |
| Floor finish | = | 1.2 $kN/m^2$ |
| Partitions | = | 0 $kN/m^2$ |
| M&E services | = | 0.5 $kN/m^2$ |
| Total Dead load, Gk | = | **7.1** $kN/m^2$ |
| Live load, Qk | = | **10** $kN/m^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **25.94** $kN/m^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 167 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **26** |
| $d_{eff}$. Reqd | 97 |
| $D_{reqd}$, mm | 138 |
| $D_{prov.}$ mm | 225 |
| **Safe in deflection** | |

| **Location** | **B.M., kN.m** | **$d_{eff}$, mm** | **K = $M/f_{cu}bd^2$** | **Lever Arm z** | **Dia of bar $\phi$** | **Required Reinforcement** As, $mm^2$ | Spacing | **Provided Reinforcement** Spacing | As, $mm^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 31.38 | 184 | 0.03 | 174.80 | **12** | 411 | **275** | **150** | **755** |
| **Short span at bottom** | 23.54 | 184 | 0.02 | 174.80 | **12** | 308 | **367** | **200** | **566** |
| **Long span at top** | 24.38 | 172 | 0.03 | 163.40 | **12** | 341 | **331** | **150** | **755** |
| **Long span at bottom** | 18.45 | 172 | 0.02 | 163.40 | **12** | 258 | **438** | **200** | **566** |

**End Condition : One short edge discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.048 | 31.38 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.036 | 23.54 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.037 | 24.38 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.028 | 18.45 kNm |

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 174.80 mm |
| Required area of the steel, $A_s$ | = | 411 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 275 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 754 mm$^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ |
| Lever arm z | = | 174.8 mm |
| Required area of the steel, $A_s$ | = | 308 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 368 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 565 mm$^2$ |
| Provided % of steel | = | 0.41 % **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 341 mm$^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 331 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **754** mm$^2$ |
| Provided % of steel | = | 0.44 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 258 mm$^2$ |
| Diameter of bar | = | **12** mm |

Required spacing of reinf. = **438** mm
Provided spacing of reinf. = **200** mm
Provided area of the steel = **566** $mm^2$
Provided % of steel = 0.33 % **Provided area of steel is OK**
**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10
Length of shorter span = 5.04 m
Actul l/d ratio = 26
Provided % of steel at bottom along short span = 0.410 %
$f_s = (2*f_y$*Req. steel)/(3*Provided steel) = 167 $N/mm^2$
$0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ = 2.00
Required effective depth = 97 mm
Available effective depth = 184 mm **Safe in Deflection**

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| **Location** | **Coeff.** | **S.Force V (kN)** | $v$ **$N/mm^2$** | **% of steel** | $v_c$, **$N/mm^2$** | **Enhanced $v_c$, $N/mm^2$** | $A_{sv}/s_v$ | **Remarks** |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.417** | 54.54 | 0.3 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.000** | 0.00 | 0 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.360** | 47.07 | 0.27 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |

**Design of Slab Panal -** S9

***Material Properties***

| | | | |
|---|---|---|---|
| Grade of the concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Grade of the steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab, D | = | 225 | mm |
| Width of slab, b | = | 1000 | mm |
| Bottom cover | = | 35 | mm |
| Short span, $l$ x | = | 3.215 | m |
| Long span, $l$ y | = | 6.5 | m |
| Span ratio, $l$ y/$l$ x | = | 2.02 | Oneway |

**Edge condition** One end continuous

***Loading***

| | | | |
|---|---|---|---|
| Self weight of the slab | = | 5.4 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **7.1** | $kN/m^2$ |
| Total Live load, Qk | = | **10** | $kN/m^2$ |
| Design load on slab, n = 1.4*Gk+1.6*Qk | = | **25.94** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| fs, $N/mm^2$ | 190 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| d eff. $_{reqd}$ | 70 |
| $D_{reqd}$, mm | 111 |
| D $_{provided}$ ,mm | 225 |
| **Safe in deflection** | |

| | | | |
|---|---|---|---|
| (-ve) moment at support, Mt = n * $l^2$/10 | = | **26.81** | kN.m |
| (+ve) moment at span, Mb = n * $l^2$/10 | = | **26.81** | kN.m |
| Minimum Area of steel (0.13% of Gross c/s area) | = | 293 | $mm^2$ |

| Location | K = $M/f_{cu}bd^2$ | $d_{eff}$, mm | bar $\phi$ | Required As, $mm^2$ | Required Spacing | Provided Spacing | Reinforcement As, $mm^2$ |
|---|---|---|---|---|---|---|---|
| **Short span at top** | 0.02 | 184 | **12** | 351 | **322** | **150** | **755** |
| **Short span at bottom** | 0.01 | 184 | **12** | 351 | **322** | **200** | **566** |
| **Distribution, 0.13*b*D** | | | **12** | 293 | **387** | **200** | **566** |

***Area of steel at support***

| | | | | |
|---|---|---|---|---|
| Available effective depth, $d_{eff}$ | | = | 184 | mm |
| **At support, Mt / $f_{cu}bd^2$** | ( K ) | = | **0.02** | |
| Z | | = | 174.8 | mm |
| Required area of the steel, $A_s$ | | = | 351 | $mm^2$ |
| Minimum Area of steel | | = | 293 | $mm^2$ |
| Diameter of bar | | = | **12** | mm |
| Required spacing of reinf. | | = | 322 | mm |
| Provided spacing of reinf. | | = | **150** | mm |
| Provided area of the steel | | = | 754 | $mm^2$ |
| Provided % of steel | | = | 0.41 | % **OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

*Area of steel at span*

| | | | | |
|---|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 | mm | |
| **At support, Mb / $f_{cu}bd^2$** | = | **0.02** | | |
| Z | = | 174.80 | | |
| Required area of the steel, $A_s$ | = | 351 | $mm^2$ | |
| Minimum Area of steel | = | 293 | $mm^2$ | |
| Diameter of bar | = | **12** | mm | |
| Required spacing of reinf. | = | 322 | mm | |
| Provided spacing of reinf. | = | **200** | mm | |
| Provided area of the steel | = | 566 | $mm^2$ | |
| Provided % of steel | = | 0.31 | % | **OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

| | | | | |
|---|---|---|---|---|
| Distribution steel, 0.13% of Gross C/A | = | 293 | $mm^2$ | |
| Diameter of bar | = | **12** | mm | |
| Required spacing of reinf. | = | **387** | mm | |
| Provided spacing of reinf. | = | **200** | mm | |
| Provided area of the steel | = | **566** | $mm^2$ | |
| Provided % of steel | = | 0.29 | % | **OK** |

**Provide 12T @ 200 c/c**

*Check for Deflection* BS8110 -Section 3.4.6.5, Table 3.10

| | | | | |
|---|---|---|---|---|
| Length of shorter span | = | 3.215 | m | |
| Actul l/d ratio | = | 23 | | |
| Provided % of steel at bottom along short span | = | 0.310 | % | |
| $f_s$ = (2*fy*Req. steel)/(3*Provided steel) | = | 190 | $N/mm^2$ | |
| M.F. for tension reinforcement, | = | 2.00 | | |
| Required effective depth | = | 70 | mm | |
| Available effective depth | = | 184 | mm | **Safe in Deflection** |

*Check for shear*

Shear force, Vsx = 41.69855 kN

| **Distance from support, m** | **av (m)** | **Vsx ( kN )** | **v ( N/mm2 )** | **$v_c$ ( N/mm2** | **Enhanced $v_c$,** | **Asv/sv** | **Remarks** |
|---|---|---|---|---|---|---|---|
| d | 0.184 | 36.93 | 0.20 | 0.67 | 1.34 | 0.00 | No. shear reinf. |
| 2d | 0.368 | 32.15 | 0.17 | 0.67 | 0.67 | 0.00 | No. shear reinf. |
| 3d | 0.552 | 27.38 | 0.15 | 0.67 | 0.67 | 0.00 | No. shear reinf. |

**Design of Slab Panal -** S10

***Material Properties***

Grade of the concrete, $f_{cu}$ = 40 $N/mm^2$
Grade of the steel, $f_y$ = 460 $N/mm^2$
Concrete Density, $\gamma_c$ = 24 $kN/m^3$

***Data***

Thickness of slab, D = 225 mm
Width of slab, b = 1000 mm
Bottom cover = 35 mm
Short span, $l$ x = 1.325 m
Long span, $l$ y = 3.325 m
Span ratio, $l$ y/$l$ x = 2.51 Oneway

**Edge condition** One end continuous

***Loading***

Self weight of the slab = 5.4 $kN/m^2$
Floor finish = 1.2 $kN/m^2$
Partitions = 0 $kN/m^2$
M&E services = 0.5 $kN/m^2$
Total Dead load, Gk = **7.1** $kN/m^2$
Total Live load, Qk = **10** $kN/m^2$
Design load on slab, n = 1.4*Gk+1.6*Qk = **25.94** $kN/m^2$

| **Deflection check** | |
|---|---|
| fs, $N/mm^2$ | 159 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| d eff. $_{reqd}$ | 29 |
| $D_{reqd,}$ mm | 70 |
| D $_{provided}$ ,mm | 225 |
| **Safe in deflection** | |

(-ve) moment at support, Mt = n * $l^2$/10 = **4.55** kN.m
(+ve) moment at span, Mb = n * $l^2$/10 = **4.55** kN.m
Minimum Area of steel (0.13% of Gross c/s area) = 293 $mm^2$

| Location | K = $M/f_{cu}bd^2$ | $d_{eff}$, mm | bar $\phi$ | Required As, $mm^2$ | Required Spacing | Provided Spacing | Reinforcement As, $mm^2$ |
|---|---|---|---|---|---|---|---|
| **Short span at top** | 0.01 | 184 | **12** | 293 | **386** | **150** | **755** |
| **Short span at bottom** | 0.01 | 184 | **12** | 293 | **386** | **200** | **566** |
| **Distribution, 0.13*b*D** | | | **12** | 293 | **387** | **200** | **566** |

***Area of steel at support***

Available effective depth, $d_{eff}$ = 184 mm
**At support, Mt / $f_{cu}bd^2$** ( K ) = **0.01**
Z = 174.8 mm
Required area of the steel, $A_s$ = 60 $mm^2$
Minimum Area of steel = 293 $mm^2$
Diameter of bar = **12** mm
Required spacing of reinf. = 386 mm
Provided spacing of reinf. = **150** mm
Provided area of the steel = 754 $mm^2$
Provided % of steel = 0.41 % **OK**

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

***Area of steel at span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm |
| **At support, Mb / $f_{cu}bd^2$** | = | **0.01** |
| Z | = | 174.80 |
| Required area of the steel, $A_s$ | = | 60 $mm^2$ |
| Minimum Area of steel | = | 293 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | 386 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 566 $mm^2$ |
| Provided % of steel | = | 0.31 % **OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

| | | |
|---|---|---|
| Distribution steel, 0.13% of Gross C/A | = | 293 $mm^2$ |
| Diameter of bar | = | **12** mm |
| Required spacing of reinf. | = | **387** mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | **566** $mm^2$ |
| Provided % of steel | = | 0.29 % **OK** |

**Provide 12T @ 200 c/c**

***Check for Deflection*** BS8110 -Section 3.4.6.5, Table 3.10

| | | |
|---|---|---|
| Length of shorter span | = | 1.325 m |
| Actul l/d ratio | = | 23 |
| Provided % of steel at bottom along short span | = | 0.310 % |
| $f_s$ = (2*fy*Req. steel)/(3*Provided steel) | = | 32 $N/mm^2$ |
| M.F. for tension reinforcement, | = | 2.00 |
| Required effective depth | = | 29 mm |
| Available effective depth | = | 184 mm **Safe in Deflection** |

***Check for shear***

| | | |
|---|---|---|
| Shear force, Vsx | = | 17.18525 kN |

| Distance from support, m | av (m) | Vsx ( kN ) | v ( N/mm2 ) | $v_c$ ( N/mm2 | Enhanced $v_c$, | Asv/sv | Remarks |
|---|---|---|---|---|---|---|---|
| d | 0.184 | 12.41 | 0.07 | 0.67 | 1.34 | 0.00 | No. shear reinf. |
| 2d | 0.368 | 7.64 | 0.04 | 0.67 | 0.67 | 0.00 | No. shear reinf. |
| 3d | 0.552 | 2.87 | 0.02 | 0.67 | 0.67 | 0.00 | No. shear reinf. |

## Design of Slab Panel - S11

***Material Properties***

| | | |
|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 N/mm$^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 N/mm$^2$ |
| Concrete Density, $\gamma_c$ | = | 24 kN/m$^3$ |

***Data***

| | | |
|---|---|---|
| Thickness of slab | = | 225 mm |
| Clear cover | = | 35 mm |
| Short span of the slab, $l_x$ | | **1.825** |
| Long span of the slab, $l_y$ | | **2.943** |
| Span ratio, $l_y/l_x$ | | **1.61** |

**TWO WAY SLAB**

**Edge Condition**: Two Adjacent Edges Discont.

***Loading***

| | | |
|---|---|---|
| Self weight of slab | = | 5.4 kN/m$^2$ |
| Floor finish | = | 1.2 kN/m$^2$ |
| Partitions | = | 0 kN/m$^2$ |
| M&E services | = | 0.5 kN/m$^2$ |
| Total Dead load, Gk | = | **7.1** kN/m$^2$ |
| Live load, Qk | = | **10** kN/m$^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **25.94** kN/m$^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, N/mm$^2$ | 38 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 40 |
| $D_{reqd}$, mm | 81 |
| $D_{prov.}$ mm | 225 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar $\phi$ | Required Reinforcement As, mm$^2$ | Required Reinforcement Spacing | Provided Reinforcement Spacing | Provided Reinforcement As, mm$^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 7.09 | 184 | 0.01 | 174.80 | **12** | 93 | **1219** | **150** | **755** |
| **Short span at bottom** | 5.34 | 184 | 0.01 | 174.80 | **12** | 70 | **1618** | **200** | **566** |
| **Long span at top** | 3.89 | 172 | 0.01 | 163.40 | **12** | 54 | **2076** | **150** | **755** |
| **Long span at bottom** | 2.94 | 172 | 0.01 | 163.40 | **12** | 41 | **2747** | **200** | **566** |

**End Condition : Two adjecent edges discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.082 | 7.09 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.062 | 5.34 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.045 | 3.89 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.034 | 2.94 kNm |

***Reinforcement along the Short Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 184 mm | |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever arm z | = | 174.80 mm | |
| Required area of the steel, $A_s$ | = | 93 $mm^2$ | |
| Diameter of bar | = | **12** mm | |
| Required spacing of reinf. | = | 1219 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | 754 $mm^2$ | |
| Provided % of steel | = | 0.41 % | **Provided area of steel is OK** |

**Provide 12T @ 150 c/c at top (Negative Reinf.)**

| | | | |
|---|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ | |
| Lever arm z | = | 174.8 mm | |
| Required area of the steel, $A_s$ | = | 70 $mm^2$ | |
| Diameter of bar | = | **12** mm | |
| Required spacing of reinf. | = | 1618 mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | 565 $mm^2$ | |
| Provided % of steel | = | 0.41 % | **Provided area of steel is OK** |

**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 172 mm | |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever Arm z | = | 163.40 mm | |
| Required area of the steel, $A_s$ | = | 54 $mm^2$ | |
| Diameter of bar | = | **12** mm | |
| Required spacing of reinf. | = | 2076 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | **754** $mm^2$ | |
| Provided % of steel | = | 0.44 % | **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.01** $N/mm^2$ |
| Lever arm z | = | 163.40 mm |
| Required area of the steel, $A_s$ | = | 41 $mm^2$ |
| Diameter of bar | = | **12** mm |

Required spacing of reinf. = **2747** mm
Provided spacing of reinf. = **200** mm
Provided area of the steel = **566** $mm^2$
Provided % of steel = 0.33 % **Provided area of steel is OK**
**Provide 12T @ 150 c/c at top (Negative Reinf.)**
**Provide 12T @ 200 c/c at bottom (Positive Reinf.)**

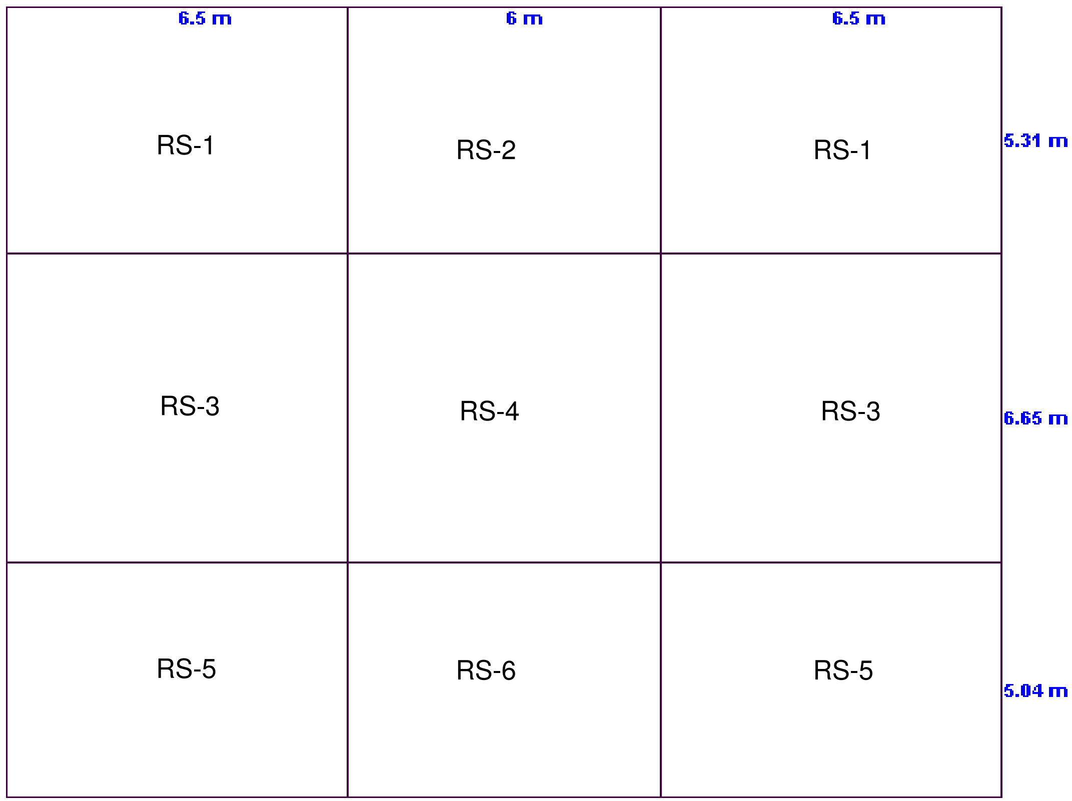
***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 1.825 m
Actul l/d ratio = 23
Provided % of steel at bottom along short span = 0.410 %
$f_s$ = (2*$f_y$*Req. steel)/(3*Provided steel) = 38 $N/mm^2$
0.55+[(477-$f_s$)/(120*(0.9+M/bd$^2$))] = 2.00
Required effective depth = 40 mm
Available effective depth = 184 mm **Safe in Deflection**

***Check for shear*** Shear coefficients from Table 3.15

$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.554** | 26.20 | 0.15 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.364** | 17.21 | 0.1 | 0.41 | 0.64 | 1.28 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.400** | 18.94 | 0.11 | 0.44 | 0.67 | 1.33 | 0 | No. shear reinf. |




Roof Slab Layout Plan

## Design of Slab Panel - RS1

***Material Properties***

| | | |
|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 N/mm$^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 N/mm$^2$ |
| Concrete Density, $\gamma_c$ | = | 24 kN/m$^3$ |

***Data***

| | | |
|---|---|---|
| Thickness of slab | = | 175 mm |
| Clear cover | = | 35 mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **5.31** |
| Long span of the slab, $l_y$ | **6.5** |
| Span ratio, $l_y/l_x$ | **1.22** |

**TWO WAY SLAB**

**Edge Condition** Two Adjacent Edges Discont.

***Loading***

| | | |
|---|---|---|
| Self weight of slab | = | 4.2 kN/m$^2$ |
| Floor finish | = | 1.2 kN/m$^2$ |
| Partitions | = | 0 kN/m$^2$ |
| M&E services | = | 0.5 kN/m$^2$ |
| Total Dead load, Gk | = | **5.9** kN/m$^2$ |
| Live load, Qk | = | **0.6** kN/m$^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **9.22** kN/m$^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, N/mm$^2$ | 178 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 116 |
| $D_{reqd}$, mm | 156 |
| $D_{prov.}$ mm | 175 |
| **Safe in deflection** | |

| **Location** | **B.M., kN.m** | **$d_{eff}$, mm** | **K = $M/f_{cu}bd^2$** | **Lever Arm z** | **Dia of bar $\phi$** | **Required Reinforcement** As, mm$^2$ | Spacing | **Provided Reinforcement** Spacing | As, mm$^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 16.76 | 135 | 0.03 | 128.25 | **10** | 299 | **263** | **150** | **524** |
| **Short span at bottom** | 12.47 | 135 | 0.02 | 128.25 | **10** | 228 | **345** | **200** | **393** |
| **Long span at top** | 11.70 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **150** | **524** |
| **Long span at bottom** | 8.84 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **200** | **393** |

**End Condition : Two adjecent edges discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.064 | 16.76 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.048 | 12.47 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.045 | 11.70 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.034 | 8.84 kNm |

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 135 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 128.25 mm |
| Required area of the steel, $A_s$ | = | 299 $mm^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 263 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 524 $mm^2$ |
| Provided % of steel | = | 0.39 % **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ |
| Lever arm z | = | 128.25 mm |
| Required area of the steel, $A_s$ | = | 222 $mm^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 353 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 393 $mm^2$ |
| Provided % of steel | = | 0.39 % **Provided area of steel is OK** |

**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 125 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 118.75 mm |
| Required area of the steel, $A_s$ | = | 225 $mm^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 348 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **524** $mm^2$ |
| Provided % of steel | = | 0.42 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ |
| Lever arm z | = | 118.75 mm |
| Required area of the steel, $A_s$ | = | 170 $mm^2$ |
| Diameter of bar | = | **10** mm |

Required spacing of reinf. = **462** mm
Provided spacing of reinf. = **200** mm
Provided area of the steel = **393** $mm^2$
Provided % of steel = 0.32 % **Provided area of steel is OK**
**Provide 10T @ 150 c/c at top (Negative Reinf.)**
**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10
Length of shorter span = 5.31 m
Actul l/d ratio = 23
Provided % of steel at bottom along short span = 0.390 %
$f_s =$ (2*$f_y$*Req. steel)/(3*Provided steel) = 174 $N/mm^2$
0.55+[(477-$f_s$)/(120*(0.9+M/bd$^2$))] = 2.00
Required effective depth = 115 mm
Available effective depth = 135 mm **Safe in Deflection**

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.477** | 23.36 | 0.18 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.315** | 15.41 | 0.12 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.400** | 19.58 | 0.16 | 0.42 | 0.71 | 1.42 | 0 | No. shear reinf. |

## Design of Slab Panel - RS2

***Material Properties***

Charactesistic strength of concrete, $f_{cu}$ = 40 N/mm$^2$

Charactesistic strength of steel, $f_y$ = 460 N/mm$^2$

Concrete Density, $\gamma_c$ = 24 kN/m$^3$

***Data***

Thickness of slab = 175 mm

Clear cover = 35 mm

Short span of the slab, $l_x$ **5.31**

Long span of the slab, $l_y$ **6**

Span ratio, $l_y/l_x$ **1.13**

**TWO WAY SLAB**

**Edge Condition** One long Edge Distcont.

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 4.2 | kN/m$^2$ |
| Floor finish | = | 1.2 | kN/m$^2$ |
| Partitions | = | 0 | kN/m$^2$ |
| M&E services | = | 0.5 | kN/m$^2$ |
| Total Dead load, Gk | = | **5.9** | kN/m$^2$ |
| Live load, Qk | = | **0.6** | kN/m$^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **9.22** | kN/m$^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, N/mm$^2$ | 178 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 116 |
| $D_{reqd}$, mm | 156 |
| $D_{prov}$. mm | 175 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar | Required Reinforcement | | Provided Reinforcement | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\phi$ | As, mm$^2$ | Spacing | Spacing | As, mm$^2$ |
| **Short span at top** | 13.29 | 135 | 0.02 | 128.25 | **10** | 237 | **331** | **150** | **524** |
| **Short span at bottom** | 9.83 | 135 | 0.02 | 128.25 | **10** | 228 | **345** | **200** | **393** |
| **Long span at top** | 9.62 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **150** | **524** |
| **Long span at bottom** | 7.28 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **200** | **393** |

**End Condition : One long edge discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.051 | 13.29 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.038 | 9.83 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.037 | 9.62 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.028 | 7.28 kNm |

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 135 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 128.25 mm |
| Required area of the steel, $A_s$ | = | 237 mm$^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 331 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 524 mm$^2$ |
| Provided % of steel | = | 0.39 % **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ |
| Lever arm z | = | 128.25 mm |
| Required area of the steel, $A_s$ | = | 175 mm$^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 448 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 393 mm$^2$ |
| Provided % of steel | = | 0.39 % **Provided area of steel is OK** |

**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, d eff | = | 125 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 118.75 mm |
| Required area of the steel, $A_s$ | = | 185 mm$^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 424 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **524** mm$^2$ |
| Provided % of steel | = | 0.42 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ |
| Lever arm z | = | 118.75 mm |
| Required area of the steel, $A_s$ | = | 140 mm$^2$ |
| Diameter of bar | = | **10** mm |

| | | | |
|---|---|---|---|
| Required spacing of reinf. | = | **560** mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | **393** $mm^2$ | |
| Provided % of steel | = | 0.32 % | **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**
**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | |
|---|---|---|---|
| Length of shorter span | = | 5.31 m | |
| Actul l/d ratio | = | 23 | |
| Provided % of steel at bottom along short span | = | 0.390 % | |
| $f_s = (2*f_y$*Req. steel)/(3*Provided steel) | = | 137 $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | |
| Required effective depth | = | 115 mm | |
| Available effective depth | = | 135 mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| **Location** | **Coeff.** | **S.Force V (kN)** | **$v$ $N/mm^2$** | **% of steel** | **$v_c$, $N/mm^2$** | **Enhanced $v_c$, $N/mm^2$** | **$A_{sv}/s_v$** | **Remarks** |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.412** | 20.17 | 0.15 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.276** | 13.51 | 0.11 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.360** | 17.62 | 0.14 | 0.42 | 0.71 | 1.42 | 0 | No. shear reinf. |

## Design of Slab Panel - RS3

***Material Properties***

| | | | |
|---|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab | = | 175 | mm |
| Clear cover | = | 35 | mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **6.5** |
| Long span of the slab, $l_y$ | **6.65** |
| Span ratio, $l_y/l_x$ | **1.02** |

**TWO WAY SLAB**

**Edge Condition** One long Edge Distcont.

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 4.2 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **5.9** | $kN/m^2$ |
| Live load, Qk | = | **0.6** | $kN/m^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **9.22** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 178 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **26** |
| $d_{eff}$. Reqd | 125 |
| $D_{reqd}$, mm | 165 |
| $D_{prov}$. mm | 175 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar $\phi$ | Required Reinforcement As, $mm^2$ | Required Reinforcement Spacing | Provided Reinforcement Spacing | Provided Reinforcement As, $mm^2$ |
|---|---|---|---|---|---|---|---|---|---|
| **Short span at top** | 16.10 | 135 | 0.03 | 128.25 | **10** | 287 | **273** | **150** | **524** |
| **Short span at bottom** | 12.23 | 135 | 0.02 | 128.25 | **10** | 228 | **345** | **200** | **393** |
| **Long span at top** | 14.42 | 125 | 0.03 | 118.75 | **10** | 278 | **283** | **150** | **524** |
| **Long span at bottom** | 10.91 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **200** | **393** |

**End Condition : One long edge discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.041 | 16.10 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.031 | 12.23 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.037 | 14.42 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.028 | 10.91 kNm |

***Reinforcement along the Short Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 135 mm |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever arm z | = | 128.25 mm |
| Required area of the steel, $A_s$ | = | 287 $mm^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 273 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | 524 $mm^2$ |
| Provided % of steel | = | 0.39 % **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**

| | | |
|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ |
| Lever arm z | = | 128.25 mm |
| Required area of the steel, $A_s$ | = | 218 $mm^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 360 mm |
| Provided spacing of reinf. | = | **200** mm |
| Provided area of the steel | = | 393 $mm^2$ |
| Provided % of steel | = | 0.39 % **Provided area of steel is OK** |

**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | |
|---|---|---|
| Available effective depth, $d_{eff}$ | = | 125 mm |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** $N/mm^2$ OK |
| K' | = | 0.156 (If redistribution not exceed 10%) |
| Lever Arm z | = | 118.75 mm |
| Required area of the steel, $A_s$ | = | 278 $mm^2$ |
| Diameter of bar | = | **10** mm |
| Required spacing of reinf. | = | 283 mm |
| Provided spacing of reinf. | = | **150** mm |
| Provided area of the steel | = | **524** $mm^2$ |
| Provided % of steel | = | 0.42 % **Provided area of steel is OK** |

| | | |
|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ |
| Lever arm z | = | 118.75 mm |
| Required area of the steel, $A_s$ | = | 210 $mm^2$ |
| Diameter of bar | = | **10** mm |

| | | | |
|---|---|---|---|
| Required spacing of reinf. | = | **374** mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | **393** $mm^2$ | |
| Provided % of steel | = | 0.32 % | **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**
**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | |
|---|---|---|---|
| Length of shorter span | = | 6.5 m | |
| Actul l/d ratio | = | 26 | |
| Provided % of steel at bottom along short span | = | 0.390 % | |
| $f_s =$ (2*$f_y$*Req. steel)/(3*Provided steel) | = | 170 $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | |
| Required effective depth | = | 125 mm | |
| Available effective depth | = | 135 mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| **Location** | **Coeff.** | **S.Force V (kN)** | ***v*** **$N/mm^2$** | **% of steel** | ***$v_c$*****, $N/mm^2$** | **Enhanced *$v_c$*, $N/mm^2$** | **$A_{sv}/s_v$** | **Remarks** |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.369** | 22.13 | 0.17 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.247** | 14.80 | 0.11 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.360** | 21.57 | 0.17 | 0.42 | 0.71 | 1.42 | 0 | No. shear reinf. |

## Design of Slab Panel - RS4

***Material Properties***

| | | | |
|---|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab | = | 175 | mm |
| Clear cover | = | 35 | mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **6** |
| Long span of the slab, $l_y$ | **6.65** |
| Span ratio, $l_y/l_x$ | **1.11** |

**TWO WAY SLAB**

**Edge Condition** Interior Panel

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 4.2 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **5.9** | $kN/m^2$ |
| Live load, Qk | = | **0.6** | $kN/m^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **9.22** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 178 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **26** |
| $d_{eff}$. Reqd | 116 |
| $D_{reqd}$, mm | 156 |
| $D_{prov.}$ mm | 175 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar | Required Reinforcement | | Provided Reinforcement | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | ϕ | As, $mm^2$ | Spacing | Spacing | As, $mm^2$ |
| **Short span at top** | 12.42 | 135 | 0.02 | 128.25 | **10** | 228 | **345** | **150** | **524** |
| **Short span at bottom** | 9.41 | 135 | 0.02 | 128.25 | **10** | 228 | **345** | **200** | **393** |
| **Long span at top** | 10.63 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **150** | **524** |
| **Long span at bottom** | 7.97 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **200** | **393** |

**End Condition : Interior panel**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.037 | 12.42 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.028 | 9.41 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.032 | 10.63 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.024 | 7.97 kNm |

***Reinforcement along the Short Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 135 mm | |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever arm z | = | 128.25 mm | |
| Required area of the steel, $A_s$ | = | 222 $mm^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 354 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | 524 $mm^2$ | |
| Provided % of steel | = | 0.39 % | **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**

| | | | |
|---|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ | |
| Lever arm z | = | 128.25 mm | |
| Required area of the steel, $A_s$ | = | 168 $mm^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 468 mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | 393 $mm^2$ | |
| Provided % of steel | = | 0.39 % | **Provided area of steel is OK** |

**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 125 mm | |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever Arm z | = | 118.75 mm | |
| Required area of the steel, $A_s$ | = | 205 $mm^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 383 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | **524** $mm^2$ | |
| Provided % of steel | = | 0.42 % | **Provided area of steel is OK** |

| | | | |
|---|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ | |
| Lever arm z | = | 118.75 mm | |
| Required area of the steel, $A_s$ | = | 154 $mm^2$ | |
| Diameter of bar | = | **10** mm | |

| | | | | |
|---|---|---|---|---|
| Required spacing of reinf. | = | **512** | mm | |
| Provided spacing of reinf. | = | **200** | mm | |
| Provided area of the steel | = | **393** | $mm^2$ | |
| Provided % of steel | = | 0.32 | % | **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**
**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | | |
|---|---|---|---|---|
| Length of shorter span | = | 6 | m | |
| Actul l/d ratio | = | 26 | | |
| Provided % of steel at bottom along short span | = | 0.390 | % | |
| $f_s = (2*f_y$*Req. steel)/(3*Provided steel) | = | 131 | $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | | |
| Required effective depth | = | 115 | mm | |
| Available effective depth | = | 135 | mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.363** | 20.05 | 0.15 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.000** | 0.00 | 0 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.330** | 18.26 | 0.15 | 0.42 | 0.71 | 1.42 | 0 | No. shear reinf. |

## Design of Slab Panel - RS5

***Material Properties***

| | | | |
|---|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab | = | 175 | mm |
| Clear cover | = | 35 | mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **5.04** |
| Long span of the slab, $l_y$ | **6.5** |
| Span ratio, $l_y/l_x$ | **1.29** |

**TWO WAY SLAB**

**Edge Condition** Two Adjacent Edges Discont.

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 4.2 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **5.9** | $kN/m^2$ |
| Live load, Qk | = | **0.6** | $kN/m^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **9.22** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 178 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 110 |
| $D_{reqd}$, mm | 150 |
| $D_{prov.}$ mm | 175 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar | Required Reinforcement | | Provided Reinforcement | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\phi$ | As, $mm^2$ | Spacing | Spacing | As, $mm^2$ |
| **Short span at top** | 16.02 | 135 | 0.03 | 128.25 | **10** | 286 | **275** | **150** | **524** |
| **Short span at bottom** | 11.85 | 135 | 0.02 | 128.25 | **10** | 228 | **345** | **200** | **393** |
| **Long span at top** | 10.54 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **150** | **524** |
| **Long span at bottom** | 7.97 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **200** | **393** |

**End Condition : Two adjecent edges discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.068 | 16.02 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.051 | 11.85 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.045 | 10.54 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.034 | 7.97 kNm |

***Reinforcement along the Short Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 135 mm | |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.03** N/mm$^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever arm z | = | 128.25 mm | |
| Required area of the steel, $A_s$ | = | 286 mm$^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 275 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | 524 mm$^2$ | |
| Provided % of steel | = | 0.39 % | **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**

| | | | |
|---|---|---|---|
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ | |
| Lever arm z | = | 128.25 mm | |
| Required area of the steel, $A_s$ | = | 211 mm$^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 372 mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | 393 mm$^2$ | |
| Provided % of steel | = | 0.39 % | **Provided area of steel is OK** |

**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Reinforcement along the Long Span***

| | | | |
|---|---|---|---|
| Available effective depth, $d_{eff}$ | = | 125 mm | |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever Arm z | = | 118.75 mm | |
| Required area of the steel, $A_s$ | = | 203 mm$^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 387 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | **524** mm$^2$ | |
| Provided % of steel | = | 0.42 % | **Provided area of steel is OK** |

| | | | |
|---|---|---|---|
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** N/mm$^2$ | |
| Lever arm z | = | 118.75 mm | |
| Required area of the steel, $A_s$ | = | 154 mm$^2$ | |
| Diameter of bar | = | **10** mm | |

| | | | |
|---|---|---|---|
| Required spacing of reinf. | = | **512** mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | **393** $mm^2$ | |
| Provided % of steel | = | 0.32 % | **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**
**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | |
|---|---|---|---|
| Length of shorter span | = | 5.04 m | |
| Actul l/d ratio | = | 23 | |
| Provided % of steel at bottom along short span | = | 0.390 % | |
| $f_s = (2*f_y$*Req. steel)/(3*Provided steel) | = | 165 $N/mm^2$ | |
| $0.55+[(477-f_s)/(120*(0.9+M/bd^2))]$ | = | 2.00 | |
| Required effective depth | = | 110 mm | |
| Available effective depth | = | 135 mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| **Location** | **Coeff.** | **S.Force V (kN)** | **$v$ $N/mm^2$** | **% of steel** | **$v_c$, $N/mm^2$** | **Enhanced $v_c$, $N/mm^2$** | **$A_{sv}/s_v$** | **Remarks** |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.497** | 23.09 | 0.18 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.328** | 15.24 | 0.12 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.400** | 18.59 | 0.15 | 0.42 | 0.71 | 1.42 | 0 | No. shear reinf. |

## Design of Slab Panel - RS6

***Material Properties***

| | | | |
|---|---|---|---|
| Charactesistic strength of concrete, $f_{cu}$ | = | 40 | $N/mm^2$ |
| Charactesistic strength of steel, $f_y$ | = | 460 | $N/mm^2$ |
| Concrete Density, $\gamma_c$ | = | 24 | $kN/m^3$ |

***Data***

| | | | |
|---|---|---|---|
| Thickness of slab | = | 175 | mm |
| Clear cover | = | 35 | mm |

| | |
|---|---|
| Short span of the slab, $l_x$ | **5.04** |
| Long span of the slab, $l_y$ | **6** |
| Span ratio, $l_y/l_x$ | **1.19** |

**TWO WAY SLAB**

**Edge Condition** One long Edge Distcont.

***Loading***

| | | | |
|---|---|---|---|
| Self weight of slab | = | 4.2 | $kN/m^2$ |
| Floor finish | = | 1.2 | $kN/m^2$ |
| Partitions | = | 0 | $kN/m^2$ |
| M&E services | = | 0.5 | $kN/m^2$ |
| Total Dead load, Gk | = | **5.9** | $kN/m^2$ |
| Live load, Qk | = | **0.6** | $kN/m^2$ |
| Design load, N= 1.4*Gk+1.6*Qk | = | **9.22** | $kN/m^2$ |

| **Deflection check** | |
|---|---|
| $f_s$, $N/mm^2$ | 178 |
| M.F. | 2.00 |
| Span/$d_{eff}$ | **23** |
| $d_{eff}$. Reqd | 110 |
| $D_{reqd}$, mm | 150 |
| $D_{prov.}$ mm | 175 |
| **Safe in deflection** | |

| Location | B.M., kN.m | $d_{eff}$, mm | K = $M/f_{cu}bd^2$ | Lever Arm z | Dia of bar | Required Reinforcement | | Provided Reinforcement | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\phi$ | As, $mm^2$ | Spacing | Spacing | As, $mm^2$ |
| **Short span at top** | 12.96 | 135 | 0.02 | 128.25 | **10** | 231 | **340** | **150** | **524** |
| **Short span at bottom** | 9.71 | 135 | 0.02 | 128.25 | **10** | 228 | **345** | **200** | **393** |
| **Long span at top** | 8.67 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **150** | **524** |
| **Long span at bottom** | 6.56 | 125 | 0.02 | 118.75 | **10** | 228 | **345** | **200** | **393** |

**End Condition : One long edge discontinous**

Coefficients from BS 8110-1, Table 3.14

| | | | |
|---|---|---|---|
| **Along short span at top** | $\beta_{sx}$ (-) | 0.055 | 12.96 kNm |
| **Along short span at bottom** | $\beta_{sx}$ (+) | 0.041 | 9.71 kNm |
| **Along long span at top** | $\beta_{sy}$ (-) | 0.037 | 8.67 kNm |
| **Along long span at bottom** | $\beta_{sy}$ (+) | 0.028 | 6.56 kNm |

| | | | |
|---|---|---|---|
| ***Reinforcement along the Short Span*** | | | |
| Available effective depth, $d_{eff}$ | = | 135 mm | |
| **At Top,K= $m_{sx}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever arm z | = | 128.25 mm | |
| Required area of the steel, $A_s$ | = | 231 $mm^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 340 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | 524 $mm^2$ | |
| Provided % of steel | = | 0.39 % | **Provided area of steel is OK** |
| **Provide 10T @ 150 c/c at top (Negative Reinf.)** | | | |
| | | | |
| **At bottom, K=$m_{sx}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ | |
| Lever arm z | = | 128.25 mm | |
| Required area of the steel, $A_s$ | = | 173 $mm^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 454 mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | 393 $mm^2$ | |
| Provided % of steel | = | 0.39 % | **Provided area of steel is OK** |
| **Provide 10T @ 200 c/c at bottom (Positive Reinf.)** | | | |
| | | | |
| ***Reinforcement along the Long Span*** | | | |
| Available effective depth, $d_{eff}$ | = | 125 mm | |
| **K= $m_{sy}$ (Top) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ OK | |
| K' | = | 0.156 (If redistribution not exceed 10%) | |
| Lever Arm z | = | 118.75 mm | |
| Required area of the steel, $A_s$ | = | 167 $mm^2$ | |
| Diameter of bar | = | **10** mm | |
| Required spacing of reinf. | = | 470 mm | |
| Provided spacing of reinf. | = | **150** mm | |
| Provided area of the steel | = | **524** $mm^2$ | |
| Provided % of steel | = | 0.42 % | **Provided area of steel is OK** |
| | | | |
| **K= $m_{sy}$ (Bot) /b*$d^2$*$f_{cu}$** | = | **0.02** $N/mm^2$ | |
| Lever arm z | = | 118.75 mm | |
| Required area of the steel, $A_s$ | = | 126 $mm^2$ | |
| Diameter of bar | = | **10** mm | |

| | | | |
|---|---|---|---|
| Required spacing of reinf. | = | **622** mm | |
| Provided spacing of reinf. | = | **200** mm | |
| Provided area of the steel | = | **393** $mm^2$ | |
| Provided % of steel | = | 0.32 % | **Provided area of steel is OK** |

**Provide 10T @ 150 c/c at top (Negative Reinf.)**
**Provide 10T @ 200 c/c at bottom (Positive Reinf.)**

***Check for Deflection*** BS8110- Section 3.4.6.5, Table 3.10

| | | | |
|---|---|---|---|
| Length of shorter span | = | 5.04 m | |
| Actul l/d ratio | = | 23 | |
| Provided % of steel at bottom along short span | = | 0.390 % | |
| $f_s$ = (2*$f_y$*Req. steel)/(3*Provided steel) | = | 135 $N/mm^2$ | |
| 0.55+[(477-$f_s$)/(120*(0.9+M/$bd^2$))] | = | 2.00 | |
| Required effective depth | = | 110 mm | |
| Available effective depth | = | 135 mm | **Safe in Deflection** |

***Check for shear*** Shear coefficients from Table 3.15
$v_c$ from Table 3.8

| Location | Coeff. | S.Force V (kN) | $v$ $N/mm^2$ | % of steel | $v_c$, $N/mm^2$ | Enhanced $v_c$, $N/mm^2$ | $A_{sv}/s_v$ | Remarks |
|---|---|---|---|---|---|---|---|---|
| **Along shorter direction at continuous edge** | **0.436** | 20.27 | 0.16 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along shorter direction at discontinuous edge** | **0.288** | 13.39 | 0.1 | 0.39 | 0.68 | 1.36 | 0 | No. shear reinf. |
| **Along longer direction at continuous edge** | **0.360** | 16.73 | 0.13 | 0.42 | 0.71 | 1.42 | 0 | No. shear reinf. |

# 7 Base Plate Design

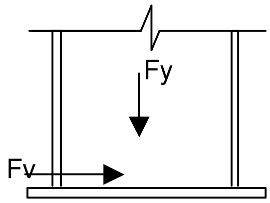
# Base Plate Design Type-I

Load case No. 214 214 1.4DL+1.4WL5 [Ref. Node on Staad = 14 ]

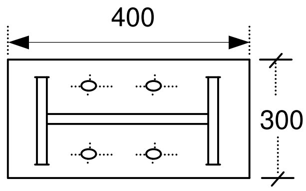
**Applied Load**

| | | **Force** (kN) | **Nature** |
|---|---|---|---|
| Axial force (P) | = | 126.439 | Compression |
| | = | 54.223 | Tension |
| Horizontal Shear (V) | = | 18.038 | Compression |
| Horizontal Shear (V) | = | 25.76 | Tension |

**Column Section**




| | | | |
|---|---|---|---|
| Depth ,( D ) | = | 300 | mm |
| Width ,( B ) | = | 200 | mm |
| Flange thickness ,( tf ) | = | 10 | mm |
| Web thickness , (tw) | = | 6 | mm |




**Bolt**

| | | | |
|---|---|---|---|
| Bolt grade | | 8.8 | |
| Bolt size | | M 16 | |
| No. of bolt required (n) | = | 4 | nos. |
| Tension capacity , $T_b$ | = | 88 | kN |
| Shear capacity ,vb | = | 59 | kN |
| Shear in each bolt | = | $\sqrt{(18.038^2+25.76^2)} / 4$ | |
| | = | 7.87 | kN |

< Shear capacity Hence OK

| | | | |
|---|---|---|---|
| Tension in each bolt | = | 54.223/4 | |
| | = | 13.56 | kN |

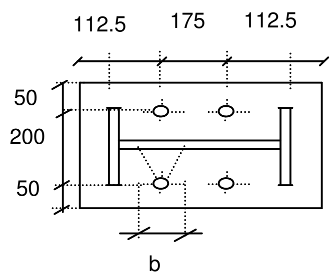
< Tension capacity Hence OK

**Combined check for tension and shear**

(Actual tension /Capacity in tension )+(Actual Shear /Capacity in Shear ) < 1.4

13.56 / 88 + 7.87 / 59 < 1.4
0.29 < 1.4, Hence Ok

**<u>Base Plate</u>**




| | | | |
|---|---|---|---|
| Width of Plate ,($B_P$) | = | 300 | mm |
| Length of Plate , ($L_P$) | = | 400 | mm |
| Thicknees of Plate ,($T_P$) | = | 20 | mm |
| $p_{yp}$ | = | 345 | N/mm2 |
| fcu | = | 30 | N/mm2 |
| 0.6 fcu | = | 18 | N/mm2 |
| Tension in one bolt | = | 13.56 | kN |
| Edge distance ( e ) | = | 47 | mm |
| Resisting width by 30 deg | = | 16 + 2x47 x Tan(30) | |
| disperssion | = | 70.28 | mm |

| | | | |
|---|---|---|---|
| Moment due to tension on bolt | = | 13.56 x 47 | |
| | = | 637.32 | kNmm |

Thickness of base plate reqrd ($t_p$) = $\sqrt{6xM_{max}/ p_y \text{ x } b}$

= $\sqrt{6x637.3210\text{^}3/(.66x345x70.28)}$

= 15.46 mm

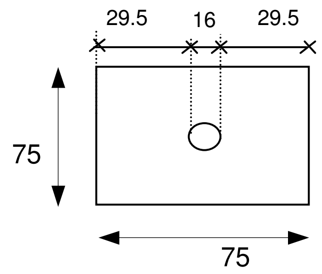
Design by the effective area method

Area of flange = 2000 mm2
x-sectional area = 5680 mm2
Column perimeter = 1388 mm
Area of base Plate = 120000 mm2
Area Required = 7024.39 mm2

Bearing Area = Hatched Area
$4c^2$ + ( column perimeter )x c + column area = Hatched Area
4c2 + 1388 x c + 5680 = 7024.39
c = 1 mm

Thickness of Base Plate ( $t_p$ ) = $c[3w/p_{yp}]^{0.5}$
Reqd. = 0.4 mm
Provide thickness of base plate = 20 mm
> Reqrd, Hence Ok !

**Design of anchor plate**




Maxm tension force on each bolt 13.56 kN

Size of anchor plate = 75 x 75
Pressure on anchor plate = 13.56/(75 x 75)
due to bolt = 2410.67 kN/mm$^2$

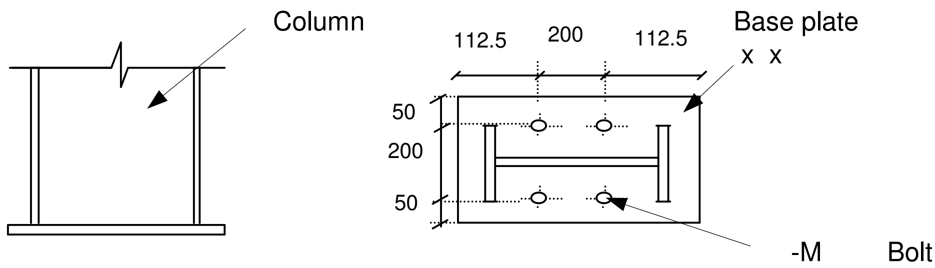
Edge dist from , e = 29.5 mm
outer face of bolt

Bending moment at = 2410.67x29.5^2/2x75
outer face of bolt = 78.68 kNm

Bending stress = 0.66 $p_y$
= 227.7 N/mm2

Thickness reqrd of anchor plate = $\sqrt{6 \text{ x } 78.68/(75 \text{ x } 227.7)}$
= 5.26 mm
Provide thick of anchor plate = 10 mm Hence OK

**Figure:**



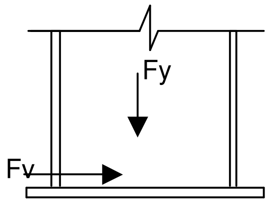
## Base Plate Design Type-2 [Roof level]    [Ref. Node on Staad = 96]

Load case No. 207    1.2DL+1.2LL+1.2WL3

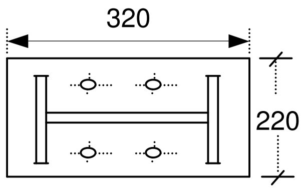
**Applied Load**

| | | Force (kN) | Nature |
|---|---|---|---|
| Axial force (P) | = | 116.52 | Compression |
| | = | 30.58 | Tension |
| Horizontal Shear (V) | = | 21.812 | Compression |
| Horizontal Shear (V) | = | 7.087 | Tension |

**Column Section**




| | | | |
|---|---|---|---|
| Depth ,( D ) | = | 300 | mm |
| Width ,( B ) | = | 200 | mm |
| Flange thickness ,( tf ) | = | 8 | mm |
| Web thickness , (tw) | = | 6 | mm |




**Bolt**

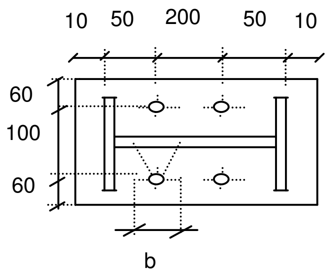
| | | | |
|---|---|---|---|
| Bolt grade | | 8.8 | |
| Bolt size | | M 16 | |
| No. of bolt required (n) | = | 4 | nos. |
| Tension capacity , $T_b$ | = | 88 | kN |
| Shear capacity ,$v_b$ | = | 59 | kN |
| Shear in each bolt | = | $\sqrt{(21.812^2+7.087^2)}$" / 4 | |
| | = | 5.74 | kN |
| | | < Shear capacity | Hence OK |
| Tension in each bolt | = | 30.58/4 | |
| | = | 7.65 | kN |
| | | < Tension capacity | Hence OK |

**Combined check for tension and shear**

(Actual tension /Capacity in tension )+(Actual Shear /Capacity in Shear ) < 1.4

7.65 / 88 + 5.74 / 59 < 1.4
0.19 < 1.4, Hence Ok

**<u>Base Plate</u>**




| | | | |
|---|---|---|---|
| Width of Plate ,($B_{P}$) | = | 220 | mm |
| Length of Plate , ($L_{P}$) | = | 320 | mm |
| Thicknees of Plate ,($T_{P}$) | = | 20 | mm |
| $p_{yp}$ | = | 345 | N/mm2 |
| fcu | = | 30 | N/mm2 |
| 0.6 fcu | = | 18 | N/mm2 |
| Tension in one bolt | = | 7.65 | kN |
| Edge distance ( e ) | = | 47 | mm |
| Resisting width by 30 deg | = | 16 + 2x47Tan(30) | |
| disperssion | = | 70.28 | mm |
| Moment due to tension on bolt | = | 7.65 x 47 | |
| | = | 359.55 | kNmm |

Thickness of base plate reqrd ($t_p$) = $\sqrt{6xM_{max}/ p_y \times b}$
= $\sqrt{6x359.5510\text{^}3/(.66x345x70.28)}$
= 11.62 mm

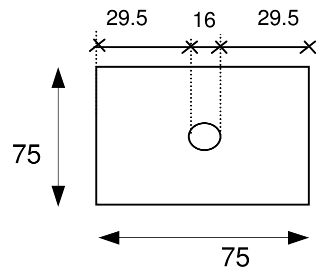
Design by the effective area method

| | | | |
|---|---|---|---|
| Area of flange | = | 1600 | mm2 |
| x-sectional area | = | 4904 | mm2 |
| Column perimeter | = | 1388 | mm |
| Area of base Plate | = | 70400 | mm2 |
| Area Required | = | 6473.34 | mm2 |

Bearing Area = Hatched Area
$4c^2$ + ( column perimeter )x c + column area = Hatched Area
4c2 + 1388 x c + 4904 = 6473.34
c = 1.2 mm

Thickness of Base Plate ( $t_p$ ) = $c[3w/p_{yp}]^{0.5}$
Reqd. = 0.5 mm
Provide thickness of base plate = 20 mm
> Reqrd, Hence Ok !

**Design of anchor plate**




Maxm tension force on each bolt 7.65 kN

Size of anchor plate = 75 x 75
Pressure on anchor plate = 7.65/(75 x 75)
due to bolt = 1360 $kN/mm^2$

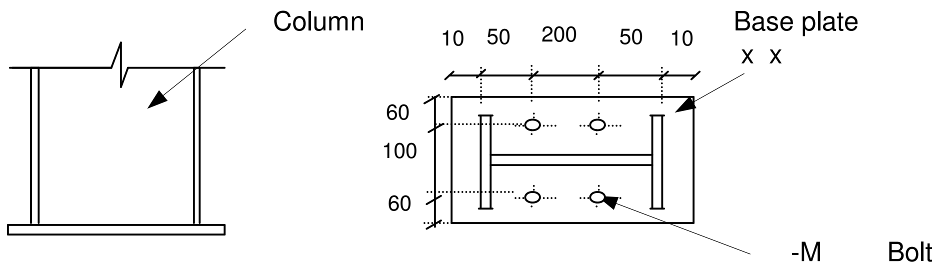
Edge dist from , e = 29.5 mm
outer face of bolt

Bending moment at = 1360x29.5^2/2x75
outer face of bolt = 44.39 kNm

Bending stress = 0.66 $p_y$
= 227.7 N/mm2

Thickness reqrd of anchor plate = $\sqrt{6 \times 44.39/(75 \times 227.7)}$
= 3.95 mm
Provide thick of anchor plate = 10 mm Hence OK

**Figure:**



## Base Plate Design Type-3 [Roof level]

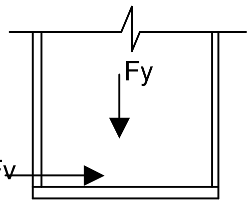
[Ref. Node on Staad = 109 ]

Load case No. 207 1.2DL+1.2LL+1.2WL3

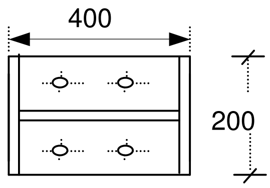
**Applied Load**

| | | Force (kN) | Nature |
|---|---|---|---|
| Axial force (P) | = | 119.699 | Compression |
| | = | 66.961 | Tension |

**Column Section**




| | | | |
|---|---|---|---|
| Depth ,( D ) | = | 400 | mm |
| Width ,( B ) | = | 200 | mm |
| Flange thickness ,( tf ) | = | 10 | mm |
| Web thickness , (tw) | = | 10 | mm |




**Bolt**

| | | | |
|---|---|---|---|
| Bolt grade | | 8.8 | |
| Bolt size | | M 16 | |
| No. of bolt required (n) | = | 4 | nos. |
| Tension capacity , $T_b$ | = | 88 | kN |
| Shear capacity ,$v_b$ | = | 59 | kN |

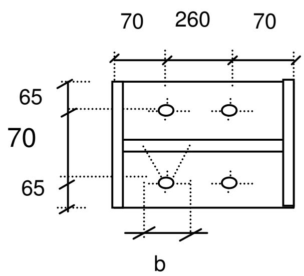
Tension in each bolt = 66.961/4
= 16.75 kN
< Tension capacity Hence OK

**Combined check for tension and shear**

(Actual tension /Capacity in tension )+(Actual Shear /Capacity in Shear ) < 1.4

16.75 / 88 < 1.4
0.2 < 1.4, Hence Ok

**Base Plate**



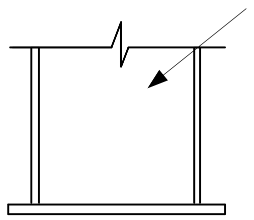

| | | | |
|---|---|---|---|
| Width of Plate ,($B_{P}$) | = | 200 | mm |
| Length of Plate , ($L_{P}$) | = | 400 | mm |
| Thicknees of Plate ,($T_{P}$) | = | 20 | mm |
| $p_{yp}$ | = | 345 | N/mm2 |
| fcu | = | 30 | N/mm2 |
| 0.6 fcu | = | 18 | N/mm2 |
| Tension in one bolt | = | 16.75 | kN |
| Edge distance ( e ) | = | 30 | mm |
| Resisting width by 30 deg | = | 16 + 2x30Tan(30) | |
| disperssion | = | 50.65 | mm |

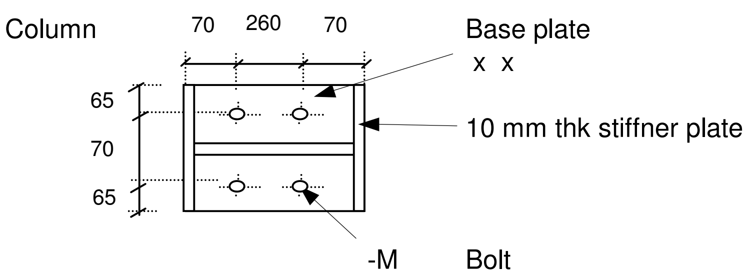
Moment due to tension on bolt = 16.75 x 30
= 502.5 kNmm

Thickness of base plate ($t_p$) reqrd = $\sqrt{6xM_{max}/\ p_y\ x\ b}$
= $\sqrt{6x502.510\text{^}3/(.66x345x50.65)}$
= 16.17 mm

Provide thickness of base plate = 20 mm
> Reqrd, Hence Ok !

**Figure:**





## Base Plate Design Type-4 [Roof level]

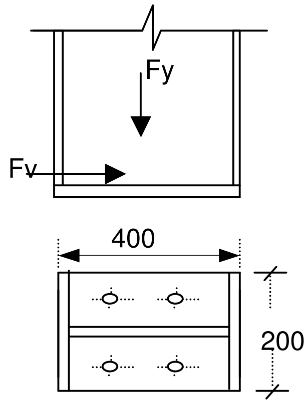
[Ref. Node on Staad = 116 ]

Load case No. 215 1DL+1.4WL5

**Applied Load**

| | | Force (kN) | Nature |
|---|---|---|---|
| Axial force (P) | = | 45.735 | Compression |
| | = | 45.782 | Tension |

**Column Section**




| | | | |
|---|---|---|---|
| Depth ,( D ) | = | 400 | mm |
| Width ,( B ) | = | 200 | mm |
| Flange thickness ,( tf ) | = | 10 | mm |
| Web thickness , (tw) | = | 10 | mm |

**Bolt**

| | | | |
|---|---|---|---|
| Bolt grade | | 8.8 | |
| Bolt size | | M 16 | |
| No. of bolt required (n) | = | 4 | nos. |
| Tension capacity , $T_b$ | = | 88 | kN |
| Shear capacity ,$v_b$ | = | 59 | kN |

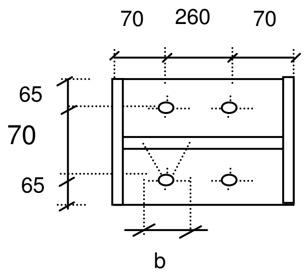
Tension in each bolt = 45.782/4
= 11.4455 kN
< Tension capacity Hence OK

**Combined check for tension and shear**

(Actual tension /Capacity in tension )+(Actual Shear /Capacity in Shear ) < 1.4

11.4455 / 88 < 1.4
0.14 < 1.4, Hence Ok

**Base Plate**




| | | | |
|---|---|---|---|
| Width of Plate ,($B_{P}$) | = | 200 | mm |
| Length of Plate , ($L_{P}$) | = | 400 | mm |
| Thicknees of Plate ,($T_{P}$) | = | 20 | mm |
| $p_{yp}$ | = | 345 | N/mm2 |
| fcu | = | 30 | N/mm2 |
| 0.6 fcu | = | 18 | N/mm2 |
| Tension in one bolt | = | 11.4455 | kN |
| Edge distance ( e ) | = | 30 | mm |
| Resisting width by 30 deg | = | 16 + 2x30Tan(30) | |
| disperssion | = | 50.65 | mm |

Moment due to tension on bolt = 11.4455 x 30
= 343.365 kNmm

| | | | |
|---|---|---|---|
| Thickness of base plate ($t_p$) reqrd | = | $\sqrt{6xM_{max}/\ p_y\ x\ b}$ | |
| | = | $\sqrt{6x343.36510\text{^}3/(.66x345x50.65)}$ | |
| | = | 13.37 | mm |
| Provide thickness of base plate | = | 20 | mm |
| | | | > Reqrd, Hence Ok ! |

**Design of anchor plate**

| | | | | |
|---|---|---|---|---|
| Maxm tension force on each bolt | = | 11.4455 | kN | |
| Size of anchor plate | = | 75 x 75 | | |
| Pressure on anchor plate | = | 11.4455/(75 x 75) | | |
| due to bolt | = | 2034.76 | $kN/mm^2$ | |
| Edge dist from , e outer face of bolt | = | 29.5 | mm | |
| Bending moment at | = | 2034.76x29.5^2/2x75 | | |
| outer face of bolt | = | 66.41 | kNm | |
| Bending stress | = | 0.66 $p_y$ | | |
| | = | 227.7 | N/mm2 | |
| Thickness reqrd of anchor plate | = | $\sqrt{6\ x\ 66.41/(75\ x\ 227.7)}$ | | |
| | = | 4.84 | mm | |
| Provide thick of anchor plate | = | 10 | mm | Hence OK |

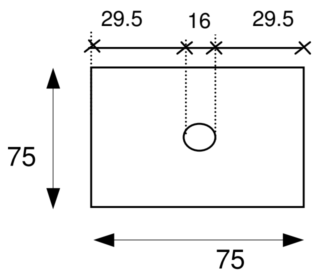
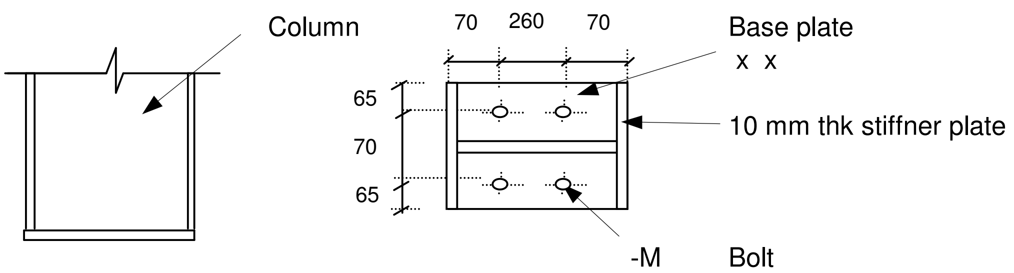



**Figure:**



# 8 Connection Design

## Moment connection of Beam to Rafter

**For Beams** **SB2** [Ref. Node on Staad 46 ]

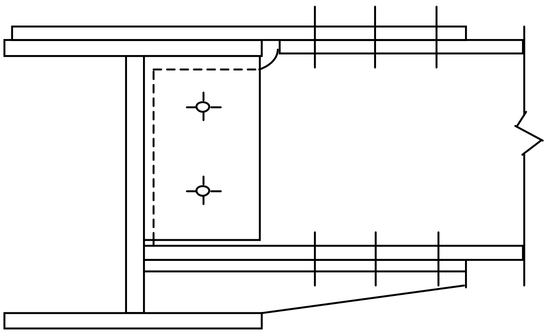
| | | | | | |
|---|---|---|---|---|---|
| Data : | Load case | 207 1.2DL+1.2LL+1.2WL3 | | | |
| Section | | Beam | SB2 Rafter | | MF1 |
| Properties of section | | h | 300 mm | | 400 mm |
| | | b | 150 mm | | 200 mm |
| | | $t_w$ | 6 mm | | 6 mm |
| | | $t_f$ | 10 mm | | 10 mm |
| | | r mm | 0 mm | | 0 mm |

| | | | |
|---|---|---|---|
| Forces : | Axial force N | 3.389 | kN |
| | Vert shear force T | 34.624 | kN |
| | Hori shear force H | 0 | kN |
| | Majr axis moment(M) | 44.724 | kN-m |

Geometrical data :

| | | |
|---|---|---|
| Bolt grade | 8.8 | |
| Shear stgth of Bolt $p_s$ | 375 | $N/mm^2$ |
| Tension stgth $p_t$ | 560 | $N/mm^2$ |
| Dia of bolt | M 20 | |
| Net Area of Bolt $A_{net}$ | 245 | $mm^2$ |
| Steel grade | S 345 | |
| Bearing stgth $p_{bs}$ | 460 | $N/mm^2$ |
| Ult tensile stgth($U_s$) | 410 | $N/mm^2$ |

Calculation :



Lever Arm = 300 mm
Bolt Tension due to M = Moment / Lev arm
( T ) = 44.724x10^6/ 300
= 149.1 kN
Total Tension in top plate = N/2 + T
= 3.389/2 + 149.1 = 150.8 kN
Therefore total shear in Bolt = 150.8 kN

Dia of Bolt for shear = M 20 $A_{net}$ = 245
Shear capacity of one bolt = $A_{net}$ x $p_s$ /1000
= 245x375 /1000
= 91.9 kN
No of bolts reqd for shear = V / Shear capacity of Bolt
= 150.7945/91.9
= 1.60
Provide no of bolts = **3** No's ok
(At Top & same at Botm)

Shear in each bolt = 50.26 kN

***Flange Plate***

Trial plate thk (top & botm) = 15 mm

Bearing strength of plate = (22x 15x460/1000

151.8 kN ok

Tensile strenght of plate = Net c/s plt area x 345/1000

Width of plate provided = 150 mm

Length of plate provided = 435 mm

Net C/s area of plate = (150-22x2)x15 = 1590 $mm^2$

Thrfore Tensile strength = (1590x345/1000)

(of plate) = 548.55 kN

> Actual tensn .: ok

***Fin Plate***

dia of bolt M 20

Net Area of Bolt $A_{net}$ 245 $mm^2$

Shear capacity of one bolt = $A_{net}$ x $p_s$ /1000

= 245x375/1000

= 92 kN

No of bolt reqrd T/ 92 = 0.4 No's

Provide NO of bolts n = 2

.:Shear in each bolt 17.312 kN

Pitch p = 100 mm

Trial thickness of plate *t* 15 mm

Provide length of plate (0.6x$d_{section}$) 280 mm

Width of plate = (2 x Edg dist + End projction)

= 2x2(20)+10 90

Provide width of plate = 90 mm

Size of plate = 90 x 280

Block shear = $0.6p_yt(L_v+K_e(L_t\text{-}kD_h))$

= 387.504 kN

> Actual shear, .: OK !

Bearing strength of plate = *t* x hole dia x $p_{bs}$

= 15x22x460/1000

= 151.8 kN

> Shear in 1 bolt .: ok

Bearing strength of supported *t* x hole dia x $p_{bs}$

beam web = 6x22x460/1000

= 60.72 KN

Force in bolt due to moment ($F_{sm}$) $F_v$ x a / $Z_{bg}$

Where $Z_{bg}$ = n(n+1)p/6 = 100

$F_{sm}$ = 8.66 kN

Force acting on bolt group $(F_{sv}^2+F_{sa}^2+F_{sm}^2)^{1/2}$

= ((34.624/2)^2 + (3.389/2^2)+8.66^2)^0.5

= 19.44 kN OK in Bearing

Thickness of fin plate > or = 0.15a .: ok in torsional bucking

where a = 2x20+10 = 50 mm

**Welding calc**

Trial Size of weld a 6 mm

Design weld stgth $p_w$ 220 N/mm$^2$

Strength of weld/mm = 0.7a x $p_w$

= 0.7x6x220

= 924

Length of weld reqrd = $(T^2+N^2)^{1/2}$ x $10^3$ /Strgth of weld/mm

= 37.7

Length of plate available for weld 280 x 2

> reqrd .: ok

## ***Check for shear capacity of supported Beam flange and web***

Block shear capacity of flange = $0.6p_yt(L_v+K_e(L_t\text{-}kD_h))$

= 0.6x345x10(45+1.2(40-0.5x22))/1000

= 165.19 kN

> Actual shear, .: OK !

Block shear capacity of web = $0.6p_yt(L_v+K_e(L_t\text{-}kD_h))$

= 155.01 kN

> Actual shear, .: OK !

## ***Local shear check of Rafter web***

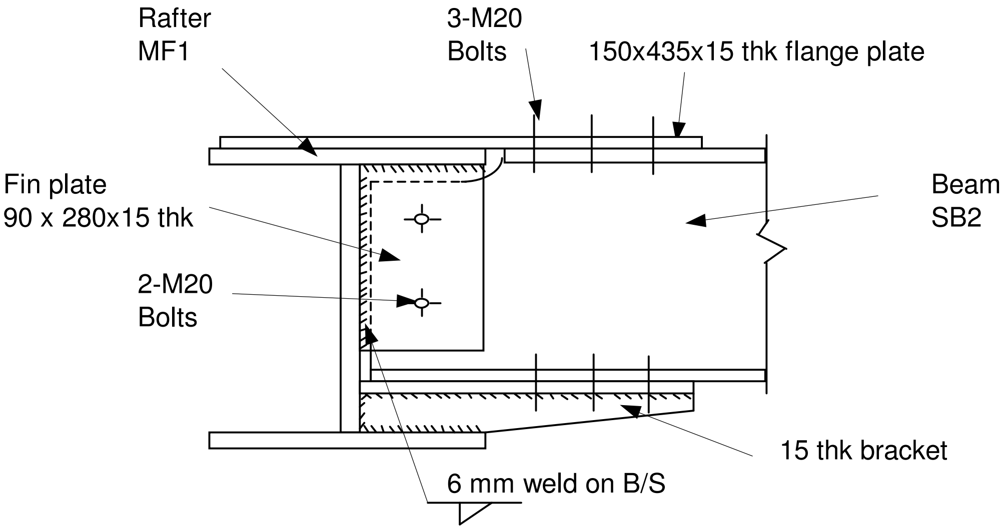
I ) Local shear failure of the rafter web

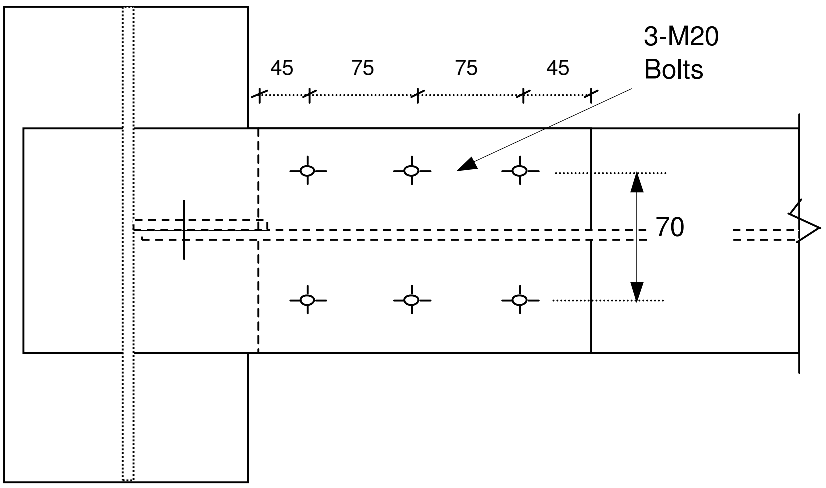
Local shear capacity = $0.6SA_v$

= 447.12 kN

> Actual beam Reaction .: Ok

Figure :




Moment connection of Beam SB2 with Rafter MF1

**_Rafter to Secondary beam connection_** **_SB2_** [Ref. Node on Staad 33 ]

Data : Load case 207 1.2DL+1.2LL+1.2WL3

Axial force N = 5.396 kN

shear T = 25.9 kN

| Section Property | Beam | SB2 | Rafter | MF11 |
|---|---|---|---|---|
| | h | 300 | mm | 400 |
| | b | 150 | mm | 200 |
| | $t_w$ | 6 | mm | 6 |
| | $t_f$ | 10 | mm | 10 |
| | r | 0 | mm | 0 |

Calculations :

| | | |
|---|---|---|
| Bolt grade | 8.8 | |
| Shear stgth of Bolt $p_s$ | 375 | $N/mm^2$ |
| Tension stgth $p_t$ | 560 | $N/mm^2$ |
| Dia of bolt | M 20 | |
| Dia of hole | 22 | mm |
| Net Area of Bolt $A_{net}$ | 245 | $mm^2$ |
| Steel grade | S 345 | $N/mm^2$ |
| Bearing stgth $p_{bs}$ | 460 | $N/mm^2$ |
| Ult tensile stgth($U_s$) | 410 | $N/mm^2$ |

1 Shear capacity of one bolt = $A_{net}$ x $p_s$ /1000
= 345x375/1000
= 92 kN

No of bolt reqrd T/ 92 = 0.3 No's
Provide NO of bolts n = 2
.:Shear in each bolt = 12.95 kN
Pitch p = = 100 mm

2 Fin plate
Trial thickness of plate *t* = 15 mm
Provide length of plate (0.6xd<sub>section</sub>) = 280 mm
Bearing strength of plate = *t* x hole dia x $p_{bs}$
= 15x22x460/1000
= 151.8 kN
> Shear in 1 bolt .: ok

Provide width of plate = 90 mm
Size of plate = 90 x 280

**_Check_**

1. Check for bearing capacity of Bolt group
Vertical force on bolt due to shear $F_{sv}$ = T/n = 12.95
Force on bolt due to axial $F_{sa}$(kN) = N/n = 2.698
Force on bolt due to moment = $F_v$ x a / $Z_{bg}$
Where $Z_{bg}$ = n(n+1)p/6 = 100 $mm^3$
Force in bolt due to moment $F_{sm}$ = 12.95
Resultant shear $F_s$ = $(F_{sv}^2+F_{sa}^2+F_{sm}^2)^{1/2}$
= 18.52 kN

Bearing capa of fin plate $P_{bs}$ = $k_{bs}dtp_{bs}$
= 138 KN
Bearing capa of beam web $P_{bs}$ = $k_{bs}dtp_{bs}$
= 55.2 KN
Adequate .: Ok

2. Check for shear & bending capacity of Beam

***For shear***

I )Shear capa of the gross section = $0.6p_yA_v$
= (0.6x345x(((300-2x(10+0))x6)))
= 347.76 kN
ii)Shear capa of the net section = $0.7p_yK_eA_{v,net}$
= 0.7x345x1.2x(Av-2x22x6)
= 410.36 kN
iii)Block shear capacity = $0.6p_yt(L_v+K_e(L_t\text{-}kD_h))$
(0.6x345x6x((2x20+(2-1)x100+1.2x((2x20)-0.5x22))))
= 217.1 kN
Therefore Beam shear capacity = 217.1 kN
> Actual Shear .: Ok

***For Bending***

Applied moment T x a = 1.295 kN-m
Moment capacity $M_{cb}$ = = pyZ = 27.05 kN-m
adequate in bending .: Ok

3. Check for shear & bending capacity of fin plate

***For shear***

I )Shear capa of the gross section = $0.6p_yA_v$ = 782.46 kN
ii)Shear capa of the net section = $0.7p_yK_eA_{v,net}$
= 904.18 kN
iii)Block shear capacity = $0.6p_yt(L_v+K_e(L_t\text{-}kD_h))$
= 542.754
Therefore fin plate shear capacity Pv = 542.76 kN
> Actual Shear .: Ok

***For Bending***

Actual shear T = 25.9 and 0.6x Pv = 325.66 kN
T < 0.6xPv
Section is subjected to low shear .: Ok
Moment capacity Mc = $p_yt[2e+(n-1)p]^2/6$
= 67.62 kN-m
fin plate adequate .: Ok

4.Thickness of fin plate > or = 0.15a .: ok in torsional bucking
where a = 2x20+15 = 50 mm

6. Local shear check of Rafter web

I ) Local shear failure of the Rafter web

Local shear capacity = $0.6SA_v$

= 312.99 KN

> Actual beam Reaction .: Ok

3 Welding calc

Trial Size of weld a 6 mm

Design weld stgth $p_w$ 220 N/mm$^2$

Strength of weld/mm = $0.7a \times p_w$

= 0.7x6x220

= 924 N/mm

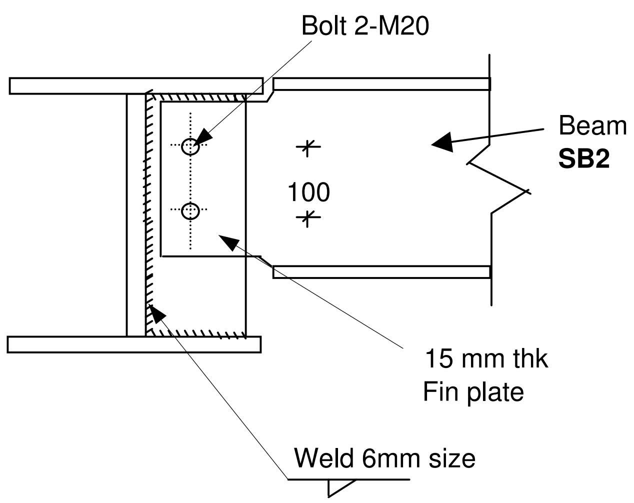
Length of weld reqrd = $(T^2+N^2)^{1/2} \times 10^3$ /Strgth of weld/mm

= 28.6 mm

Length of plate available for weld = 280 x 2 mm

> reqrd .: ok

Figure :



## Cap Plate Design Type-I

Load cse No.206 1DL+1.4WL2 [Ref. Node on Staad 8 ]

**Applied Load**

Data :

| | | | |
|---|---|---|---|
| Section | Stub col. | ST1 Beam | MF1 |
| Properties of section | h | 300 mm | 400 mm |
| | b | 200 mm | 200 mm |
| | $t_w$ | 6 mm | 6 mm |
| | $t_f$ | 8 mm | 10 mm |
| | r mm | 0 mm | 0 mm |

| | | | |
|---|---|---|---|
| Forces : | Axial force N | 21.701 | kN |
| | Vert shear force T | 3.95 | kN |
| | Hori shear force H | 2.6 | kN |
| | Majr axis moment(My) | 1.28 | kN-m |
| | Minor axis moment(Mz) | 1.98 | kN-m |

Geometrical data :

| | | |
|---|---|---|
| Bolt grade | 8.8 | |
| Shear stgth of Bolt $p_s$ | 375 | $N/mm^2$ |
| Tension stgth $p_t$ | 560 | $N/mm^2$ |
| Dia of bolt | M 16 | |
| Net Area of Bolt $A_{net}$ | 157 | $mm^2$ |
| Steel grade | S 345 | |
| Bolts Nos in Tension | 8 | = n |
| Tension capacity | 88 | kN |

Calculation :

**1. Bolt**

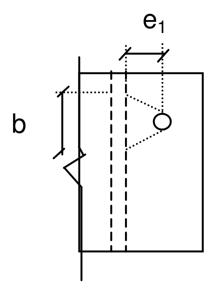
Tension in each bolt

| | | | |
|---|---|---|---|
| Due to axial force | = | 21.701/8 | |
| | = | 2.72 | kN |
| Due to moment | = | 1.28/(100 x 4)+1.98/200 | |
| | = | 13.1 | kN |
| Total tension in each bolt | = | 15.82 | kN |
| | | < 88 OK | |

**2. Cap plate**

| | |
|---|---|
| Trial thickness of plate *t* | 20 mm |
| Provide length of plate (0.6xd$_{section}$) | 500 mm |
| Width of Plate | 200 mm |

edge dist $e_1$ = 47 mm




Moment due to tension in each bolt M

= 15.82x47

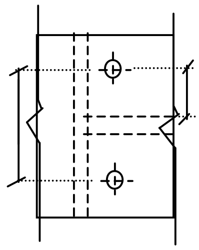
= 743.54 kNmm

Resisting width b = 16+2(47tan30)

By 30deg dispertion = 70.28 mm

Thickness required = $\sqrt{6xM/(p_y xb)}$
= $\sqrt{(6x743.54x10^3/(.66 x345x70.28))}$
= 16.697 mm
< Plate thk OK !

### 3. Check for Beam Flange




edge dist from ($e_2$) = 47 mm

Moment due to tension in each bolt M
= 15.82x47
= 743.54 kNmm

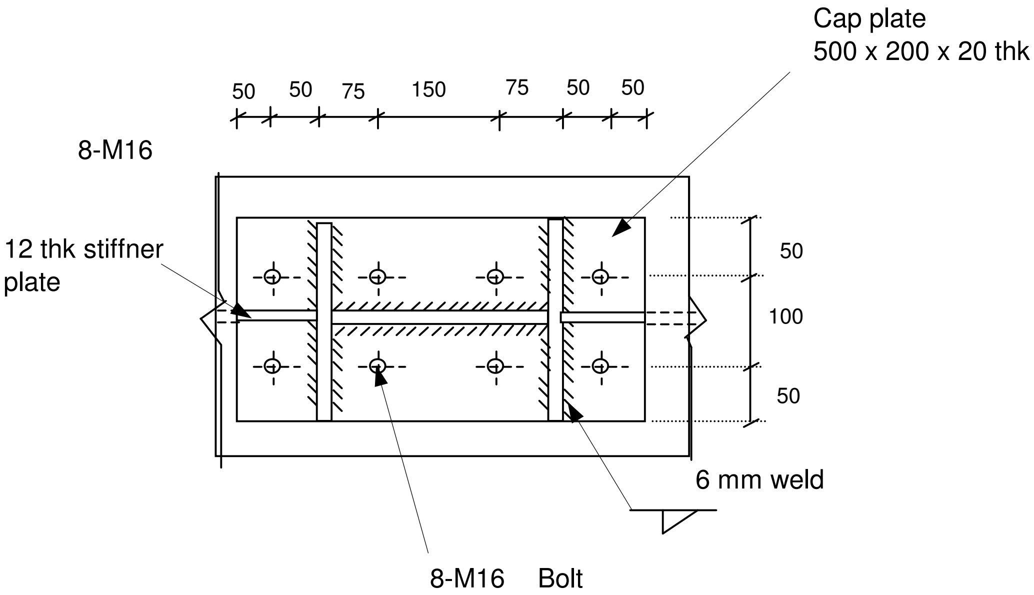
Resisting width b = 16+2(47tan30)
By 30deg disspertion = 70.28 mm
By three way distribution = 3 x 70.28

Thickness required = $\sqrt{6xM/(p_y xb)}$
= $\sqrt{(6x743.54x10^3/(.66x345x3 x 70.28))}$
= 9.64 mm
< Flange thk OK !

### 4. Welding Calculation

Trial Size of weld a = 6 mm
Design stgth of weld $p_w$ = 220 N/mm$^2$
Strength of weld/mm = 0.7a x $p_w$
= 0.7x6x220
= 924 N/mm
Resultant force acting on weld (Fr) = 56.86 kN
Length reqd for weld = 56.86/924
= 61.54 mm
Length available = 1356 mm
> reqd, OK !

**Figure:**


Cap plate
500 x 200 x 20 thk
50
50
75
150
75
50
50
8-M16
12 thk stiffner plate
50
100
50
6 mm weld
8-M16 Bolt

**Purlin Design:** ( Roof )

| | | | | | | |
|---|---|---|---|---|---|---|
| | Select a Size | = | Z 200 25 | | | |
| | Self wt. | = | 0.08 | kN/m | | |
| | Moment of Inertia | = | 507.8 | $cm^4$ | | |
| Loadings: | | | | | | |
| | | (from frame 2 analysis) | | | | |
| | DL (upper bound) | = | 0.4 | kPa | | |
| | DL (lower bound) | = | 0.2 | kPa | | |
| | LL | = | 0.6 | kPa | | |
| | Maximum Downward wind load, WL3 | = | 0.93 | kN/m | | |
| | Maximum Upward wind load, WL4 | = | 1.83 | kN/m | | |
| | Span length of purlin | = | 3.325 | m | | |
| | For Purlin spaced at | = | 1.5 | m c/c | | |
| Load Comb. 1 | 1.4DL + 1.6LL | = | 1.52 | kN/m | | |
| Load Comb. 2 | 1.7DL + 1.4WL3 | = | 1.982 | kN/m | | |
| Load Comb. 3 | 1.2DL+1.2LL+1.2WL3 | = | 2.316 | kN/m | | |
| | Maximum Downward ultimate load | = | 0.902 | kN/m | | |
| | Maximum Upward ultimate load | = | 2.162 | | 1.4WL4 - 1.0DL | |
| | From LYSAGHT catalogue, for | | 8.1 | m | double lapped span | |
| | Purlins with | | 2 | rows of bridging, | | |
| | Allowable Downward load | = | 3.57 | kN/m | > | 0.902 |
| | Allowable Upward load | = | 3.57 | kN/m | > | 2.162 |
| Deflection Check: | | | | | | |
| | 0.8(LL+WL3) | = | 1.224 | kN/m | | |
| | WL4 - DL | = | 1.43 | kN/m | | |
| | Design service load | = | 1.84 | kN/m | | |
| | Span of purlin | = | 3325 | mm | | |
| | Permissible Deflection | = | span/200 | | | |
| | | = | 16.625 | mm | | |
| | Deflection | = | $5*Load*Span^4/384*E*I$ | | | |
| | | = | 2.88 | mm | | |
| | | < | 16.625 | mm | | |

Hence O.K.

**Purlin Design:** ( Side )

| | | | | |
|---|---|---|---|---|
| | Select a Size | = | Z 200 25 | |
| | Self wt. | = | 0.08 | kN/m |
| | Moment of Inertia | = | 507.8 | $cm^4$ |
| Loadings: | | | | |
| | | (from frame 2 analysis) | | |
| | DL (upper bound) | = | 0.4 | kPa |
| | DL (lower bound) | = | 0.2 | kPa |
| | LL | = | 0 | kPa |
| | Maximum Sidewall wind load, WL3 | = | 1.7 | kN/m |
| | Maximum Upward wind load, WL4 | = | 0.94 | kN/m |
| | Span length of purlin | = | 3.5 | m |
| | For Purlin spaced at | = | 1.5 | m c/c |
| Load Comb. 1 | 1.4WL3 | = | 2.38 | kN/m |
| | Maximum Sidewall ultimate load | = | 2.38 | kN/m |
| | Maximum Upward ultimate load | = | 1.316 | 1.4WL4 - 1.0DL |
| | From LYSAGHT catalogue, for | | 8.1 | m |
| | Purlins with | | 2 | rows of bı double lapped span |
| | Allowable Downward load | = | 3.57 | kN/m > 2.38 |
| | Allowable Upward load | = | 3.57 | kN/m > 1.316 |
| Deflection Check: | | | | |
| | WL3 | = | 1.7 | kN/m |
| | Design service load | = | 1.84 | kN/m |
| | Span of purlin | = | 3500 | mm |
| | Permissible Deflection | = | span/200 | |
| | | = | 17.5 | mm |
| | Deflection | = | $5*Load*Span^4/384*E*I$ | |
| | | = | 3.27 | mm |
| | | < | 17.5 | mm |

Hence O.K.

**Bracing Connection** **UA70x70x6**

Load case no. 219 — 1.2DL+0.5LL+1.0EQX+0.3 EQZ — [Ref. Member on Staad = 522 ]

Axial force N — Tension 13.992 kN

Comp 19.704 kN

**safety factor**

| | | | | | |
|---|---|---|---|---|---|
| Section Property | Bracing | UA70x70x6 | $\gamma_m$ | 1.00 | material |
| | h or b | 70 | $\gamma_{m2}$ | 1.20 | Net area |
| | $A_g$ | 820 $mm^2$ | $\gamma_{Mb}$ | 1.00 | Bolts |
| | t | 6 mm | $\gamma_{Ov}$ | 1.00 | Overstrength |
| governing radius | r | 13.7 mm | | | |

Calculations :

| | | | |
|---|---|---|---|
| Bolt grade | | 8.8 | |
| Shear stgth of Bolt $p_s$ | | 375 | $N/mm^2$ |
| Tension stgth $p_t$ | | 560 | $N/mm^2$ |
| Steel grade | | S 245 | $N/mm^2$ |
| Bearing stgth $p_{bs}$ | | 460 | $N/mm^2$ |
| Tensile strength $f_U$ | | 430 | $N/mm^2$ |
| Ult tensile stgth($U_s$) | | 410 | $N/mm^2$ |
| Dia of bolt | | M 16 | |
| Net Area of Bolt $A_{net}$ | | 157 | $mm^2$ |
| Dia of hole | | 18 | mm |
| Provide NO of bolts n | = | 2 | nos. |
| Thk of Gusset plate (t) | = | 10 | mm |

1 **Bolt Capacity**

Shear capacity of one bolt = $A_b$ x $p_s$ /1000

= 157x375/1000

= 58.88 kN

.:Shear in each bolt = 9.85 kN

< 58.88, .:OK !

Bearing stng of 10 mm thk gusset plate

= 18x10x460/1000

= 82.8 kN

2 **Checking for compression capacity**

Member Length L = 4450 mm

$\lambda$ = 4450/13.7

= 324

Pc = 16.6 $kN/mm^2$ (From curve C )

Compression resistance = 820x16.6/1000

= 27.23 kN

> 19.704, .:OK !

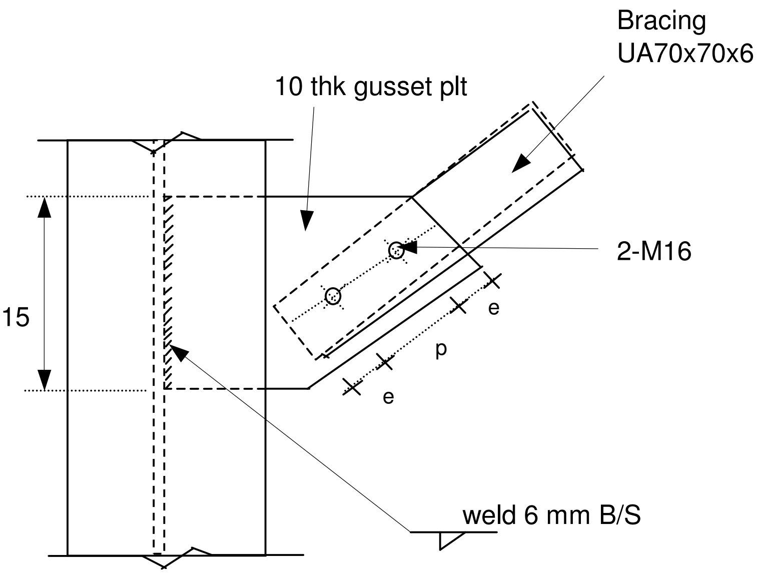
3 **Checking for connection design capacity**

| | | |
|---|---|---|
| Shear stng of member | = | 58.88x2 |
| | = | 117.76 kN |
| | | < Axial load |
| Bearing stng of plate | = | 82.8x2 |
| | = | 165.6 kN |
| | | < Axial load |

4 Welding calc

| | | |
|---|---|---|
| Trial Size of weld a | = | 6 mm |
| Design stgth of weld $p_w$ | = | 220 N/mm$^2$ |
| Strength of weld/mm | = | 0.7a x $p_w$ |
| | = | 0.7x6x220 |
| | = | 924 |
| Length of weld reqrd | = | Fr x 10$^3$ /Strgth of weld/mm |
| | = | 19.704x1000/924 |
| | = | 21.3 mm |
| | = | 10.66 x 2 |
| Provide | = | 15 x 2 |

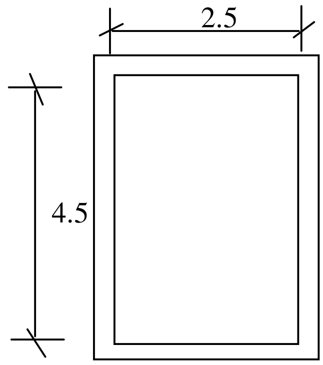
Figure :




| | | | |
|---|---|---|---|
| e | = | 50 | mm |
| p | = | 100 | mm |

# 9 Cable Pit Design

***Cable Pit Design -*** Type -1 ( panel 2.5 x 4.5 m )



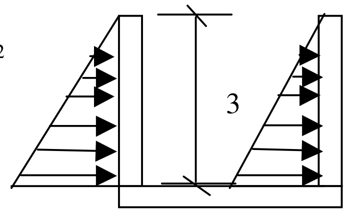

Dimension

| | | |
|---|---|---|
| Width of Pit (B) | 2.5 | m |
| Length of Pit (L) | 4.5 | m |
| Depth of Pit | 3 | m |
| Thickness of wall | 200 | mm |
| Thk of Base slab | 200 | mm |
| Height of wall | 3 | m |
| Unit weight of soil | 20 | $kN/m^3$ |

| | |
|---|---|
| $f_{cu}$ | 40 |
| $f_y$ | 460 |

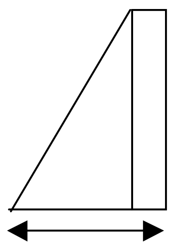
Angle of repose 30°

Soil pressure = 30.00 $kN/m^2$




Most Critical load case -
Due To Soil pressure

As L/B ratio = 1.80 < 2



Total design pressure outside = Soil pressr

Total design pressure outside = 30.00 $kN/m^2$

**Long wall Design**

| *Vertical direction* | | *For* | *Outside* | | |
|---|---|---|---|---|---|
| Moment Coefficient at support | = | | 0.06 | | for span ratio = 1.5 |
| Moment Coefficient at span | = | | 0.018 | | |
| Max moment at base | = | | 16.2 | | kN-m/m |
| Max moment at span | = | | 4.9 | | kN-m/m |
| Eff thk of wall | = | | 155.0 mm | | |

| | | For outside prssure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.017 | 0.006 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 147.25 | 147.25 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 10 | 10 |
| Spacing reqrd (mm) | = | 196 | 196 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

| *Horizontal direction* | | *For* | *Outside* | | |
|---|---|---|---|---|---|
| Moment Coefficient at support | = | | 0.019 | | for span ratio = 1.5 |
| Moment Coefficient at span | = | | 0.008 | | |
| Max moment at wall | = | | 11.5 | | kN-m/m |
| Max moment at span | = | | 4.9 | | kN-m/m |
| Eff thk of wall | = | | 154.0 mm | | |

| | | For outside prssure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.013 | 0.006 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 146.3 | 146.3 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 12 | 12 |
| Spacing reqrd (mm) | = | 283 | 283 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

**Short wall Design** (As a twoway supported pannel )

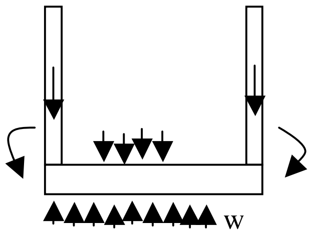
| *Vertical direction* | | *For* | *Outside* | | |
|---|---|---|---|---|---|
| Moment Coefficient at support | = | | 0.03 | | for span ratio = 0.83 |
| Moment Coefficient at span | = | | 0.009 | | |
| Max moment @ suprt | = | | 8.1 | | kN-m/m |
| Max moment at span | = | | 2.4 | | kN-m/m |
| Eff thk of wall | = | | 154.0 mm | | |

| | | For outside pressure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.009 | 0.003 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 146.3 | 146.3 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 12 | 12 |
| Spacing reqrd (mm) | = | 283 | 283 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

| *Horizontal direction* | | *Outside* | | |
|---|---|---|---|---|
| Moment Coefficient at support | = | 0.033 | | for span ratio = 0.83 |
| Moment Coefficient at span | = | 0.016 | | |
| Max moment @ suprt | = | 6.19 | | kN-m/m |
| Max moment at span | = | 3.0 | | kN-m/m |
| Eff thk of wall | = | 154.0 mm | | |

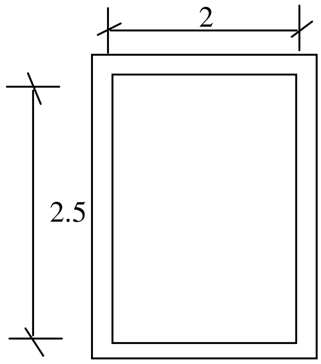
| | | For outside pressure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.007 | 0.004 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 146.3 | 146.3 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 12 | 12 |
| Spacing reqrd (mm) | = | 283 | 283 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

***Base Slab Design***




| | | | |
|---|---|---|---|
| Vertical wall load | = | 14.4 kN | |
| Selfweight of slab | = | 13.92 kN | |
| Upword reaction w (kN/m) | = | 11.52 | |
| Max moment at top | = | 31.81 kN-m | $d_{eff}$ = 115 |
| $K = M/bd^2 f_{cu}$ | = | 0.061 | |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 106.59062 | |
| Area required at base | = | 1024.36 | Min$^m$ steel = 400 |
| Dia of bar | = | 12 | |
| Spacing required | = | 110 | |
| Provide | = | 100 | |
| | | Ok | |

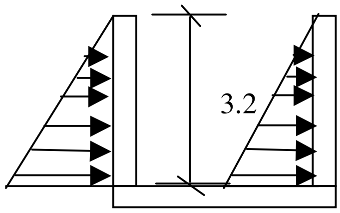
***Cable Pit Design -*** Type -1 ( panel 2.0 x 2.5 m )




Dimension
Width of Pit (B) 2 m
Length of Pit (L) 2.5 m
Depth of Pit 3.2 m
Thickness of wall 200 mm
Thk of Base slab 200 mm
Height of wall 3.2 m
Unit weight of soil 20 $kN/m^3$

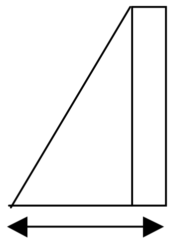
| | |
|---|---|
| $f_{cu}$ | 40 |
| $f_y$ | 460 |

Angle of repose 30°
Soil pressure = 32.00 $kN/m^2$




Most Critical load case -
Due To Soil pressure

As L/B ratio = 1.25 < 2



Total design pressure outside = Soil pressr
Total design pressure outside = 32.00 $kN/m^2$

**Long wall Design**

| *Vertical direction* | | *For* | *Outside* | | |
|---|---|---|---|---|---|
| Moment Coefficient at support | = | | 0.028 | | for span ratio = 0.78 |
| Moment Coefficient at span | = | | 0.008 | | |
| Max moment at base | = | | 9.2 | | kN-m/m |
| Max moment at span | = | | 2.6 | | kN-m/m |
| Eff thk of wall | = | | 155.0 mm | | |

| | | For outside prssure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.01 | 0.003 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 147.25 | 147.25 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 10 | 10 |
| Spacing reqrd (mm) | = | 196 | 196 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

| | | | | |
|---|---|---|---|---|
| *Horizontal direction* | *For* | *Outside* | | |
| Moment Coefficient at support | = | 0.034 | | for span ratio = 0.78 |
| Moment Coefficient at span | = | 0.016 | | |
| Max moment at wall | = | 6.8 | | kN-m/m |
| Max moment at span | = | 3.2 | | kN-m/m |
| Eff thk of wall | = | 154.0 mm | | |

| | | For outside prssure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.008 | 0.004 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 146.3 | 146.3 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 12 | 12 |
| Spacing reqrd (mm) | = | 283 | 283 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

**Short wall Design** (As a twoway supported pannel )

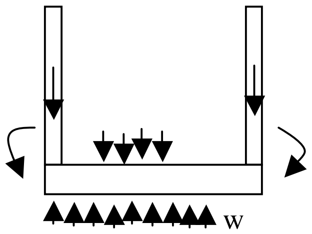
| | | | | |
|---|---|---|---|---|
| *Vertical direction* | *For* | *Outside* | | |
| Moment Coefficient at support | = | 0.02 | | for span ratio = 0.63 |
| Moment Coefficient at span | = | 0.005 | | |
| Max moment @ suprt | = | 6.6 | | kN-m/m |
| Max moment at span | = | 1.6 | | kN-m/m |
| Eff thk of wall | = | 155.0 mm | | |

| | | For outside pressure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.007 | 0.002 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 147.25 | 147.25 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 10 | 10 |
| Spacing reqrd (mm) | = | 196 | 196 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

| *Horizontal direction* | | *Outside* | | |
|---|---|---|---|---|
| Moment Coefficient at support | = | 0.038 | | for span ratio = 0.63 |
| Moment Coefficient at span | = | 0.019 | | |
| Max moment @ suprt | = | 4.86 | | kN-m/m |
| Max moment at span | = | 2.4 | | kN-m/m |
| Eff thk of wall | = | 154.0 | mm | |

| | | For outside pressure | |
|---|---|---|---|
| | | At support | At span |
| $K = M/bd^2 f_{cu}$ | = | 0.006 | 0.003 |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 146.3 | 146.3 |
| Area required at base($mm^2$) | = | 400 | 400 |
| Dia of bar (mm) | = | 12 | 12 |
| Spacing reqrd (mm) | = | 283 | 283 |
| Provide (mm) | = | 100 | 100 |
| | | Ok | Ok |

***Base Slab Design***




| | | | | | | |
|---|---|---|---|---|---|---|
| Vertical wall load | = | 15.36 | kN | | | |
| Selfweight of slab | = | 11.52 | kN | | | |
| Upword reaction w (kN/m) | = | 15.36 | | | | |
| Max moment at top | = | 14 | kN-m | $d_{eff}$ = | | 115 |
| $K = M/bd^2 f_{cu}$ | = | 0.027 | | | | |
| Z = d (0.5+Sqrt(0.25-K/0.9)) | = | 109.25 | | | | |
| Area required at base | = | 439.86 | | Min$^{m}$ steel | = | 400 |
| Dia of bar | = | 12 | | | | |
| Spacing required | = | 257 | | | | |
| Provide | = | 100 | | | | |
| | | Ok | | | | |